

Oracle® Flow Manufacturing

User's Guide

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Oracle Flow Manufacturing User's Guide, Release 12.2

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Preface

Intended Audience

Welcome to Release 12.2 of the *Oracle Flow Manufacturing User's Guide*.

See Related Information Sources on page xii for more Oracle E-Business Suite product information.

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Structure

1 Overview of Flow Manufacturing

This chapter will give you an overview of the features contained in Oracle Flow Manufacturing.

2 Setting Up Flow Manufacturing

This chapter describes the processes involved in setting up Oracle Flow Manufacturing.

3 Line Design and Balancing Procedures

This chapter describes the steps to design Flow manufacturing lines.

4 Graphical Line Designer Workbench

5 Mixed Model Map Workbench

This chapter describes the features of the Mixed Model Map Workbench, a graphical interface used to balance your production lines.

6 Flow Sequencing

This chapter describes Flow sequencing, the process of determining when items are built on the production line to create Flow schedules.

7 Line Scheduling Workbench

8 Feeder Line Synchronization

9 Outbound Broadcast Sequencing

This chapter describes Outbound Broadcast Sequencing—a process that enables the creation of replenishment signals on a Flow line where the supplier has special requirements, preferences, or conditions.

10 Flow Execution Workstation

This chapter describes the Flow Execution Workstation—used to track the flow of shop floor work, view schedule and line operation information, and complete schedules.

11 Graphical Kanban Workbench

12 Reports and Processes

A Windows and Navigation Paths

B Line Scheduling Calculation Examples

This appendix contains examples of the calculations used to create flow schedules in the Line Scheduling Workbench.

C Mixed Model Map Workbench Calculations

D Oracle Flow Manufacturing Client Extensions

This appendix provides information about the Flow Manufacturing client extensions.

Glossary

Related Information Sources

- *Oracle Advanced Planning Implementation and User's Guide*

This guide describes how to use Oracle's supply chain planning performance for forecasting and managing both supply and demand for your items. You plan your requirements, and execute and release the plan for discrete jobs, repetitive schedules, and flow schedules.

- *Oracle Bills of Material User's Guide*

This guide describes how to create various bills of materials to maximize efficiency, improve quality and lower cost for the most sophisticated manufacturing environments. By detailing integrated product structures and processes, flexible product and process definition, and configuration management, this guide enables you to manage product details within and across multiple manufacturing sites.

- Oracle E-Business Suite Documentation Set

This documentation set provides planning and reference information for the Oracle

E-Business Suite System Administrator. Oracle E-Business Suite Setup Guide contains information on system configuration steps, including defining concurrent programs and managers, enabling Oracle Applications Manager features, and setting up printers and online help. Oracle E-Business Suite Maintenance Guide provides information for frequent tasks such as monitoring your system with Oracle Applications Manager, administering Oracle E-Business Suite Secure Enterprise Search, managing concurrent managers and reports, using diagnostic utilities including logging, managing profile options, and using alerts. Oracle E-Business Suite Security Guide describes User Management, data security, function security, auditing, and security configurations.

- *Oracle Cost Management User's Guide*

This guide describes how to use Oracle Cost Management in either a standard costing or average costing organization. Cost Management can be used to cost inventory, receiving, order entry, and work in process transactions. It can also be used to collect transaction costs for transfer to Oracle Projects. Cost Management supports multiple cost elements and multiple subelements. It also provides comprehensive valuation and variance reporting.

- *Oracle General Ledger User's Guide*

This guide explains how to plan and define your chart of accounts, accounting period types and accounting calendar, functional currency, and set of books. It also describes how to define journal entry sources and categories so you can create journal entries for your general ledger. If you use multiple currencies, use this manual when you define additional rate types, and enter daily rates. This manual also includes complete information on implementing Budgetary Control.

- *Oracle Inventory User's Guide*

This guide describes how to define items and item information, perform receiving and inventory transactions, maintain cost control, plan items, perform cycle counting and physical inventories, and set up Oracle Inventory.

- *Oracle Order Management User's Guide*

This guide describes how to enter sales orders and returns, copy existing sales orders, schedule orders, release orders, create price lists and discounts for orders, run processes, and create reports.

- *Oracle Purchasing User's Guide*

This guide describes how to create and approve purchasing documents, including requisitions, different types of purchase orders, quotations, RFQs, and receipts. This guide also describes how to manage your supply base through agreements, sourcing rules and approved supplier lists. In addition, this guide explains how you can automatically create purchasing documents based on business rules through integration with Oracle Workflow technology, which automates many of the key

procurement processes.

- *Oracle Quality User's Guide*

This guide describes how Oracle Quality can be used to meet your quality data collection and analysis needs. This guide also explains how Oracle Quality interfaces with other Oracle Manufacturing applications to provide a closed loop quality control system.

- *Oracle Work in Process User's Guide*

This guide describes how Oracle Work in Process provides a complete production management system. Specifically this guide describes how discrete, repetitive, assemble-to-order, project, flow, and mixed manufacturing environments are supported.

Integration Repository

The Oracle Integration Repository is a compilation of information about the service endpoints exposed by the Oracle E-Business Suite of applications. It provides a complete catalog of Oracle E-Business Suite's business service interfaces. The tool lets users easily discover and deploy the appropriate business service interface for integration with any system, application, or business partner.

The Oracle Integration Repository is shipped as part of the Oracle E-Business Suite. As your instance is patched, the repository is automatically updated with content appropriate for the precise revisions of interfaces in your environment.

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Oracle provides powerful tools you can use to create, store, change, retrieve, and maintain information in an Oracle database. But if you use Oracle tools such as SQL*Plus to modify Oracle E-Business Suite data, you risk destroying the integrity of your data and you lose the ability to audit changes to your data.

Because Oracle E-Business Suite tables are interrelated, any change you make using an Oracle E-Business Suite form can update many tables at once. But when you modify Oracle E-Business Suite data using anything other than Oracle E-Business Suite, you may change a row in one table without making corresponding changes in related tables. If your tables get out of synchronization with each other, you risk retrieving erroneous information and you risk unpredictable results throughout Oracle E-Business Suite.

When you use Oracle E-Business Suite to modify your data, Oracle E-Business Suite automatically checks that your changes are valid. Oracle E-Business Suite also keeps track of who changes information. If you enter information into database tables using

database tools, you may store invalid information. You also lose the ability to track who has changed your information because SQL*Plus and other database tools do not keep a record of changes.

Overview of Flow Manufacturing

This chapter will give you an overview of the features contained in Oracle Flow Manufacturing.

This chapter covers the following topics:

- Overview of Flow Manufacturing
- Demand Management
- Line Design and Balancing
- Line Scheduling and Sequencing
- Production Execution
- Kanban Planning and Execution

Overview of Flow Manufacturing

Flow Manufacturing is a manufacturing approach employing the following practices:

- Building to customer demand rather than building to a forecast
Reduces finished goods inventories, reduces cost, and increases customer satisfaction.
- Linear manufacturing rather than batch production
Reduces work in process inventory, rework, and scrap and increases product quality as quality and assembly problems are immediately addressed.
- Mixed model production rather than process layout
Enables a mix of products to be produced on one production line in the same day, which allows the manufacturer to respond to the demand generated by customer orders.
- In-process quality rather than inspecting quality into the product after assembly

Builds quality into the product during the production process resulting in less rework, less scrap, lower cost, and higher customer satisfaction.

- JIT Replenishment of component materials rather than receipts of MRP forecasted demands.

Reduces the amount of component inventory and reduces costs.

- Uses rate-based production rather than work orders

Simplifies shop floor activities and reduces cost.

- Labor flexibility rather than labor specialization

Increases productivity and reduces costs.

- Backflush material and costs upon completion rather than at each operation

Simplifies shop floor activities and reduces costs.

Oracle Flow Manufacturing includes a comprehensive set of features that support the entire flow process from line design and balancing to production execution. It enables implementation of the demand pull system using kanbans for both raw material and in-process assemblies. The planning process is streamlined using Flow schedules to sequence and schedule mixed model production.

The features provided by Oracle Flow Manufacturing are classified into the following categories:

- Demand Management, page 1-2
- Line Design and Balancing, page 1-3
- Line Scheduling and Sequencing, page 1-5
- Production Execution, page 1-7
- Kanban Planning and Execution, page 1-9

Demand Management

Demand management is a factor during both the initial design of flow production lines and the daily execution of the rate-based schedules. During the design phase, the objectives of demand management are to group similar products into families, which allows for planning at an aggregate level and to develop a peak daily volume for all products that will be used for line balancing and kanban planning. During production execution, the objectives are to increase responsiveness to customer demand and to maximize resource use.

Oracle Flow Manufacturing uses the demand management tools provided in Oracle

MPS/MRP, Supply Chain Planning and Advanced Planning and Scheduling to plan production volumes.

Line Design

Line design is based upon the anticipated demand for a product or product family. This anticipated demand represents the peak daily volume. All production resources, machines, and labor are then calculated based on these volumes. Additionally, these demand figures and factory schedule hours are used to derive the takt time, or rate that products must be produced to meet the schedule. This figure represents the rate that products must leave the end of the line in order to meet customer demand. See: Product Synchronization, page 3-3.

Kanban Planning

Oracle Flow Manufacturing uses a kanban pull replenishment system to signal material requirements and pull material from its defined source as needed to meet daily customer demand. The objective of the kanban replenishment system is to continuously improve the production with zero stock outs, shorter lead times, reduced inventory, and minimal supervision. The statement of demand used to balance flow lines to peak daily demand is generally used to size kanbans of components. The use of an MPS or MRP for kanban items gives visibility to the long term plan. This plan is then communicated to trading partners, including marketing and suppliers.

In Oracle Flow Manufacturing, any number of baseline and simulation kanban plans can be created in which kanban quantities for each item or kanban location can be calculated and stored. Optimal kanban quantities can be calculated for the desired demand schedule.

Production Execution

In a flow manufacturing environment, daily production rates tend to be based on customer demand, including marketing orders. Flow lines are balanced with machines and resources to produce at the takt needed to meet the peak daily volume. By design, the daily customer demand should be less than that peak, ensuring that customer demand will be met. However, the mixed and demand represented in daily customer orders are not necessarily in the same ratios as the peak daily volumes are designed for. To manage the demand fluctuations, Oracle Flow Manufacturing uses simple flow schedules to schedule and sequence sales orders.

Line Design and Balancing

Line design and balancing is the foundation of a flow production line. The objective is to balance lines for mixed model production at the expected peak demand that take the least possible time to execute. This is accomplished through flat product structures and simple processes which can be communicated visually to meet customer demand daily.

Line design includes grouping products into product families, defining the processes and events required to produce each product, and re-grouping events into line operations to approximate takt time.

Effective flow line design and balanced production smooths production and eliminates bottlenecks. Flow lines are designed to reduce production time and resource costs by identifying value added events and eliminating non-value events. Simplified processes are designed to reduce overhead costs by optimizing floor space utilization. This simplified design also helps increase communication on the production floor through visual management tools. Quality is improved by implementing quality into the process, instead of inspecting at the end.

Product Synchronization

Product synchronization defines the processes used to make each product and the sequence of events within each process. Events are granular tasks within the process that define the physical activities on the line. Each event is assigned the machine and labor times used to perform the event.

Product synchronization tasks are performed in Oracle Flow Manufacturing using flow routings. Flow routings support both processes and sequence of events. In addition flow routings can be used to define the network of processes including feeder lines, rework loops, and both primary and alternate processes required to produce an item. This will help model the line as is to the system. Flow routings and the routing network are also used to calculate the total product cycle time. Total product cycle time is the longest time path on the production line, including feeder lines, required to make the product. See:

Creating a Routing, *Oracle Bills of Material User's Guide*

Product Synchronization, page 3-3

Mixed Model Map Workbench

The Mixed Model Map Workbench is used to achieve balanced flow lines. The Mixed Model Map Workbench combines the information from the products assigned to a specific line—including their routing, product family grouping, and forecasted demand. This information is displayed in a graphical interface. The takt time is calculated from the daily production requirements (demand) and the available production hours in a day. The summary portion of the Mixed Model Map displays the machine and labor resources needed to meet the takt time. Since takt time establishes the rhythm of the line, all the processes along the line are streamlined to complete within the takt time. This ensures that each product moves from process to process within the takt time, which maintains the steady production rate necessary to meet the demand. See: Overview of Mixed Model Map Workbench, page 5-1

Line Balancing

While engineering provides information on product routings in the form of processes and sequence of events, manufacturing faces the challenge of organizing work on a production line so that bottlenecks are minimized and work cells are balanced. As a result, manufacturing evaluates the feasibility of reorganizing the events into groups that approximate takt time. Managing the imbalances leads to analysis for the following:

- Reducing or eliminating non-value added work
- Using smaller units for the work, or grouping into larger units
- Increasing resource (machine, labor) capacity
- Increasing in-process inventory
- Adding additional lines

Once the current line has been set up with items, organizations, the flow line, a forecast, and the product routings defined, the line can be balanced through iterative processes using the Graphical Line Designer Workbench. See: Accessing Graphical Line Designer Workbench Windows, page 4-8

Operation Method Sheets

Operation Method Sheets (OMS) provide the needed information to shop floor personnel for performing an operation on the line. This information can include pictures of the assembly process, required materials, and written work instructions. These are the primary tools used to perform the operations and are especially critical on lines making many models.

Oracle Flow Manufacturing supports Operation Method Sheets using the Attachments feature. Multimedia documents can be attached to, and maintained for, each line operation on the flow routing of each item.

Line Scheduling and Sequencing

In flow manufacturing, line scheduling is equal to line design and balancing in terms of importance. A line well designed to meet average daily demand that is scheduled improperly can cause excessive changeover, large peaks in demand to kanbans, or unbalanced demand on resources. Each of these will undermine the line's ability to perform to takt. Oracle Flow Manufacturing allows for the creation and use of simple scheduling rules to schedule either sales or planned orders for the flow line. The objective of line scheduling is to sequence production to maximize resource utilization and minimize delays.

Line scheduling allows for:

- Creation of a scheduling rule based on predefined sequencing criteria and scheduling algorithms
- Viewing of all unscheduled orders for your line and choose which orders you want to schedule
- Scheduling the line and viewing the final flow schedule.

Scheduling rules determine the logic used to create flow schedules from orders. They are a combination of sequencing criteria and scheduling algorithms that can help you pull demand forward or backward in time, in order to meet takt for each day.

Scheduling Algorithms

Scheduling algorithms are used to smooth demand by restricting the number of each assembly that is scheduled in any given day. Demand smoothing is important because the line is designed based on an average daily mix. However, seldom will a line produce exactly the mix for which it was designed. Demand smoothing ensures that the mix is maintained as closely as possible, which helps the line to operate more efficiently and achieve takt time on a daily basis.

Note: Line Scheduling does not schedule a portion of a product: the algorithm rounds decimal quantities up to whole units before scheduling.

Sequencing Criteria

Sequencing criteria are used to determine the priority in which sales orders are scheduled.

Flow Manufacturing Sequencing and Execution

Sequencing is the process of determining the order in which assembly items are built on a production line. Attributes, constraints, rules are used in your production lines to create sequencing tasks. The sequence in which assembly items are built determines the efficiency of the line, and therefore, the productivity of the manufacturing organization. The Sequencing pages enable you to set up task components, generate tasks, and view the results. The Flow Sequencing concurrent program takes the sequencing task as input, and creates Flow schedules to meet production demands.

Related Topics

Overview of the Line Scheduling Workbench, *Oracle Master Scheduling/MRP and Oracle Supply Chain Planning*.

Line Scheduling Workbench Options Window, *Oracle Master Scheduling/MRP and Oracle Supply Chain Planning*.

How Scheduling is Calculated, *Oracle Master Scheduling/MRP and Oracle Supply Chain Planning*.

Overview of Flow Sequencing, page 6-2

Creating Sequencing Tasks, page 6-17

Production Execution

As throughput time and inventory levels are driven down during implementation of Oracle Flow Manufacturing, the need for and value of detailed production information becomes less critical. This creates the opportunity to eliminate or streamline production execution transactions. One of the tools provided by Oracle Flow Manufacturing to help in this process is the Work Order-less Completion transaction.

Mixed Model Map Workbench

The Mixed Model Map Workbench is the tool that Oracle Flow Manufacturing uses to achieve balanced flow lines. After lines have been designed and during the production cycle, the Mixed Model Map becomes a tool to evaluate the impact of the current production schedule on the resources, operational times and in-process kanbans. In this mode, the summary portion of the Mixed Model Map displays the machine and labor resources needed to meet the takt time for the current production load. The baseline comparison highlights where labor resources can be removed to allow for flexing and where extra resources or overtime may be needed to produce the demand. See: Overview of Mixed Model Map Workbench, page 5-1

Work Order-Less Completions

Production is recorded with the Work Order-less Completion transaction against flow schedules created with the Line Scheduling Workbench. Completions can be either unscheduled or scheduled against a flow schedule. The system backflushes all components and performs resource and overhead transactions upon recording completion of the finished product. Additionally, Oracle Flow Manufacturing allows assembly completions to be recorded without having to create work orders, a job or repetitive schedule, or a flow schedule.

Work order-less completions do the following:

- Backflush pull and push components
- Charge resources and overhead based on the flow routing

See: Work Order-less Completions, *Oracle Work in Process*

Manufacturing Costing

The costing method of the organization determines whether mixed manufacturing methods can be used. Costing in a flow environment is usually done using standard costing methods. Average costing may also be used, particularly when used in conjunction with project manufacturing. Cost variances are collected and posted during the Work Order-less Completion transaction.

Component Picking in a Flow Environment

You can perform component picking transactions for flow schedules using the rules based system in Oracle Warehouse Management. Flow manufacturing environments use kanban signals to control the replenishment of components on the line. However, the picking functionality can be used for slower moving or randomly used planned components to create move orders for supply type Push components. The requirements in component picking for flow schedules include:

- The material picked is based on Oracle Inventory allocation rules entered on the Picking Rules window, or Oracle Warehouse Management rules based picking recommendations.
- Material must be defined at a valid line operation, and the line is open.
- The supply type of the component is Push.
- Move orders are generated only when the flow schedule components can be filled. If the components are not on hand, backorders are not created.

There are two methods for accessing the Component Pick Release program in a Flow environment. You can use the Component Pick Release window to select requirements, or you can specify a Pick Release Operation in the Flow Execution Workstation preferences.

- The Component Pick Release window is used to select requirements and create move orders. This window is a graphical workbench that displays fields applicable to the job or schedule type accessed.
- You can specify an operation in the Flow Execution Workstation parameters to call the Component Pick Release program. In the Pick Release Operation parameter field, specify a standard operation code. When you enter a value in this field, the Component Pick Release program creates move orders at that point in the flow line. When the Line Operation completion transaction occurs, the Picking API is called for the schedule. This functionality does the same action as entering a schedule in the Schedule Number field of the Component Pick Release window.

Overview of Component Picking, *Oracle Work in Process*

Using the Component Pick Release Window, *Oracle Work in Progress User's Guide*

Flow Execution Workstation

The Flow Execution Workstation provides features for tracking the flow of work, completing assemblies, and viewing detail information. You can view Flow schedules, line operations and their corresponding event sequences, components, resources, and instructions. The workstation enables you to complete schedules and line operations, record lot and serial information, enter quality plan data, customize your display of information, and check for eligible engineering change orders (ECOs). See: Initializing the Flow Execution Workstation, page 10-9

Kanban Planning and Execution

A key objective of flow manufacturing and JIT production is to minimize inventory and increase inventory turns. Raw material is pulled into production as needed to meet demand and the pulled material is replenished using a kanban signal. The minimum amount of material possible, based on replenishment time is held at the line in kanban bins. As each bin is emptied, a signal is issued to replenish the bin and the next bin is used to pull the material.

There are many kanban systems in use today but the most commonly used are the two-bin (two card) system and the multibin (multicard) system. Oracle Flow Manufacturing supports both these systems for planning and execution during production.

Calculation of Kanban Size and Number of Cards

Oracle Flow Manufacturing calculates the optimal number of kanbans needed in each kanban location. It can calculate the number of kanbans needed when a bin quantity is specified or it can calculate the kanban size if the number of bins are specified. Oracle Flow Manufacturing allows the simulation of multiple kanban plans. Kanbans currently in use can be compared and adjusted against kanban requirements for consistently varying demands. This process helps in both maintaining minimal material inventory and continuity of material flow from the supplier to the production line.

Pull Sequences

Oracle Flow Manufacturing helps you define the kanban locations for each item along with the supply source information. The supply source can be suppliers, other production lines, other inventory organizations in the enterprise, or other kanban locations in the same organization. A complete chain of demand-supply pull sequences can be defined for kanban planning.

Non-replenishable Kanbans

Non-replenishable kanbans are used to meet unexpected peaks in demand. For demand

variations that are infrequent and unplanned, Oracle Flow Manufacturing allows non-replenishable kanban cards. These cards are created for specific items in specific quantities as needed. Non-replenishable kanbans cycle through the system only once, after which they are removed from production.

Kanban Execution

Oracle Flow Manufacturing generates appropriate events for each kanban signal. For kanbans that are sourced through suppliers, the system can generate requisitions and purchase orders. Sourcing rules can also be used to create blanket releases against purchase contracts. Internal requisitions can be generated for interorganization replenishments.

Oracle Flow Manufacturing supports different statuses of kanban cards to help track their release cycle. Supported statuses include Full, Empty, In-Process, In-Transit, Hold, and others. Kanban Cards can also be accumulated and released together when a minimum order quantity greater than the kanban size is specified. A number of modifiers such as safety stock days and lot multiplier allow you to create a smooth signaling and replenishment system.

Pull Sequence Window, page 11-12

Modifying and Creating Pull Sequences on the Graphical Network, page 11-10

Setting Up Flow Manufacturing

This chapter describes the processes involved in setting up Oracle Flow Manufacturing.

This chapter covers the following topics:

- Setting Up Objectives
- Setup Steps
- Fixed Component Usage

Setting Up Objectives

You first need to perform related product setup steps before implementing Oracle Flow Manufacturing setup steps.

Products and Parts

The objectives of products and parts setup are to create both finished goods and component items, and to group the products and parts in logical product families. This facilitates demand forecasting and production assembly. To implement Oracle Flow Manufacturing it is necessary to set up:

- Items

Item templates are defined sets of attributes that you can use over and over to create many similar items. Templates make initial item definition easier. Oracle recommends that you use templates—either those Oracle provides or those you define. Oracle provides two templates for Flow manufacturing:

- Flow Finished Goods or Flow Assembly template—used to define the standard template for creation of Finished Goods items.
- Flow Component Template—used to define the standard template for creation of Flow Components items.

- Product Families and Product Family Members

Since production planning occurs at the aggregate level, creating a production plan involves grouping products into product families. These groupings are based on product's similarities in design, manufacturing process, and resource usage.

- Categories and Category Sets
- Bill of Materials

The objective of bill of materials setup is to create a bill of material for the products and parts, and to assign a subinventory to each item in the bill of material.

Work in Process

Set up parameters in the following tabbed regions of the Work in Process Parameters window:

- Discrete
 - Default Discrete Class: Flow
 - Default Lot Number Type: Based On Inventory Rules
 - Respond To Sales Order Changes: Always
- Move Transactions
 - Require Scrap Account: No
 - Allow Creation Of New Operations: Yes
 - Allow Moves Over Move Shop Floor Status: Yes
- Backflush Defaults
 - Supply Subinventory: user defined
 - Supply Locator: user defined
 - Lot Selection Method: Expiration Date
 - Lot Verification: Exceptions Only
- Other
 - Component ATP Rule: Standard
 - Default Over Completion Tolerance Percent: n/a

Demand Management

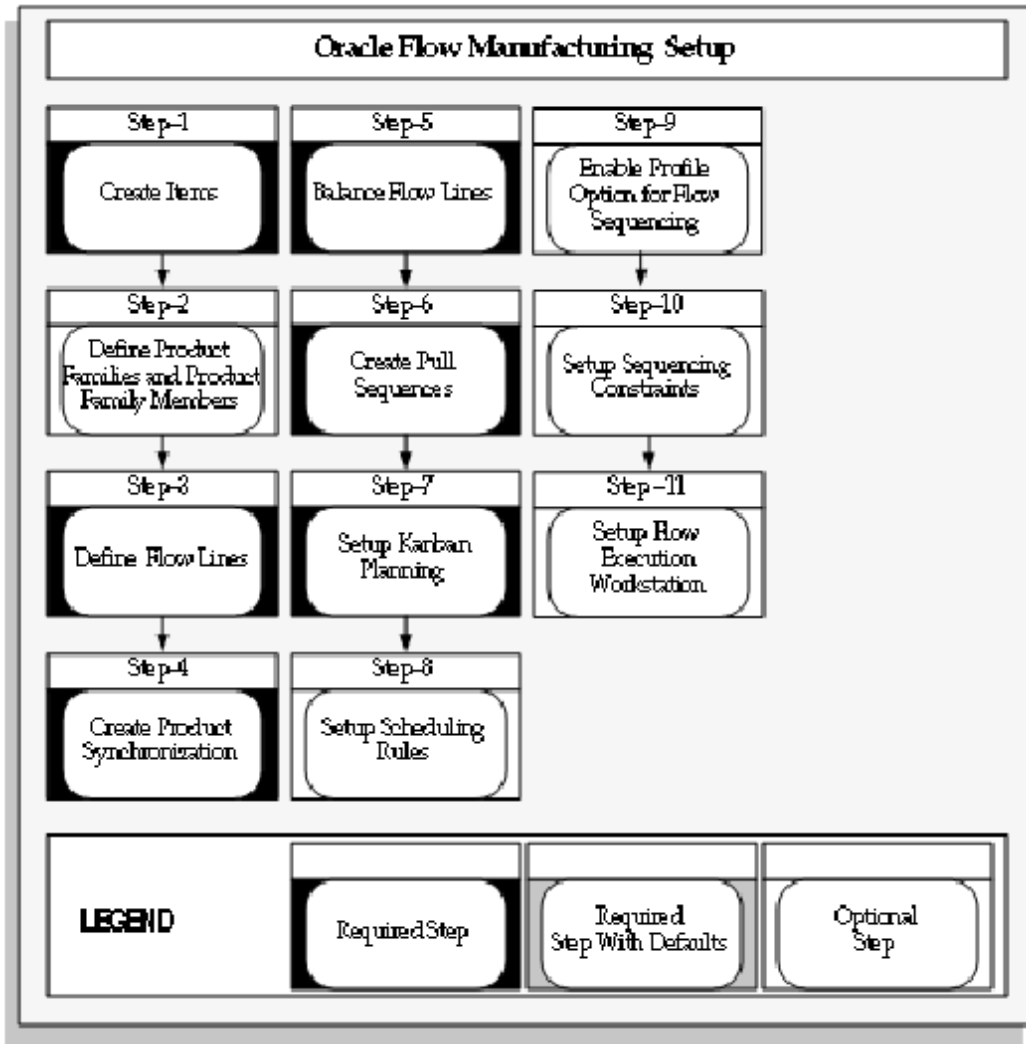
In designing a flow line, a company analyzes the market forecast for the products and takes a look at the long term forecast. In demand management you can assign planning percentages to members of the product family, and use the relationship between a product family item and its members in:

- Forecast explosion
- Consumption
- Master Scheduling
- Capacity and Materials Planning

Sales orders for member items consume the forecast for the member items as well as for the product family.

Setup Steps

The following flowchart depicts the steps that you need to perform to set up Flow Manufacturing. Some of the steps outlined in this flowchart are Required and some are Optional. Required Step with Defaults refers to setup functionality that comes with pre-seeded, default values. You should review the defaults and decide whether or not to change them to suit your business needs. You need to perform Optional steps only if you plan to use the related feature or complete certain business functions.



Flow Manufacturing Prerequisite Steps

1. Create Flow Assemblies and Components

In Oracle Inventory you can define inventory organizations and the items to be produced. You need to define both finished goods and component item inventory records. Once defined, assign items to organizations. Templates are used to create similar items, and they simplify item definitions. See: *Defining an Item, Oracle Inventory User's Guide*, and *Item Templates, Oracle Inventory User's Guide*

2. Define Product Families and Product Family Members

You can plan based on the planning percentages and effectivity of the product family members. Product families are optional. Typically, a flow line is designed for one specific product family containing a mix of products. However, a flow line can be designed for multiple product families. See: *Creating Product Families, Oracle*

3. Define Flow Lines

Flow lines are manufacturing work areas where you produce a flow assembly, usually associated with a routing. You can build many different assemblies on the same line, at the same time. See: *Defining Production Lines, Oracle Work in Process User's Guide*

4. Create Product Synchronization

The flow line is an organization of resources and departments. Once the flow line is defined, processes and activities (events) are applied to create operations on the flow line. Product Synchronization defines the processes used to make each product including flow routings, events, processes, and line operations. The Graphical Line Designer Workbench is used to create these records and network connections, and key calculations. See: *Accessing Graphical Line Designer Workbench Windows, page 4-8*

5. Balance Flow Lines

Oracle Flow Manufacturing provides a planning tool, the Mixed Model Map Workbench, to calculate process times and volumes to balance resources and events, and optimize flow line efficiency. The objective is to balance lines at the expected peak demand. See: *Overview of the Mixed Model Map Workbench, page 5-1*

6. Create Pull Sequences

Flow Manufacturing uses a kanban pull replenishment system to signal material requirements and pull material from its defined source for demand. Kanban pull sequences define the relationship between an item, its point of use and its point of supply. The attributes of a pull sequence define the kanban size, number of kanban containers, replenishment time, safety stock, lot multiplier and method of replenishment. See: *Pull Sequence Window, page 11-12*

7. Set Up Kanban Planning

Kanban setup and planning uses the Graphical Kanban Workbench for transactions. The Graphical Kanban Workbench combines kanban setup, kanban planning, pull sequence definition, planning simulation, and kanban transactions. An item is pulled through the kanban system, rather than pushed by the planner. The kanban location for a kanban item is the designated location where that item is stored, and where replenishment is delivered. For every kanban item, there is a pull sequence—a series of kanban locations that models the actual replenishment network on the shop floor or external suppliers, specifying the sequence to follow to obtain the kanban item. See: *Accessing the Production Region, page 11-6* and *Accessing the Planning Region, page 11-19*

8. Set Up Scheduling Rules

The primary objective of Line Scheduling is to synchronize production with customer demand. This objective is achieved through the Line Scheduling Workbench. You can use the system provided rules, or set up your own custom scheduling rules. See: *Defining Scheduling Rules, Oracle Flow Manufacturing User's Guide*

9. Enable Profile Option for Flow Sequencing

The Flow Manufacturing profile option, FLM:Enable Flow Sequencing is required for new customers to Oracle Flow Manufacturing if you have Flow Sequencing installed. The profile option is not required for customers upgrading from previous versions of the software. See: *Profile Option, page 2-6*

10. Set Up Sequencing Constraints

Sequencing is used to consider your demands and determine a sequence for building assembly items. You use attributes, constraints, and group them into rules for assigned production lines:

- Attributes are a characteristic of your demand
- A constraint is a restriction enforced on an attribute
- Rules are a grouping of prioritized constraints defined for the attributes

See: *Overview of Flow Sequencing, page 6-2*

11. Set Up Flow Execution Workstation

The primary objective of setting up production execution is to meet daily production requirements. This objective is achieved through Flow Execution Workstation. You can use the Flow Execution Workstation to perform flow line transactions and track the flow of work throughout the shop floor. The Flow Execution Workstation provides a comprehensive set of features for completing assemblies, and viewing details of component and resource information. See: *Overview of the Flow Execution Workstation, page 10-1*

Profile Options

During implementation, you set a value for each user profile option to specify how to access and process data. You are required to provide a value for a Required profile option. An Optional profile option provides a default value, which you alone can change if you do not want to accept the default.

FLM: Enable Flow Sequencing

This site level profile option is required for new customers to Oracle Flow Manufacturing if Flow Sequencing is installed. Enable Flow Sequencing can legally be

set to Yes for customers licensed for Oracle Flow Sequencing.

The profile option is not required for customers upgrading from previous versions of the software. The default value is No.

FLM: Enable E-Kanban

Set this profile option to Yes to display the Kanban setup forms, Graphical Kanban Workbench, and Kanban Planning Workbench. The default value is No.

FLM: Consider Non Matching Subinv Demand

Set this profile option to Yes, to create and distribute demand for pull sequences that do not have matching BOM subinventories and that have allocation percentage not equal to 100 percent. The default value is No.

FLM:Include Buy Items in Kanban Plan

Set this profile option to Yes to include the top-level assemblies in the E-Kanban planning process when the Inventory Item Attribute Make or Buy flag is set to Buy.

FLM:Include ECOs in Kanban Planning

Set this profile option to Yes to include Engineering Change Orders (ECOs) in Kanban Planning explosion. The default value is No.

Fixed Component Usage

Flow Manufacturing supports fixed component usage for quantity calculations. You can define a fixed usage on a bill of material. For these components, the total quantity required remains the same without regard to the build quantity, and consumption quantity does not vary with the assembly quantity produced. This feature is used when a quantity of a component is used for manufacturing setup, production startup, or during the course of the manufacturing process.

Enable this feature in the Bills of Material window by setting the Basis field to Lot. For lot type basis, component usage is fixed per lot. See: *Creating a Bill of Material, Oracle Bills of Material User's Guide*

The fixed component usage feature is not available for components of:

- Phantom assemblies
- Option items, and sub-models on an option class or models.

It is available for mandatory (included) components on an option class or model.

Fixed component usage requirements affect kanban planning, replenishment, sequencing, and feeder line synchronization calculations:

- Kanban Planning

The kanban engine explosion process considers the item basis type for component calculations. Component usage is calculated by multiplying the higher level

component demand by the quantity per assembly, planning percent, and yield.

For basis type of lot, the fixed order quantity of the parent assembly is used to estimate the lot size:

- If the parent assembly is a phantom, the next prior level assembly, that is not a phantom, is used to get the fixed order quantity.
 - The minimum order quantity is used if the fixed order quantity is not defined. If minimum order quantity is not defined, maximum order quantity is used.
 - If the minimum and maximum order quantity have not been defined, the fixed order quantity is assumed to be 1.
- Feeder Line Synchronization

For the basis type item, the feeder subassembly requirement is calculated by multiplying the parent schedule quantity by the component usage. For the basis type of lot, the component usage is not multiplied by the parent schedule quantity—the component usage quantity is assumed to be the required quantity.
 - Outbound Broadcast Sequencing

For the basis type item, the component requirement is calculated by multiplying the flow schedule quantity by the component usage. For the basis type of lot, the component usage is not multiplied by the parent schedule quantity—the component usage quantity is assumed to be the required quantity.
 - Flow Sequencing

Item basis type determines the calculation of component requirements using the Component Availability constraint.

Line Design and Balancing Procedures

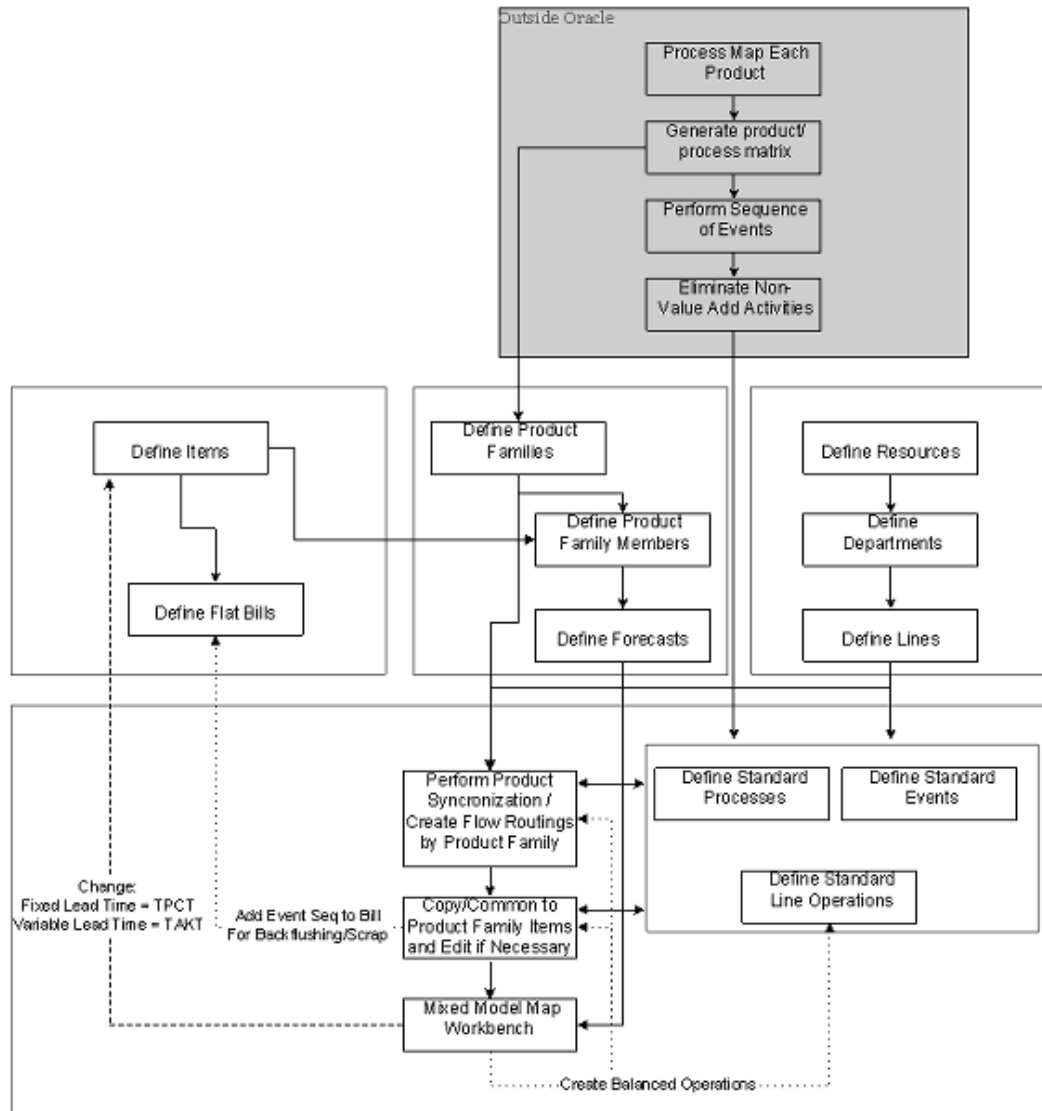
This chapter describes the steps to design Flow manufacturing lines.

This chapter covers the following topics:

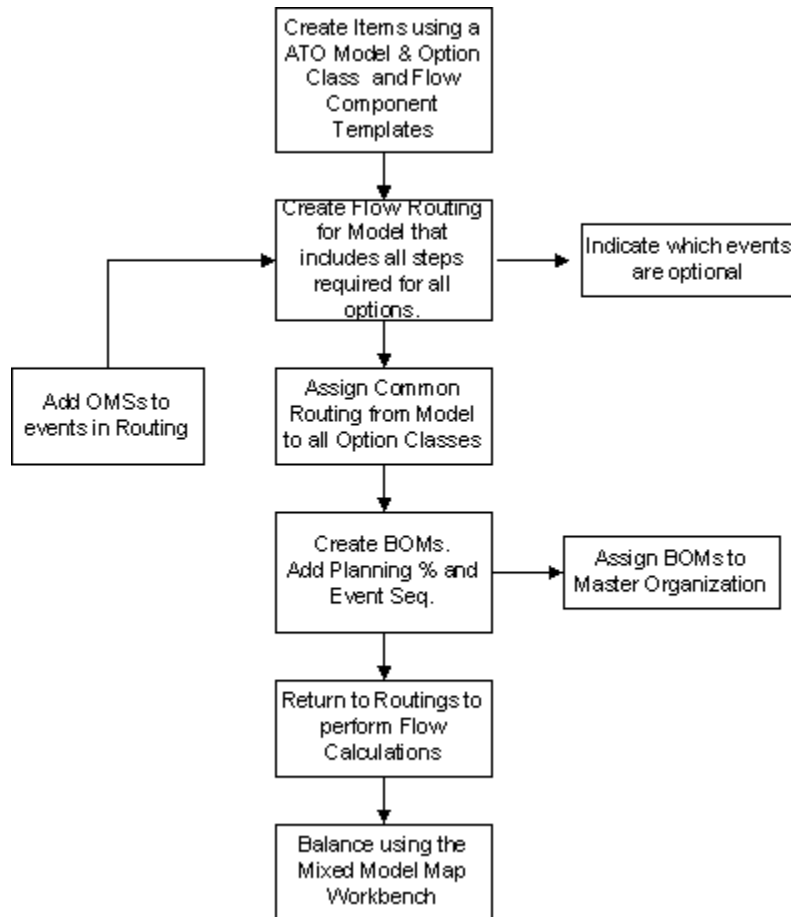
- Overview of Line Design and Balancing
- Product Synchronization
- Defining Flow Manufacturing Standard Processes
- Defining Flow Manufacturing Standard Events
- Defining Flow Manufacturing Standard Line Operations
- Flow Manufacturing Line Balance
- Creating a Flow Routing
- Creating a Routing Network
- Calculating Operation Times
- Operation Times Calculations
- Calculating Total Product Cycle Time
- Calculating Operation Yields
- Operation Yields Calculations

Overview of Line Design and Balancing

The following illustrations show the process flow of line design and balancing in Flow Manufacturing, including the process flow for assemble-to-order (ATO) final assemblies.



Assemble-to-Order (ATO) environments are used to create final assembly orders according to customer orders. ATO items enable you to apply attributes for standard, model, and option class items.



Operation Sequence: Add the event sequence number to each of your items on the bill of material. For Flow Manufacturing, the, operation sequence on the bill of material indicates the event sequence. The events you assign to your optional items are used to create a routing specific to the options a customer orders. A single optional component can be tied to multiple events.

Planning Percentages: Assign planning percentages to each of your optional items in the material control region. The program uses these percentages to calculate the average Operational Cycle Times and Total Product Cycle Times for your ATO Model.

Important: If you do not assign planning percentages, cycle times will be inflated because they will be a sum of all events in the line, instead an average based on option chosen.

Product Synchronization

Product Synchronization is the process of defining your product's process flow. This is done through the Flow routing where you are able to define processes, events and

operations for your flow line. You can then define the process network, add process yield information, and calculate operation times, yields and total product cycle time.

Standard processes and standard operations must be created before adding them to your flow routing. They are created either on the Graphical Line Designer Workbench, or in the Standard Processes or Standard Operations windows. See: Creating the Graphical Network, page 4-28

To create product synchronization for standard items:

- Create a Flow routing for an item, or product family and its members.
To reduce data entry, you can copy the product family Flow routing to the product level Flow routings. You can then modify as needed. Processes and line operations that are copied from the family to the member will retain any references to corresponding standard processes or line operations.
- Assign the processes and events that you have defined.
- Assign resources to nonstandard events.

Note: To edit the resources for a standard event, deselect the referenced check box.

- You have the option to create the calculations for Operation Times, Total Product Cycle Time, Operation Yields.
- Optionally, create a forecast, master demand schedule, or master production schedule with demands for the items. This step is not part of the product synchronization procedure and can be performed at any time before generating the mixed model map.

Defining Flow Manufacturing Standard Processes

Standard Processes can be created and used only with Oracle Flow Manufacturing. Processes are generic activities that often comprise multiple events which are performed in a specific sequence.

Note: Processes and standard processes can also be created on the Graphical Line Designer Workbench. See: Creating the Graphical Network, page 4-28

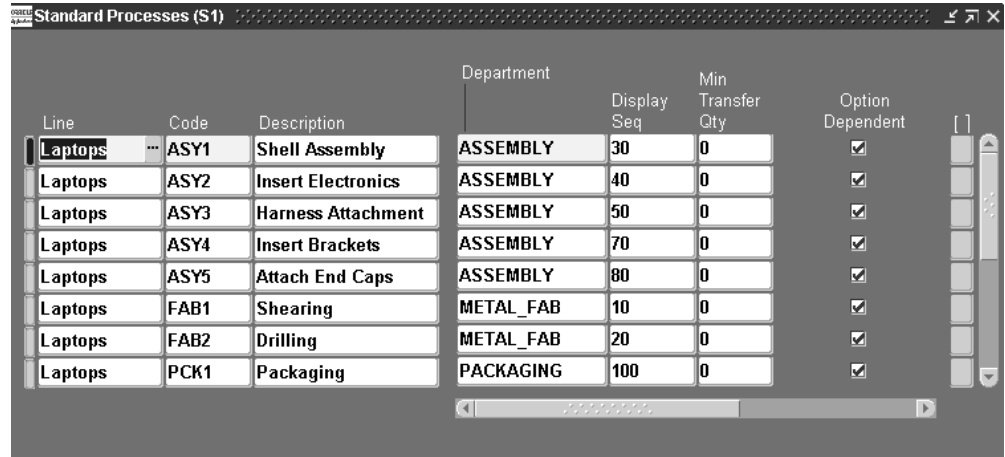
Prerequisites

- ☐ To define flow manufacturing processes, you must first define a line in Oracle Work

in Process.

To define a Standard Process for a flow manufacturing line:

1. Navigate to the Standard Processes window.



The screenshot shows the 'Standard Processes (S1)' window. It contains a table with the following data:

Line	Code	Description	Department	Display Seq	Min Transfer Qty	Option Dependent
Laptops	ASY1	Shell Assembly	ASSEMBLY	30	0	☒
Laptops	ASY2	Insert Electronics	ASSEMBLY	40	0	☒
Laptops	ASY3	Harness Attachment	ASSEMBLY	50	0	☒
Laptops	ASY4	Insert Brackets	ASSEMBLY	70	0	☒
Laptops	ASY5	Attach End Caps	ASSEMBLY	80	0	☒
Laptops	FAB1	Shearing	METAL_FAB	10	0	☒
Laptops	FAB2	Drilling	METAL_FAB	20	0	☒
Laptops	PCK1	Packaging	PACKAGING	100	0	☒

2. Enter the line identifier, sequence number, code, and description.
3. Select a department.
4. Optionally, enter a display sequence and minimum transfer quantity.
5. Save your work.

Defining Flow Manufacturing Standard Events

Standard Events can be created and used only with Oracle Flow Manufacturing. Events and standard events can only be created and used with Oracle Bills of Material. Events and Standard Events can also be created on the Graphical Line Designer Workbench. See: Creating the Graphical Network, page 4-28

Note: When using Oracle Flow Manufacturing, the Operation Sequence on the Bill of Materials refers to the Event Sequence.

Prerequisites

- ☐ To define Flow manufacturing line operations, you must first define a line in Oracle Work in Process. See: Defining Production Lines, *Oracle Work in Process User's Guide*

To define a Standard Event for a flow manufacturing line:

1. Navigate to the Standard Events window.

The screenshot shows the 'Standard Events (S1)' window. It contains a table with the following columns: Line, Code, Description, Department, Display Seq, Min Transfer Qty, and Option Dependent. The table lists eight events for 'Laptops'.

Line	Code	Description	Department	Display Seq	Min Transfer Qty	Option Dependent
Laptops	ASY1	Shell Assembly	ASSEMBLY		0	☒
Laptops	ASY2	Attach Harness to Dis	ASSEMBLY		0	☒
Laptops	ASY3	Attach End Caps & Lo	ASSEMBLY		0	☒
Laptops	FAB1	Shearing	METAL_FAB		0	☒
Laptops	FAB2	Drilling	METAL_FAB		0	☒
Laptops	FAB3	Press Brake	METAL_FAB		0	☒
Laptops	INS1	Insert Display Panel	ASSEMBLY		0	☒
Laptops	INS2	Insert Mounting Brack	ASSEMBLY		0	☒

At the bottom right of the window is a button labeled 'Event Resources'.

2. Enter the code and description.
3. Choose the Event Resources button to define resources.

When defining events, you must choose whether a resource is scheduled or not scheduled. In Flow Manufacturing, a nonscheduled resource is a resource that is used in parallel to a scheduled resource. Unscheduled resources are included in the total time for the resource type (labor or machine), but is not included in the total elapsed time for the process. Scheduled resources are included in both.

Therefore, in the case of parallel processes, the activity with the longest resource time should be scheduled because the system elapsed time calculation should always consider the longest resource time.

4. Save your work.

Related Topics

Calculating Operation Times, page 3-12

Defining Flow Manufacturing Standard Line Operations

Standard operations for Flow Manufacturing enable you to create line operations for a regular manufacturing line and a flow manufacturing line. In Oracle Flow Manufacturing, operations are used to group events into balanced work groups. They are generally defined and assigned after viewing the mixed model map, although they

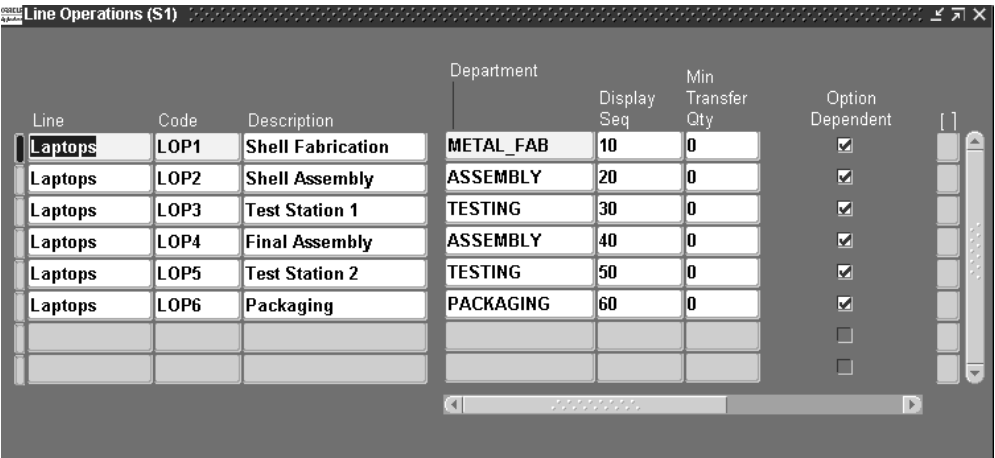
can be entered at any time.

Note: Operations and standard operations can also be created on the Graphical Line Designer Workbench. See: Creating the Graphical Network, page 4-28

Note: When using Flow Manufacturing, the operation sequence on the bill of material refers to the event sequence.

To define a Standard Line Operation for a flow manufacturing line:

- 1. Navigate to the Line Operations window.



- 2. Enter the line identifier, sequence numbers, code, and description.
- 3. Select a department and optionally choose the remaining options.
- 4. Save your work.

Flow Manufacturing Line Balance

To balance a Flow manufacturing line:

- 1. Generate a mixed model map for a line, product family, or forecast. You can also assign a boost percentage, number of days, and number of hours per day for the line.

The Mixed Model Map Workbench displays the line takt time, process takt times, and weighted average times. See: Overview of the Mixed Model Map Workbench, page 5-1

2. Create line operations for the current line.

See: Creating the Graphical Network, page 4-28

3. Assign line operations to the product level routings.

Note: You can copy or reference all the line operations for the current line into the routing to reduce data entry effort.

4. Regroup events on each product level routing into line operations to approximate takt time.

You can do this by assigning the line operation in the Parent Line Op field of the event.

5. Create a routing network for operations.

6. Optionally, add yields to operations.

7. Save your work.

8. Regenerate the mixed model map by line operation to review and confirm that the line is balanced. The mixed model map will also show the resource (machine/labor) and IPK requirements. See: Overview of the Mixed Model Map Workbench, page 5-1

Creating a Flow Routing

To create a product synchronization you can design a flow routing including events, processes, and line operations. You can create a routing manually, copy an existing routing, or reference a common routing.

Prerequisites

- ☐ Define at least one department before creating a routing.
- ☐ Define a line.
- ☐ BOM Allowed parameter must be set to Yes for the item you are creating a routing.

To create a flow routing:

1. Navigate to the Routings window.
2. Select the item, or product family or product family members, for creating a flow routing.

Additional Information: For ATO Items, the ATO Model Flow Routing will be used for line balancing, and will be used to create the configured routings during autcreate config. Therefore, define a routing for the model that contains all possible events, processes and operations for all possible options in your model. Then assign the model routing as a common routing for all option classes.

Seq	Code	Department	Process Seq	Process	Line Op Seq	Line Op	Effective	Di
10	FAB1	METAL_FAB	10	FAB1	10	LOP1	20/JAN/1998	
20	FAB2	METAL_FAB	10	FAB1	10	LOP1	20/JAN/1998	
30	ASY1	ASSEMBLY	30	ASY1	20	LOP2	20/JAN/1998	
40	ASY2	ASSEMBLY	40	ASY2	30	LOP3	20/JAN/1998	
50	TST1	TESTING	40	ASY2	30	LOP3	20/JAN/1998	

3. You can assign an existing routing or copy a routing to this Item:
 - To assign a common routing, choose Assign Common Routing from the Tools menu and select an item to copy.
 - To copy an existing flow routing, choose Copy Routing From in the Tools menu and select an item to copy.
4. If you are creating an alternate flow routing, enter an alternate. For a primary flow routing, do not enter an alternate.

Note: To create a standard (nonflow) routing for the same item, use the Bills of Material responsibility.

5. Optionally, select the Capable to Promise.

Capable to Promise describes an available to promise calculation that considers both available material and capacity of manufacturing and distribution resources.

You are able to define one and only one CTP routing for each item.

6. Enter the line for the flow routing.
7. If you are updating an existing flow routing, enter the routing revision and the effective date.
8. Optionally, enter a value in the Total Cycle time field.

The total product cycle time can be manually input now, or can be calculated by the application after you have defined processes and events.

9. Select a display option to display All, Current, or Future and Current operations effective as of the revision date you specify.
10. There are several ways to set up events, processes, and operations:
 - Use the Graphical Line Designer, see Creating the Graphical Network, page 4-28
 - In the list of values, choose Processes and enter the first process sequence and code. Choose the All Events and enter all events for that process. Close and repeat for all processes. When you are ready to balance your line, choose Line Operations to enter operation sequences and codes. Then assign events to the operations in the All Events region.

Note: The sequence number determines the order in which the processes are displayed in the Mixed Model Map Workbench.

- Choose Processes, and enter in all process sequences and codes. Optionally, choose the Line Operations, and enter all operation sequences and codes. Choose All Events tab, and enter in each event in sequence, assigning each to a process, and optionally operation.
 - Choose the Process tab, and enter in the desired processes. Optionally, choose the Operations tab and enter the desired operations. Choose All Events, and enter the desired events.
11. If you are using nonstandard events or editing standard events, choose Event Resources and define resource values. For an event, you can also define a department, parent process, and parent line operation.

Note: Processes and line operations must already be assigned to the routing before you assign them to events.

12. Enter effective date ranges.

13. For operations in ATO model and option class flow routings, indicate if the event is option dependent. An event is option dependent when the event is dependent on the choice of an optional component. A configured routing will contain all selected components and their related events, all mandatory components and their related events, and all events which are not assigned to a specific component.

Note: All processes and operations will always appear on the configured routing, regardless of whether or not any events assigned to them are included.

14. Choose the WIP tab to indicate whether to backflush components on shop floor moves at this operation. A backflush transaction automatically pulls operation pull components from inventory. Work in Process also pulls all Operation pull components at nonbackflush operations preceding this operation using the previous completed backflush operation.
15. Choose the Operation Times tab to define or view operation times and yields.
16. To view rolled-up operation times, select Calculate Operation Times from the Tools menu. Optionally, enter user defined times. Either can be used in Mixed Model Map Workbench calculations.
17. Switch to the Operation Times tabbed region and optionally enter operation yield values for processes and/or line operations.

Note: Operation yield values are required to perform calculation of cumulative and reverse cumulative operation yields.

18. Create a routing network. See: Creating the Graphical Network, page 4-28
19. Once the routing network is created, calculate cumulative yield, reverse cumulative yield, and net planning percent, by selecting Calculate Operation Yields from the Tools menu. Choose Update Events to add these yields and calculations to events. This is required if you want kanban planning to consider process yields when calculating kanban sizes.

Note: You can also manually insert these values.

20. To calculate Total Product Cycle Time, choose Calculate Total Product Cycle Time from the Tools menu.
21. To enter completion subinventory and locator information, view a common flow routing, or enable the routing for use in the Mixed Model Map Workbench, choose the Routing Details.

22. In the Routing Details window, optionally, enter a priority. Lowest priority routing will be used for backflushes and during autocreate config on ATO items.
23. Save your work.

Creating a Routing Network

In defining product synchronization you must specify the network of processes through which the item is routed. This includes specifying alternate processes, planning percent, rework loops, rework percentage, and feeder lines. The routing network is created in the Graphical Line Designer Workbench.

Related Topics

Creating the Graphical Network, page 4-28

Modifying the Line Operation Tree and Process Tree, page 4-33

Calculating Operation Times

You can calculate machine time, labor time, and elapsed time for your processes, line operations, and events defined in your flow routing. You can also enter your own machine times, labor times and elapsed times in the Flow Routing window.

Prerequisites

- ☐ You must first define any events, processes and line operations to be included in your flow routing.

To calculate operation times:

1. Navigate to the Flow Routing window.
2. Place your cursor in the Total Cycle Time field in the header.
3. Select Calculate Operation times from the Tools menu. Choose to calculate either process times or operation times.

Note: In the Flow Routing window the unit of measure for all times is hours.

Note: You must re-perform the operation times and total product cycle time calculations any time you make a change to events,

processes or operations.

Note: You can also enter your own machine times, labor times, and elapsed times in the Flow Routing window. To do so, manually enter this information in the User Machine Time, User Labor Time, and User Elapsed Time fields. You determine when creating a mixed model map, whether the calculations are based on rolled-up or user entered times.

Operation Times Calculations

The operation times calculations are based on resources in the Flow Routing which have the resource unit of measurement (UOM) in the UOM class associated with the BOM profile option, BOM: HOUR UOM.

Calculated Labor Time

Calculated labor time per event is the sum of both scheduled and unscheduled labor resource usage rates for the event. Calculated Labor Time per process or operation is the sum of the labor times for all the events underneath each process or operation. If you enter events by lot, it will temporarily convert the number to an item basis for the purposes of this calculation.

Note: For ATO model routings: The program goes to the multilevel BOM for the model, and searches for the planning percentage on items that are assigned to specific routing events. It then multiplies the calculated labor time for this event by the planning percent to get a weighted value for the event time. If an event is NOT assigned to any part on the multilevel BOM, planning percent is assumed to be 100%.

Note: This will work only if you define a routing for the ATO model, and then use it as a common routing for all the option classes that follows it.

Note: Each Option on the routing must belong to a unique event for operation calculations to be successful.

Calculated Machine Time

Calculated Machine Time is the same as Calculated Labor time, only it uses the scheduled and unscheduled machine resources for each event.

Calculated Elapsed Time

Calculated Elapsed Time for each event is the sum of the scheduled Labor and Machine usage rates for each event. Elapsed time per process or operation is the sum of the elapsed times for all the events that fall under each process or operation. If you enter events by lot, it will temporarily assume that the time entered is per item for the purposes of this calculation.

Times for ATO model routings are adjusted based on the planning percent of the item assigned to an event, as described in calculated labor times.

Calculating Total Product Cycle Time

You can calculate the Total Product Cycle Time (TPCT) for your line. TPCT is the sum of all elapsed event times along the longest path on your Flow Line.

Prerequisites

- ☐ You must first define the events, processes and/or line operations to be included in your flow routing.
- ☐ You must first create a routing network for either processes or operations.
- ☐ You must first calculate operation times

To calculate Total Product Cycle Time:

1. Navigate to the Flow Routing window.
2. Place your cursor in the Total Cycle Time field in the header.
3. Select Calculate Total Cycle Time from the Tools menu. Choose to calculate either process times or operation times.

Total Product Cycle time is always calculated using the rolled-up (system calculated) elapsed times, even if you enter manual elapsed times. You can manually enter a Total Cycle Time as well.

You must re-perform the operation times and total product cycle time calculations any time you make a change to events, processes or operations.

Once the routing network and the Total Cycle Time for a routing are finalized for each item on your flow line, return to the organization item master and update the fixed lead time to equal your total cycle time. The takt time (which is displayed on the MMM) for the line should be entered in the variable lead time.

Note: In the Flow Routing window, the unit of measure for all times is hours.

Calculating Operation Yields

The system will calculate the cumulative yield, reverse cumulative yield and net planning percent for each process and line operation. This information is used in Mixed Model Map Workbench calculations and kanban planning.

Prerequisites

- ☐ Optionally, enter a value for yield at some or all processes or line operations.
- ☐ You must define a routing network for either the processes or the line operations.

To calculate operation yields:

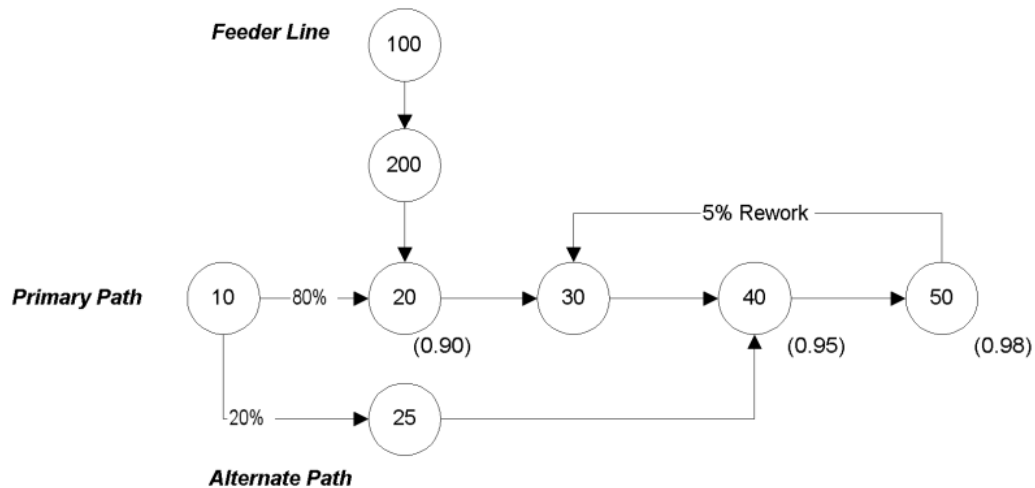
1. Navigate to the Flow Routing window.
2. Place your cursor in the Total Cycle Time field in the header.
3. Select Calculate Operation Yields from the Special menu.
4. Select process or operation. Choose update events to translate yields and net planning percent to events.

Important: You must update events (manually or by choosing to update events when you perform the calculations) if you want kanban planning to consider yields when sizing kanbans.

Additional Information: If you have yields in your line, you must either calculate or manually enter operation yields for each process/line operation for the mixed model map results to be accurate.

Operation Yields Calculations

The system will calculate the cumulative yield, reverse cumulative yield and net planning percent for each process and line operation. This information is used in Mixed Model Map Workbench calculations and kanban planning. The following diagram displays an illustration of the setup used in the example calculations.



In the above example, the primary path is operation 10 through 50. The numbers in parentheses are the yields for the operation number above it, as entered by the user. The percentages on the network paths represent the network percent assigned to each path. Blank paths assume 100% network percentages. (The incoming network percentage for the first operation and the outgoing network percentage for the last operation are always 100%.)

Note: Any alternate path must skip at least one operation (or process) on the main line for the yield and planning percent calculations to work properly.

Net Planning Percent

The system calculates the net planning percent by using the following formula.

$$N_c = S (P_p * I_c) + S_r (I_o * W_o),$$

where

N = Net Planning Percent, in decimal format

S = sum over all paths

P = Planning Percent = $(P_p * I_c)$ (system calculated - not displayed)

Note: Planning percent for the first operation is always 100%.

I = Incoming Network Percent, in decimal format (from Routing Network)

Note: Incoming network percent for the first operation is always 100%.

p = previous operation

c = current operation

Sr = Sum over all rework loops within which is this operation

W = Rework percent (from the routing network)

o = Operation originating the rework

In other words, the system calculates the net planning percent by taking the sum over all primary and alternate paths of the product of the network percentages along that path, then adds a rework calculation if the operation is within the rework loop. The planning percent and incoming network percent for the first operation on the primary path are always 1. The planning percent for the Feeder line is always the same as the operation to which they feed.

The rework calculation is equal to the planning percent of the rework's originating operation, times the rework percent coming from that operation (in this example, OP50 is the originating operation for the rework, and its planning percent is 100%, and the rework loop is 5%. Therefore, the rework calc is $100\% * 5\% = 5\%$. This value is added to all operations in the loop (30, 40 and 50).

The number of paths is determined by working backwards along primary and alternate paths from the end of the line (so in the example, operations 30, 40 and 50, each have two paths due to the alternate path).

Caution: If a rework loop returns to an operation that is also fed by a feeder line, the feeder line will include the rework loop percent in the Net Planning Percent.

Operation	Incoming Network Percent	Planning % Calculation + Rework Calculation	Net Planning Percent
10	100	$1 * 1 + 0$	1
20	80	$1 * 0.80 + 0$	0.84
25	20	$1 * 0.20 + 0$	0.20
30	100	$(1 * 0.80 * 1) + 1 * .05$	1.05
40	100	$(1 * 0.80 * 1 * 1 + 1 * 0.20 * 1) + 1 * .05$	1.05

Operation	Incoming Network Percent	Planning % Calculation + Rework Calculation	Net Planning Percent
50	100	$(1 * 0.80 * 1 * 1 * 1 + 1 * 0.20 * 1 * 1) + 1 * .05$	
100	feeder - uses network% from 20	0.80 + 0	0.80
200	feeder - uses network% from 20	0.80 + 0	0.80

Cumulative Yield

The cumulative yield for the first operation along the main line and any feeder lines is the yield entered by the user. The system calculates the cumulative yield for all other operations using the following formula:

$$C_c = Y_c * S (C_p * P_p * I_c / P_c),$$

where

C = Cumulative Yield

Y = Yield (User entered)

S = sum of all paths

P = Planning Percent, in decimal format (P = Net Planning% - Rework Calc)

I = Incoming Network Percent, in decimal format (from Routing Network)

p = previous operation

c = current operation

For example, operation 20 in the example above:

$$C_{20} = Y_{20} * S(C_{10} * P_{10} * I_{20} / P_{20}) = 0.90 * (1 * 1 * 0.8 / 0.8) = 0.9$$

Starting from the front of the line, the results for the example are:

Operation	Yield Entered	Incoming Network %	Plan %	Cumulative Yield Calculation	Cumulative Yield
10	null	100	1	1	= 1
20	0.90	80	0.8	0.90 * (1 * 1 * 0.8 / 0.8)	= 0.90
25	null	20	0.2	1 * (1 * 1 * 0.2 / 0.2)	= 1
30	null	100	0.8	1 * (0.9 * 0.8 * 1 / 0.8)	= 0.9
40	0.95	100	1	0.95 * (0.9 * 0.8 * 1 / 1) + (1 * 0.2 * 1 / 1)	= 0.874
50	0.98	100	1	0.98 * (0.874 * 1 * 1 / 1)	= 0.8565
100	null	100	1	1	= 1
200	null	100	1	1 * (1 * 1 * 1 / 1)	= 1

Reverse Cumulative Yield

The reverse cumulative yield for the last operation along the main line is the yield entered by the user. The system calculates the reverse cumulative yield for all other operations using the following formula:

$$R_c = Y_c * S(O_c * R_n),$$

where

R = Reverse Cumulative Yield

Y = Yield (user entered)

O = Outgoing Network Percent, in decimal format (from Routing Network)

Note: Outgoing network percent for the first operation is 100%.

n = next operation

c = current operation

S = the sum over all outgoing paths

For example, operation 20 in the example above:

$$R20 = Y20 * S(O 20 * R 30) = 0.90 * (1 * 0.931) = 0.8379$$

Operation	Yield	Outgoing Network %	Reverse Cumulative Yield Calculation	Reverse Cumulative Yield
50	0.98	100	0.98	
40	0.95	100	$0.95 * (1 * 0.98)$	0.98
30	null	100	$1 * (1 * 0.931)$	0.931
20	0.9	100	$0.9 * (1 * 0.931)$	0.8379
25	null	100	$1 * (1 * 0.931)$	0.931
10	null	80 and 20	$1 * [(0.8 * 0.8379) + (0.2 * 0.931)]$	0.85652
200	null	100	$1 * (1 * 0.838)$	0.8379
100	null	100	$1 * (1 * 0.838)$	0.8379

Graphical Line Designer Workbench

This chapter covers the following topics:

- Overview of the Graphical Line Designer Workbench
- Navigating on the Graphical Line Designer Workbench
- Accessing Graphical Line Designer Workbench Windows
- Adding a New Item Routing
- Deleting a Routing
- Copying and Common Flow Routings
- Copying a Routing to a Different Flow Line
- Viewing Routing Revisions
- Using the Routing Details Window
- Creating the Graphical Network
- Defining a Network Connection
- Modifying the Line Operation Tree and Process Tree
- Modifying Line Operations, Processes, Events, and Connections
- Creating Routings in the Template Routings View
- Mass Changes on Item Routings
- Component Assignments on the Bill of Material
- Specifying Cost Rollup Resource Information

Overview of the Graphical Line Designer Workbench

The Graphical Line Designer Workbench offers an integrated user interface for performing product synchronizations. This workbench enables the automatic creation of standard processes, standard events, and standard operations by using defaulted

information. The features in the Graphical Line Designer Workbench include:

- Ability to drag and drop icons onto a canvas to add processes, events and line operations to the item routing
- Processes, events, and line operations added to a routing are also automatically added to the line as a standard entity
- Routing calculations are automatically performed every time a change is made to the routing
- Information fields for Quality, value added, and process efficiency
- Ease of use features for drawing size changes, and graphical view of network planning percentage connections under 100 percent

Events, Processes, and Line Operations

Events represent the detailed work content of your line. You can assign resources and time to events.

Processes are generally logical groupings of events that represent what is being done on the shop floor before going to flow. They can equal the discrete routing operations (assembly, welding, machining, test, and so forth). They can also be at an even lower level than that—especially in an assembly environment (assemble fan, add engine, inspect welds, and so forth). You may want to define processes first to give you a chance to take a look at the balance of your shop before moving to flow, using the Mixed Model Map Workbench. It gives you a starting point to divide or group work to equal takt. You are not required to define processes.

Operations are balanced groupings of events, and represent the workstations on your flow line. They may be more detailed than processes such as when one process is divided into several operations. Or, operations may be more generic such as when several processes are grouped into a single operation. This may be the case in a low volume assembly shop. Operations must be defined in order to be able to scrap assemblies through work order-less completions, to use the flow workstation, and to perform feeder line synchronization.

Related Topics

Product Synchronization, page 3-3

Navigating on the Graphical Line Designer Workbench, page 4-3

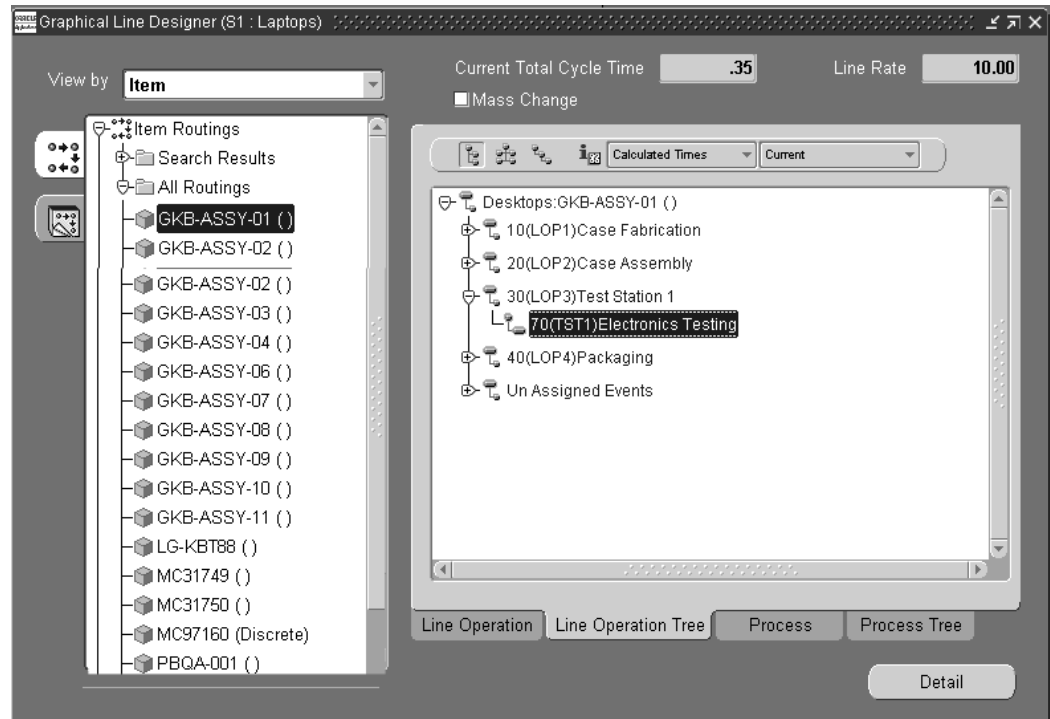
Accessing Graphical Line Designer Workbench Windows, page 4-8

Creating the Graphical Network, page 4-28

Navigating on the Graphical Line Designer Workbench

The Graphical Line Designer Workbench comprises two panes. The left pane is the tree hierarchy and the right pane displays the graphical network.

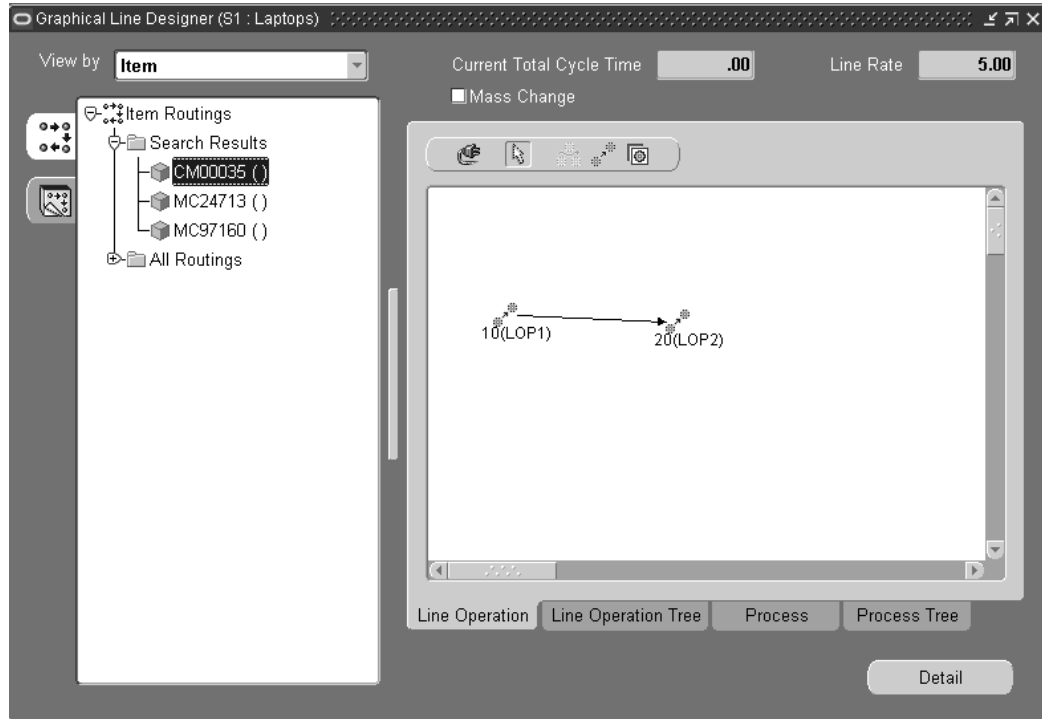
The View By list box allows you to change the way you view data in the tree—your choices are by Item, Product Family, or Common Routing.



There are two tree tabs located to the left of the tree hierarchy that enable you to toggle between the Item Routings and Template Routings views on the right pane. The Item Routings tab displays the routings for all items assigned to the designated line, while the Templates tab gives access to product family routings that can be used as templates for copying information into routings. The right pane contains the data that is associated with the selected item on the tree. When an item is selected, four tabs at the bottom of the window provide access to windows displaying Line Operation network, Line Operation Tree, Process network, and Process Tree.

Resizing the Detail Pane

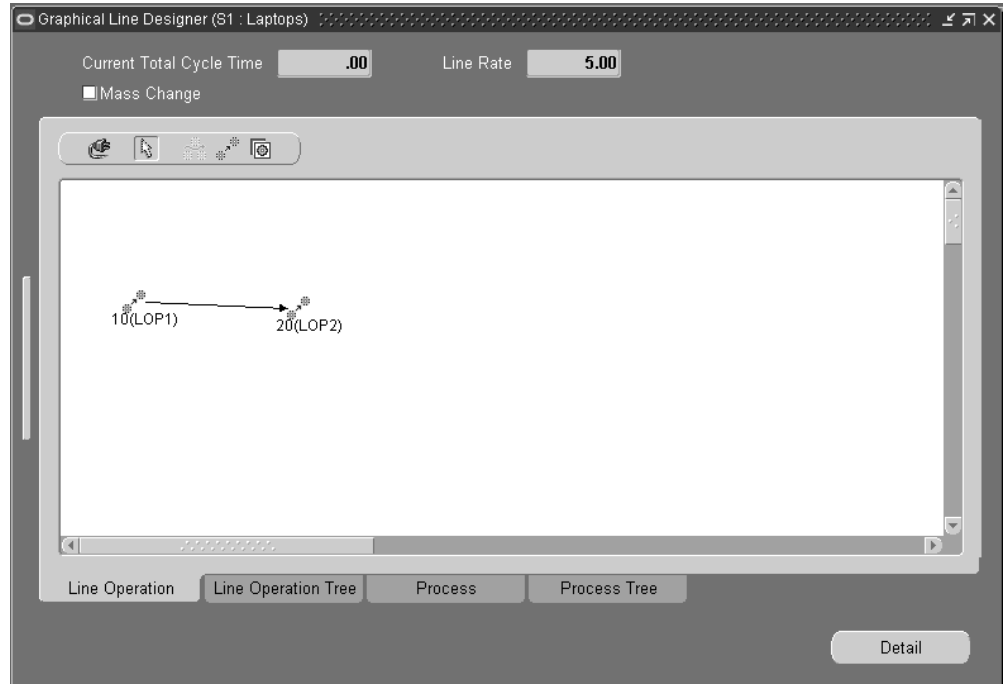
You can change the size of the detail pane, enabling you to have a larger working canvas area when creating and modifying the graphical network.



To resize the Graphical Line Designer Workbench window:

- Select the button between the navigation tree and the detail pane, to maximize the window.

When you maximize the window size in this method, the navigation tree is not displayed. This button enables you to toggle between the full screen and divided pane sizes.



- You can also resize the detail pane by resizing the entire workbench. Select the resize icon.

This method maximizes the window size, and also displays the navigation tree. This icon enables you to toggle between the full screen and smaller pane sizes.

Workbench Toolbars

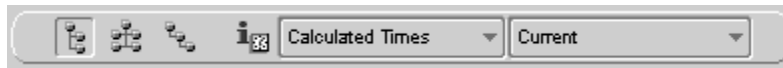
In the right pane canvas, there are two toolbars containing icons used for creating and modifying your graphical network. One toolbar displays when you are in the Line Operation or Process window.



This toolbar contains icons for:

- *Connector*: Enables you to create links between entities on the canvas
- *Pointer*: Allows you to move existing entities on the canvas
- *Process*: Enables you to add a new process to the routing
- *Line Operation*: Enables you to add a new line operation to the routing
- *Event*: Enables you to add a new event to the routing

The other toolbar displays when you are in the Operation Tree or Process Tree window.



It contains three icons that allow you to change the display view of the tree including:

- *Vertical Style*: The tree is displayed in an up-and-down format by routing, line, line operation code, and description.
- *Organizational Chart Style*: The tree is displayed in a hierarchical box format. The routing is at the top level, the lines, line operation codes, and descriptions are graphically displayed below the routing.
- *Interleaved Style*: The tree is displayed in an indented hierarchical box format. The routing is at the top level and displayed on the left of the pane; the line, line operation codes, descriptions are graphically displayed to the right of the routing.
- *Showtime*: Toggles the view of the event, line operation and process times when viewing the data in a tree format.

This toolbar also contains two menu boxes. One box allows you to choose views on the pane by either current, future and current or all events on the routing.

The Times box has two selections: Calculated Times and User Times. This enables you to choose either to view the operation time that is calculated by the system or entered in manually.

Tools Menu and Pop-up Menus

The Graphical Line Designer Workbench enables you to access other manufacturing functionalities. These options are available, depending upon the current active window and other conditions, in the Tools menu and pop-up menus.

The Tools Menu

Standard Events: Accesses the Standard Events window. It displays all standard events for a particular line. See: Defining Flow Manufacturing Standard Events, page 3-5.

Standard Processes: Accesses the Standard Processes window. It displays all standard processes for a particular line. See: Defining Flow Manufacturing Standard Processes, page 3-4.

Standard Line Operation: Accesses the Standard Line Operations window. It displays all standard line operations for a particular line. See: Defining Flow Manufacturing Standard Line Operations, page 3-6.

Mixed Model Map: Accesses the Mixed Model Map Workbench enabling you to generate a mixed model map. See: Overview of the Mixed Model Map Workbench, page 5-1.

Resource Where Used: Accesses the Resource WhereUsed window in Oracle Bills of Material for viewing the operation and routings that use a particular resource. See: Viewing a Resource Usage, *Oracle Bills of Material User's Guide*.

Routing Revisions: Accesses the Routing Revisions window to view the changes for a particular routing. See: Viewing Routing Revisions, page 4-25.

Switch to Primary: Allows you to switch the selected alternate with the primary routing.

Items: Accesses the Master Item window in Oracle Inventory where you can define and update items and the attributes associated with them. See: Defining Items, *Oracle Inventory User's Guide*.

Product Family Members: Accesses the Product Family window, enabling you to group products that have a similarity in resource usage, design, and manufacturing processes. See: Assigning Product Family Members, *Oracle Bills of Material User's Guide*.

Bills of Material: Accesses the Bills of Material window that contains information on the parent item, components, attachments, and descriptive elements. See: Creating a Bill of Material, *Oracle Bills of Material User's Guide*.

Indented Bills: Accesses the Indented Bill of Material window in Oracle Bills of Material where you can view an indented (multilevel, exploded) manufacturing or engineering bill of material. See: Viewing an Indented Bill of Material, *Oracle Bills of Material User's Guide*.

Lines: Accesses the Production Lines window in Oracle Work in Process. See: Defining Production Lines, *Oracle Work in Process User's Guide*.

Departments: Accesses the Departments window in Oracle Bills of Material. See: Defining a Department, *Oracle Bills of Material User's Guide*.

Item Selection List: Accesses the Apply Changes window for creating mass changes, edits, and deletions on multiple flow lines. See: Mass Changes on Item Routings, page 4-40.

Set Resource Cost Rollup: Accesses the Set Resource Cost Rollup window used to specify if an event is included in the cost rollup. See: Specifying Cost Rollup Resource Information, page 4-49.

Pop-up Menus

You can access windows and functions in the pop-up menus by right-clicking on an icon in the right pane, or a node in the tree hierarchy.

The following choices are available, depending upon the currently active window and other conditions, by right-clicking on an operation, process, event, or connection icon in the right pane:

- *Cut:* Allows you to remove an operation, process, event, or any connection to it from the item routing.
- *Paste:* Allows you to restore an operation, process, event, or connection you have

just cut from the item routing.

- *Delete*: Allows you to permanently remove the operation, process, event, and any connections to it from the item routing.
- *Properties*: Accesses the Operation, Process or Event window. For example, if you select an event, the Event window opens.
- *Standard*: Accesses the Standard Line Operations, Standard Processes or Standard Events window, depending on the entity that you have selected. For example, if you select a process, the Standard Processes window opens.

The choices available in the pop-up menu in the tree hierarchy include the applicable Tools menu, Workbench menu options, and the following additional menu options in the tree hierarchy:

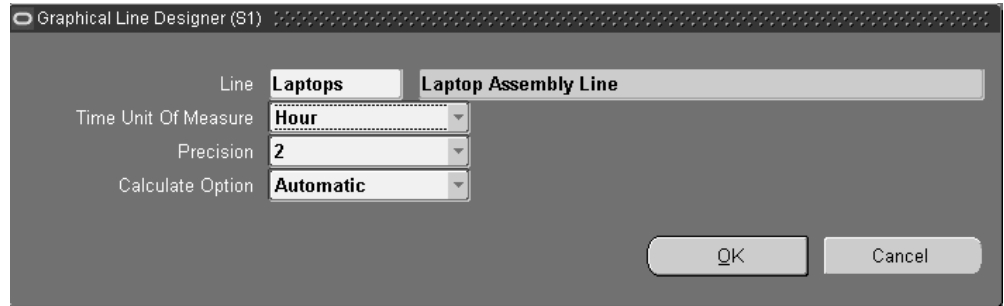
- *New*: Accesses the Add Item Routing window enabling you to create new routings. See: Adding a New Item Routing, page 4-17
- *Routing Details*: Accesses the Routing Details window enabling you to view specifics or to add information to a routing. See: Using the Routing Details Window, page 4-27
- *Copy/Common Routing To*: Accesses the Copy Routing window used to copy that item's routing to other items, or create routings that are common to it. See: Copying and Common Flow Routings, page 4-21
- *Copy Routings To*: Accesses the Copy Routings window enabling you to copy a routing to a different flow line. See: Copying a Routing to a Different Flow Line, page 4-23
- *Component Assignments*: Accesses the Update Component window, where you can associate the components on the bill of material with the routing events where they are consumed. See: Component Assignments on the Bill of Material, page 4-47
- *Delete*: Accesses the Delete Routing window to delete a routing from your navigation tree. See: Deleting a Routing, page 4-19

Accessing Graphical Line Designer Workbench Windows

To access windows for creating product synchronizations:

1. Select the Graphical Line Designer from the Flow menu.

The Select window displays.



2. In the Line field, select the assembly line you want to view on the workbench.

3. In the Time Unit of Measure field, select the unit of measure you are using for routing calculations.

Your choices are hour, minute, or second.

4. In the Precision field, select a value for the number of decimal places you want to display in routing calculations.

Your choices are 1, 2, 3, 4. For example, if you enter 2—two decimal places display.

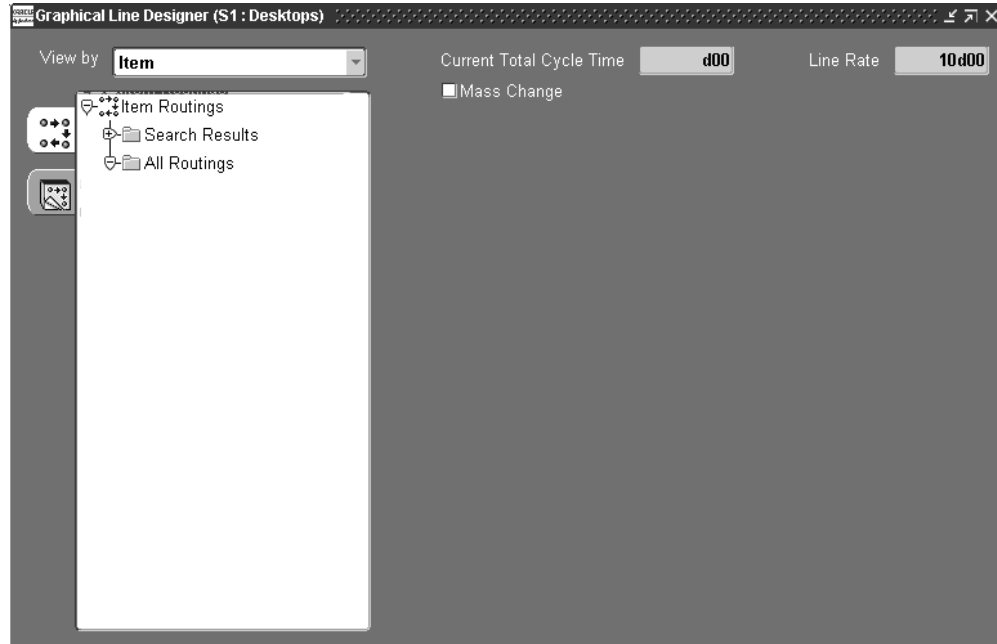
5. Select a Calculate Option for calculating the Current Total Cycle Time. This value displays on the workbench. Your choices are:

- Automatic—Current Total Cycle Time is automatically calculated every time there is a change or update.
- Deferred—Current Total Cycle Time calculation is only initiated manually, when you select Calculate on the workbench.

6. Choose OK.

The Graphical Line Designer displays.

In the initial view, the Item Routing tree tab is selected and the tree hierarchy is collapsed. The two tree tabs located to the left of the tree hierarchy allow you to toggle between the Item Routings and Template Routings views on the right pane.



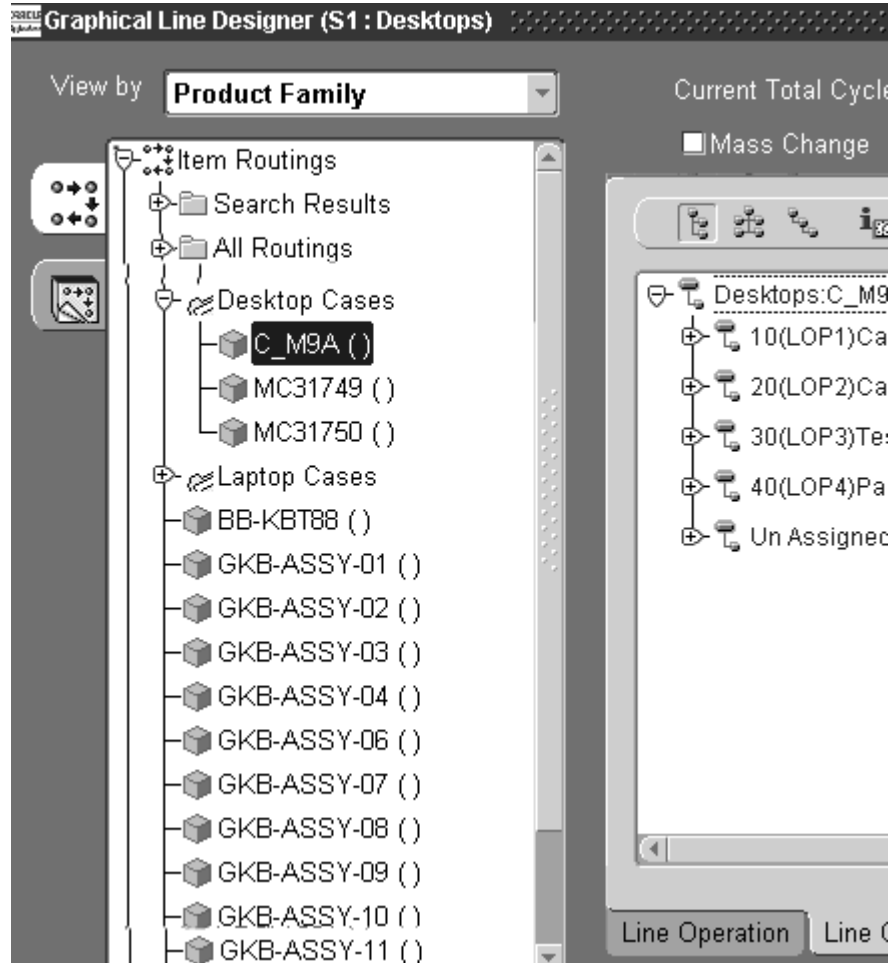
The Item Routings tab displays the routings for all items assigned to the line designated. In the initial view the right pane is blank, when you select an item, the graphical network displays.

- The View By list box is used to select the way you want to view data in the tree.
- The Current Total Cycle Time field displays for this item. It is calculated automatically every time you change tabs, add entities on the routing, or change items. See: Calculating Total Product Cycle Time, page 3-14
- The Line Rate value defaults from the line definition.

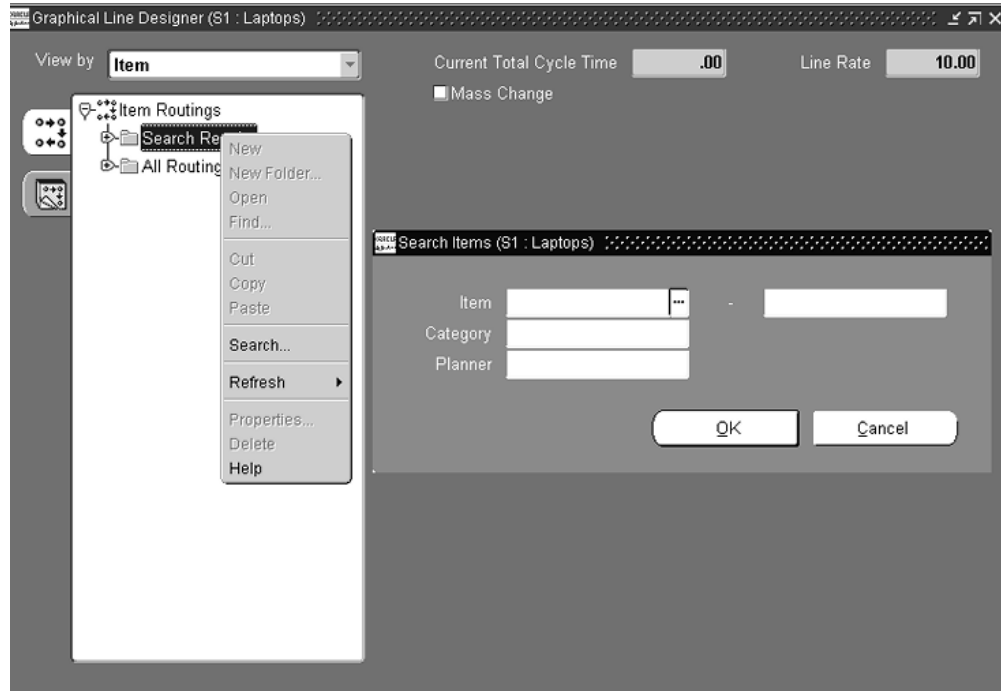
In the Item Routing node, you can search for routings using the Search Results node, or view all the routings in the All Routings node.

7. In the View By list box, select to view data in the tree by Item, Product Family, or Common Routing.

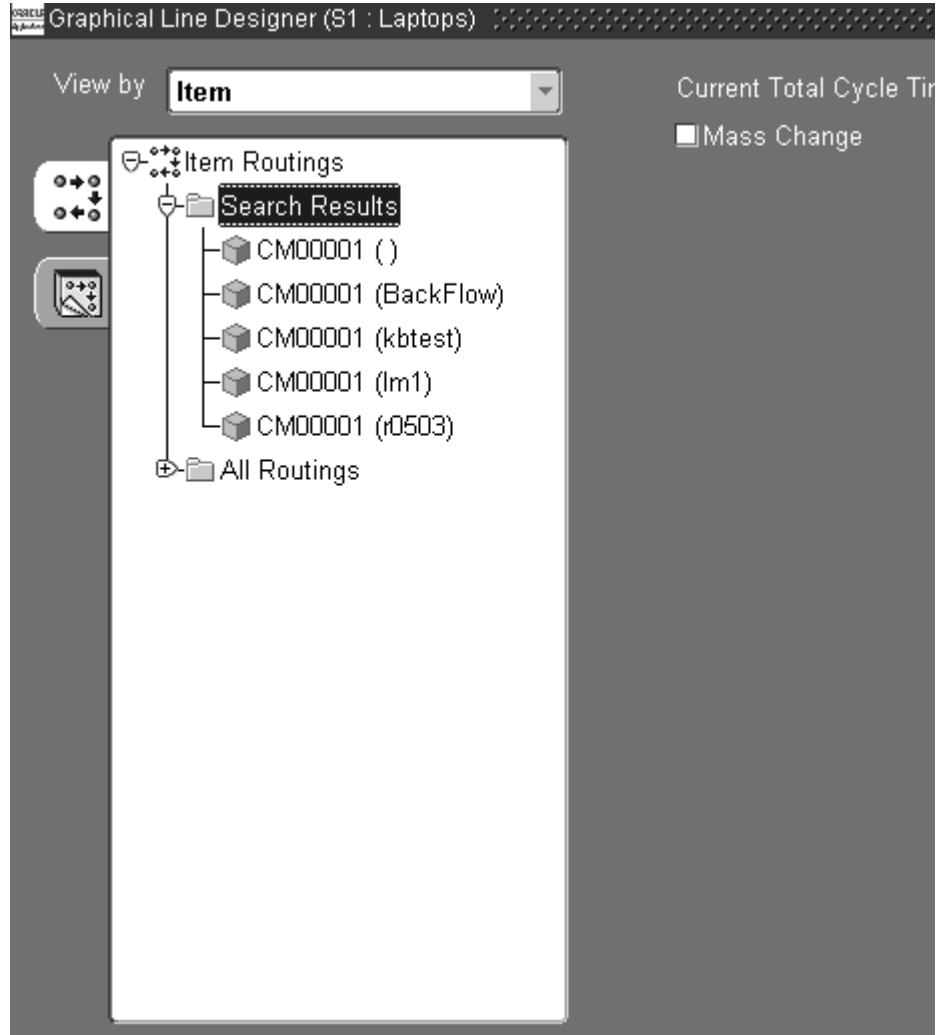
If you choose product family and the items are assigned to a product family, the product family displays in between the line and item.



8. If you want to find a specific record, or range of records, select the Search Results node and right-click with your mouse to display the pull-down menu.
9. Select Search to display the Search Items window.



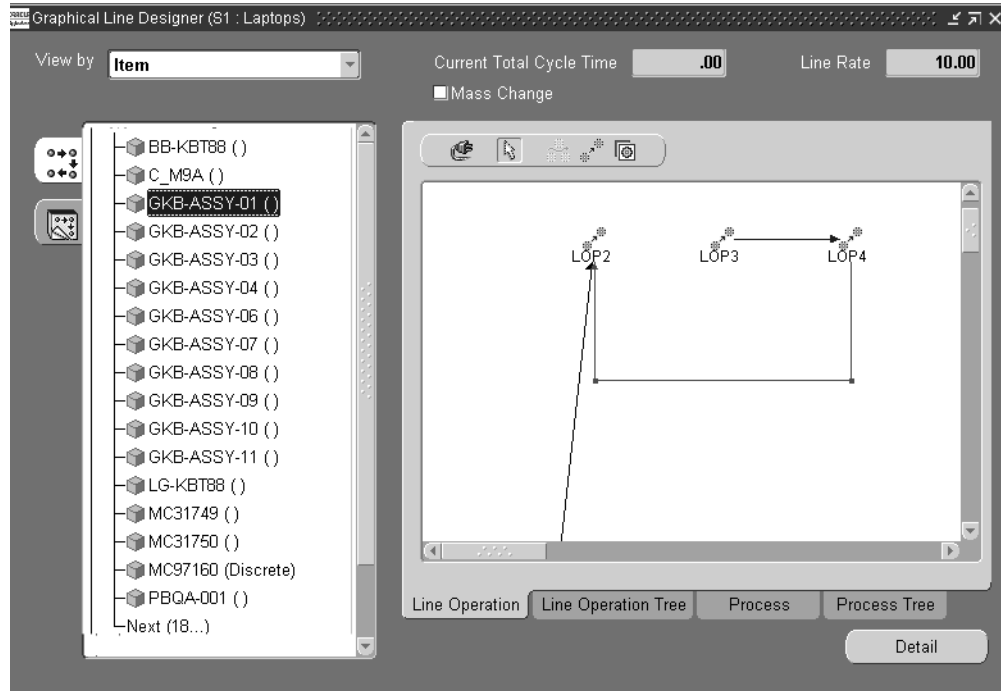
10. Enter search criteria; you can search by item, category, planner, or any combination of these values.
11. Choose OK to display the results of your query under the Search Results node.



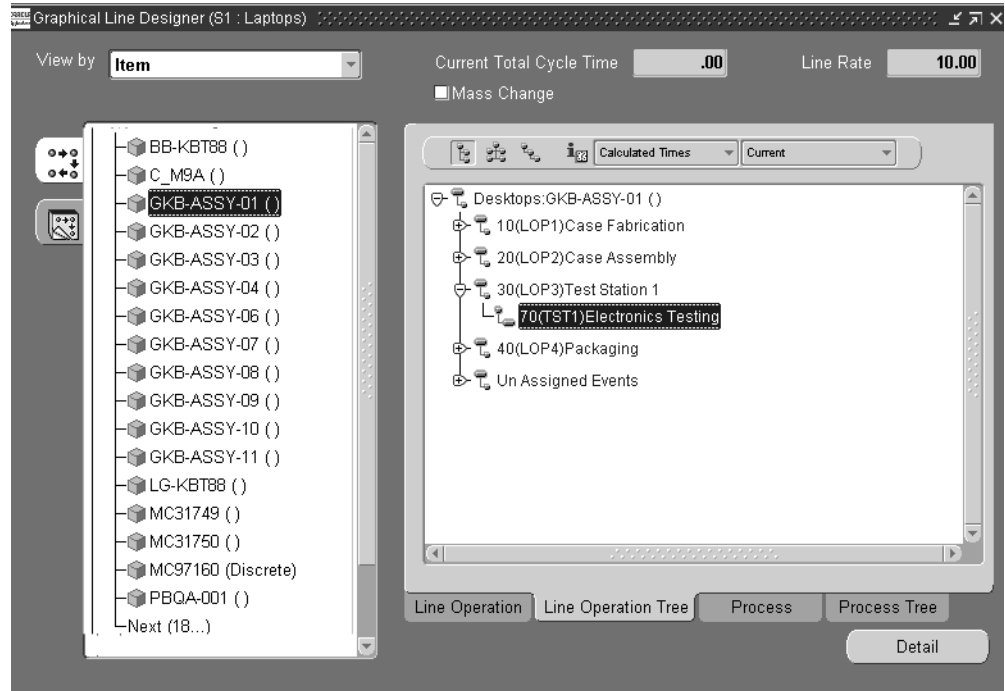
12. To display information in the right pane, select a node detail on the tree.

The graphical network displays a view of the line operation network in the first window tab, Line Operation.

You can add, change, delete, or move the icons in the graphical network. See: Creating the Graphical Network, page 4-28



13. Select the Line Operation Tree tab.
14. Left-click with your mouse to select a node detail on the tree hierarchy.
In the right pane the top level of the routing for this item displays.



15. Expand the items on the routing to display the operations in the routing.

You can change the tree format to one of three styles. See: Workbench Toolbars, page 4-5 The icons in the tree will change to match those on the toolbar for the three entities. The last node at the bottom of the tree is named Unassigned Events. This is a placeholder for all the events that are on the routing but are not assigned to any line operation. For example, an event may be unassigned because you defined processes first, and have not assigned the events to the operations.

You can add, change, delete, or move the icons in the tree. See: Modifying the Line Operation Tree and Process Tree , page 4-33

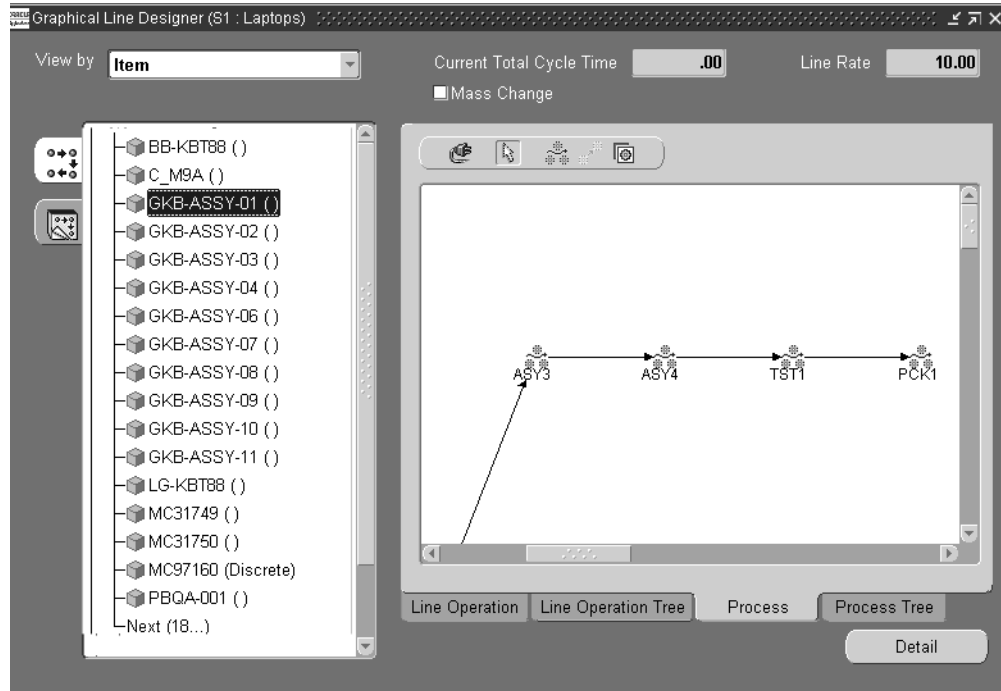
The code and description are listed for each line operation and event.

The toolbar enables you to view the line operations and events on the routing by current, future and current, or all events. You can also see the times associated with each line operation and event.

16. Select the Process tab.

The graphical network displays a view of the process.

You can add, change, delete, or move the icons in the graphical network. See: Creating the Graphical Network, page 4-28



17. Select the Process Tree tab.
18. Select a node detail on the tree hierarchy, and left-click with your mouse.
19. Expand the items on this routing to display the processes in the routing.

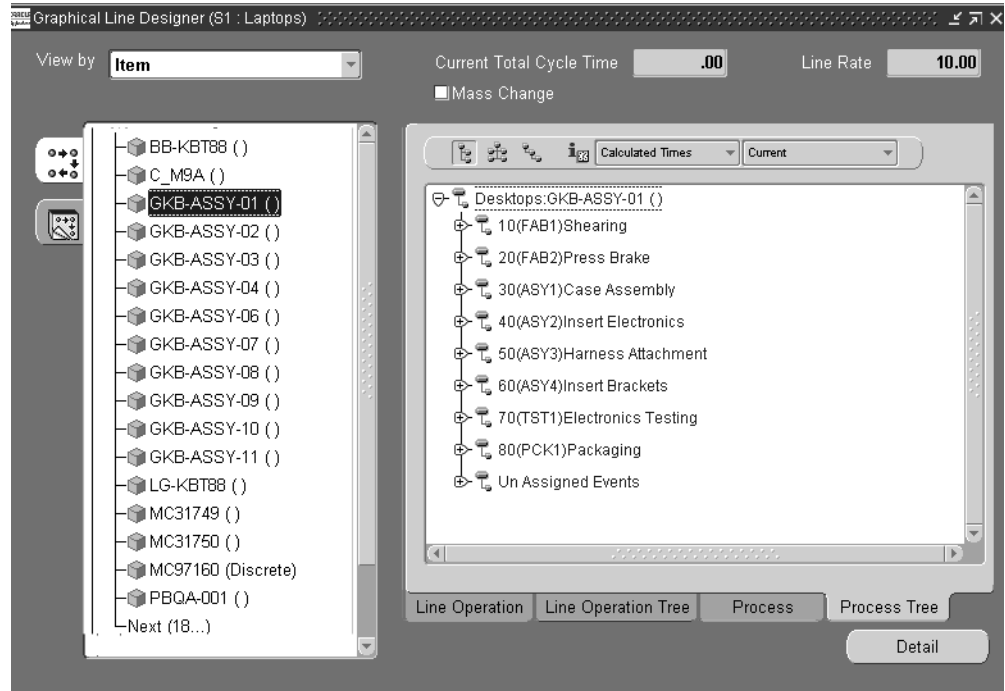
You can change the tree format to one of three styles. See: Workbench Toolbars, page 4-5 The icons in the tree will change to match those on the toolbar for the three entities.

The last node at the bottom of the tree is named Unassigned Events. This is a place holder for all the events that are on the routing but are not assigned to any process. For example, an event may be unassigned because you defined the operations first, and have not assigned the events to the processes.

You can add, change, delete, or move the icons in the tree. See: Modifying the Line Operation Tree and Process Tree , page 4-33

The code and description are listed for each process and event.

The toolbar has an option that enables you to see the times associated with each process and event.



Adding a New Item Routing

The Add Item Routing window enables you to create new routings on the Graphical Line Designer Workbench. You have three options in creating routings. You can create a routing as an entirely new routing, copy it from another item or a template, or assign a common routing as a reference for the new routing.

To add a new item routing:

1. Navigate to the Graphical Line Designer Workbench.
2. Select the top node on the tree hierarchy, Item Routings, and right-click with your mouse.

The pop-up menu displays.

3. Choose New.

The Add Item Routing window displays.

4. To create a new routing, select an item from the list of values in the Item field, and enter any appropriate information in the Alternate, Capable to Promise, Revision, and Date fields.

See: Creating a Routing,, *Oracle Bills of Material User's Guide*

5. To create a routing by copying one from another item or a template, choose Copy From Item/Template in the Routing Source region.

Your current Organization information displays.

In the Item field, select the routing to be used as a template from the list of values.

See: Creating Routings in the Template Routings View, page 4-38

Enter the any appropriate information in the Alternate, Revision, Engineering Routing, and Date fields.

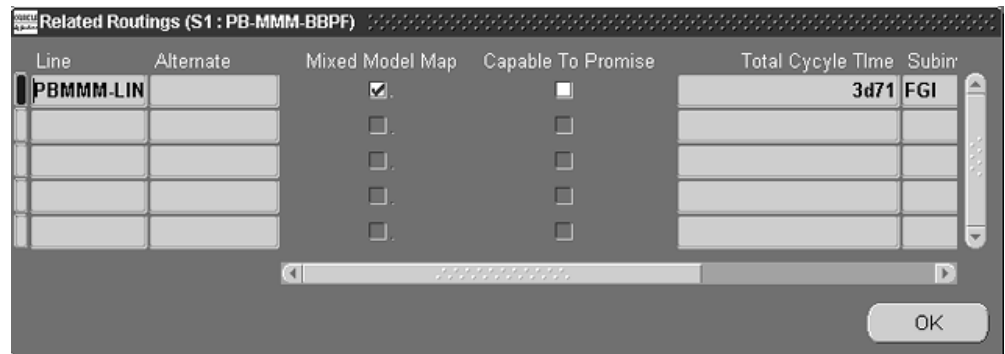
Select a Copy Option, your choices are:

- *Current*: operations effective on the revision date
- *Future and Current*: includes current and all future operations

6. To create a routing from a common routing, choose Assign Common Routing.

Any two items that are of the same bill type can share common bills and routings. See: Referencing Common Bills and Routings, *Oracle Bills of Material User's Guide*.

7. Select the routing you want to use from the list of values.
8. Optionally, you can choose Related Routings to view a list of the other routings that have been defined for this item on all the lines in this organization.



9. Choose OK to close the Related Routings window.
10. Choose OK to close the Add Item Routing window.

The new routing appears at the bottom of the routings list on the tree hierarchy, and it is highlighted. If this is a new routing, the right pane canvas is blank. If the routing was created from a template or a common routing, the right pane displays the inherited routing information.

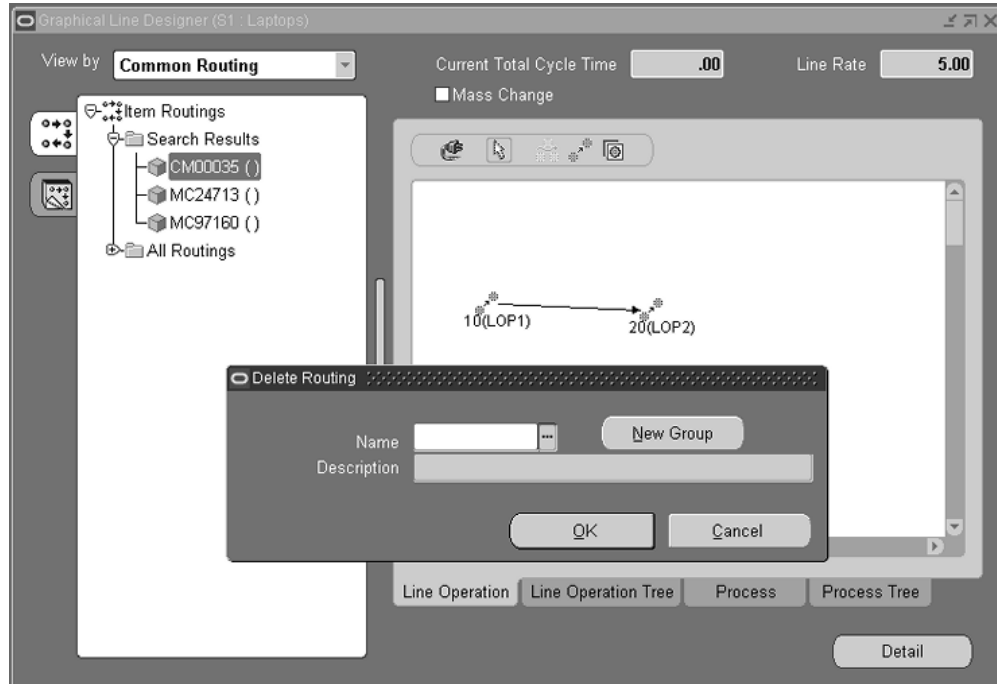
11. Save your work.

Deleting a Routing

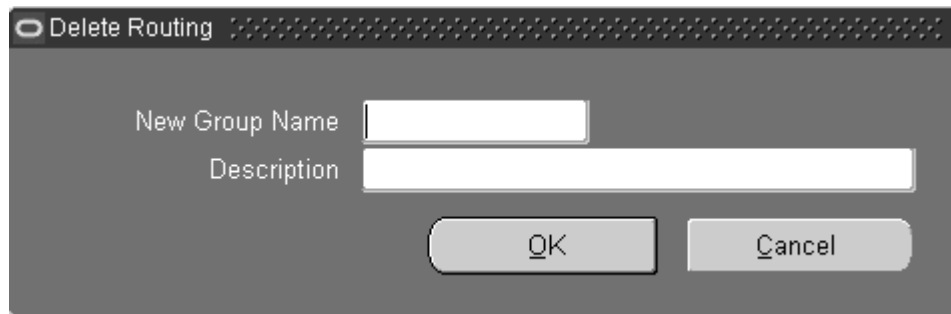
You have the option to delete a routing from the navigation tree by selecting a default delete group, or creating a new one. All operations are deleted when a routing is deleted. Any links to attachments are also severed.

To delete an item routing:

1. Navigate to the Graphical Line Designer Workbench.
2. Select the top node on the tree hierarchy, Item Routings, and right-click with your mouse.
3. Choose Delete from the pop-up menu to display the Delete Routing window.



4. In the Name field, select the name of a Delete Group. The description displays for this group.
5. You also have the option to create a new Delete Group. Choose New Group to display fields for creating a new delete group.



6. Enter a value in the New Group Name and Description fields.
7. Choose OK to save the New Delete Group Name.
8. Choose OK to save your work.
9. Navigate to the Deletion Groups window to delete the routing. See: Deleting Items, Bills, Routings, Components, and Operations, *Oracle Bills of Material User's Guide*

Copying and Common Flow Routings

You can select an item on the navigation tree, and then copy that item's routing to other items—or create routings that are common to it.

Note: If an item already has a primary routing, the copy common transaction is not created.

You can also copy a routing to another flow line. When you select items and copy the routing to another flow line—all relevant standard operations, processes, and events are duplicated. See: Copying a Routing to a Different Flow Line, page 4-23

To copy or common a flow routing:

1. Select the node detail for the item you want to copy to other items, then right-click with your mouse.

The pop-up menu displays.

2. Select Copy/Common Routing To.

The Copy Routing window displays. The Source region displays the Item name and description.

Copy Routing (S1 : Corrola)

Source

Item:

Option

☒ Copy ☐ Common

Current

	Item	Alternate	Description
☐			
☐			
☐			
☐			
☐			
☐			
☐			
☐			
☐			
☐			
☐			
☐			

Retrieve Copy Cancel

3. In the Option region, choose either Copy or Common.
 - Use Copy for copying this item's routing to one or more items. Select Current, or Future and Current to indicate the effective date. The system uses the effective date on the Routings window for the operation code.
 - Use Common to create one or more routings that are common to this routing.
4. In the multi-row Item region, select the items and, if applicable, alternate designators, where the Source item is copied.
5. To add multiple items, choose Retrieve.

Add Item Retrieve window displays. This window restricts your search for items according to your specifications.
6. Select your search criteria.

You can search by any combination of values including item, product family, category, planner, or alternate designator.

If this routing is for a Configure to Order model you can search for items by option class. Choose Common in the Option region, and enable the Retrieve Only Option Classes box. The system retrieves items for all option classes under the model and criteria specified.

7. Choose OK to display the list on the Copy routings window.
8. Choose Copy.

You will receive a confirmation message when the transaction is completed. You can view the copied routing for the item on the hierarchical tree and right pane of the Graphical Line Designer Workbench.

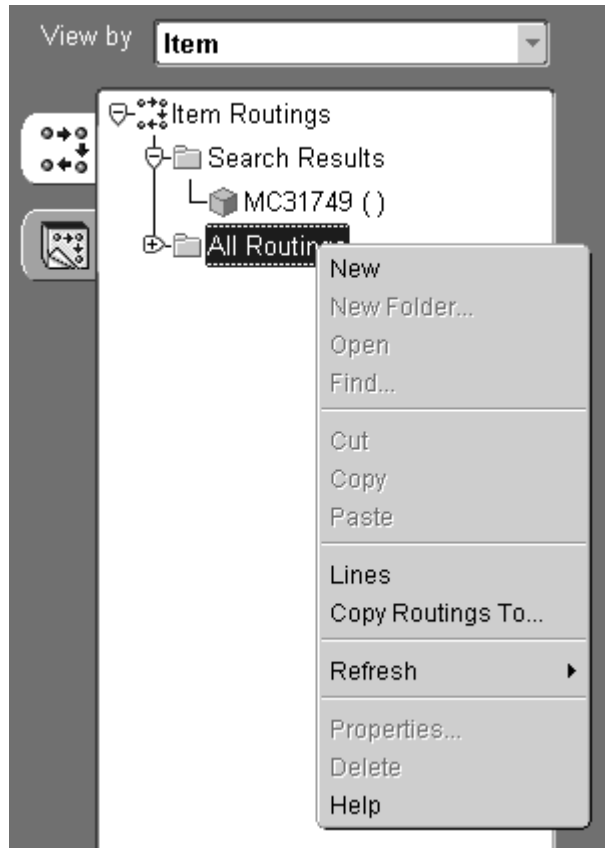
Copying a Routing to a Different Flow Line

You can copy a routing to a different flow line, enabling you to maintain multiple design models. All standard operations, processes, and events on the selected routing are copied to the line. When you perform the copy function:

- Both flow lines must be within the same inventory organization
- You have the option to duplicate the bill of material
- Only current and future versions of the routing and bill of material are copied
- When copying a referenced routing, the original routing referenced is copied

To copy a routing to another flow line:

1. Navigate to the Graphical Line Designer Workbench.
2. Select the All Routings node, and right-click with your mouse to display the pop-up menu.



3. Select Copy Routings To.
The Copy Routings window displays.

Copy Routings (S1 : Desktops)

Destination

Line **Metal Fab** ... Alternate

☒ Copy BOM

Source Criteria

Assembly -

Total Cycle Time -

Line Operation Process

Alternate

4. In the Destination region, select the line where the data will be copied.

5. In the Alternate field, enter the alternate destination designator.

Flow routings names must be unique; the alternate designator is the new identifier for this copied routing. Either select an existing alternate designator from the list of values, or enter a new alternate name.

6. Optionally, you can copy the bill of material information by checking the Copy BOM check box.

7. In the Source Criteria region, specify the assembly to be copied in the Assembly field. Or select a range of criteria for copying.

You can specify a range by assembly name, total product cycle time, alternate designator, or line operation and process.

8. Choose Copy

The selected routing information is copied to the destination line with the given alternate designator and includes standard line operations, processes, events, and event resources.

Viewing Routing Revisions

The Routing Revisions enables you to view all the revisions and changes for a particular routing.

To view the revisions for a specific routing:

1. Select a routing detail on the tree hierarchy
2. From the Tools menu or the pop-up menu, select Routing Revisions.

The Routing Revisions window displays. This window lists all the revisions and information for a specific routing. See: *Creating a Routing Revision, Oracle Bills of Material User's Guide*.

Routing Revisions (S1 : Laptops)

Item: **VD-RESTRICT SI** **VD ITEM TO RESTRICT SUB-INVENTOR**

Routing Revisions

Revision	Effective Date	Time	Implement Date	ECO	[]
A	2001-04-18	09:41:26	2001-04-18		

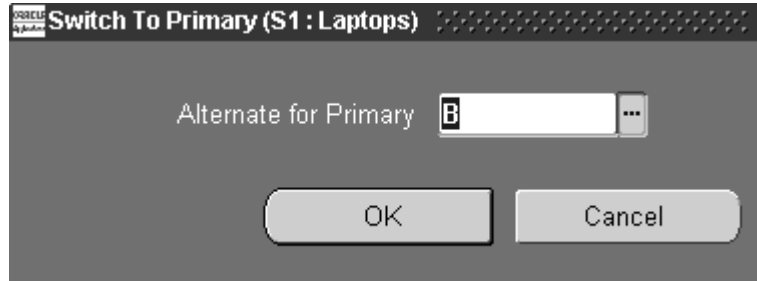
To change your routing from an alternate routing to a primary routing

1. Select a routing on the tree hierarchy that is an alternate routing.

The tree hierarchy displays alternate routings by designating the alternate name in parenthesis on the tree.

2. From the Tools menu or the pop-up menu, select Switch to Primary.

The Switch to Primary window appears with the alternate designation displaying. You can choose this routing, or select another routing from the list of values.



3. Save your work.

See: Switching Between Primary and Alternate Routings, *Oracle Bills of Material User's Guide*

Using the Routing Details Window

You can access the Routing Details window for a particular routing to view specifics or to modify information.

To view or modify routing details:

1. Navigate to the Graphical Line Designer Workbench.
2. Left-click with your mouse to select a node detail on the tree.

The graphical network displays a view of the line operation in the first window tab, Line Operation.

3. Choose Detail.

The Routing Details window displays.

Code	Seq	Department	Yield	Cumulative Yield	Reverse Cumulative Yield
LOP1	10	METAL_FAB		1d00	
LOP6	11	PACKAGING			
LOP2	20	ASSEMBLY		1d00	
LOP3	30	TESTING		1d00	

You can change several of the fields on this window including: Date, Capable to Promise, Mixed Model Map, Priority, Subinventory, Locator, Comment, and Display. See: Creating a Routing Revision, *Oracle Bills of Material User's Guide* .

In the tabbed region, you can also modify Yield, User Labor, or User Machine Times for any line operation, process, or event.

4. Save your work.

The routing calculations are performed automatically.

Creating the Graphical Network

Product Synchronization is the process of defining your product's process flow. This is done through the flow routing where you are able to define processes, events and operations for your flow line. The Graphical Line Designer Workbench provides icons in the toolbar that you can drag and drop onto the right pane canvas to create the product synchronization. This graphical interface allows you to create, delete, cut, and paste line operations, processes, events—and their connections. You can also add new item routings, see: Adding a New Item Routing , page 4-17

To add a new line operation on the Graphical Line Designer Workbench:

1. Select the Line Operation tab to display this window in the detail pane.

2. Select the Operation icon from the toolbar.

This icon is only available in the Line Operation window of the workbench.

3. Left-click with your mouse on the right pane, in the section of the canvas where you want to locate this line operation.

The Line Operation window displays.

4. Enter the Display Sequence and Code values.

You have the option of entering a new code in the Code field, this creates a new standard line operation code which is added to the routing. Or you can select a code from the list of values, which adds the existing standard code to the routing.

Line Operation (S1 : Laptops)

Display Sequence: 10 Code: LOP1

Description: Shell Fabrication

Department: METAL_FAB

Subinventory: [] Locator: []

Routing Specific Details

☒ Referenced

Yield: [] Net Planning Percent: 100.00

Cumulative Yield: 1.00 Reverse Cumulative Yield: 1.00

User Machine Time: [] Machine Process Efficiency: 0

User Labor Time: [] Labor Process Efficiency: 0

Elapsed User Time: [] Total Process Efficiency: 0

OK Cancel

5. Subinventory and Locator values default from the referenced standard operation. You have the option to enter values if there is no referenced standard operation.

6. Enter values in the Routing Specific Details region.

If the Referenced check box is checked, the information displayed for the Descriptions field is referenced from the standard line operation. To edit, deselect the referenced check box.

Information displays for Yield, Cumulative Yield, User Machine Time, Elapsed User Time, Net Planning Percent, Cumulative Yield, Reverse Cumulative Yield, and Process Efficiency values for machine and labor

See: Operation Yields Calculations, page 3-15

You can access the Line Operation window through the pop up menu if you want to modify this data for a standard line operation.

7. Choose OK to close the window and perform routing calculations.

The new line operation displays on the canvas.

8. Save your work.

To add a new process on the Graphical Line Designer Workbench:

1. Select the Process tab to display this window in the right pane.
2. Select the Process icon from the toolbar.

This icon is only available in the Process window of the workbench.

3. Left-click with your mouse on the right pane, in the section of the canvas where you want to locate this process.

The Process window displays.

The screenshot shows the 'Process (S1: Laptops)' window. At the top, there are fields for 'Display Sequence' (value: 10), 'Code' (value: FAB1), 'Description' (value: Shearing), and 'Department' (value: METAL_FAB). Below these is a 'Routing Specific Details' section. It contains a 'Referenced' checkbox which is checked. Under this checkbox, there are five fields: 'Yield' (empty), 'Cumulative Yield' (value: 1d00), 'User Machine Time' (value: d00), 'Elapsed User Time' (value: d00), and 'Net Planning Percent' (value: 100d00). To the right of the 'Referenced' checkbox, there are three more fields: 'Reverse Cumulative Yield' (value: 1d00) and 'User Labor Time' (value: d00). At the bottom right of the window are 'OK' and 'Cancel' buttons.

The information in the top region of the window is defaulted from where you positioned the icon on the workbench. You can change Display Sequence and Description information.

4. Enter values in the Routing Specific Details region.

If the Referenced check box is checked, the information displayed for the Description field is referenced from the standard process. To edit, deselect the referenced check box.

For information on Net Planning Percent, Cumulative Yield, and Reverse Cumulative Yield, see Operation Yields Calculations, page 3-15

You can access the Standard Processes window through the pop up menu if you want to modify this data for a standard process.

5. Choose OK to close the window and perform routing calculations.

The new process displays on the canvas.

6. Save your work.

To add a new event on the Graphical Line Designer Workbench:

1. Select either the Line Operation or Process tab to display one of these windows in the right pane.

2. Select the Event icon from the toolbar.

This icon is only available in the Line Operation and Process windows of the workbench.

3. Left-click with your mouse on an existing line operation or process in the right pane.

The Event window displays.

Event (S1 : Corrola)

Display Sequence Code

Description

Department **PAINTING** []

Process Line Operation **PNT**

Routing Specific Details

☒ Referenced ☒ Option Dependent

Yield Net Planning Percent

Cumulative Yield Reverse Cumulative Yield

Effectivity Date **28-JUL-2004 15:44:56** Disable Date

☐ Critical To Quality ☐ Value Added

☐ Include Resource(s) in Cost Rollup

Seq	Resource	UOM	Basis	Usage	Available 24 Hours	Schedule
1			Item		☒	Yes
					☐	
					☐	
					☐	

OK Cancel

4. Enter values in the Routing Specific Details region including Yield, Effectivity Date, and Disable Date.

If the Referenced check box is checked, the information displayed for the Description field is referenced from the standard event. To edit, deselect the referenced check box.

For information on Net Planning Percent, Cumulative Yield, and Reverse

Cumulative Yield, see Operation Yields Calculations, page 3-15

You can access the Standard Events window through the pop up menu if you want to modify this data for a standard process.

5. Use the Option Dependent box to indicate if this event is dependent on the choice of an optional component.
6. Enable the Critical to Quality box if this is a quality related task.
7. Enable the Value Added box if this event adds value to the Process. Resource Efficiency calculations for the process or line operation consider this setting.
8. The Include Resource in Cost Rollup box is used to specify if the event is included in the cost rollup. This flag can be updated here, or from the Set Resource Cost Rollup window. See: Specifying Cost Rollup Resource Information, page 4-49
9. Optionally, indicate whether the resource is available 24 hours a day. You cannot assign shifts to a resource that is available 24 hours a day.
10. Choose OK to close the window and perform routing calculations.
The new process displays on the canvas.
11. Save your work.

Defining a Network Connection

The Network Connection window is used when adding or modifying a connection on the Graphical Line Designer. This window is accessed through the pop up menu on the right pane.

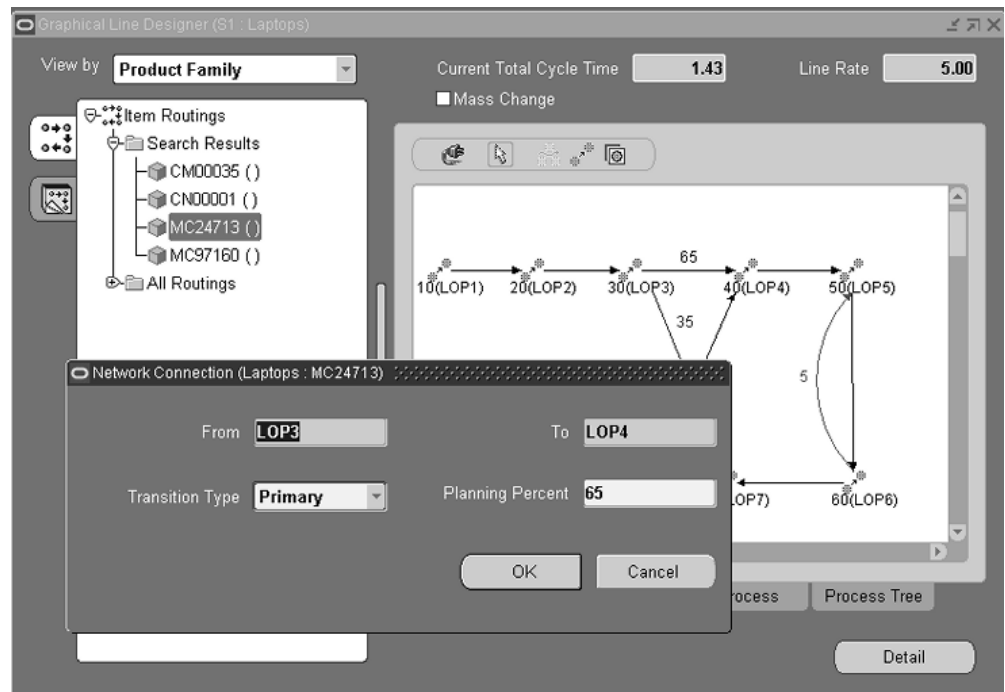
Note: Network planning percentage connections under 100 percent display on the detail pane of the graphical network.

To add a network connection on the Graphical Line Designer Workbench:

1. Navigate either to the Line Operation network or Process network window on the right pane of the Line Designer.
2. Select the Connections icon from the toolbar.
This icon is only available in the Line Operation and Process windows of the workbench.
3. Left-click and hold with your mouse on the right pane, on the process or line operations where you want to start this connection—and drag the line to the

process or line operation you want to it to join.

The Network Connection window displays. In the Operation From and To fields, the operation codes that you connected on the canvas automatically display.



4. You can use the defaulted values in the Transition Type and Planning Percent fields, or enter other values.

Transition type values are primary alternate, and rework.

The planning percent is the percentage of product that follows this network path. If the planning percent for the network connection is less than 100 percent, the value displays on the graphical network detail pane.

5. Choose OK to close the window and perform routing calculations.

The new connection displays on the canvas.

6. Save your work.

Modifying the Line Operation Tree and Process Tree

The Line Operation Tree and Process Tree windows display your product synchronizations for your routing network in a tree hierarchy format. The Graphical Line Designer Workbench enables you to modify the routing network tree—you can create, delete, cut, and paste line operations, processes, and events.

To modify components on the Line Operation Tree and Process Tree:

1. Select either the Line Operation Tree or Process Tree tab to display the hierarchical view of the routing network.

2. Left-click with your mouse to select a node detail on the tree hierarchy.

In the right pane the top level of the routing for this item displays.

3. Expand the items on the routing to display the operations or processes in the routing.

The code and description are listed for each line operation and event.

You can change the tree format to one of three styles. The toolbar has an option that enables you to see the times associated with each line operation and event. See: Workbench Toolbars, page 4-5

The last node at the bottom of the tree is named Unassigned Events. This is a placeholder for all the events that are on the Routing but are not assigned to any line operation or process. For example, an event may be unassigned because you defined processes first, and have not assigned the events to the operations.

4. Select a detail on the tree and right-click to display the pop up menu.

The pop up menu displays, depending on the conditions, several options: Cut, Paste, Delete, Properties, and Standard.

5. Select an event and drag it to another line operation.

Routing calculations are automatically performed.

6. Save your work.

Modifying Line Operations, Processes, Events, and Connections

The Line Operation window is used when adding or modifying a line operation on the Graphical Line Designer. It contains routing specific details. This window is accessed through the pop up menu on the right pane.

To modify a line operation on the Graphical Line Designer Workbench:

1. Navigate to the Line Operation Tree or Line Operation network window on the right pane of the Line Designer.

2. Select the line operation you want to modify on the right pane canvas.

If you are adding a new line operation, see: Creating the Graphical Network, page 4-28

3. Right-click on the operation icon in the right pane to display the pop up menu.
See: Tools Menu and Pop-up Menus, page 4-6
4. Select Properties.

The Line Operation window displays. The information is defaulted from the icon on the workbench. You can change Display Sequence, Yield, User Machine Time, User Labor Time information. To change the description, uncheck the Referenced box. See: Creating the Graphical Network, page 4-28

5. Choose OK to close the window and perform routing calculations.
6. Save your work.

The Process window is used when adding or modifying a process on the Graphical Line Designer. It contains routing specific details. This window is accessed through the pop up menu on the right pane.

To modify a process on the Graphical Line Designer Workbench:

1. Navigate to the Process Tree or Process network window on the right pane of the Line Designer.
2. Select the process you want to modify on the right pane canvas.
If you are adding a new process, see: Creating the Graphical Network, page 4-28
3. Right-click on the process icon in the right pane to display the pop up menu.
See: Tools Menu and Pop-up Menus, page 4-6
4. Select Properties.

Process (S1: Laptops)

Display Sequence: 10 Code: FAB1

Description: Shearing

Department: METAL_FAB

Routing Specific Details

☒ Referenced

Yield: Net Planning Percent: 100d00

Cumulative Yield: 1d00 Reverse Cumulative Yield: 1d00

User Machine Time: d00 User Labor Time: d00

Elapsed User Time: d00

OK Cancel

The Process window displays. The information is defaulted from the icon on the workbench. You can change Display Sequence, Yield, User Machine Time, User Labor Time information. To change the description, uncheck the Referenced box. See: Creating the Graphical Network, page 4-28

5. Choose OK to close the window and perform routing calculations.
6. Save your work.

The Event window is used when adding or creating an event on the Graphical Line Designer. It contains routing specific details. This window is accessed through the pop up menu on the right pane.

To modify an event on the Graphical Line Designer Workbench:

1. Navigate either to the Line Operation Tree or Process Tree window on the right pane of the Line Designer.
2. Select the event you want to modify on the right pane canvas.
If you are adding a new event, see: Creating the Graphical Network, page 4-28
3. Right-click on the event in the right pane to display the pop up menu.
See: Tools Menu and Pop-up Menus, page 4-6
4. Select Properties.

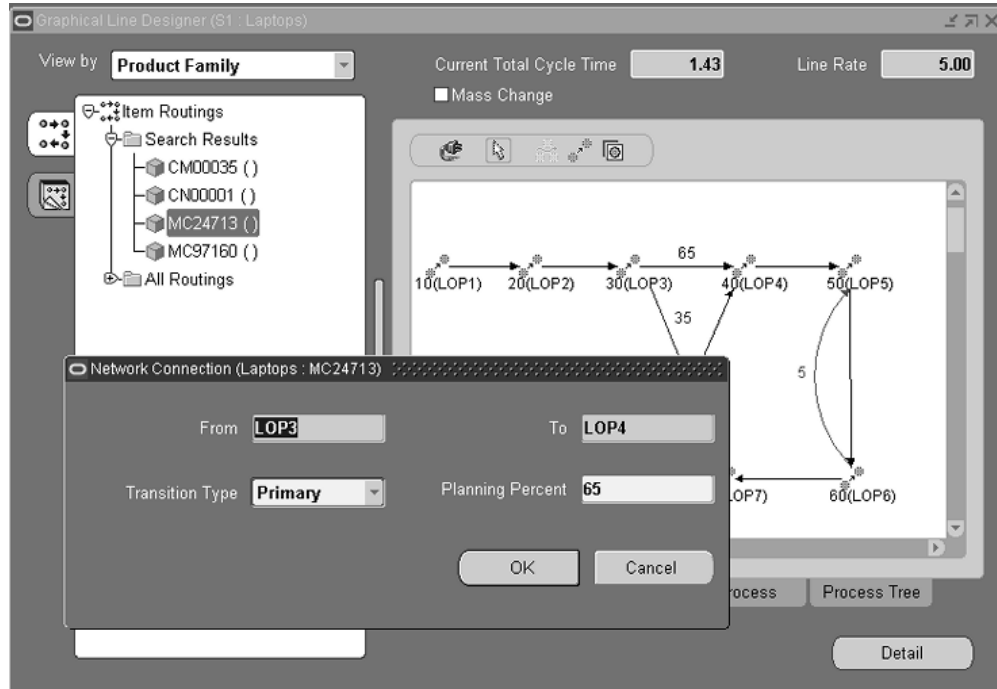
The Event window displays. The information is defaulted from the icon on the workbench. You can change Display Sequence, Effectivity Date, and Disable Date information. If you uncheck the Referenced box, you can change description and resource information. See: Creating the Graphical Network, page 4-28

5. Choose OK to close the window and perform routing calculations.
6. Save your work.

The Network Connection window is used when adding or creating a connection on the Graphical Line Designer. This window is accessed through the pop up menu on the right pane.

To modify a network connection on the Graphical Line Designer Workbench:

1. Navigate either to the Line Operation network or Process network window on the right pane of the Line Designer.
2. Select the connection you want to modify on the right pane canvas.
3. Right-click on the icon in the right pane to display the pop up menu. See: Tools Menu and Pop-up Menus, page 4-6
4. Select Properties.



The Network Connection window displays and you can change the values, see: Defining a Network Connection, page 4-32

Note: Network planning percentage connections under 100 percent display on the detail pane of the graphical network.

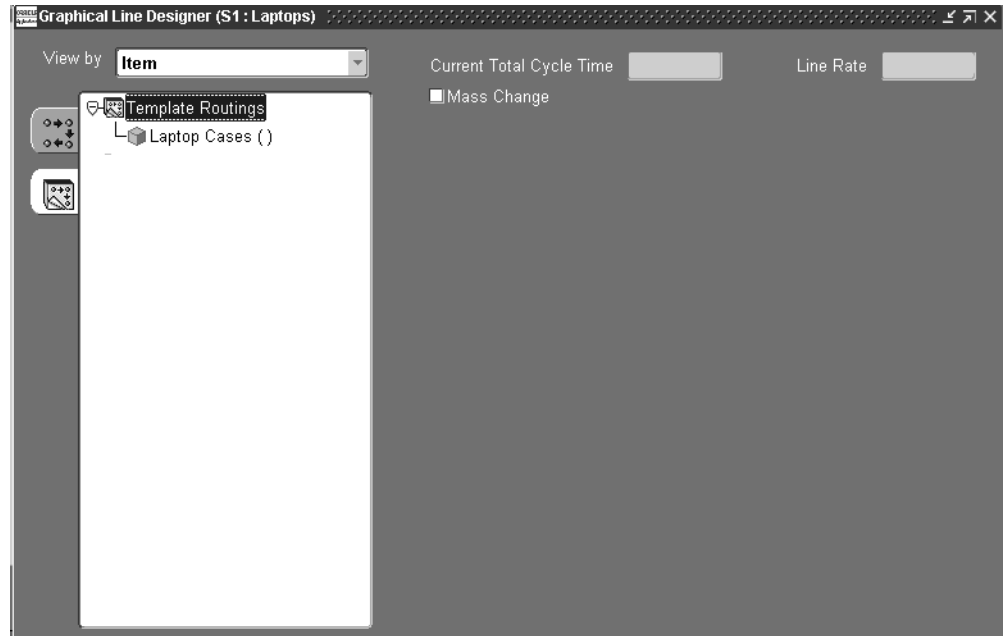
5. Save your work.

Creating Routings in the Template Routings View

The Template Routings tree tab provides access to Product Family Routings that can be used as templates for copying elements to an item routing you are creating. You can define a routing for a product family the same way as performed in the Item Routings view of the workbench, and can perform all the same functions as in the routings region.

Using product family routings as templates:

1. Choose the Template Routings tree tab.



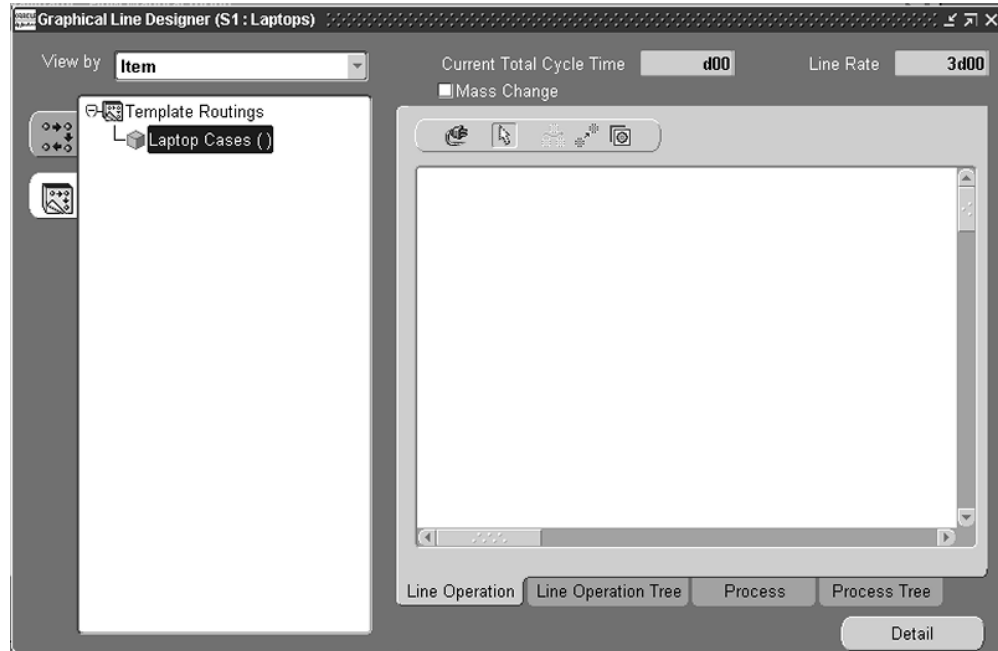
2. Select Template Routings node detail on the tree, and left-click with your mouse.

The Template Routings tab displays the routings for all Product Families assigned to the line designated. In the initial view the right pane is blank.

3. Select New from the popup menu.

Or you can create a new product family routing. In the Add Item Routing window, use the option to create an entirely new routing. See: Adding a New Item Routing, page 4-17

After you select a routing, a blank Line Operation window displays in the right pane. This blank window is the canvas used as a template.



4. Use the toolbar to add icons to create a routing network. See: Creating the Graphical Network, page 4-28
5. Save your work.
This routing is now available to use when creating a new routing, see: Adding a New Item Routing , page 4-17

Mass Changes on Item Routings

You can simultaneously make changes, edits, and deletions to multiple flow routings. Work performed in the graphical pane can be updated for all items on the line, or in the product family. To enable this functionality, check the Mass Change check box on the graphical pane of the Graphical Line Designer Workbench. Mass change functionality is only available in the Line Operation and Process windows of the Graphical Line Designer Workbench:

- In the View by Item mode, you can mass change all item routings on that production line.
- In the View by Product Family mode or the Template Routing tree tab view, you can mass change routings that belong to the same product family.
- In the View by Common Routing mode or the Template Routing tree tab view, you can mass change routings that belong to the same common routing.

Changes are still created by selecting an icon or menu item, and making the changes on

the related line operation, process, event, or connection property window. However, in the mass change mode, you can also:

- Populate changes to a range of other item routings after you save the transaction
- Update operation information for user machine time, user labor time, and yield
- Revise outgoing network connections from the node for planning percentages

To make mass changes to item routings:

1. Navigate to the Graphical Line Designer Workbench.
2. Check the Mass Change check box on the graphical pane.
3. Select the line operation, process, event, or connection icon, and make the changes in the related window. See: Creating the Graphical Network, page 4-28
4. Choose OK on the related line operation, process, event, or connection window to save the transaction and display the Apply Changes window.

When first viewed in a session, this window does not display data. The applied values from your query default for the subsequent operation changes.

You can also access this window from the Tools menu by selecting Item Selection List.

	Item	Alternate	Description
☒	MC31749		Standard Steel Case
☒	MC31750		Heavy Steel Case
☐			
☐			
☐			
☐			
☐			
☐			
☐			
☐			
☐			

Retrieve Clear Apply Cancel

5. Enter the specific items you want to change, and any applicable alternate designators.
6. You can also search for item routings by choosing Retrieve to display the Retrieve Routings window.

Item -

Product Family

Category

Planner

☐ Alternate Designator

Alternate

☐ Retrieve Only Applicable Item Routings

Find Cancel

7. Select search criteria in Retrieve Routings window.
You can search by item or range of items, product family, category, planner, and

alternate designator.

8. Check the Retrieve Only Applicable Items check box to display only eligible item routings.

Eligible item routings are in context to the changes:

- For event changes or deletions, routings with line operations or processes referencing that specific event. When adding the event, routings with line operations or processes that do not already have that event.
- For line operation and process changes, routings with those specific line operations and processes.

If you do not select this check box, and all search criteria fields are empty—when you choose Find all items routings display.

Note: Even though you can display all routings, updates occur only on applicable routings.

9. Choose Find to display your search on the Apply Changes window.

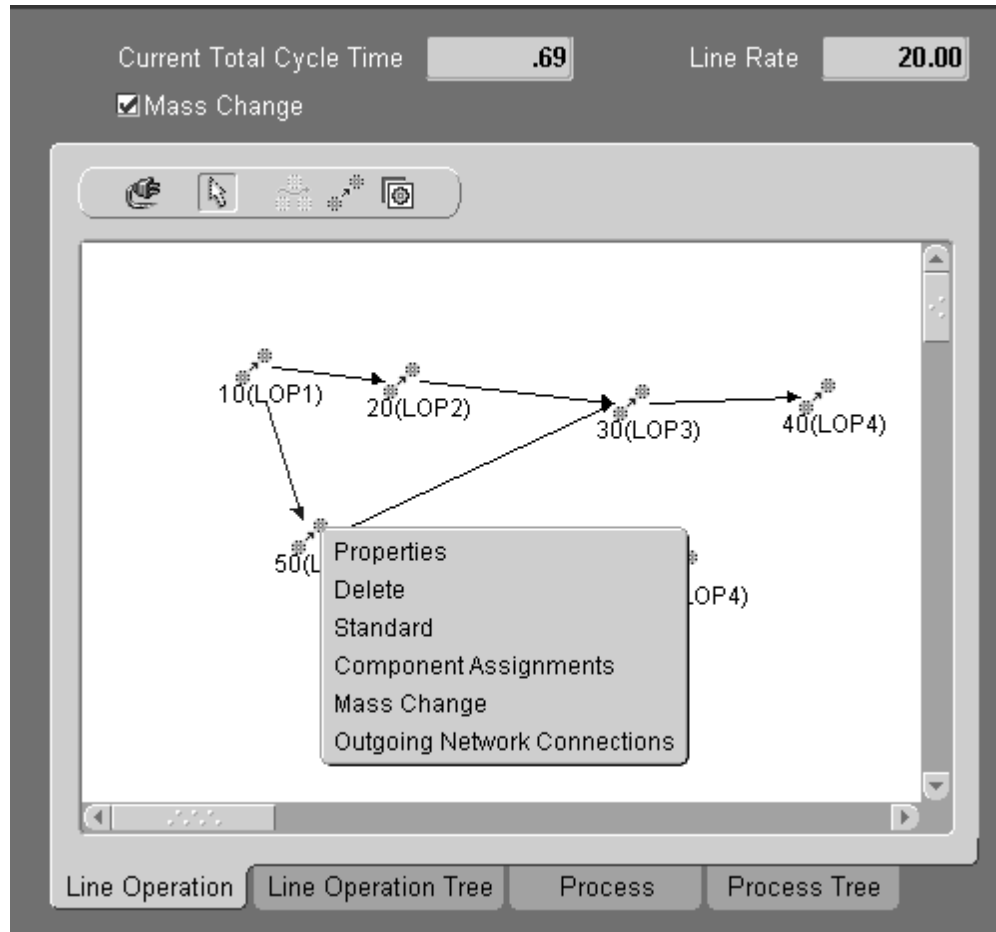
Uncheck the items you do not want included in the mass update.

10. Choose Apply to create the mass update transaction.

You can view the routing changes on the graphical pane of the Graphical Line

To update operation user time and yield information:

1. Navigate to the Graphical Line Designer Workbench, and check the Mass Change check box.
2. Select an operation, and right-click with your mouse to display the pop-up menu.



3. Select Mass Change to display the Operations window.

Operations (S1 : Desktops)

Code **LP50** **Line Operation 50**

Department **ASSEMBLY**

User

	Machine Time	User Labor Time	Yield
Current Value	1.26	.80	.99

Change By

Value			
+/- Percent			
+/- Value			

OK Cancel

4. In the Change By region, enter applicable changes for the Value fields.

You can enter a new value in the Value field for User Machine Time, User Labor Time, and Yield.

Alternately, for the User Machine Time and User Labor Time columns, you can enter either a value or percentage to add or subtract from the current value. For example:

If the Current Value for User Machine Time on operation LP50 is 1.26 And you enter .4 in the +/- Value field The system adds .4 to 1.26, to equal 1.66

When you apply your change, the system updates the User Machine Time on operation LP50 to 1.66—in all instances where it occurs on the selection list of item routings.

5. Choose OK to display the Apply Changes window.
6. Select your edit list criteria.

You can choose Retrieve to display the Retrieve Routings window for your search criteria; and use the Retrieve Only Applicable Items check box for eligible items. Choose Find to display your search on the Apply Changes window, and uncheck any items you do not want included in the mass update.

7. Choose Apply to update your changes.

Eligible items are those with values in the applicable fields for the selected operation.

Note: Even though you can display all routings, updates occur only

on applicable routings.

To update outgoing network connections:

1. Navigate to the Graphical Line Designer Workbench, and check the Mass Change check box.
2. Select a node on the graphical pane, and right-click with your mouse to display the pop-up menu.
3. Select Outgoing Network Connections to display the Network Connections window.

To Operation	Transition Type	Planning Percent
LOP2	Primary	85
LP50	Alternate	15

OK Cancel

4. Enter changes in the Planning Percent fields.

Network planning percentage connections under 100 percent display on the detail pane of the graphical network. The total planning percent must equal 100 in order to update item routings.

5. Choose OK to display the Apply Changes window.
6. Select your edit list criteria.

You can choose Retrieve to display the Retrieve Routings window for your search criteria; and use the Retrieve Only Applicable Items check box for eligible items.

Choose Find to display your search on the Apply Changes window, and uncheck any items you do not want included in the mass update.

7. Choose Apply to update your changes.

Eligible item routings are those with same the geometry match including:

- The same specified network connection
- The same outgoing network connections defined for the To Operation
- Transition type and destination must match the current item routing
- If you check the Match Planning Percentage check box, the planning percents must match and not exceed 100

Note: Even though you can display all routings, updates occur only on applicable routings.

Component Assignments on the Bill of Material

You can associate the components on the bill of material with the routing events where the components are consumed. This association links one or more sub-inventory and locator combinations to line operations—and then automatically updates the material control portion of the parent bill of material.

These transactions are performed on the Update Components window. This window is accessed from the Graphical Line Designer Workbench by selecting the item routing in the hierarchical tree; or the line operation, process, or event in the graphical pane. A bill of material must exist for the selected routing in order to display this window.

To assign components to a bill of material:

1. Navigate to the Graphical Line Designer Workbench.
2. Select either an item routing on the hierarchical tree; or a line operation, process, or event in the graphical pane.
3. Right-click with your mouse to display the pop-up menu, and select Component Assignments.

The Update Component window displays.

Information defaults on the Components region of the window depending on your selection. For example, if you access the window from the line operation or process—the line operation and any associated events display. If you access the window from the event—the line operation and all instances of this event display.

And if you select the item routing, all lines and events display.

Update Components (S1 : Desktops : MC31749)

Display Option: **Assigned**

Line Operation: -

Event: -

Component: - **Find**

Components

Event	Line Operation	Item Seq	Component	Subinventory	Locator	Mass Assignment Selection
50 (AS31)	20 (LOP2)	100	CM23596	KB4		☐
70 (TST1)	10 (LOP1)	10	CM23596	KB4	KB4.1.1..	☐
70 (TST1)	10 (LOP1)	20	CM23597	KB4		☐
70 (TST1)	10 (LOP1)	30	CM31542	KB1	1.1.1..	☐
70 (TST1)	10 (LOP1)	40	SC19312	KB1	1.1.1..	☐
70 (TST1)	10 (LOP1)	50	CM93501	KB1	1.1.1..	☐
70 (TST1)	10 (LOP1)	50	CM96713	KB1	1.1.1..	☐

☒ Mass Assignments

Event: Subinventory: Locator:

Select All **Unselect All** **Assign**

OK **Cancel**

4. You can also use the upper region of the window to enter criteria for a new query.
 - In the Display Option field, choose the display mode, within the context of other specified parameters, for the Components region of the window. Your display choices are:
 - Assigned—retrieves components assigned to any event for this specified routing.
 - Unassigned—retrieves components that are not currently assigned to any event in this specified routing. This mode enables you to assign components to the routing.
 - All—retrieves all components both assigned and unassigned.
 - Optionally select a value, or range of values in the Line Operation field.
 - Optionally select a value in the Event field.
 - Optionally select an item number, or range of components in the Components field.
5. Choose Find to displays the results of your query. on the Components region.

6. Optionally, you can manually change the Event, Subinventory, and Locator values for any component listed.

The component attributes are displayed for event, line operation, item sequence, component name, subinventory location, and locator.

7. Check the Mass Assignments check box if you want update the records in your query in mass update mode.

Several different mass update transactions are available including:

- Activates the Mass Assignment Selection check boxes for each component on the Components region. Select components by checking the Mass Assignment Selection check boxes in individual rows. This field is disabled when the Mass Assignments check box is not checked.
 - Choose Select All to activate all the records on the window.
 - Choose Unselect All to deactivate all the records on the window.
8. The Mass Assignments check box also enables the Mass Assignments region on the window. You can enter values in the Event, Subinventory, and Locator fields.
 9. Check the records in the Components region you want to update.
 10. Choose Assign to update the applicable records and fields on the Components region.
 11. Choose OK to update the component assignment to the bill of material for the item routing selected

All specified components on the window are updated on the item routings's bill of material. The Graphical Line Designer Workbench displays again.

If you selected Mass Change on the Graphical Line Designer Workbench—the Apply Changes window displays. See: Mass Changes on Item Routings, page 4-40 on page 4-43
 12. Save your work.
 13. Select Bills of Material from the Tools menu to view the update in the Bills of Material window.

Specifying Cost Rollup Resource Information

The Set Resource Cost Rollup window is used to specify if an event's resources are included in the cost rollup. If you include an event in the rollup, the checkbox on the Event's window is enabled. See: Creating the Graphical Network, page 4-28 This window provides a method for creating a mass change for events on a particular item

routing.

To set resources for cost rollup:

1. Navigate to the Graphical Line Designer Workbench.
2. Select an item routing on the hierarchical tree, and choose Set Resource Cost Rollup from the Tools menu.

The Set Resource Cost Rollup window displays, showing all the events and line operations for this routing.

[illegible]

3. Enable the Include box for the event and line operations you want included in the cost rollup. Use the Exclude box for events you do not want calculated in the cost rollup.
4. Choose OK.
5. Save your work.

Related Topics

Rolling Up Supply Chain Costs, *Oracle Cost Management User's Guide*

Mixed Model Map Workbench

This chapter describes the features of the Mixed Model Map Workbench, a graphical interface used to balance your production lines.

This chapter covers the following topics:

- Overview of the Mixed Model Map Workbench
- Viewing, Creating, and Updating Plans
- Viewing Plan Production Line and Line Operation Information
- Viewing Plan Resource Information
- Resource Assignment
- In Process Kanban Assignment
- Filtering and Sorting Information

Overview of the Mixed Model Map Workbench

The Mixed Model Map Workbench is a graphical interface for monitoring line balancing—the process of defining production lines by including the products made, operations performed, and resources and materials consumed at each step. It is used to balance a line against specified demand—and calculate process volume, machine and labor requirements, TAKT times, and In-Process Kanban (IPK) requirements. The Mixed Model Map Workbench provides:

- A tree navigation graphical workbench displaying information at different detail levels for plans, lines, operations, resources.
- Detail panes for each node selected on the navigation tree.
- The ability to analyze and balance lines, or run simulations, considering both the assigned resources and the In Process Kanbans (IPK) available.
- Both a line operation and process view enabling the assignment and analysis of

resource demand across multiple production lines.

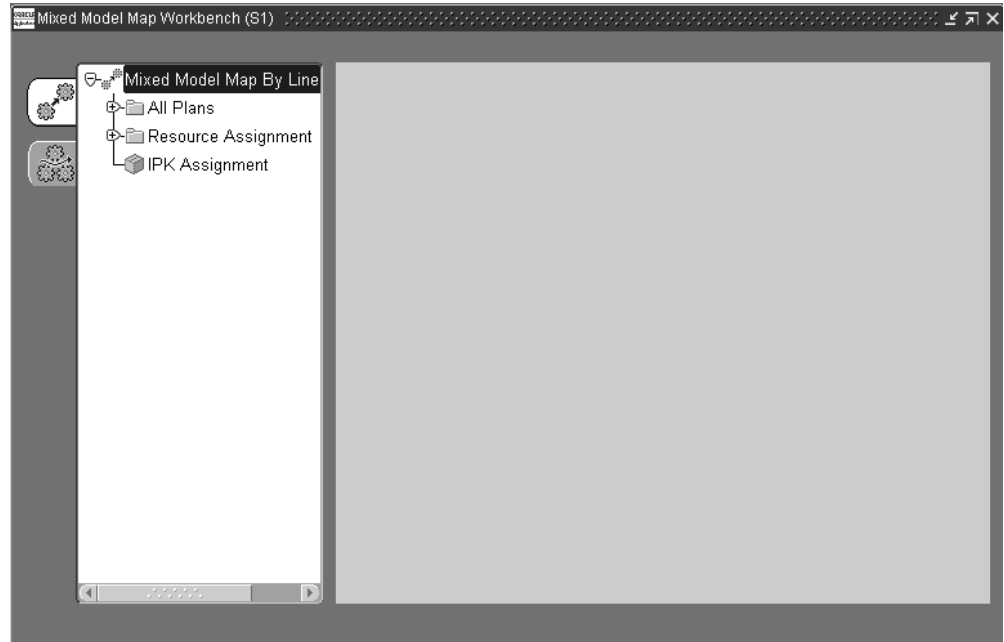
- Support for multiple plans for simulation studies.
- Identification of resource usage and problem operations.
- The ability to use unscheduled planned orders and sales orders as a type of demand.
- An engine using demand as input to calculate the resource/IPK usage as needed for each resource, at each operation.
- Display and calculation of resource efficiency and resource total units for a plan.
- Display of line demand and line capacity in summary view.
- Update of item variable lead time using line TAKT.

The workbench displays a navigation tree, and a detail pane for each node selected on the tree. There are two tree tables enabling you to toggle between the Line Operation and Process regions. Both regions display a line and resource view:

- The Line Operation region displays operations specific to the line, as derived and defined during line balancing. This region shows the resulting line information after mixed model map calculations.
- The Process region displays the activities performed in defined processes on the line. This region shows the result of mixed model map runs for all the processes.

The Navigation tree contains three nodes—when a node is selected the corresponding detail pane displays:

- All Plans—All the plans defined in the organization selected.
- Resource Assignment—The total available units from applicable departments, and total assigned units to operations for a resource.
- IPK Assignment—In Process Kanbans assigned, a summary of all the IPK master assignments.



Note: Mixed Model Map refers to Mixed Model Map Workbench.

Related Topics

Viewing, Creating, and Updating Plans, page 5-3

Viewing Plan Production Line and Line Operation Information, page 5-6

Viewing Plan Resource Information , page 5-13

Resource Assignment, page 5-17

In Process Kanban Assignment, page 5-18

Viewing, Creating, and Updating Plans

The Plans node displays all the plans in the selected organization. This enables you to view multiple plans using different line balancing techniques. You can view the details of a specific plan by selecting the node in the navigation tree, the data displays in the canvas.

To view or update a plan:

1. Select the All Plans node, and expand the navigation tree to display the plans.
2. Select a plan to display the pane for this plan's definition.

3. Update the fields you want to change.

4. Choose Calculate or Replan

If you choose Calculate, the mixed model map calculation is performed based on the parameters specified for this plan. Calculations use master assigned values for resources and IPKs. A confirmation displays if the process is successful.

If you choose Replan, the plan assigned values for resource and IPKs are used for recalculation, along with the parameters specified in this plan. A confirmation displays if the process is successful.

5. Save your work.

To create a new plan:

1. Select the All Plans node, and right click to display the Menu.
2. Choose New to display the pane for entering your plan definition.
3. Enter a unique name in the Plan Code field, and optionally any description for this plan.
4. Optionally, you can select values in the Product Family, a group of products with similar characteristics, and product Line fields for this plan.
5. In the Demand Type field, select a value representing the source of this demand. Your choices are:

- Forecast—an estimate of future demand
 - MDS—master demand schedule, the anticipated forecasted ship schedule
 - MPS—master production schedule, the anticipated forecasted build schedule
 - Actual Production—demand generated from existing Flow schedules
 - Sales Orders
 - Planned Orders
6. If the value in the Demand Type fields is Forecast, MDS, or MPS, you are required to select a Demand Name value.
 7. The Start and End Date values are derived from the forecasted demand, and can be changed.

Demand Days for this Demand Type display. Demand days are the number of working days in between specified demand start and end date.
 8. The Hours Per Day field displays the working hours for this line. This value can be changed.
 9. You can enter a Boost Percent value. This specifies an increase percentage of the demand in the plan.
 10. Choose a Calculation Option. This region is used to select how operational cycle times, resource requests, and IPK values are calculated. Your choices are:
 - No IPK—resource requirements are calculated using the IPK value as zero on the line operation.
 - Based on one resource—the IPK needed on the line operation is calculated assuming one resource at each operation.
 - Based on assigned IPK—resources are calculated using the assigned IPK at the line operation.
 - Based on assigned resource—the IPK needed on the line operation is calculated using all the assigned resources.
 11. The Display Options region is used for selecting viewing options for the generated mixed model map. Indicate how you want to display time in the Time Uom field. Your choices are Hour, Minute, or Second.
 12. In the View By field, indicate the sort option, on the navigation tree, for this plan's process or line operation display. Your choices are Display Sequence, Operation

Code, or Description.

13. The Precision field is the decimal setting for your calculation values. Your choices are 1,2,3, or 4 decimal places.
14. Choose Calculate to run the mixed model map calculation based on the parameters specified for this plan.

Calculations use master assigned values for resources and IPKs. A confirmation displays if the process is successful.
15. Optionally, choose Replan to use the plan assigned values for resource and IPKs in recalculation, along with the parameters specified in this plan. A confirmation displays if the process is successful.
16. Optionally, choose Update Assignment to copy the assigned resources and IPKs in this plan to the master resource and IPK assignments. A confirmation displays if the process is successful.
17. Save your work.

After you create or update information in this window, you can perform three processes.

Viewing Plan Production Line and Line Operation Information

Production line information is available on the workbench in summary, line detail and line operation, and graphical views. You can use the summary view to determine if there is an imbalance in the line, resource overload, or IPK assignment. You can then view the details of a specific line and line operation.

Line Operation views enable you to access resource and labor information.

The production line views enable you to copy the values that are required in the Needed fields to the Assigned fields. In the navigation tree, right click to display the menu, and choose. Select Copy Needed To Assigned.

You can also update item variable lead times by copying the line TAKT time. In the navigation tree, right click to display the menu, and choose Update Items Variable Lead Time. This function copies the value of the plan's:

- (Line TAKT / Line hours per day) for each line to the corresponding Items' Variable Lead Time

Related Topics

Line Summary View, page 5-7

Line Detail - Summary View, page 5-8

Color Representation on the Mixed Model Map Workbench, page C-5

The Line Summary pane displays fields for Line name, Hours, Daily Demand, line TAKT time, Assigned TAKT time, Daily Capacity—and two regions—Resource Percentage and IPK Percentage. These regions display maximum Over and Under values for the line.

[illegible]

Color coding provides a visual display of these comparisons. See: Color Representation on the Mixed Model Map Workbench, page C-5

In the Resource Percentage region, Over signifies a resource overload. That is, the resource needed is greater than the resource assigned in the line. Under signifies that the resource is not sufficiently used.

In the IPK Percentage region, Over means that there is IPK overload. That is, the IPK needed is greater than the IPK assigned in any of the line operations or processes. Under signifies that IPK are not sufficiently used.

Detail information for these values and modifications created here is available in tabbed regions for Summary, IPK Table, and Graph.

Line Detail - Summary View

The Line Detail panes are view only windows. In the Line Detail view, the Summary tabbed region shows line operation detail information in a table format. The line name, line hours, and line TAKT display.

The screenshot displays the Mixed Model Map Workbench (S1) interface. On the left, a tree view shows the hierarchy: All Plans, HW-TEST1, LRMMM1, TS1, Lines, Corrola, and Resources. The Corrola line is selected. The main panel shows the Line Detail for Corrola Line. It includes fields for Line (Corrola), Line Hours (10.0), and Line TAKT (8.0). Below these is a table of operations with columns for Operation, OP TAKT, and OP TAKT As Assigned. The table lists four operations: ASSY (Assembly), PNT (Painting), PTST (Pre Testing), and TST (Testing). The ASSY, PNT, and TST operations have an OP TAKT of 7.8, while PTST has an OP TAKT of 133.3. The OP TAKT As Assigned column shows 7.8 for ASSY, PNT, and TST, and 0.0 for PTST. A Filter/Sort button is located to the right of the table. At the bottom, there are tabs for Summary, Resource/IPK Table, and Graph. The Summary tab is currently selected.

Operation	OP TAKT	OP TAKT As Assigned
ASSY : Assembly Operation :	7.8	7.8
PNT : Painting Operation :	7.8	7.8
PTST : Pre Testing Operation :	133.3	0.0
TST : Testing Operation :	8.0	8.0

The table displays the operation information including Operation name, Operation TAKT, Operation TAKT as Assigned, Resource Capacity, and Process Efficiency. Both over-assigned and under-assigned resources display in the Resource Capacity column. Process Efficiency is a statement of value-added activities. It is equal to the line operation efficiency multiplied by the demand.

Color coding provides a visual display for comparisons of values needed and exceeded. See: Color Representation on the Mixed Model Map Workbench, page C-5

You can filter or sort the data displayed in the table by choosing Filter/Sort. See: Filtering and Sorting Information, page 5-20

You can view item demand details of the line operation and operation TAKT for the line by choosing Item Details from the Summary tabbed region. See: Viewing Item Details for a Line, page 5-13

You can copy the values that are in the required in the Needed fields to the Assigned fields. In the navigation tree, right click to display the menu, and choose. Select Copy Needed To Assigned.

You can also update item variable lead times by copying the line TAKT time. In the navigation tree, right click to display the menu, and choose. Update Items Variable Lead Time.

Line Detail - Resource/IPK Table View

The Resource/IPK Table tabbed regions contain two table areas—Resources and In Process Kanbans regions.

Mixed Model Map Workbench (S1)

Mixed Model Map By Li

- All Plans
 - HW-TEST1
 - LRMMM1
 - TS1
 - Lines
 - Corrola
 - ASSY :
 - PNT : P
 - PTST : F
 - TST : Te
 - Resources
 - LBR-ASSY
 - LBR-ASSY
 - LBR-ASSY
 - LBR-TEST1
 - LBR-TEST2
 - MCH-ASSY

Resources

Resource	Type	Needed	Assigned
LBR-ASSY1	Person	3.1	3.1
LBR-ASSY2	Person	2.3	2.3
LBR-ASSY3	Person	0.8	0.8
LBR-TEST1	Person	3.0	3.0
LBR-TEST2	Person	0.8	0.8

In Process Kanbans

Operation	Needed	Assigned
ASSY : Assembly Oper	0	0
PNT : Painting Operat	0	0
PTST : Pre Testing Op	0	0
TST : Testing Operatic	0	0

Summary Resource/IPK Table Graph

The Resources region displays line resource details including fields for Resource, resource Type, resources Needed, and resources Assigned. You can filter or sort the data displayed in the table by choosing Filter/Sort. See: Filtering and Sorting Information, page 5-20

The In Process Kanban regions displays a table showing the line operation, and values needed and assigned.

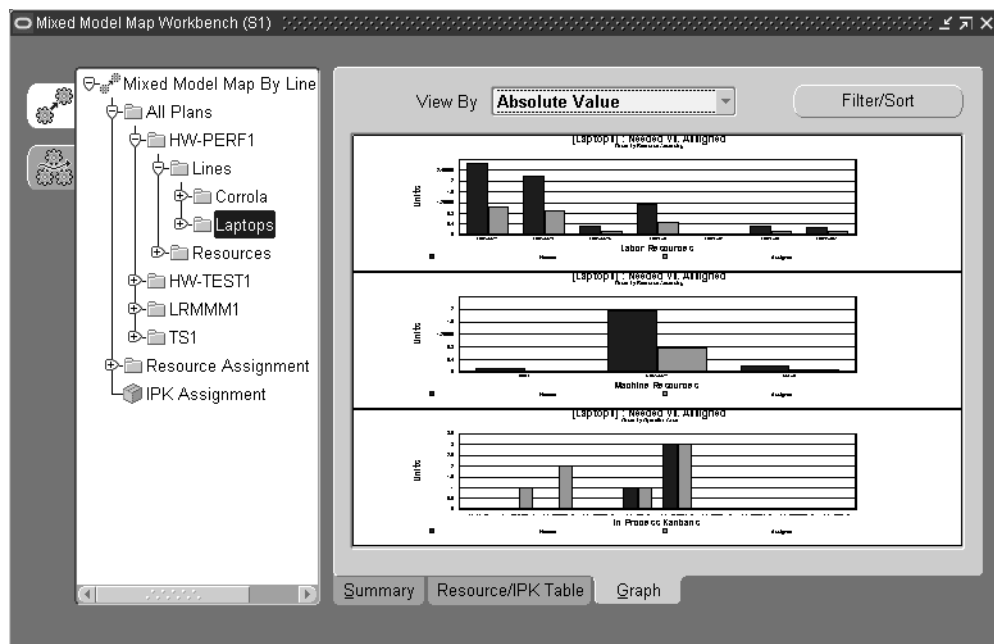
The production line views enable you to copy the values that are in the needed fields to the assigned fields. In the navigation tree, right click to display the menu, and choose. Select Copy Needed To Assigned.

You can also update item variable lead times by copying the line TAKT time. In the navigation tree, right click to display the menu, and choose. Update Items Variable Lead Time.

Line Detail - Graphical View

The Graph tabbed region of the line detail displays a graphical view of the line resource and IPK values. In the View By field, you can change the graph information from percentage values or absolute values. There is a choice of three views including:

- **Percentage Needed:** the percentage difference between Assigned and Needed values—divided by the Needed value. Over-assigned and under-assigned resources display on this view.
- **Percentage Assigned:** the percentage difference between Assigned and Needed values—divided by the Assigned value. Over-assigned and under-assigned resources display on this view.
- **Absolute Value:** Needed and Assigned values display for comparison.



You can modify the display by filtering or sorting the data by choosing Filter/Sort. See: Filtering and Sorting Information, page 5-20

You can enlarge the graph by double-clicking on the graph. Double click again to return to the original view. You can also use the cursor to drag over a region to enlarge a section of the graph. Use your cursor to rollover text details on the graph to display tooltip information.

Color coding provides a visual display for comparisons of values needed and exceeded.

See: Color Representation on the Mixed Model Map Workbench, page C-5

Line Operation - Detail View

Line Operation Detail views display specific data for the operation node selected. The Detail tabbed region displays fields for this line operation for Operation TAKT, Line TAKT, IPK Needed and IPK Assigned.

Mixed Model Map Workbench (S1)

Operation TAKT: 29.41 Line TAKT: 30.00
IPK Needed: 0 IPK Assigned: 0

Labor Resources

Resource	Usage	Needed	Assigned
LBR-ASSY1	0.41	0.41	1.00
LBR-ASSY2	0.20	0.20	1.00
LBR-ASSY3	0.20	0.20	1.00
ASSEMBLER	0.64	0.64	1.00

Machine Resources

Resource	Usage	Needed	Assigned
MCH-ASSY2	0.20	0.20	1.00

Item Detail Recalculate (R)

Detail Graph

There are two regions—Labor Resources and Machine Resources region. Both regions display fields for Resource name, Usage, Needed, and Assigned.

These fields enable you to compare and assign resource capacity values. Color coding provides a visual display for comparisons of values needed and exceeded. See: Color Representation on the Mixed Model Map Workbench, page C-5

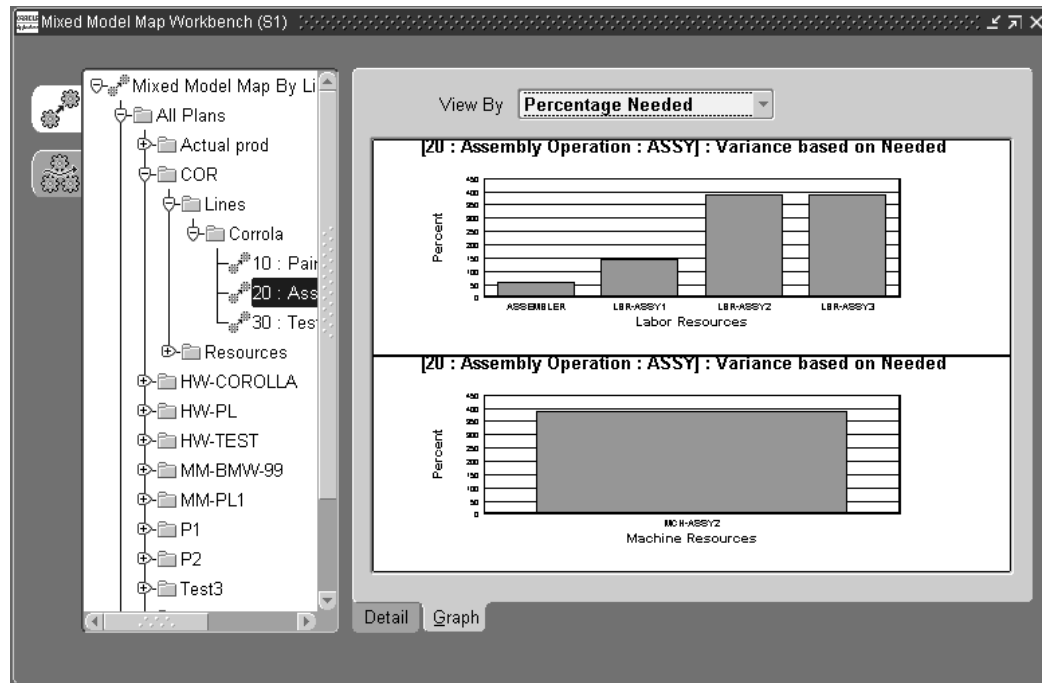
You can view item demand details of the line operation and operation TAKT for the line operation choosing Item Details from the Summary tabbed region. See: Viewing Item Details for a Line, page 5-13

You can copy the values that are in the required in the Needed fields to the Assigned fields. In the navigation tree, right click to display the menu, and choose. Select Copy Needed To Assigned.

Values can be changed in the Assigned fields and recalculated for a line operation. Enter the value you want to change, and choose Recalculate. Resources are calculated and new values display in the usage and needed fields. The plan assigned values for resource and IPKs are used for recalculation, along with the parameters specified in this plan.

Line Operation - Graphical View

The Graph tabbed region of the line operation detail displays a graphical view of labor resources and machine resources.



In the View By field, you can change the graph information from percentage values or absolute values. There is a choice of three views including:

- **Percentage Needed:** the percentage difference between Assigned and Needed values—divided by the Needed value. Over-assigned and under-assigned resources display on this view.
- **Percentage Assigned:** the percentage difference between Assigned and Needed values—divided by the Assigned value. Over-assigned and under-assigned resources display on this view.
- **Absolute Value:** Needed and Assigned values display for comparison.

You can copy the values that are in the required in the Needed fields to the Assigned fields. In the navigation tree, right click to display the menu, and choose. Select Copy Needed To Assigned.

You can enlarge the graph display by double-clicking on the graph. Double click again to return to the original view. You can also use the cursor to drag over a region to enlarge a section of the graph. Use your cursor to rollover text details on the graph to display tooltip information.

Color coding provides a visual display for comparisons of values needed and exceeded.

See: Color Representation on the Mixed Model Map Workbench, page C-5

Viewing Item Details for a Line

The Item Details window display line operation and operation TAKT details for a specific line. This is a view only window that displays fields for Product, Process volume, Machine Time, labor time, and Elapsed Time. The Weighted Time values also display for the time fields.

This window is accessed from the Line Detail view, the Summary tabbed region—and the Line Operation Detail views by choosing Item Details.



The screenshot shows the 'Item Details (S1)' window. At the top, there are input fields for 'Operation' (ASSY), 'Operation TAKT' (0.0274), 'Product Family', and 'Demand'. Below these is a table with the following columns: Product, Process Volume, Machine Time, Labor Time, and Elapsed Time. The table contains seven rows of data for different product variants, followed by a 'Weighted Time' summary row.

Product	Process Volume	Machine Time	Labor Time	Elapsed Time
C-LX-GRN	40.4077	0.1000	0.9000	0.8000
C-LX-WHT	40.4077	0.1000	0.6500	0.5500
C-LX-BLK	40.7310	0.1000	0.9000	0.8000
C-DX-BLK	81.1386	0.1000	0.6500	0.5500
C-DX-GRN	81.1386	0.1000	0.6500	0.5500
C-DX-WHT	81.1386	0.1000	0.6500	0.5500
Weighted Time		0.1000	0.7056	0.6056

Viewing Plan Resource Information

You can view resources needed and assigned for a particular plan—and filter and sort the displayed information. Resources are displayed in summary and individual resource detail views.

Resources Summary Views

The Resource Summary views are a view only panes displaying Summary and Graph regions. Updates can be made on the detail views.

The Summary tabbed region has two regions—Labor Resources and Machine Resources region. Both regions display fields for Resource name, Needed, Assigned, and Relocation Needed.

These fields enable you to compare resource capacity values. Color coding provides a visual display for comparisons of values needed and exceeded. See: Color Representation on the Mixed Model Map Workbench, page C-5

The Relocation Needed field signifies if resources are being allocated sufficiently. For

example, some line operations have extra resources while others need resources. The values in this field are:

- Yes—there are extra resources assigned to this line operation, while other operations have less assigned resources than needed.
- No—there are no extra resources assigned to this line operation, and no other operations need this resource.
- Null—resources needed is greater than resources assigned.

Labor Resources

Resource	Needed	Assigned	Relocation Needed
LBR-ASSY1	3.1	3.1	No
LBR-ASSY2	2.3	2.3	No
LBR-ASSY3	0.8	0.8	No
LBR-TEST1	3.0	3.0	No

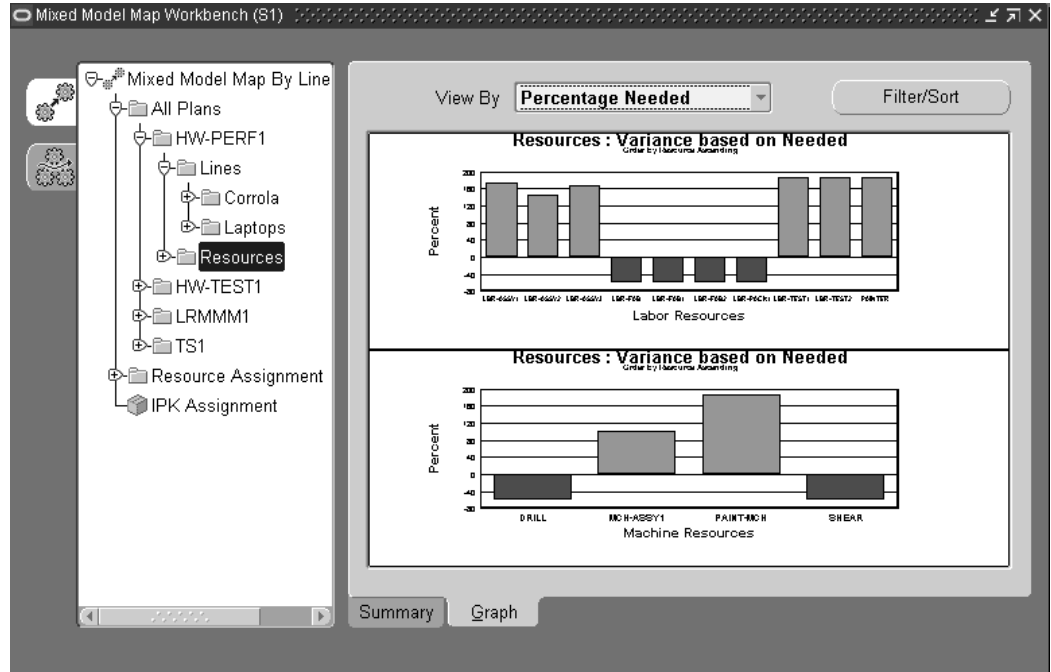
Machine Resources

Resource	Needed	Assigned	Relocation Needed
MCH-ASSY1	0.8	0.8	No
PAINT-MCH	3.8	3.8	No

Summary Graph

You can modify the display by filtering or sorting the data by choosing Filter/Sort. See: Filtering and Sorting Information , page 5-20

The Graph tabbed region displays Labor Resources and Machine Resources regions.



In the View By field, you can change the graph information from percentage values or absolute values. There is a choice of three views including:

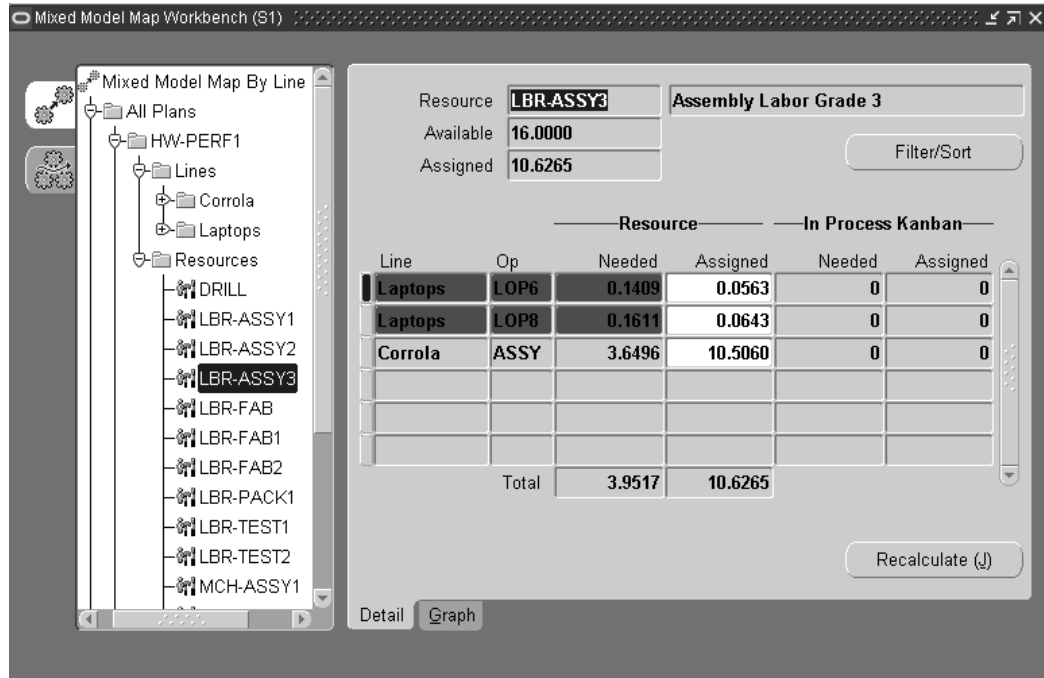
- **Percentage Needed:** the percentage difference between Assigned and Needed values—divided by the Needed value. Over-assigned and under-assigned resources display on this view.
- **Percentage Assigned:** the percentage difference between Assigned and Needed values—divided by the Assigned value. Over-assigned and under-assigned resources display on this view.
- **Absolute Value:** Needed and Assigned values display for comparison.

You can enlarge the graph display by double-clicking on the graph. Double click again to return to the original view. You can also use the cursor to drag over a region to enlarge a section of the graph. Use your cursor to rollover text details on the graph to display tooltip information.

Color coding provides a visual display for comparison. See: Color Representation on the Mixed Model Map Workbench, page C-5

Resource Detail Views

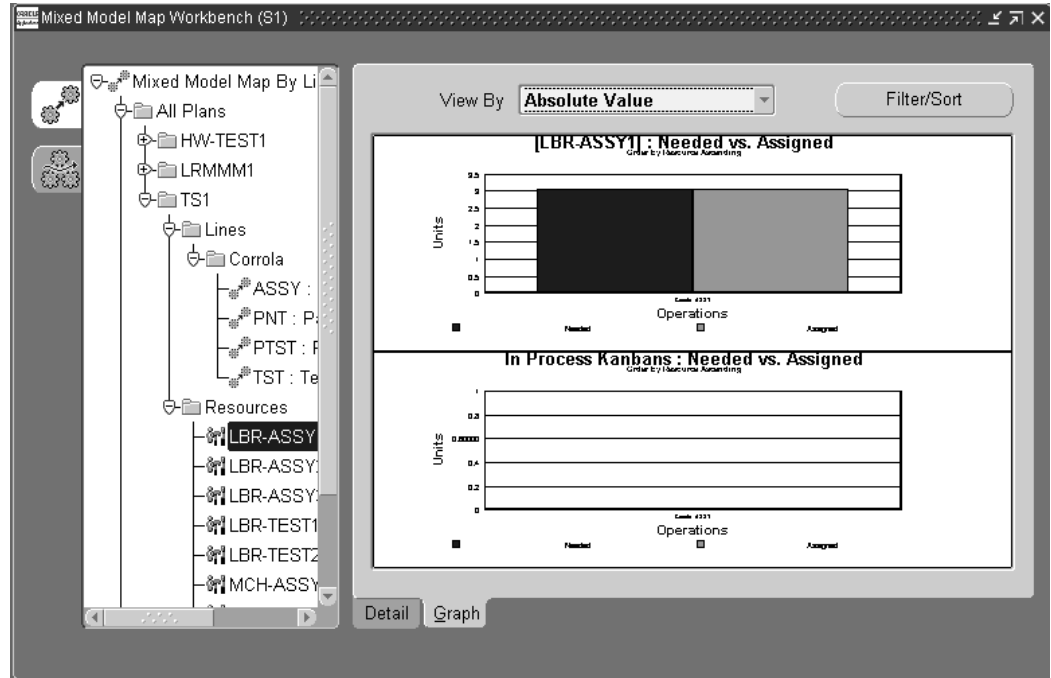
The Resource Detail view shows the resource and IPK needed and assigned in the line operation across line operations. You can filter or sort the data displayed by choosing Filter/Sort. See: Filtering and Sorting Information, page 5-20



The resource Assigned can be modified and recalculated. Choose Recalculate to update resource and IPK values. The plan assigned values for resource and IPKs are used for recalculation, along with the parameters specified in this plan.

You can copy the values that are in the required in the Needed fields to the Assigned fields. In the navigation tree, right click to display the menu, and choose. Select Copy Needed To Assigned.

The Graph Resource view displays resource details in a graphical view. In the Absolute Value view, the graph displays the resource and IPK needed and assigned in the line operation—across line operations. You can filter or sort the data displayed by choosing Filter/Sort. See: Filtering and Sorting Information, page 5-20



In the View By field, you can change the graph information from percentage values or absolute values. There is a choice of three views including:

- **Percentage Needed:** the percentage difference between Assigned and Needed values—divided by the Needed value. Over-assigned and under-assigned resources display on this view.
- **Percentage Assigned:** the percentage difference between Assigned and Needed values—divided by the Assigned value. Over-assigned and under-assigned resources display on this view.
- **Absolute Value:** Needed and Assigned values display for comparison.

You can enlarge the graph display by double-clicking on the graph. Double click again to return to the original view. You can also use the cursor to drag over a region to enlarge a section of the graph. Use your cursor to rollover text details on the graph to display tooltip information.

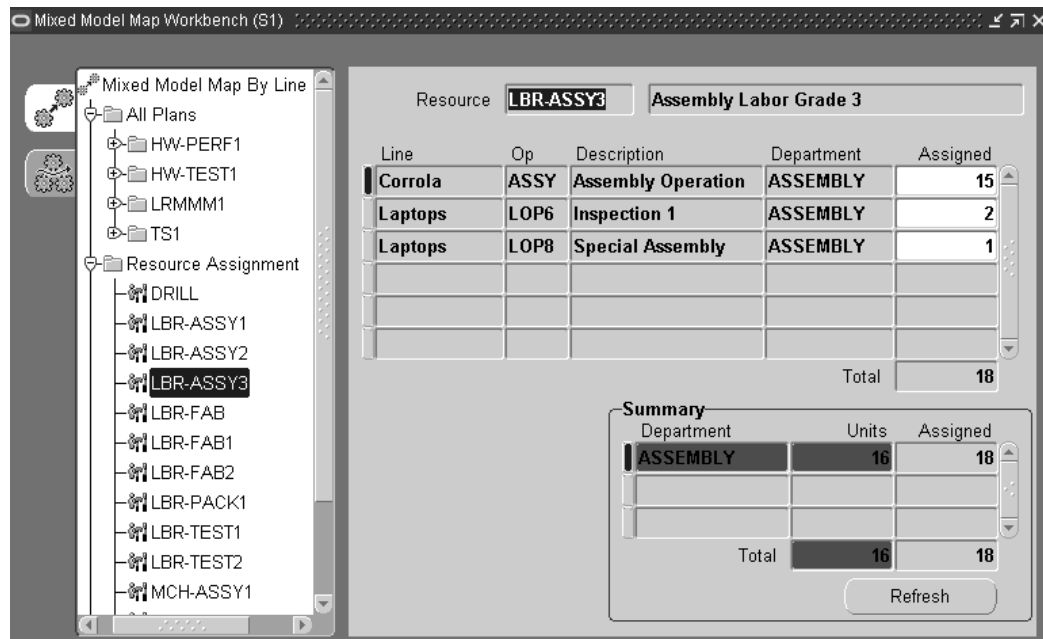
Color coding provides a visual display for comparisons of values needed and exceeded. See: Color Representation on the Mixed Model Map Workbench, page C-5

Resource Assignment

The expanded Resource Assignment node enables you to view all your resources—and display the details for the line, line operation, department, and assigned values. This pane displays a detail of each line the resource is assigned to, a total Assigned value. There is Summary region on this window displaying the department, units, assigned,

and totals.

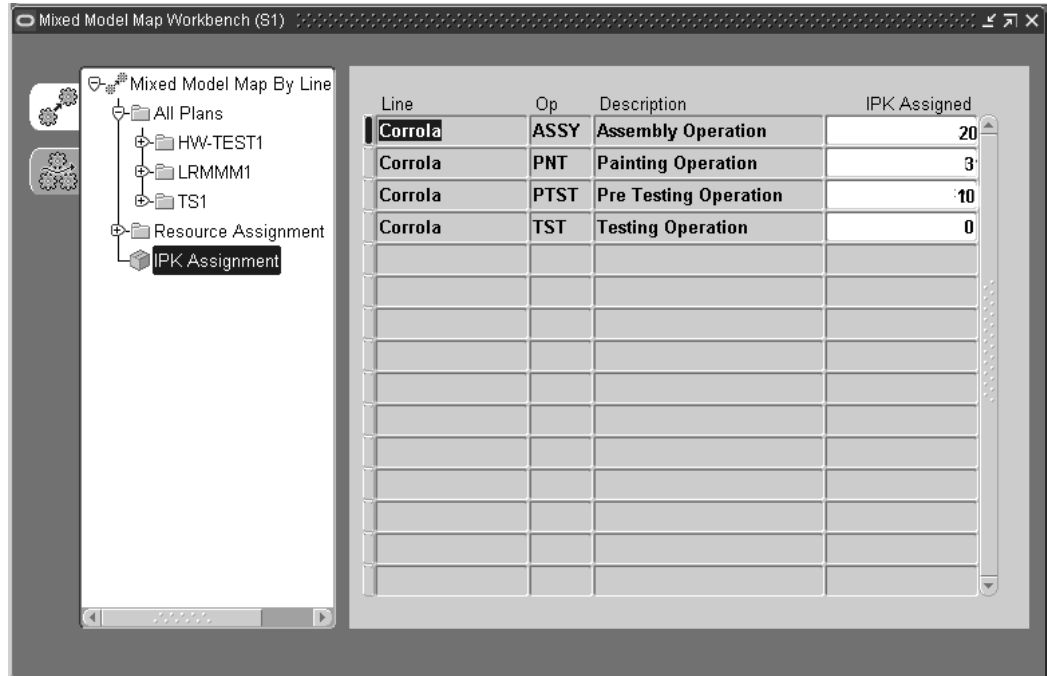
You can update the Assigned field value, and choose Refresh to update the resource values.



In Process Kanban Assignment

The IPK Assignment pane displays a summary of all the IPK master assignments for your lines. It displays the IPK assignment for the line and line operation combination. This data is updated when you choose Update Assignment for a plan, or when you change values in the IPK Assignment field on this pane.

Note: The Bills of Material Workday calendar is used during kanban planning.



In Process Kanbans are designated as a work in process holding area for work in between operations. If the operation time for one operation is shorter in length than the next operation—the IPK is where the item is designated until the next operation is available. The outputs are used by the next operation to keep the line flowing. For example:

- A Line with 2 working hours, consists of 2 continuous operations—Operation1 and Operation2
- Operation1 and Operation2 use one resource each—R1 and R2
- R1 completes an operation every 30 minutes, while R2 completes an operation every 15 minutes

In order for the line to produce items every 15 minutes, and fulfill 8 demands, IPK is needed for Operation1 because it takes more time than Operation2.

- R1 in Operation1 only produces 4 items in 2 working hours
- R2 needs 8 unfinished items from R1 to produce 8 final items

Therefore, extra items—or IPK—are needed at the end of R1 in Operation 1, so R2 in Operation 2 can keep producing at its capacity level. IPK can be fulfilled by using extra line operation resources at the operation.

In Process Kanban Calculation

IPKs are calculated as follows:

$$\frac{(\text{elapsed weighted time} - \text{line takt time}) * \text{hours/day} * \text{time conversion factor}}{\text{elapsed weighted time} * \text{line takt time}}$$

The Calculation Options setting in the Plan detail pane is used to select an operational cycle time setting for the number of resources assigned to the operation.

Related Topics

Viewing, Creating, and Updating Plans, page 5-3

Mixed Model Map Workbench Calculations, page C-1

Filtering and Sorting Information

You have the option to filter or sort table information in the Line Detail, Resource, and Resource Detail views. This enables you to use sort criteria and ordering information, or filter criteria to limit your display of records. You can filter and sort on the Summary, Resource IPK, and Graph tabbed regions.

To filter or sort line detail records:

1. On the Line Detail pane of the selected line, choose Filter/Sort.
The Filter/Sort Criteria window displays.
2. In the Sort Criteria region, choose ordering options for line operation details. In the Order By field, your choices are Operation, or Operation TAKT.
3. In the Order field, choose the order of display for records—Ascending or Descending.

Filter/Sort Criteria

Sort Criteria

Order By **Operation** Order **Ascending**

Filter Criteria

Operation TAKT

Resource Overcapacity

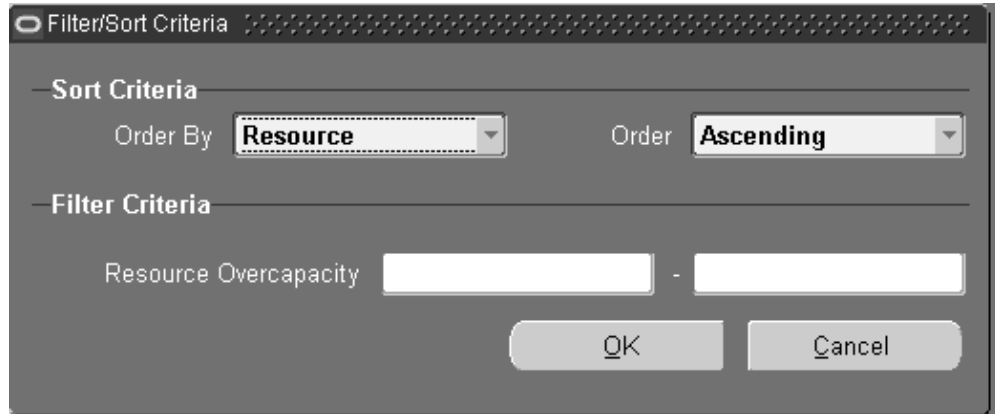
IPK Status

OK **Cancel**

4. In the filter Criteria region, you have the option to enter values for filtering your records. You can enter a value, or range of values, in the Operation TAKT and Resource Overcapacity fields.
5. You can select an IPK Status value. Your choices are either Over or Under
6. Choose OK to filter and sort your records. The results display on the Line Detail views.

To filter or sort resource and resource detail records:

1. On the Resource or Resource Detail pane of the selected resource, choose Filter/Sort. The Filter/Sort Criteria window displays.
2. In the Sort Criteria region, choose ordering options for line operation details. In the Order By field, your choices are Resource or Overcapacity.
3. In the Order field, choose the order of display for records—Ascending or Descending.



The image shows a 'Filter/Sort Criteria' dialog box. It has a title bar with a close button and the text 'Filter/Sort Criteria'. The dialog is divided into two main sections: 'Sort Criteria' and 'Filter Criteria'. In the 'Sort Criteria' section, there are two dropdown menus: 'Order By' set to 'Resource' and 'Order' set to 'Ascending'. In the 'Filter Criteria' section, there is a label 'Resource Overcapacity' followed by two text input fields separated by a minus sign. At the bottom right, there are two buttons: 'OK' and 'Cancel'.

4. In the filter Criteria region, you have the option to enter a value, or range of values, in the Resource Overcapacity field.
5. Choose OK to filter and sort your records. The results display on the Resource and Resource Detail views.

Flow Sequencing

This chapter describes Flow sequencing, the process of determining when items are built on the production line to create Flow schedules.

This chapter covers the following topics:

- Overview of Flow Sequencing
- Setting Up Flow Sequencing
- Viewing and Defining Sequencing Attributes
- Viewing Sequencing Rules
- Defining Sequencing Rules and Constraints
- Viewing and Updating Production Lines
- Assigning Production Lines and Rules
- Creating Sequencing Tasks
- Entering Sequencing Task Parameters
- Selecting Production Lines for a Task
- Filtering Demands for a Sequencing Task
- Viewing Results and Submitting a Sequencing Task
- Updating Line Task Parameters
- Moving Demand
- Updating Demands
- Splitting Demands
- Resequencing Tasks
- Viewing Sequencing Tasks
- Sequencing Task Details and Results

Overview of Flow Sequencing

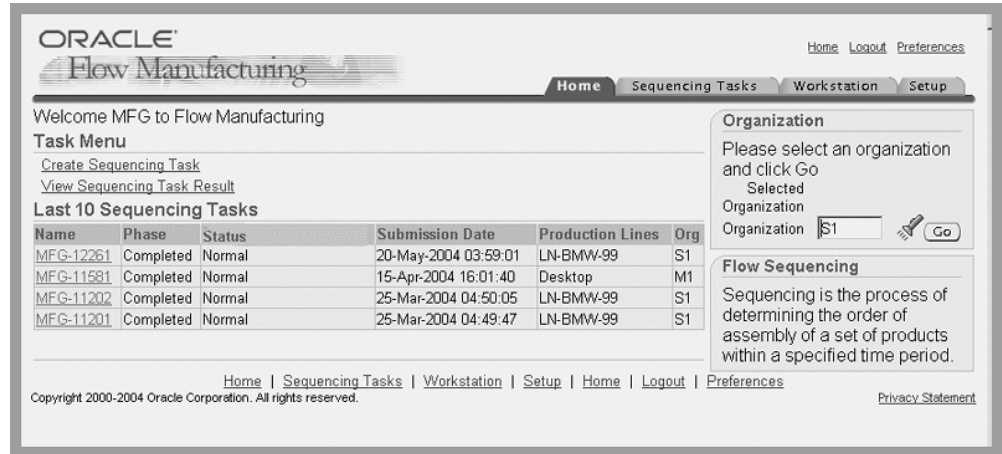
Sequencing is the process of determining the order assembly items are built on a production line, in a time horizon. Flow manufacturing environments comprise a mixture of different assembly items, or mixed models, built on the same production line. The sequence assembly items are built determines the efficiency of the line, and the efficiency determines the productivity of the manufacturing organization. Flow Sequencing is a module for considering your demands and determining a sequence for building assembly items. You use attributes, constraints, and group them into rules for assigned production lines.

In sequencing tasks, constraints are used to decide the best sequence of production. Depending on the production mix—constraints are a combination of many factors including resources, components, and physical dimensions. The Flow Sequencing concurrent program takes the sequencing task as input, and creates Flow schedules to meet production demands. Flow sequencing process begins with the initial data setup. This includes setting up attributes and constraints into rules, and assigning to production lines. After the setup, the process consists of two tasks:

- Specifying parameters including:
 - Date horizon, schedule group, and source of data
 - Production lines
 - Demand
 - Viewing and adjusting demand
- Submitting the sequencing task

Flow Manufacturing Sequencing is setup, tasks created, and submitted in the Flow Manufacturing Sequencing and Execution pages. Attributes, constraints, rules, and production lines are set up in the Setup tabbed region. Tasks are created in the Sequencing Tasks tabbed region, and later submitted when your adjustments are finalized. You first select an organization in order to navigate to the other pages. The Home page enables you to:

- View up to the last ten sequencing tasks generated. These are displayed in a table displaying showing the task Name, Phase, Status, Submission Date, and the Production Line assigned, and Organization. See: Viewing Sequencing Tasks, page 6-29
- Navigate to the Task Summary detail page. See: Sequencing Task Summary, page 6-31
- Create a new task. See: Creating Sequencing Tasks, page 6-17



Setting Up Flow Sequencing

The Setup tabbed region of the Flow Manufacturing Sequencing and Execution page is used to set up your attributes, constraints, rules, and assign them to production lines.

Attributes

Attributes are a characteristic of your demand. For example—on an automobile production line, you can specify an attribute type as Component, and name it Color.

Attribute types can be one of several classes including entity, property, and date attribute types.

Note: You have to specify category set for item category attribute type.

Entities are their names such as Item or Line Operation—where the name of the item or line operation is the attribute type.

Property types require data attributes from the source table. For example, the Flow Schedule Property attribute requires an attribute from the Flow Schedule Property table—such as ALLOCATED_FLAG or DEMAND_CLASS.

Date types also require data attributes from the source table—the specific date type needed for the attribute. For example, the Sales Order Due Date attribute requires an attribute from the Sales Order table—such as PROMISE_DATE or REQUEST_DATE.

The types of attributes in Flow Sequencing are:

- Component
- Flow Schedule Property
- Item

- Item Category
- Item Customized
- Item Property
- Line Operation
- Model
- Planned Order Customized
- Planned order Due Date
- Planned Order Property
- Product Family
- Sales Order Customized
- Sales Order Due Date
- Sales Order Property

Constraints

A constraint is a restriction enforced on an attribute. For example—in an automobile production line, you have an attribute specified as Color. A constraint is created on the Color such as the required transition—first red is manufactured, then black.

Constraints have priorities attached to them, and the priority is unique within a rule. The sequencing process always attempts to honor constraints from the lowest to the highest priority.

Note: It is good practice to define priorities in increments of 10.

A value of zero indicates a hard constraint. If these constraints are broken—or violated—Flow Sequencing stops.

Note: Constraint types do not apply to every attribute type.

Category list shows categories that are assigned to category set, which is defined while creating the attribute.

Constraints types are:

- Required Transition: For the constrained attribute, a certain value must be

immediately followed by another certain value on production line. For example, on a product assembly line—constraint A must be followed by B; otherwise it is a violation.

- **Forbidden Transition:** For the constrained attribute, a certain value must not be followed by another certain value. For example, on a product assembly line—constraint A must not be followed by B; otherwise, it is a violation.
- **Grouping:** For the constrained attribute, a certain value is on the production line as a group of certain size (or multiple of that size).
 - **Attribute-level grouping:** If an attribute value is not specified, the constraint applies to every value of the constraint attribute.
 - **Unlimited grouping:** The group size is not specified, the constraint is a different value in one group.
- **Spacing:** For the constrained attribute, a certain value must be separated by other values and a specified distance.
 - **Attribute-level spacing:** if an attribute value is not specified, the constraint applies to every value of the constraint attribute. Spacing distance is always required.
- **Capacity:** Within any day, or the whole horizon, the number of occurrences of a certain attribute value cannot exceed a certain limit. There are three types of capacity constraints:
 - **Daily limit** – the limit is defined for each every day.
 - **Horizon limit** – the limit is defined for the whole sequencing horizon.
 - **Component Availability** – the limit is from checking ATP of a certain component. In this case, the attribute can only be component,
- **Ordering:** Within the sequencing horizon, the values of one or more attributes appear in either ascending, or descending, order. For example, constraint Order price descending, engine power ascending—will first order price in descending order, and engine power in ascending order.
- **Due Date:** All demands must be scheduled before the corresponding due date (for example, sales order promise date).
- **Conditional Transition:** For the constrained attribute, a certain value must be immediately followed by itself, or another certain value on production line. For example, constraint A must be followed by itself or B—means that on a production line—every A must be followed by a A or B assembly; otherwise it is a violation.

Rules

Rules are a grouping of prioritized constraints defined for the attributes. Rules are assigned to specific production lines. For example, Rule 1 consists of two prioritized constraints:

Priority	Constraint	Attribute	Description
10	Required Transition	Assembly Item	Assembly 1 must be followed by Assembly 2
20	Group	Assembly Item	Group assembly 3

On an automobile production line, you have an attribute specified as Color, a constraint created for the required transition. Then you can create another constraint for spacing—one of every 4 automobiles is red, the rest are black. Then group the constraints into a rule, and name it Rule 1.

Production Lines

A rule is a collection of constraints, and associated with one or more production lines.

For example, on an automobile production line, you have an attribute specified as Color, a constraint created for the required transition. Then you can create another constraint for spacing—one of every 4 automobiles is red, the rest are black. Then group the constraints into a rule, and named Rule 1—and associate it with a specific production line.

Viewing and Defining Sequencing Attributes

To view sequencing attributes:

1. In the Flow Manufacturing Sequencing and Execution Home page, select a value in the Organization field and choose Go.
2. Select the Setup tabbed region.
3. Select Sequencing Attributes to display the Sequencing Attributes page.

ORACLE®
Flow Manufacturing

Home Logout Preferences

Home Sequencing Tasks Workstation Setup

Sequencing Attributes | Sequencing Rules | Production Lines | Workstation Preferences

Sequencing Attributes

Search

Name

Type

Source

Go

Create Attribute (1)

Name	Description	Type	Source	Delete
No search conducted.				

- To view existing attributes, enter values in the Search region to filter your search, or leave these fields blank to display all defined attributes.

You have the option to enter:

- Name—a unique identifier for the attribute
 - Type—such as Item, Model, or Product Family. See: Attributes, page 6-3
 - Source—this is a table column field for property type attributes.
- Choose Go to display the results of your search criteria.

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Sequencing Attributes

Search

Name

Type

Source

Go

Create Attribute (1)

Name	Description	Type	Source	Delete
Car Color	Color Of The Car	Component		
Color	Car Color	Component		
SO Due Date	Sales Order due date	Sales Order Due Date	SCHEDULE SHIP DATE	
Assembly	Assembly Item	Item		

To define sequencing attributes:

- Choose Create Attribute to display the Create Attribute page.

2. In the Type field, select an attribute type.
See: Attributes, page 6-3
3. Enter a unique Name, and optionally a Description for this attribute.
4. If you are creating more than one attribute, choose Create Another Attribute to define another record.
5. Choose Apply to save your work.
A confirmation page displays when all your records are saved.

Viewing Sequencing Rules

To view sequencing rules:

1. In the Flow Manufacturing Sequencing and Execution Home page, select an organization in the Organization region, and choose Go.
2. Select the Setup tabbed region, and choose Sequencing Rules to display the Sequencing Rules page.
3. Enter values in the Search region to filter your search, or leave these fields blank to display all defined sequencing rules.

You have the option to enter Name and Description values.

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[Sequencing Attributes](#) | [Sequencing Rules](#) | [Production Lines](#) | [Workstation Preferences](#)

Sequencing Rules (S1)

Search

Name

Description

Name	Description	Assign Lines	Update	Delete
No data exists.				

4. Choose Go to display the results of your search criteria.

This page displays a list of rules according to your search criteria. You can also view details of a rule, assign lines, and update the rule with new information. See: Assigning Production Lines and Rules, page 6-16, and Constraints, page 6-4

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Sequencing Rules (S1)

Search

Name

Description

Name	Description	Assign Lines	Update	Delete
Laptops Rule				
Rule 1	Rule 1			

5. Select a rule in the Name field to display the Rule detail page.

This page displays the Constraints attached to the rule, and the assigned production lines. See: Rules, page 6-6

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[Setup: Sequencing Rules](#) >

Rule: Rule 1 (S1)

Header

Name **Rule 1**
Description **Rule 1**

Constraints

Priority	Type	Attribute	Definition
10	Grouping	Assembly	Group MC31749 by 2
20	Grouping	Assembly	Group MC97160 by 3
30	Required Transition	Assembly	MC31750 must be followed by MC97160
50	Spacing	Assembly	Separate MC31750 by 5
60	Daily Limit	Assembly	Daily limit of C-DX-BLK is 1

Assigned Production Lines

Name	Description
Corrola	Corrola Line
Desktops	Desktop Assembly Line

[Return to Sequencing Rules \(0\)](#)

Defining Sequencing Rules and Constraints

Rules are a grouping of prioritized constraints, the constraints use priorities to determine the sequence of production.

To define a sequencing rule:

1. Choose Create Rule to display the Create Rule page.

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Sequencing Attributes | **Sequencing Rules** | Production Lines | Workstation Preferences

Setup: Sequencing Rules

Create Rule (S1)

Rule is a prioritized collection of constraints applied to attributes.
* Indicates required field

Cancel Apply

Header

* Name Rule 14

Description New constraints added

Constraints

Select Constraint Component Availability ▼

Create Constraint (1)

*Priority	Type	Attribute	Definition	Update	Delete
No data exists.					

Cancel Apply

2. Enter a unique value for this rule in the Name field, and optionally a description of this rule.
3. In the Constraints region, select constraint values from the Select Constraint type box.
4. Choose Create Constraint to display the Create Constraint page.
5. Enter a value in the Priority field, and select an attribute.

When you select an attribute, the Details field displays on the page.

Note: The lower the priority number indicates a high priority. A value of zero indicates a hard constraint that must not be violated. It is good practice to define priorities in increments of 10. See: Constraints, page 6-4

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[Sequencing Attributes](#) | [Sequencing Rules](#) | [Production Lines](#) | [Workstation Parameters](#)

[Setup: Sequencing Rules](#) > [Create Rule \(M3\)](#) >

Create Constraint for Rule Rule 14: Enter Details (S1)

* Indicates required field

Constraint Type **Component Availability**

* Priority

* Attribute

Details

* Item

[Cancel](#) [Finish](#)

6. Select a component item number for this constraint.

7. Choose Finish to save your work.

The new constraint displays in the Constraints region of Create Rule page.

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[Setup: Sequencing Rules](#) >

Create Rule (S1)

Rule is a prioritized collection of constraints applied to attributes.

* Indicates required field

Header

* Name

Description

Constraints

Select Constraint Type [Create Constraint \(1\)](#)

*Priority	Type	Attribute	Definition	Update	Delete
20	Component Availability	Component	Check availability of AS-ASY1		

[Cancel](#) [Apply](#)

8. To modify constraints displayed in this list, select a constraint and choose Update. The Update Constraint for Rule page displays.

You have the option to modify the Priority and Attribute fields.

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Sequencing Attributes Sequencing Rules Production Lines Workstation Preferences

Setup : Sequencing Rules > Create Rule (S1)

Update Constraint for Rule Rule 14: Specify Priority and Attribute(s) (S1)

Specify priority and attribute(s) of the constraint. The lower the priority number, the higher the priority. A value 0 indicates a hard constraint that must not be violated. We suggest to define priority in increments of 10.

* Indicates required field

	Constraint Type	Component Availability
* Priority		20
* Attribute	Color	

Cancel Continue

9. Choose Continue to display the Update Constraint Details page.
10. You have the option to change the item on this page.
11. Choose Finish to save your work. The updated constraint displays in the Constraints region of Create Rule page.
12. If you want to delete any of the constraints created for this rule, choose Delete for the selected rule.
13. Choose Apply to save your work

Viewing and Updating Production Lines

Flow manufacturing environments comprise a mixture of different items built on the same production line. You consider your demands and determining a build sequence. Rules are associated and assigned to one or more production lines.

To view production lines:

1. In the Flow Manufacturing Sequencing and Execution Home page, select a value in the Organization field and choose Go.
2. Select the Setup tabbed region.
3. Select Production Lines to display the Production Lines page.
4. You have the option to enter values in the Search region to filter your search. Enter values in the Name or Description fields. You can also leave these fields blank to display all defined production lines.

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Sequencing Attributes | Sequencing Rules | **Production Lines** | Workstation Preferences

Production Lines (S1)

Search

Name

Description

Name	Description	Start Time	Stop Time	Hourly Rate	Daily Rate	Update
Camry	Camry Line	10:00:00	20:00:00	5	50	
Corrola	Corrola Line	07:00:00	17:00:00	2	20	
Desktops	Desktop Assembly Line	07:00:00	23:00:00	50	800	
Laptops	Laptop Assembly Line	07:00:00	23:00:00	100	1600	
Metal Fab	Metal Fabrication Production Line	07:00:00	15:30:00	45	382.5	
PCB-ASM	PCB - ASSEMBLY LINE	07:00:00	15:00:00	100	800	
Plastic #1	Plastic #1 Production Line	07:00:00	15:30:00	140	1190	
Plastic #2	Plastic #2 Production Line	07:00:00	15:30:00	140	1190	

To update production line information:

1. Navigate to the Production Lines page.
2. Search for the record to update.
3. Select the production line and choose Update.

The Update Production Line page displays.

4. In the Line Definition region, you are required to enter values in the Hourly Rate, Start Time, and Stop Time fields.

Note: These values are defaulted from the Production Lines window.

5. In the Sequencing Default Values region, you can enter default values for when running the Sequencing Concurrent program for this line. In the Scheduling Direction box, you have the option to change the scheduling from Forward or Backward.

ORACLE® Flow Manufacturing Return to Portal Logout Preferences Personalize Page

Home Sequencing Tasks Workstation **Setup**

Sequencing Attributes | Sequencing Rules | **Production Lines** | Workstation Preferences

Setup: Production Lines >

Update Production Line: Laptops (S1) Cancel Apply

* Indicates required field

Line Definition

* Name **Laptops** * Start Time
(example: 23:59:59)
 * Hourly Rate * Stop Time

Sequencing Default Values
 Enter the sequencing default values to be used when running the sequencing engine for the selected production line.

Scheduling Direction Fix Sequence ☒ None
 Connect Sequence ☐ Time Hours
 Combine Schedule ☐ Quantity

Assigned Rules
 Select Rule:

Select Name	Description	Default
☐ Rule 1	Rule 1	
☐ Laptops Rule		☒

Cancel Apply

6. The Connect Sequence field is used to connect this sequencing task to a previous sequencing task. Your choices are Yes or No.

 This option enables you to sequence between days within the horizon, and gives you the option to connect the sequences with existing sequences without violating constraints.
7. The Combine Schedules field is used to combine similar concurrent demand into one schedule. Your choices are Yes or No.
8. The Fix Sequence field is used to prevent the resequencing of existing Flow schedule when new demand is added to the schedules, or a portion of the schedules are resequenced. Your choices are:
 - None—Indicates all existing schedules are resequenced.
 - Time—Existing schedules with a completion time less than or equal to this value are not resequenced.
 - Quantity—Flow schedules up to the specified quantity are not resequenced.
9. In the Assigned Rules region, you can update rule information for this production line. If you want to change the default rule, select the rule in the table, and choose Set as Default. The default symbol displays in the Default field.
10. To delete a rule assigned to this line, select the rule and choose Delete.
11. To assign another rule to this production line, choose Assign Rule.

The Assign Rule to Production Line page displays. See: Assigning Production Lines and Rules, page 6-16

12. Choose Apply to save your work.

Assigning Production Lines and Rules

Rules are used for calculating the best sequence for schedule creation. You can assign production lines to specific rules, or assign rules to production lines.

To assign a rules to a production line:

1. Navigate to the Production Lines page, and Search for the record to update.
See: Viewing and Updating Production Lines, page 6-13
2. Select the production line, choose Update to display the Update Production Line page.

The screenshot shows the Oracle Flow Manufacturing web interface. At the top, there's a header with the Oracle logo and 'Flow Manufacturing' text. To the right of the header are links: 'Return to Portal', 'Logout', 'Preferences', and 'Personalize Page'. Below the header is a navigation bar with tabs: 'Home', 'Sequencing Tasks', 'Workstation', and 'Setup'. The 'Setup' tab is selected. Under the 'Setup' tab, there's a sub-navigation bar with links: 'Sequencing Attributes', 'Sequencing Rules', 'Production Lines', and 'Workstation Preferences'. The 'Production Lines' link is selected. Below the navigation bar, the page title is 'Assign Rule To Production Line: Laptops (S1)'. There's a note: '* Indicates required field'. Below this, there's a form field labeled '* Rule Name' with a search icon to its right. At the bottom right of the form, there are two buttons: 'Cancel' and 'Apply'.

3. In the Assigned Rules region, choose Assign Rule.
Assign Rule to Production Line page displays.
4. Enter a search string in the Rule Name field and choose Search.
The Search and Select page displays with the results of your search.
5. Select a record. The record displays on the Assign Rule to Production Line page.
6. Choose Apply to assign this rule to your production line.
The rule displays in the Assigned Rules region of the Update Production Line page.
7. Choose Apply to save your work.

To assign production lines to a rule:

1. In the Flow Manufacturing Sequencing and Execution Home page, select a value in the Organization field and choose Go.

2. Select the Setup tabbed region.
3. Choose Sequencing Rules.
4. Display the appropriate rules. See: Viewing Sequencing Rules, page 6-8
5. Choose Assign Lines to display the Assign Production Lines to Rule page.

In the Rule region, the name and description of the rule displays.

6. In the Assign Production Lines region, select production lines from the Available box, and move them to the Assigned box.

You can move all the production lines, remove lines, or remove all the production lines in the Assigned box.

7. Choose Apply to save your work.

Creating Sequencing Tasks

The Sequencing Task pages display a chronology of steps, or a train, showing where you are in the process. The train displays Enter Parameters, Select Lines, Filter Demands, and Adjust and Submit:

- The first step is to enter parameters for the sequencing run. You create a name for the task, and specify the horizon dates, any applicable schedule group, and data used to create new schedules—or if the task is resequencing existing schedules.

- The next step is to select the production lines used in this run.
- In the Filter Demands step, you specify demand dates. The Filter Demands page has two regions—the first is for selecting demand date. The Optional region is used to add additional filter information such as assembly, customer, and attribute. For example, you may want to consider all sales orders of a specific due date, and then specify the particular customer and assembly number. This page is not used if you are resequencing existing schedules.
- The Adjust and Submit Sequencing page enables you to view the relevant production lines and sales orders, along with the remaining capacity and quantity to sequence. You can update information, and delete production line demand attached to this task. The last step is submitting the task. The sequencing concurrent request is triggered when a task is submitted and Flow schedules are generated.

Results of the sequencing run display on the Flow Manufacturing Sequencing and Execution Home, Task Summary, and Task Detail pages.

Note: Flow schedules are viewable on the Line Scheduling Workbench.

Related Topics

Entering Sequencing Task Parameters, page 6-18

Selecting Production Lines for a Task, page 6-19

Filtering Demands for a Sequencing Task, page 6-20

Viewing Results and Submitting a Sequencing Task, page 6-22

Sequencing Concurrent Program, page 6-24

Viewing Sequencing Tasks, page 6-29

Sequencing Task Summary, page 6-31

Sequencing Task Details and Results, page 6-31

Entering Sequencing Task Parameters

The Sequencing Task pages display the train of steps to complete. When you navigate to the Entering Sequencing Task Parameters page, Enter Parameters is highlighted. This page is used to specify information needed to create a unique task. You can also select a previous task's parameters on the Sequencing Task summary page, rename the task, and create a new submission. See: Viewing Sequencing Tasks, page 6-29

To enter sequencing task parameters:

1. Navigate to the Entering Sequencing Task Parameters page.

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Enter Parameters Select Lines Filter Demands Adjust and Submit

Enter Sequencing Task Parameters (M1)

* Indicates required field

Cancel Step 1 of 4 Next

* Name MFG-12321

* Schedule Date From 25-May-2004 00:00:00
(example: 24-May-2004 19:45:00)

* Schedule Date To 25-May-2004 23:59:59

Schedule Group

Type ☒ Create New Schedules based on Sales Order ☐ Resequence Existing Schedules

Cancel Step 1 of 4 Next

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2. Select a Name for this sequencing task.
This must be a unique value, the default value is User Name and the sequential number following the last task created.
3. Select values in the Schedule Date To and From fields.
4. Optionally, you can enter a Schedule Group
5. In the Type region, choose whether you are Creating New Schedules, or Resequencing Existing Schedules.
If you are creating new schedules, select the data used to create new schedules. Your choices are either Sales Order or Planned Order.
6. Choose Next to display the Select Production Lines page.

Selecting Production Lines for a Task

When you navigate to the Enter Production Lines page, Select Lines is highlighted in the step train.

To select production lines for a sequencing task:

1. Navigate to the Enter Production Lines page by choosing Next from the Enter Parameters page.

2. Select lines from the Available box, and move to the Selected box.

You can move all the production lines, remove lines, or remove all the production lines in the Assigned box.

3. Choose Next to display the Filter Demands page.

Filtering Demands for a Sequencing Task

You specify demands dates for creating Flow schedules, and any other criteria used for this task. This can include assembly part numbers, customer, and sales order attributes.

Note: This page is not used if you are resequencing existing schedules because that information was previously entered.

Honor Planning Recommendations and Include Alternate Routings Check Boxes

Flow Manufacturing can be configured to honor the lines recommended by Planning for the planned orders. Planned orders are not treated in the same way as sales orders and they do not appear on all the lines that have routing for the assembly. You can see only those planned orders as demand that are recommended by planning on the selected lines. You will not be able to see planned orders if the assembly has routing on the selected line, unless planning recommends them on the line.

When you create schedules from Planned Orders from Flow Sequencer, you can filter planned orders based on planning recommendations. As a result planned orders of an assembly will not be shown for all the lines on which the assembly has routing. This helps you to avoid selecting two or more lines having routing for the same assembly.

Thereby it prevents you from creating schedules for the same demand that is scheduled twice for double the quantity than the actual planned order quantity.

Include Alternate Routings check box filters demands for planned orders. This is also selected along with Honor Planning Recommendations check box for the planned orders. This will be the same from both the Line Scheduling Workbench (LSWB) and Sequencer.

The following list presents the combined behavior of these check boxes:

- **Honor Planning Recommendations unchecked/Include Alternate Routings checked**

By default shows planned orders of all the assemblies that have routing on the selected line.

- **Honor Planning Recommendations checked/Include Alternate Routings checked**

Shows only planned orders that planning has recommended on the selected line.

- **Honor Planning Recommendations unchecked/ Include Alternate Routings unchecked**

Shows planned orders of assemblies that have primary routing on the selected line.

- **Honor Planning Recommendations checked/Include Alternate Routings unchecked**

Shows all the planned orders which planning has recommended on the selected line if the assembly of the planned order has primary routing on the selected line.

The screenshot displays the Oracle Flow Manufacturing Sequencing Tasks Workstation Setup page, specifically the 'Filter Demands' step. The page is titled 'ORACLE Flow Manufacturing' and includes navigation links for Home, Logout, Preferences, Help, Personalize Page, and Diagnostics. The main content area is divided into several sections:

- Filter Demands: MFG-14826 (M1)**: This section contains a text box for 'Enter Parameters' and a 'Select Lines' button. Below this, there is a note: 'Enter the demand dates to be considered for sequencing. Demand filter criteria can be entered in the Optional Filter Criteria section. * Indicates required field'.
- Filter Demands**: This section contains three checkboxes: 'Honor Planning Recommendations' (unchecked), 'Include Alternate Routings' (checked), and 'Adjust Detail Demands' (unchecked). Below these are two date pickers: '* Demand Date From' (16-Feb-2009 00:00:00) and '* Demand Date To' (16-Feb-2009 23:59:59).
- Optional Filter Criteria**: This section contains a text box for 'To add a filter criteria, select the Attribute and Operator from the pulldown list, and click Add. Then supply the values on the newly added filter criteria. Click Remove to delete the filter criteria.' Below this are two rows of filter criteria: 'Assembly between' and 'Customer Equals'. Each row has a 'Remove' button.
- Add Criteria**: This section contains a dropdown menu for 'Attribute' (set to 'Assembly'), a dropdown menu for 'Operator' (set to 'Equals'), and an 'Add' button.

The page also includes a 'Cancel' button and a 'Back' button. The bottom of the page features a footer with 'About this Page', 'Privacy Statement', and 'Copyright © 2008, Oracle. All rights reserved.'

To filter demands for a sequencing task:

1. Navigate to the Filter Demands page by choosing Next from the Enter Production Lines page.
2. Select Honor Planning Recommendations check box to see only those planned orders as demand type that are recommended by planning on the selected lines.

This enables you to avoid planned orders getting scheduled more than once. If you do not want to honor planning recommendations, you can uncheck the Honor Planning Recommendations check box.
3. If you want to display sales order information on the Adjust and Submit Sequencing Task page, enable the Adjust Detail Demands box.
4. Select dates in the Demand Date From and To fields.
5. In the Optional Filter Criteria region, you can select additional sales order attributes with a formula operator. The operator values are Equals or Between. In the Add Criteria region, select the attributes in the Attribute box, and the operator in the Operator box.
6. Choose Add to display these sales order attributes on this page.

For example, you can select the attribute, Assembly Part Number, and the operator Between. This enables you to list a range of assembly part numbers in your filter criteria.

Or you can select the attribute, Customer, and the operator Equals. This enables you to add a specific Customer's sales orders.

Viewing Results and Submitting a Sequencing Task

The Adjust and Submit Sequencing page is used to view the remaining capacity and demand for this sequencing task. If you enabled the Adjust Detail Demands box, on the Filter Demands page, demand data displays in the Selected Orders.

This page enables you to update task parameter lines, delete production line demand from this sequencing task, and update sales order demand information. You can also delete, move, or split the sales orders attached to this task.

To edit sequencing task results:

1. Navigate to the Adjust and Submit Sequencing Task page by choosing Next from the Select Production Lines or Filter Demands pages.

Note: If you are resequencing existing schedules, the Filter

Demands page is not used, and this page displays after the Select Production Lines page.

The Production Lines region displays Production Line, Description, Remaining Capacity, and Quantity to Sequence fields:

- Remaining capacity is the capacity for the line. The value is calculated from the existing Flow schedules and the line capacity total.
- The Quantity to Sequence value is calculated from the filter criteria you selected.

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Enter Parameters Select Lines Filter Demands **Adjust and Submit**

Adjust and Submit Sequencing Task: MFG-12369 (S1)

To adjust production line parameter, click on the Production Lines Update icon. Click Submit to run the sequencing process. [Cancel](#) [Back](#) Step 4 of 4 [Submit](#)

Production Lines

Select Production Line	Description	Remaining Capacity	Qty to Sequence	Update	Delete
☐ KC-TRUCK	TRUCK ASSEMBLY LINE	336	25		
☒ LN-BMW-99	LINE FOR BMW 99	206	5		
☐ LN-BMW-HW	HW BMW PRODUCTION LINE	630	0		
Total		1172	30		

Sales Orders : LN-BMW-99

All the sales order displayed are the sales order having assembly items that can be built on the selected production line. You can delete one or more sales order(s) so that the particular sales order(s) will not be scheduled. Also you can move and split the sales order to be built on different production line.

Select Sales Orders: [Delete](#) [Move](#) [Update](#) [Split](#)

[Select All](#) | [Select None](#)

Select	Order Number	Line Number	Order Date	Assembly	Assembly Description	Open Qty	Qty to Sequence	UOM	Alternate Designator
☐	58040	1	01-Apr-2004	BMW5-BLK-99	BMW 5 SERIES BLACK 99	5	5	Ea	

2. To update line task parameters, select a production line. Then choose Update to display the Update Line Task Parameters page.
See: Updating Line Task Parameters, page 6-24
3. To delete a production line from this task, select a production line and choose Delete.
4. In the Select Sales Order region, you have the option to delete, move, update, or split sales orders that are attached to the task. To remove a sales order from consideration, select the order number and choose Delete.
5. To move a sales order to another production line, select the order number and choose Move. See: Moving Demand, page 6-26

6. To update the sales order demand or change the routing, select the order number and choose Update. See: Updating Demands, page 6-27
7. To divide the demand between multiple Flow schedules and other production lines, choose Split. See: Splitting Demands, page 6-28
8. To submit this sequencing task to create Flow schedules, choose Submit.
The sequencing engine concurrent request is triggered when a task is submitted.
See: Sequencing Concurrent Program, page 6-24

Sequencing Concurrent Program:

The last step is submitting the task. The sequencing engine is a concurrent request. The request takes all information specified in the sequencing task, and generates Flow schedules.

The result of the sequencing is a series of Flow schedules—build sequences indicate the order of production.

Updating Line Task Parameters

After you enter sequencing task information, and view information on the Adjust and Submit Sequencing Task page—you have the option to change the original parameters. You can modify the time, rate, default sequencing direction, and constraints on this page. You can also choose to combine schedules, and to connect with existing schedules.

To update line task parameters:

1. On the Adjust and Submit Sequencing Task page, in the Production Lines region, select a line.
2. Choose Update to display the Update Line Task Parameters page.
3. Enter a Start and Stop Time range. These fields are required.
4. Enter the Hourly Rate of this line. This field is required.

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Enter Parameters Select Lines Filter Demands Adjust and Submit

Update Line Task Parameters: LN-BMW-99 (S1)

* Indicates required field

Cancel Apply

Parameters

Production Line	LN-BMW-99	Description	LINE FOR BMW 99
* Start Time	-08:00:00 (example: 23:59:59)	Remaining Capacity	3
* Stop Time	02:00:00	Quantity to Sequence	5
* Hourly Rate	4	Fix Sequence	☒ None
Daily Rate	40	☐ Quantity	
Scheduling Direction	Forward	☐ Time	 (example: 27-May-2004 19:45:00)
Connect Sequence	Yes	☐ Build Sequence	
Combine Schedule	Yes		

5. In the Scheduling Direction box, you have the option to change the scheduling from Forward or Backward.
6. The Connect Sequence field is used to connect this sequencing task to a previous sequencing task. Your choices are Yes or No.

This option enables you to sequence between days within the horizon. When sequencing for a specified horizon, this flag gives you the option to connect the sequences with existing sequences without violating constraints.
7. The Combine Schedules field is used to combine similar concurrent demand into one schedule. Your choices are Yes or No.
8. The Fix Sequence field is used to prevent the resequencing of existing Flow schedule when new demand is added to the schedules, or a portion of the schedules are resequenced. Your choices are:
 - None—Indicates all existing schedules are resequenced.
 - Quantity—Flow schedules up to the specified quantity are not resequenced.
 - Time—Existing schedules with a completion time less than or equal to this value are not resequenced.
 - Build Sequence—Flow schedules up to the specified build sequence are not resequenced.
9. Choose Apply to save your work.
10. In the Rules and Constraints region, you can use the Copy Constraint from Rule box

to select a rule, and use that rule's constraints. Choose Copy to copy the constraints to your current process.

The copied constraint displays in the Constraints region. The Constraints region displays all constraints in the current sequencing task. You can change priority—and create, modify, or delete constraints.

Rule and Constraints

To copy constraints of the rule to be used for the current sequencing process, select the rule from the Copy Constraint pulldown list.

Copy Constraint from Rule: **R-BMW99**

Constraints

All constraints displayed will be used by the current sequencing process. They can be added, modified, or deleted.

Select Constraint Type: **Component Availability** Previous **1-10 of 14** Next **4**

*Priority	Type	Attribute	Definition	Update	Delete
10	Spacing	Car Type	Separate PF-BMW7-99 by 2		
20	Daily Limit	Car Type	Daily limit of PF-BMW5-99 is 30		
30	Forbidden Transition	Car Color	CMP-PAINT-BLK-99 can not be followed by CMP-PAINT-WHT-99		
40	Daily Limit	Car Color	Daily limit of CMP-PAINT-BLK-99 is 15		
50	Ordering	Assembly	Order Assembly descending		
60	Spacing	Car Type	Separate PF-BMW7-99 by 2		
10	Required Transition	Assembly	MC24713 must be followed by MC97160		

Select Constraint Type: **Component Availability** Previous **.** Next **4**

11. To change the priority level of the constraint, enter the new priority value in the Priority field. Choose Apply to save your changes.
12. To create a new constraint for this task, choose Create Constraint to display the Update Constraint page. See: Defining Sequencing Rules and Constraints, page 6-10
13. To delete a constraint, select the constraint and choose Delete.
14. To update a constraint listed on this page, choose Update.

Moving Demand

You may want to move demand to another production line. You can only move demand to an alternate routing, for a production line selected, when you created the sequencing task. See: Selecting Production Lines for a Task, page 6-19

To move demands on a sales order:

1. On the Adjust and Submit Sequencing Task page, in the Order region, select an order.
2. Choose Move to display the Move Demands page.

This page displays the sales order information including fields for Order Number, Line Number, Assembly number, Assembly Description, and the Current Line for this demand.

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Enter Parameters Select Lines Filter Demands Adjust and Submit

Move Demands (S1) Cancel Apply

Order Number	Line Number	Assembly	Assembly Description	Current Line
54714	5	MC97160	Plastic Cover - Black	Laptops

Move to Production Line : Desktops

Cancel Apply

3. In the Move to Production Line box, select the production line where you want to move this line.
4. Choose Apply to save your work.

You will return to the Adjust and Submit Sequencing Task page displaying the updated information.

Updating Demands

Demand is based on the assembly's primary routing. The Update Demands page enables you to change either one or both of the values for the quantity to sequence, and the routing. The alternate routing must exist for the same line as the primary routing.

To update demand and routing information:

1. On the Adjust and Submit Sequencing Task page, in the Order region, select an order.
2. Choose Update to display the Update Demands page.
This page displays the order information including fields for Order Number, Line Number, Assembly number, Assembly Description, Open Quantity, and Other Quantity to Sequence.
3. To change the demand for this order, enter a new value in the Quantity to Sequence field.

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Enter Parameters Select Lines Filter Demands **Adjust and Submit**

Update Demands {S1}

Cancel Apply

Order Number	Line Number	Assembly	Assembly Description	Open Qty	Other Qty to Sequence	Qty to Sequence	Alternate Designator
58040	1	BMW5-BLK-99	BMW 5 SERIES BLACK 99	5	0	5	

Cancel Apply

4. To change the routing to an alternate routing, select a value in the Alternate Designator field.
 5. Choose Apply to save your work.
- You will return to the Adjust and Submit Sequencing Task page displaying the updated information.

Splitting Demands

You can optimize production by dividing sales order line quantities to create multiple schedule, or splitting a quantity and moving it to another production line. You can only split demand to an alternate routing, for a production line selected, when you created the sequencing task. See: Selecting Production Lines for a Task, page 6-19

To split demands on a sales order:

1. On the Adjust and Submit Sequencing Task page, in the Order region, select an order.
2. Choose Split to display the Split Demands page.
This page displays the sales order information including Order Number, Line Number, Assembly number, Assembly Description, and Quantity to Sequence fields.
3. In the Split Quantity box, enter the quantity you want to put on a new Flow schedule.

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Enter Parameters Select Lines Filter Demands Adjust and Submit

Split Demands (S1)

Order Number	Line Number	Assembly	Assembly Description	Qty to Sequence	Split Qty
58040	1	BMW5-BLK-99	BMW 5 SERIES BLACK 99 5		0

Split to Production Line : LN-BMW-99

Cancel Apply

Cancel Apply

4. In the Split to Production Line, you have the option to select another production line for this new Flow schedule.

The alternate routing must exist first to change the routing path

5. Choose Apply to save your work.

You will return to the Adjust and Submit Sequencing Task page displaying the updated information.

Resequencing Tasks

The sequencing concurrent program considers existing flow schedules within the specified sequencing horizon—the From Start Date Time to the To End Date Time—and re-sequences them. The resequencing of existing schedules is used in the following situations:

- Correct existing schedules to modify the number of hours in production, or the quantity.
- Relieve overloaded Flow schedules.

Flow schedules with scheduled Completion Date equal to the Start Date Time are not considered to be within the horizon. A Flow schedule is within the horizon if the Completion Date is greater than horizon Start Date Time, and no later than horizon End Date Time.

You should not resequence if the concurrent program for that schedule is still running, the broadcast sequence is sent, the traveler is printed, or the feeder line has started.

Viewing Sequencing Tasks

The Sequencing Tasks Search page displays a table with the last ten sequencing tasks from the Home page, for the organization selected. This page enables you to filter your search, and access pages to create a new task or delete a task. You can navigate to the summary and detail pages for the search results.

To search for sequencing tasks:

1. Navigate to the Sequencing Tasks Search page.
2. Select search criteria to filter your search.

You can select criteria in the following fields: sequencing task Name, Submission Dates From and To, concurrent request Status, concurrent request Phase, Production Line, and User Name.

3. Choose Go to display the results of your search.

If you do not select any criteria, all sequencing task records display. Records display in a table view.

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Sequencing Tasks (S1)

* Indicates required field

Search

Name Status

Submission Date From (example: 19-May-2004 19:45:00) Phase

Submission Date To Production Line

* Maximum Requests Retrieved 10 User Name

Go

Create Sequencing Task (1)

Sequencing Task Name	Request Id	Phase	Status	Submission Date	User	Production Lines	Create Similar Task	Delete
FLM-10563	3257904	Completed	Normal	02-Jun-2004 12:02:18	FLM	Laptops		
FLM-10562	3257865	Completed	Normal	02-Jun-2004 11:53:04	FLM	Laptops		
FLM-10560	3256672	Completed	Normal	02-Jun-2004 09:46:20	FLM	Laptops		

4. Choose Create Sequencing Task to create a new task. See: Creating Sequencing Tasks , page 6-17 .
5. You can select a task in the Sequencing Task Name field to view the summary information of the task. See: Sequencing Task Summary, page 6-31 The summary page navigates to the details page, see: Sequencing Task Details and Results, page 6-31
6. If you want to create a task using some or all of the parameters of an existing sequencing task, choose Create Similar Task.

This navigates to the Entering Sequencing Tasks Parameter page. The default name displays, you can modify the necessary information. See: Entering Sequencing Task

Sequencing Task Summary:

The Task Summary page displays summary data for an individual task and production line—including the available capacity for the line, the quantity fulfilled—and the percentage of resequencing, demand, and constraint accomplished. Choose View to see the details, see: Sequencing Task Details and Results, page 6-31

Production Line	Available Capacity	Total Fulfilled Quantity	Resequenced Satisfaction (%)	Demand Satisfaction (%)	Constraint Satisfaction (%)	View
Desktop	62	0	100	100	100	

Sequencing Task Details and Results

The Task Detail page is a view only, printable page displaying the results from your sequencing run. This page contains regions for:

- Sequencing Task Parameters
- Constraints
- Results Summary
- Unfulfilled Demand
- Violated constraints
- Violated Flow Schedules
- Other Exceptions

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[Sequencing Tasks](#) > [Task Summary: MFG-10522 \(M1\)](#)

Task Detail: Desktop (M1) [Printable Page](#)

Sequencing Task Parameters

Name	MFG-10522		
Production Line	Desktop	Type	Resequence
Daily Rate	100	Demand Date Range	-
Referenced Rule	ss-rule	Schedule Date Range	31-May-2004 14:30:00 - 05-Jun-2004 14:29:59

▼ **Constraints**

Priority	Type	Attribute	Definition
5	Ordering	Assembly	Order Assembly ascending
10	Grouping	Assembly	Group MC24713

▼ **Result Summary**

Available Capacity	400	Quantity Before First Violation	300
Constraint Satisfaction (%)	100	Total Violations	0

Type	Satisfaction (%)	Fulfilled Quantity	Unfulfilled Quantity	Requested Quantity
Resequenced Schedules	100	300	0	300
Demands	100	0	0	0
Total		300	0	300

The Sequencing Task Parameters region displays setup details for the production line, and parameters entered for the task. This includes production line, daily rate, rule, constraints, demand type, and dates for demand and schedule ranges. The Constraints region displays a list of constraints for this rule and can be used in comparing results in the Violated Constraints list.

The Results Summary region provides a view of production line actual capacity considered on this specific run. It excludes the capacity fixed. For example, if the option is to fix schedules starting at the time 10:00 am, and the line time is 8:00 am to 4:00 pm, it considers the capacity from 10:00 am to 4:00 pm.

The Constraint Satisfaction percentage displays. For example, 80 percent signifies that of a total of 100 lines, the Flow schedules created on the first 80 lines do not have any violations.

The Unfulfilled Demand region provides the details for the Demand Satisfaction listed in the summary view. For example, 80 percent signifies that out of 100 quantity demands, 80 of them are scheduled.

▼ Unfulfilled Demands

Order Number	Line Number	Assembly	Assembly Description	Customer	Requested Quantity	Sequenced Quantity
54712	3	MC24713	Plastic Base - Black	Computer Service and Rentals	168	100
54714	1	MC24713	Plastic Base - Black	Total Internet	15	0
54715	1	MC24713	Plastic Base - Black	A. C. Networks	10	0
54715	2	MC24713	Plastic Base - Black	A. C. Networks	20	0
54716	4	MC24713	Plastic Base - Black	Computer Service and Consulting	3	0

▼ Violated Constraints

Priority	Type	Attribute	Definition	Fulfilled Up to Quantity	Violations
0	Daily Limit	Assembly	Daily limit of MC24713 is 100	470	1

▼ Violated Flow Schedules

Schedule Number	Build Sequence	Number of Violated Constraints	Violated Constraint Priorities
No data exists.			

▼ Other Exceptions

Description
No data exists.

The Violated Constraints region lists the constraints broken in this sequencing run. Constraint violations are pushed to the end of the sequencing horizon.

The Violated Flow Schedule region displays a list of schedule numbers, their build sequences, the number of constraints violated, and the priorities of the constraints. The sequencing program fails when a hard constraint is violated on any newly created schedules. Violations are ignored on existing schedules. The priority level designates if the constraint is classified as a hard constraint—a constraints that must be fulfilled. If these constraints are broken—or violated—Flow Sequencing stops. A value of zero indicates a hard constraint.

▼ Violated Flow Schedules			
		☐ Previous ☒ 1-25 ☐ Next 25 ☐	
Schedule Number	Build Sequence	Number of Violated Constraints	Violated Constraint Priorities
123116	1681	2	30, 50
123117	1682	2	30, 50
123118	1683	2	30, 50
123119	1684	2	30, 50
123120	1685	2	30, 50
123121	1686	2	30, 50
123122	1687	2	30, 50
123123	1688	2	30, 50
123124	1689	2	30, 50
123125	1690	2	30, 50

The Other Exceptions region displays any other explanations for Flow schedules not created. For example, the exception message of Invalid Build Sequence Range indicates that no build sequences available in order to create Flow schedules.

Line Scheduling Workbench

This chapter covers the following topics:

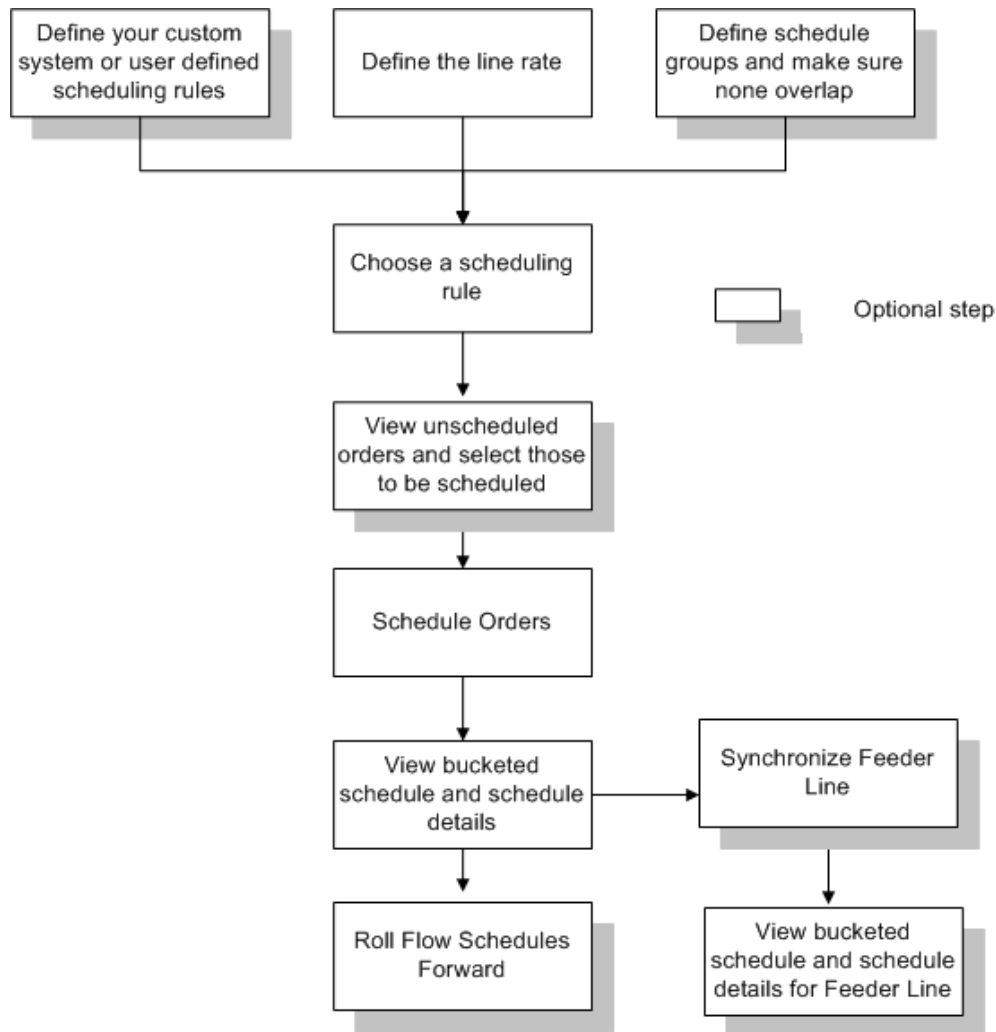
- Overview of Line Scheduling Workbench in Flow Manufacturing
- Defining Scheduling Rules
- Define Scheduling Rules Window
- Line Scheduling Workbench Options Window
- Viewing Unscheduled Orders
- Viewing, Creating, and Modifying Flow Schedules
- Creating Flow Schedules
- Flow Schedule Summary Window
- Flow Schedule Details Window
- How Scheduling is Calculated
- Line Scheduling Workbench Tools Menu
- Roll Flow Schedules
- Deleting Flow Schedules
- Component Availability and Purging Flow Schedules

Overview of Line Scheduling Workbench in Flow Manufacturing

Line scheduling is the process of sequencing and scheduling work to be done on your balanced flow lines. You create flow schedules to represent the volume and mix of products to be produced. The objective of line scheduling is to synchronize production as tightly as possible to customer demand, as well as establishing an achievable pace and consistent flow of products throughout the production process and supply chain.

The Line Scheduling Workbench is used by schedulers to create and manage flow schedules. You can simulate the assignment of work to production lines, and schedule

work on specific lines. While you can view all scheduled orders (such as discrete jobs, repetitive schedules, and flow schedules) through the workbench windows, the only type of work you can create and schedule in the workbench is a flow schedule. The following illustration shows the process flow for line scheduling.



The Line Scheduling Workbench Options window is the interface to the other planning windows in line scheduling. The functions available through the workbench are:

- Viewing Unscheduled Orders
You can view unscheduled orders in the Line Scheduling Workbench Unscheduled Orders window. You have the option to display the unscheduled planned orders or sales orders. These unscheduled orders can be converted into flow schedules.
- Creating and Maintaining Flow Schedules
You can create and modify schedule quantities and dates of flow schedules.

- **Prioritizing and Sequencing Flow Schedules**
You can schedule your flow schedules either by using the scheduling rules provided, or ones that you have defined.
- **Flow Schedule Completion**
You can perform completion transactions for scheduled flow schedules when production is finished.
- **Mixed Model Map**
The Mixed Model Map is used to monitor your line design by balancing a line against a specified statement of demand.
- **Roll Flow Schedules**
You can roll flow schedules forward, based on under completions and overcompletions, in order to adjust future production.
- **Deleting Flow Schedules**
You can delete flow schedules in order to provide a clean slate for new schedule generation.
- **Checking the Availability of Components**
You can access the ATP Results window to view component availability.
- **Viewing Resource Load in Relationship to Available Capacity**
You can view a graphical representation of the required load of the line and its associated resources.

Defining Scheduling Rules

A scheduling rule determines the logic used to set the timing of the flow schedules. The rule is comprised of an algorithm and the criteria used in sequencing the flow schedules. Oracle Flow Manufacturing provides several types of scheduling rule options. Pre-defined scheduling rules combine a system algorithm with sequencing criteria. In addition, there are two methods to create your own rules.

Scheduling Rule Options

- Pre-defined scheduling rules combine a system algorithm and one sequencing criteria. Five pre-defined rules are provided.
- **System-Defined Rules**
User-defined scheduling rules allow you to combine a system algorithm with one

or more of the system provided criteria.

- User-Defined

You can also create your own rules defined by modifying the custom line scheduling procedure that is executed at the time of scheduling flow lines.

Scheduling Algorithms

Three algorithms are provided for line scheduling:

No Level Loading

Flow schedules are sequenced according to criteria, and the demand is placed in the days of the scheduling horizon. Starting with the first available day, demand is scheduled until capacity is reached.

Criteria Order is used to determine the sequence of the flow schedules. Beginning with the demand with lowest sequence (or highest priority), demand scheduled in ascending order.

Level Daily Rates

Flow schedules on each day are sequenced based on the demand ratio. In this algorithm the production rate for each item is calculated for each day. Therefore, items with higher demand have a higher production rate. That rate is then used on each day of the scheduling horizon.

Production rate for an item on a particular day is determined by multiplying the available capacity on that day by the demand ratio for the item.

Mixed Models

In this algorithm the production rate for different items is calculated in order to establish a production rule pattern for a mix of items. The rule pattern is repeated within a day and across the scheduling horizon.

Sequencing Criteria

You can sequence flow schedules using one or more of the following values:

- Order Entry Date
- Order Priority
- Order Promise Date
- Order Request Date

- Order Schedule Date
- Production Sequence Forward
- Production Sequence Reverse

The criteria is listed in the order of priority. For example, you can sequence by Order Request Date and Order Entry Date. The Order Request Date has priority 1, the Order Entry Date has priority 2. If two different flow schedules have the same order request date, then the order with the earliest order entry date is scheduled first.

Criteria are only applied to unscheduled sales orders. For unscheduled planned orders, the generated Planned Order Due Date value is the only criteria used for sequencing.

Related Topics

Define Scheduling Rules Window, page 7-5

Define Scheduling Rules Window

You can view scheduling rules and sequencing criteria, as well as define your own custom rules in this window.

Define Scheduling Rules

Name

Description

☒ Enabled ☐ Default []

Definition Type

☐ System ☒ User

Leveling **Mixed Model**
Level Daily Rate
No Level Loading

Criteria Name	Criteria Order

To view existing scheduling rules:

1. Navigate to the Define Scheduling Rules window.
2. From the Query menu choose Enter, and then choose Run.
Use the Arrow keys to page through the list of scheduling rules.

To define system-defined or user-defined scheduling rules:

1. Navigate to the Define Scheduling Rules window.
2. Enter the name of the scheduling rule in the Name field.

If you are creating a custom rule, this value is user-defined. The following rules are provided in the system:

- Mixed Model - mixed model by order request date
- Level Daily Rate- level daily rate by order request date
- Request Date- order request date with no leveling

- Schedule Date - order schedule date with no leveling
 - Promise Date - order promise date with no leveling
 - Production Sequence Forward - order schedule date, order production date sequence in ascending order with no level loading
 - Production Sequence Reverse - order schedule date, order production date sequence in descending order with no level loading
3. Enter the description of the rule in the Description field.
- For rules you create, it is beneficial to have the name and description reflect the algorithm being used.
4. Uncheck the Enabled check box if you are disabling an existing rule, otherwise leave it marked.
- This is a Yes/No flag indicating whether this rule is currently in use.
5. Select the Default check box if this rule is to be your default scheduling rule.
- This is Yes/No flag indicating if this is the default rule. The default rule is used unless it is overridden; there can only be one default rule for scheduling.
6. If this is a system-defined rule, choose the System Defined option button. Otherwise, if this rule is to be defined by a user procedure, proceed to Step 9.
- This option button enables the region for rules using system provided algorithms and sequencing criteria.
7. Select an algorithm in the Leveling box.
- The system algorithm values in this box are:
- No Level Loading—sequencing is calculated only by priority criteria
 - Level Daily Rate—the calculation is the ratio of total demand for each item, multiplied by the line rate
 - Mixed Model—daily demand is calculated from different products and different lot sizes
8. In the Criteria Name box, select criteria values from the list of values.
- Enter one or more values in order of priority. For example, if you select Order Request Date first, and then Order Entry Date—the order with the earliest request date and earliest order entry date is scheduled first.
- Order Entry Date

- Order Priority
 - Order Promise Date
 - Order Request Date
 - Order Schedule Date
 - Production Sequence Forward
 - Production Sequence Reverse
9. For user procedures, choose the User Defined option button.
- This option button sets the rule to user-defined. To add custom line scheduling logic, modify the Custom_Schedule procedure (MRPPCLSS.pls/MRPPCLSB.pls).
10. Save your work.

Line Scheduling Workbench Options Window

The Line Scheduling Workbench Options window is the starting window from which to access line scheduling options. From this window you can access the other windows to view unscheduled orders, create flow schedules, view and modify existing schedules.

Honor Planning Recommendations and Primary Routing Only Check Boxes

The following list presents the combined behavior of these check boxes:

- **Honor Planning Recommendations checked/Primary Routing Only unchecked**
Shows only planned orders that planning has recommended on the selected line.
- **Honor Planning Recommendations unchecked/Primary Routing Only checked**
Shows only planned orders of assemblies that have primary routing on the selected line.
- **Honor Planning Recommendations unchecked/Primary Routing Only unchecked**
By default shows planned orders of assemblies that have routing on the selected line.
- **Honor Planning Recommendations checked/Primary Routing Only checked**
Shows all the planned orders which planning has recommended on the selected line if the assembly of the planned order has primary routing on the selected line.

Line Scheduling Workbench Options (M1)

Line: Desktop Desktop Flow Line

Schedule Group:

Daily Production Rate: 500

Unscheduled Orders

Type: Planned Orders

Order Dates: 03-FEB-2009 00:00:00 - 03-FEB-2009 23:59:59

☐ Honor Planning Recommendations

☐ Primary Routing Only

Scheduling

Rule: Mixed Model

Schedule Dates: 03-FEB-2009 00:00:00 - 03-FEB-2009 23:59:59

☐ Flex Tolerance

Unscheduled Orders Create Flow Schedules

All Flow Schedule Details Flow Schedule Summary

To view unscheduled orders, create flow schedules, or modify existing schedules:

1. Navigate to the Line Scheduling Workbench Options window by selecting an organization and production line.

The Line Scheduling Workbench Options window displays information on the selected line. You can access the other windows using the defaulted information, or you can enter other values.

The Daily Production Rate is information for this production line. It is the calculation of the Hourly Production Rate multiplied by the number of work hours for this line.

2. Optionally select a Schedule Group value.

The Schedule Group is defined in Work in Process. The value entered here is used as a default for any schedules created.

3. In the Unscheduled Orders region, select the Type and Order Date ranges for unscheduled orders.

You can select Sales Orders or Planned Orders. Sales orders are created from Order Management transactions; Planned Orders are created from all of your planning production.

If you select Planned Orders, then select Primary Routing Only check box to see planned orders for assemblies that have primary routing on the selected line.

Select Honor Planning Recommendations along with Primary Routing Only check box to filter planned orders based on planning recommendations. As planning can recommend lines that have assemblies with alternate routings, this helps you to filter those orders with demand type as Planned Orders.

Note: If you create multiple plans marked as production – the Production flag is checked on the Master Production Schedules window or the MRP Names window– you can create duplicate flow schedules.

Note: Flow schedules are created from planned orders in the Line Scheduling Workbench—not releases from the Planner Workbench.

The Order Dates range determines which orders are displayed on the Unscheduled Orders window, and signifies the period of time to consider demand for scheduling. It refers to either schedule ship date of sales orders, or due dates of planned orders.

4. In the Scheduling region select a scheduling rule.
5. Select beginning and ending date values.

Flow schedules are created for the date ranges that you select. See: How Scheduling is Calculated, page 7-22

6. Check the Flex Tolerances check box if you want to enable flexible tolerance fences in your scheduling calculations.

Flexible tolerance fences provide temporary increases to capacity to handle increases in orders. The line rate is adjusted by the tolerance fence percentage before scheduling rules are applied. Tolerance fences are set up when creating line definition. See: Defining Production Lines and Tolerance Fences, *Oracle Work in Process User's Guide*

7. Choose one of the following for line scheduling functions:

Unscheduled Orders: Displays the Unscheduled Orders window showing all demand that has not been converted into Flow schedules. See: Viewing Unscheduled Orders, page 7-11

- *All Flow Schedule Details:* Displays the Flow Schedule Details window enabling you to view and modify flow schedules at the detail level of the assembly. This function displays all open Flow schedules without regard to the schedule dates. See: Flow Schedule Details Window, page 7-20

- *Create Flow Schedules*: Creates new Flow schedules using the values selected on the Line Scheduling Workbench Options window. See: How Scheduling is Calculated, page 7-22
- *Flow Schedule Summary*: Displays the Flow Schedule Summary window for viewing existing schedules. See: Flow Schedule Summary Window, page 7-18

Viewing Unscheduled Orders

The Unscheduled Orders window is used to view unscheduled demand - demand that has not been converted into Flow schedules. You can change certain values and create Flow schedules directly from this window. You can also use criteria to filter your unscheduled orders, and create Flow schedules according to that criteria. See: How Scheduling is Calculated, page 7-22.

To view unscheduled orders:

1. Navigate to the Line Scheduling Workbench Options window and enter values, or use the default information.
2. Choose Unscheduled Orders to display the Find Unscheduled Orders window.
3. Enter search criteria, or leave all fields blank to retrieve all records that fit the criteria set in the Line Scheduling Options window.

Field	Condition	From	To

Find Clear

4. To filter your search, enter a search string by selecting Field values and an operator in the Condition field.

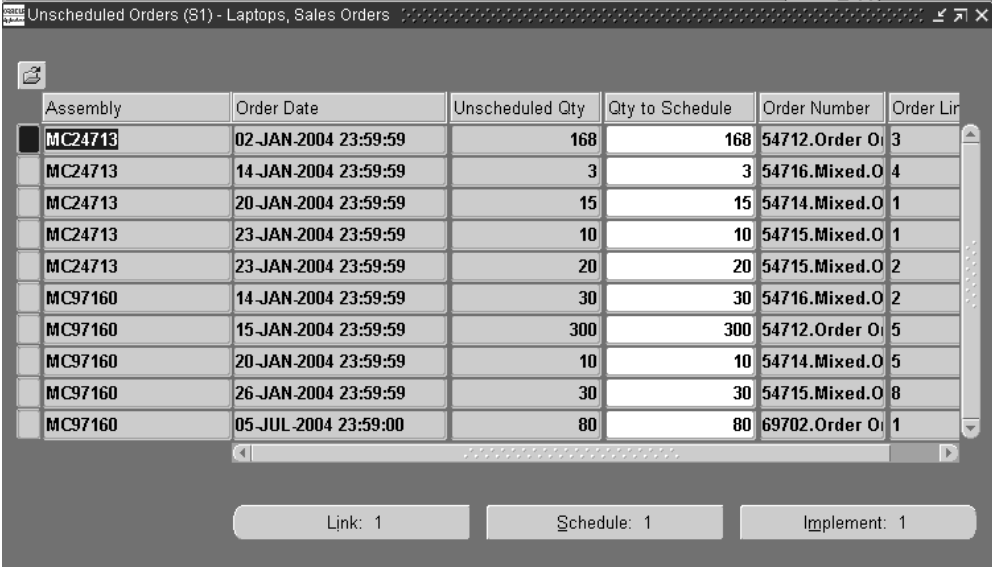
You can use one string or several strings.

You can also select the Folder menu to customize your display and save your Find criteria.

See: Customizing the Presentation of Data in a Folder, *Oracle E-Business Suite User's*

5. Choose Find.

The Unscheduled Orders window shows all orders for the selected order type, date range, and filter criteria.



Assembly	Order Date	Unscheduled Qty	Qty to Schedule	Order Number	Order Line
MC24713	02-JAN-2004 23:59:59	168	168	54712.Order O	3
MC24713	14-JAN-2004 23:59:59	3	3	54716.Mixed.O	4
MC24713	20-JAN-2004 23:59:59	15	15	54714.Mixed.O	1
MC24713	23-JAN-2004 23:59:59	10	10	54715.Mixed.O	1
MC24713	23-JAN-2004 23:59:59	20	20	54715.Mixed.O	2
MC97160	14-JAN-2004 23:59:59	30	30	54716.Mixed.O	2
MC97160	15-JAN-2004 23:59:59	300	300	54712.Order O	5
MC97160	20-JAN-2004 23:59:59	10	10	54714.Mixed.O	5
MC97160	26-JAN-2004 23:59:59	30	30	54715.Mixed.O	8
MC97160	05-JUL-2004 23:59:00	80	80	69702.Order O	1

Link: 1 Schedule: 1 Implement: 1

This window displays fields for assembly number, order date, the unscheduled quantity, the quantity available to schedule, sales order number, and corresponding sales order lines.

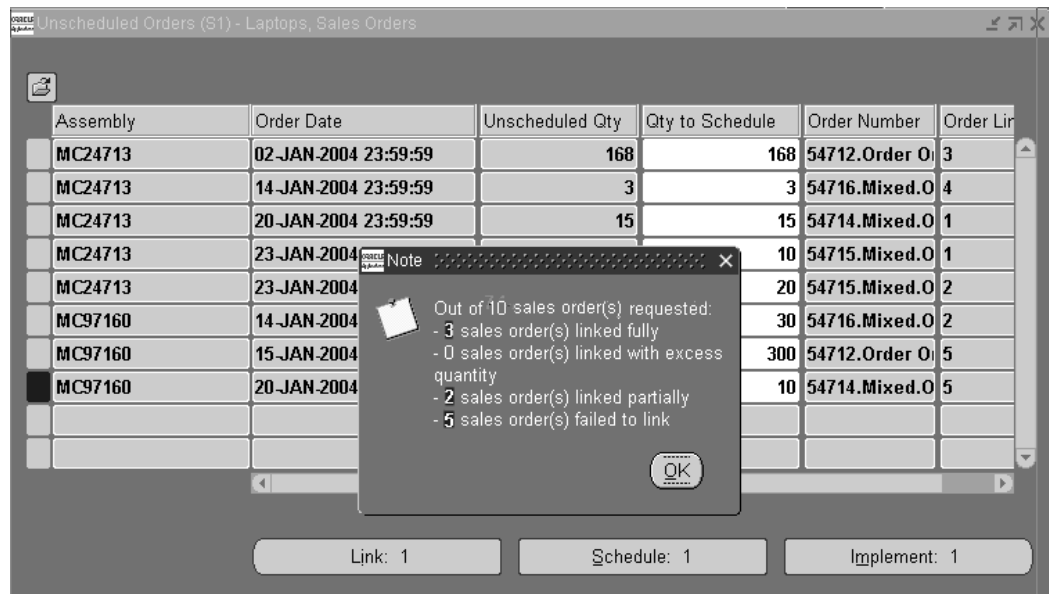
6. Choose Link to link Flow schedules to sales order lines. See: Linking Sales Orders to Schedules, page 7-12
7. Choose Schedule to create schedules for records selected. The Flow Schedule Summary window displays to view schedule information of planned and completed quantities at an aggregate level. This method enables you to create Flow schedules according to filtering criteria, schedule rule, and line capacity.
8. Choose Implement to create schedules for all unscheduled orders selected. These orders are scheduled without regard to the line capacity or scheduling rules. See: How Scheduling is Calculated, page 7-22

Linking Sales Orders to Schedules

Flow schedules are created from sales order demand, planned order demand, or created manually. Manually created schedules or schedules that have had the demand type removed do not reference a sales order. The linking functionality enables you to establish a reference to sales orders during Flow schedule creation - if one does not already exist, or if it is linked to a planned order. This functionality is accessed from the Unscheduled Orders window.

Note: To consume the sales order demand, select the Consume Order check box. This will consume the sales order into flow schedules created from planned orders and these planned orders will not appear as unscheduled orders. But if you want to schedule it separately, then do not select the check box and the planned orders whose flow schedules are linked to the sales order appear as unscheduled orders.

By default the Consume Order check box is not displayed, as it is a folder field. To display the field, select the field from the folder. By default, the check box is unchecked



When you choose Link from the Unscheduled Orders window, sales order quantities are matched to selected schedule quantities that fall within the date range specified. A Note window displays the linking results:

- Sales orders linked fully

The quantity in the sales order line matches total quantity in the Flow schedules.

- Sales orders linked with excess quantity

The quantity in the sales order line is less than the total quantity in the Flow schedules.

- Sales orders linked partially

The quantity in the sales order line is greater than the total quantity in the Flow schedules. Flow schedules fulfill partial quantity of the sales order line.

- Sales orders failed to link

Sales order is not linked to any Flow schedule.

For example, you may have sales order and unscheduled order quantities as demonstrated in the following table. The linking process matches sales order demand with unscheduled demand resulting in some linked orders, and others not linked.

Unscheduled Sales Order Demand	Flow Schedule Quantities not Linked	Matched Quantities
Order1, 6 each	FS1, 2 each	Order1, FS1. 2
-	FS2, 2 each	Order1, FS2, 2
-	FS3, 3 each	Order1, FS3, 3
Order2, 4 each	FS4, 2 each	Order2, FS4, 2
-	FS5, 2 each	Order2, FS5, 2
Order3, 6 each	FS6, 2 each	Order3, FS6, 2
Order4, 8 each	-	Order4, 0
Order1, 6 each	FS1, 2 each	Order1, FS1. 2

In this example, the Note window displays these results:

- 1 sales order linked fully (Order2)
- 1 sales order inked with excess quantity (Order1)
- 1 sales order linked partially (Order3)
- 1 sales order failed to link (Order4)

Viewing, Creating, and Modifying Flow Schedules

You access both flow schedule summary and detail information from the Line Scheduling Workbench Options window. This allows you to view, create, and change flow schedules. The Flow Schedule Summary window shows schedule information grouped by assembly and date, showing the planned and completed quantities at an aggregate level. You can then navigate to the Flow Schedule Details window for more information on the assembly and its components.

Note: Oracle Project Manufacturing feature of model/unit effectivity is supported in the Line Scheduling Workbench. If this feature is enabled, the Unit Number field is visible when viewing unscheduled orders and passed on to new flow schedules created. See: Model/Unit Effectivity, *Oracle Project Manufacturing Implementation Manual*.

Related Topics

Creating Flow Schedules, page 7-15

Flow Schedule Summary Window, page 7-18

Flow Schedule Details Window, page 7-20

Creating Flow Schedules

You can create flow schedules the following ways:

- Use the Create Flow Schedules button on the Line Scheduling Workbench Options window. When you choose this button, flow schedules are created for all unscheduled orders that fall within the date ranges if the capacity is adequate.

Note: If the scheduling rule is set to Do Not Schedule, unscheduled orders are converted to flow schedules according to order dates. All unscheduled orders in the specified date range are converted to flow schedules – schedule dates, capacity, and sequencing criteria are ignored.

- Use the Implement button on the Unscheduled Orders window. Flow schedules are created for all unscheduled orders that you have selected on this window. These orders are scheduled without regard to the line capacity or scheduling rules.
- Choose Schedule on the Unscheduled Orders window. This method enables you to create Flow schedules according to filter criteria.
- Modify the schedule on the Flow Schedules Summary window. If a schedule quantity is added to a day that does not already have a flow schedule, one is created.
- You can add a new record on the Flow Schedules Details window.

To create flow schedules from the Line Scheduling Workbench Options window:

1. **Note:** Do not use FLM-INTERNAL as prefix for schedule number when you manually create flow schedules from line scheduling workbench.

Navigate to the Line Scheduling Workbench Options window.

2. You can use the default information on this window, or enter values. See: Line Scheduling Workbench Options Window.

Flow schedules for a given line are sequenced within a Schedule Group, schedules without a schedule group are sequenced together.

The Order Date ranges are used for the period of time to consider demand for scheduling. Flow schedules are created for unscheduled orders that fall within this date range – if the capacity is adequate.

3. Choose Create Flow Schedules.

Flow schedules are created from unscheduled orders by considering the following elements:

- The demand for the order type within the specified date range in the Unscheduled Orders window.
- The orders topmost in priority, based on sequencing criteria
- The line capacity for the date range. If there is not enough capacity, the order remains unscheduled.

The results of this process are shown in the Flow Schedule Summary window.

If you choose Unscheduled Orders, you can see the orders that were not converted into flow schedules.

4. Save your work.

To convert unscheduled orders to flow schedules:

1. Navigate to the Unscheduled Orders window.
2. Select the records to implement into flow schedules.

You can make any modifications or adjustments to the details of your records. You can select multiple assemblies on this window for viewing on the detail window.

Flow schedules are created for all unscheduled orders that you have selected on this

window. These orders are scheduled without regard to the line capacity or scheduling rules

3. Choose Implement.
4. Choose OK. Otherwise, to cancel this transaction, choose Cancel.
5. Choose Yes to view the Flow Schedule Details window with the new schedule record. Choose No to view the Unscheduled Orders window.

After the new schedule is created, the demand does not appear on the Unscheduled Orders window.

6. Save your work.
7. Choose Schedule to create schedules for records selected. The Flow Schedule Summary window displays to view schedule information of planned and completed quantities at an aggregate level. This method enables you to create Flow schedules according to filtering criteria, schedule rule, and line capacity.

To create flow schedules by modifying information on the Flow Schedules Summary window:

1. Navigate to the Flow Schedules Summary window.

Only schedules without details (such as schedule group, project, or task) can be modified at the summary level. If details exist, the Flow Schedule Details window appears.

2. Enter the new planned quantity, or scroll to the date you to modify and update quantity. You can only modify the Total Planned Quantity, not the completed or variance value.
3. Save your new record.

If a schedule quantity is added to a day that does not already have a flow schedule, a new record is created for that day.

Note: If the quantity is changed to 0, the flow schedule is deleted – but you cannot delete flow schedules with partial or full completion status.

To create flow schedules by adding a new record on the Flow Schedule Details window:

1. Navigate to the Flow Schedule Details window.

2. Enter the record information and save your record.

You can add flow schedules only for the line you have specified on the Line Scheduling Workbench Options window.

Flow Schedule Summary Window

The Flow Schedule Summary window is used to view and manage flow schedules at an aggregate level. This window shows all flow schedules, discrete jobs, and repetitive schedules at a summary level.

Note: Only flow schedules can be maintained through this window. Repetitive schedules and discrete jobs are displayed but cannot be modified.

To view or modify flow schedules at a summary level:

1. Navigate to the Line Scheduling Workbench Options window.
2. You can use the default information on this window, or enter values. See: Line Scheduling Workbench Options Window.

Schedule quantities can be modified in both the summary and detail windows. However, only schedules that do not have details (such as schedule group, project, or task) can be modified at the summary level.

3. Choose Flow Schedule Summary

The Flow Schedule Summary window shows schedules, in a bucketed format, from the start dates selected on the Line Scheduling Workbench Options window. You can scroll back or forward in this window.

Three types of order information can be viewed. You can select the types displayed on this window through Preferences in the Tools Menu. See: Preferences, page 7-25

- Planned quantity is calculated when the flow schedule is created, this amount can be changed.
- Completed quantity is updated when you perform a completion on this schedule.
- Variance is the difference between Planned and Completed quantities.

Flow Schedule Summary (S1) - Laptops

Week Of **14-JUN-2004**

Assembly	Type	Total	14-JUN-2004	15-JUN-2004	16-JUN-2004	17-JUN-2004
MC24713	Planned	2	0	0	2	
	Completed	1	0	0	1	
	Variance	1	0	0	1	
MC97160	Planned	2	0	0	2	
	Completed	0	0	0	0	
	Variance	2	0	0	2	
Total Planned Quantity		4	0	0	4	
Total Completed Quantity		1	0	0	1	
Total Variance		3	0	0	3	

Description: **Plastic Base - Black**

Unscheduled Orders Details: 1

4. To modify a flow schedule quantity, change the Planned quantity in the date column selected. However, only schedules that do not have details (such as schedule group, project, or task) can be modified at the summary level.

The Total and Total Planned Quantity for this assembly reflect the changes you just made.

This window offers the feature of spreading weekly quantities across the valid workdays. Enter a value in the Total field, and choose Tab.

The new total quantity will spread even across all the workdays and display in the window.

5. Save your work.
6. You can access the following windows:
 - *Unscheduled Orders*: Displays the Unscheduled Orders window. See: Viewing Unscheduled Orders, page 7-11
 - *Details*: Displays the Flow Schedule Details window for the assemblies and dates selected. The number after the word *Detail* indicates how many records are selected. You can select multiple assemblies on this window for viewing on the detail window. See: Flow Schedule Details Window, page 7-20

Flow Schedule Details Window

The Flow Schedule Details window enables you to view and modify Flow schedules at the detail level of the assembly. You can access this window from the Line Scheduling Options or Flow Schedule Summary windows. If you are navigating from the Line Scheduling Options window, you can filter your search by entering field and operator values. All details within criteria specified display on the Flow Schedule Details window. If you are accessing this window from the Flow Schedule Summary window, only the schedules selected display.

When Oracle Flow Manufacturing is integrated with Oracle Advanced Supply Chain Planning, manually entered Build Sequence values cannot exceed 9 digits. This restriction does not apply to Flow Manufacturing when not integrating with these products.

Note: Only flow schedules can be maintained through this window. Repetitive schedules and discrete jobs are displayed but cannot be modified.

To enter flow schedule details:

1. Navigate to the Line Scheduling Workbench Options window.
2. Enter values, and choose All Flow Schedule Details. See: Line Scheduling Workbench Options Window
The Find Flow Schedule Details window displays.
3. Enter search criteria, or leave all fields blank to retrieve all records.
4. To filter your search, enter a search string by selecting Field values and an operator in the Condition field

You can use one string or several strings.

You can also select the Folder menu to customize your display and save your Find criteria.

Field	Condition	From	To
Quantity	Greater Than	5	

Find Clear

Customizing the Presentation of Data in a Folder, *Oracle E-Business Suite User's Guide*, and Searching for Data, *Oracle E-Business Suite User's Guide*

5. Choose Find.

The Flow Schedule Details window displays information according to your search criteria.

To select specific Flow schedule details from the summary window:

1. Navigate to the Flow Schedule Summary window, select one or more assemblies.
2. Choose Details.

The Flow Schedule Details window displays information on the assemblies selected in the summary window.

To view or modify Flow schedule details:

1. Navigate to the Flow Schedule Details window.

This window displays fields for Assembly, Build Sequence, Completion Date, Quantity, Completed Quantity, Order Number, Order Line, Order Requested Date, and Unit Number.

Assembly	Build Seq	Completion Date	Quantity	Completed Quantity	Order Number
MC24713		12-JUN-1998 16:47:00	0	0	
MC24713		13-JUL-1998 00:00:00	160	0	36348.Standa
MC24713	1	13-JUL-1998 07:00:36	1	0	36348.Standa
MC24713	2	13-JUL-1998 07:01:12	1	0	36348.Standa
MC24713	4	13-JUL-1998 07:02:24	1	0	36348.Standa
MC24713	5	13-JUL-1998 07:03:00	1	0	36348.Standa
MC24713	7	13-JUL-1998 07:04:12	1	0	36348.Standa
MC24713	8	13-JUL-1998 07:04:48	1	0	36348.Standa
MC24713	10	13-JUL-1998 07:06:00	1	0	36348.Standa
MC24713	11	13-JUL-1998 07:06:36	1	0	36348.Standa

Unscheduled Orders Component Availability: 1

2. You can modify some values including Build Sequence, Completion Date, and Quantity fields.

Note: When Oracle Flow Manufacturing is integrated with Oracle Advanced Supply Chain Planning, manually entered Build Sequence values cannot exceed 9 digits.

3. Save your work.
4. You can access the following windows:
 - *Unscheduled Orders*: Displays the Unscheduled Orders window. See: Viewing Unscheduled Orders, page 7-11
 - *Component Availability*: Displays the ATP Results window to show the status of material used on your lines. See: Component Availability, page 7-30

Component availability checking is permitted only for items with the Check ATP item attribute set to Yes. See: Order Management Attribute Group, *Oracle Inventory User's Guide*

How Scheduling is Calculated

Scheduling rules determine the timing of the flow schedules using the following logic in the calculation routine:

- Sequence is determined by the criteria in the rule.

Demand is sequenced using the criteria in the scheduling rule. The priority of each criteria determines the order in which it is applied.

- The first flow schedule created goes to the first available date in the scheduling horizon.

Flow schedules are created for orders starting with the first available due date that has capacity. Flow schedules are not created for orders that cannot be scheduled within the scheduling horizon.

- This process of attaining the line rate for each day continues until the end of the scheduling horizon.

Capacity is calculated for the day using the line rate, subtracting the existing workload (flow schedules, discrete jobs, repetitive schedules) for that day. When available capacity for the day is attained, schedules are slated for the next day until the end of the scheduling horizon.

- If you are using flexible tolerance fences, scheduling begins after the last day indicated from the Days in Advance value.

Tolerance fences provide temporary increases to capacity above their normal operating rate to handle rises in orders.

Calendar Shifts Crossing Non-Work Days

Oracle Flow Manufacturing supports shifts that cross calendar days, as long as the start of the shift falls on a working day. To accomplish this, ensure that the Oracle Bills of Material Workday Calendar, the line definition, and the scheduling window you use in the Line Scheduling Workbench are synchronized. For example, you can specify the workday calendar is setup for the days Monday - Friday, and three shifts:

- Shift 1—7 am to 3 pm
- Shift 2—3 pm to 11 pm
- Shift 3—11 pm to 7 am

The line definition is then set up to work from 7 am to 7 am. To schedule all shifts for Friday, ensure that the current schedule on the Line Scheduling Workbench is set to Saturday at 7 am. The Line Scheduling Workbench then schedules through 7 am, Saturday—even if Saturday is a non-work day.

You cannot start a shift on a non-work day. For example, if you have a shift starting at 11 pm on Sunday, the Line Scheduling Workbench cannot schedule for the first hour of the shift. See: *Using the Workday Calendar, Oracle Bills of Material User's Guide*

Algorithms

The scheduling rule is comprised of an algorithm and the criteria used in sequencing the flow schedules. Oracle Flow Manufacturing provides the following three algorithms:

- No Level Loading

If you select a scheduling rule that has no level loading, the criteria defined in the scheduling rule is used to prioritize and sequence the orders. Orders are scheduled with capacity considerations beginning with the first available date of the scheduling date range. If the entire quantity of the sales order cannot be satisfied on a date, the remaining quantity is scheduled on subsequent days where capacity is available.

- Level Daily Rate

If you choose to schedule for each day using the level daily rate, each item is scheduled at a rate in regard to the demand ratio of the item. The demand ratio of an item equals the total demand of the item divided by the grand total demand.

- Mixed Model

This algorithm establishes a pattern to manufacture items in order to meet demand. The pattern distributes units according to the demand of each item for each day.

Line Scheduling Workbench Tools Menu

The following are functions available in the Tools Menu for maintaining flow schedules. This menu is available through any of the windows in the Line Scheduling Workbench.

- Completions
- Mixed Model Map
- Preferences
- Roll Flow Schedules
- Delete Flow Schedules

Completion of Flow Schedules

You can complete flow schedules through the Work Order-less Completions window. Choose Completions from the Tools menu to display the window. The details of the schedules you selected are defaulted in the Work Order-less Completions window.

Related Topics

Performing Work Order-less Completions, *Oracle Work in Process User's Guide*

Mixed Model Map

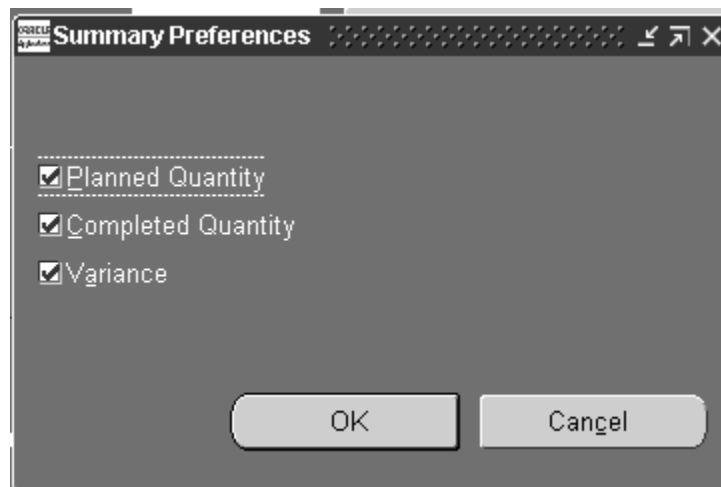
You can access the Mixed Model Workbench from the Tools menu. The Mixed Model Map Workbench helps you to create and monitor your line design by balancing a line against a specified forecast.

You can use the Mixed Model Map to compare line information with demand requirements. This provides information to adjust resources on the line to meet the demand. The output of the Mixed Model Map will give you the line rate.

Preferences

The Preferences window is used to change the quantity types that show on the Flow Schedule Summary window.

The Planned quantity is originally calculated when the flow schedule is created using the rule and available capacity. The Completed quantity is updated when you perform a completion on this schedule. The Variance is the difference between Planned and Completed quantities.



Roll Flow Schedules

You must maintain flow schedules so that production schedule information is accurate; and planning information such as Available-To-Promise and demand for components is current. The Roll Flow Schedules program is designed to allow you to adjust future schedules based on actual performance in the past. You adjust the quantity on future production schedules based on the completion quantities and past production. You can

change schedules for a line, a range of lines, or for all lines at once.

Note: Planning calculations do not consider flow schedules with past due dates. Use the Roll Flow Schedules window to update your flow schedules in order to keep planning accurate.

Note: The Feeder Line Synchronization program ignores flow schedules created from the Roll Flow Schedules program. You must roll schedules on both the parent and feeder lines.

Three options are available for adjusting schedules: Close All Past Schedules, Roll With Details, and Spread Evenly Without Details.

Close All Past Schedules

This option closes all open past due schedules.

Note: Schedules are automatically closed when the completion quantity is greater than or equal to the planned quantity.

Closed flow schedules cannot be changed or rolled with details, and they are denoted with gray fields. You can close individual or a group of schedules. For example, a customer might reduce the order quantity because a delivery date can not be made. The original schedule is closed and the program does not roll the excess quantity no longer needed.

Roll With Details

This option rolls any open schedules within the specified date range to a single date. The new schedules maintain the original schedule details including the order information, build sequence, and alternate bill of material or routing information. The rolled schedules are scheduled at specific times if no schedules exist on the rolled to date.

Note: The original schedules are closed after the roll. Therefore the same schedule cannot be rolled multiple times, nor can you change the original schedule once it has been rolled. The schedule number from the original schedule is rolled to the new schedule.

Spread Evenly Without Details

You can spread schedules without order information evenly across multiple days, for use in a repetitive environment, allowing you to:

- Calculate variance and roll by line and item
- Consider both the planned quantity and the completed quantity when rolling

overcompletions

- Schedule specific times for the rolled schedules if no schedules exist on the specific rolled day

Note: When rolling flow schedules without details, if you roll the same flow schedules multiple times, it is possible to create excess supply.

To roll flow schedules forward:

1. Select Roll Flow Schedules from the Tools menu.

The Roll Flow Schedules Parameters window appears.

2. Select an Option.

Your choices are:

- *Close All Past Due Schedules*
- *Roll With Details*
- *Spread Evenly Without Details*

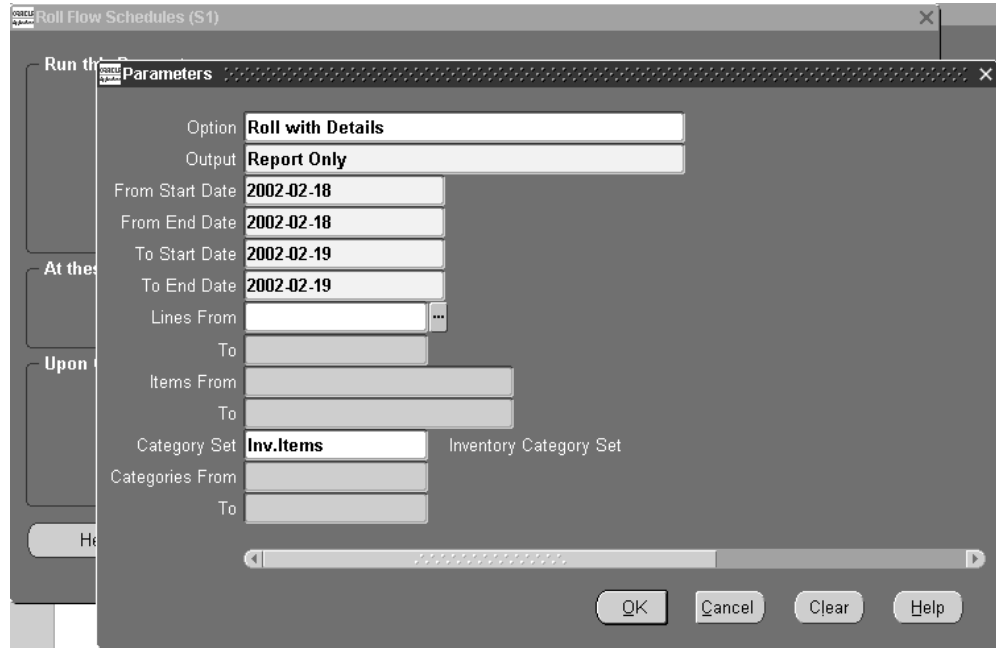
3. Select an Output option.

The values for Output are:

- *Report Only:* creates a report but does not update your flow schedules. This could be used to simulate the effects on your schedule.
- *Report and Update* creates a report and rolls the dates forward on your schedules.

4. Select a range of Start and End Dates From.

Flow schedules that do not have completion dates within this range are not rolled forward. You cannot enter a date in the From Start Date greater than today.



5. Select a range of Start and End Dates To.
6. You can select a range of flow schedule lines to consider in the Lines From and To fields.

Note: If you do not specify lines, item numbers, or categories all flow schedules that fall within the specified date ranges are rolled forward.

7. Select a range of assembly items to consider in the Items From and To fields.
8. Select a range of Category Sets From and To.
9. Choose OK.
10. Choose Submit Request.

Related Topics

Submitting a Request, *Oracle E-Business Suite User's Guide*

Deleting Flow Schedules

You can delete flow schedules in order to provide a clean slate for new schedule generation. Flow schedules can be deleted as long as the following conditions are true:

- There has been no completion performed, that is, the quantity completed must equal 0.
- The schedule is open.
- There is no pending completion against it in Material Transactions Interface.

If the deleted flow schedule is associated with a sales order, then that portion of the sales order is again considered as an unscheduled sales order.

To delete flow schedules

1. Select records on the Detail or Summary windows, and choose Delete.
2. You can also select Delete Flow Schedules from the Tools menu.
The Delete Flow Schedules Parameters window appears.
3. Select a range of Dates Start and From.
Flow schedules that do not fall within this range of dates are not deleted.
4. Optionally, select a range of Lines From and To.

Note: If you do not specify lines, item numbers, or categories all eligible flow schedules that fall within your specified date ranges are deleted.

5. Optionally, select a range of Items From and To.
6. To delete flow schedules by category, select a value in the Category Set field, and enter a range of categories Category From and To.
7. Choose OK.
8. Choose Submit Request.

Related Topics

Running Reports and Programs, *Oracle E-Business Suite User's Guide*

Defining Category Sets , *Oracle Inventory User's Guide*

Component Availability and Purging Flow Schedules

Component Availability

You can access Available To Promise information through the ATP Results window and view the status of material used on your flow schedules. Component availability checking is permitted only for items with the Check ATP item attribute set to Yes. See: Order Management Attribute Group, *Oracle Inventory User's Guide*

To display the ATP Results window

1. Navigate to the Flow Schedule Details window.

ATP Results

Calendar Organization **S1** Demand Class

View Results Group Available Date **05/APR/2001**

Required and Projected ATP and Early ATP Description

Org	Item/ Resource	Type	UOM	Required Quantity	Required Date	Available Quantity	Projected Transactable
S1	CM93501	Item	Ea	150	22/FEB/2001	0	
S1	CM93502	Item	Ea	150	22/FEB/2001	0	
S1	CM23594	Item	Ea	150	22/FEB/2001	0	0
S1	CM23593	Item	Ea	150	22/FEB/2001	0	
S1	CM23595	Item	Ea	300	22/FEB/2001	4900	4900
S1	CM31542	Item	Ea	150	22/FEB/2001	0	

Request Item **CM93501** **Mounting Bracket Left Hand**

Supply/Demand Period ATP Open

2. Select assemblies for viewing component information.

3. Choose Component Availability.

The ATP Results window appears. The components for each flow schedule are listed individually on separate rows.

Purging Flow Schedules

Closed flow schedules should be purged on a periodic basis to remove unneeded records and maintain database performance. A concurrent program purges according to parameters specified.

Flow schedules are purged if they meet the following criteria:

- Status of the flow schedule is closed
- Scheduled completion date of the flow schedule is in a closed accounting period
- All material transactions have been purged

To purge flow schedules

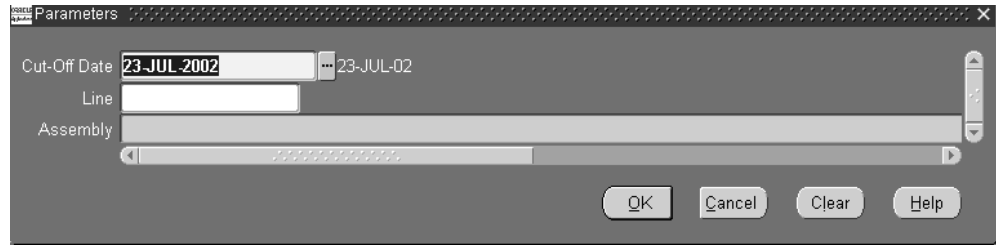
1. Navigate to the Purge Flow Schedules window.

The Parameters window displays.

2. Select the effective date to use for purging schedules in the Cut-Off Date field.

Flow schedules with a completion date prior to the cutoff date are purged.

3. Optionally, you can select a value in the Line field to purge specific flow lines.



4. You can also select a specific assembly in your purging criteria.
5. Choose OK to display the Purge Flow Schedule request window.
6. Choose Submit to create this request.

Feeder Line Synchronization

Feeder Line Synchronization

In manufacturing environments where many types of items are produced, the total amount for different configurations of a product may not be sufficient to support a Kanban replenishment system. Oracle Flow Manufacturing resolves this situation by enabling you to create schedules for lines that are directly derived from parent assembly line schedules. These feeder lines use sequence information from the parent assembly schedules so that you can synchronize the higher and sub-assembly material with production.

The feeder line synchronization program calculates the date and time the subassembly is delivered to the particular line operation. This date is then be used as the completion date for the subassembly line. The completion date of the feeder line flow schedule is calculated as the difference between the parent assembly's scheduled completion date and the sum of the operation times. This is the sum of the operation times from the consuming operation to the end of the parent line.

Note: The basis type of the subassembly is considered in the calculation. See: Fixed Component Usage, page 2-7

The start time for the subassembly is calculated by subtracting the lead time from the completion date and time. Lead time is obtained from fixed lead time and variable lead time on the organization item. This allows you to synchronize subassembly lines to deliver product exactly when needed by the main line.

You cannot create new synchronized schedules if corresponding schedules already exist on the feeder line. You also cannot re-synchronize rolled schedules. This eliminates potentially creating over production of the subassembly.

Prerequisites

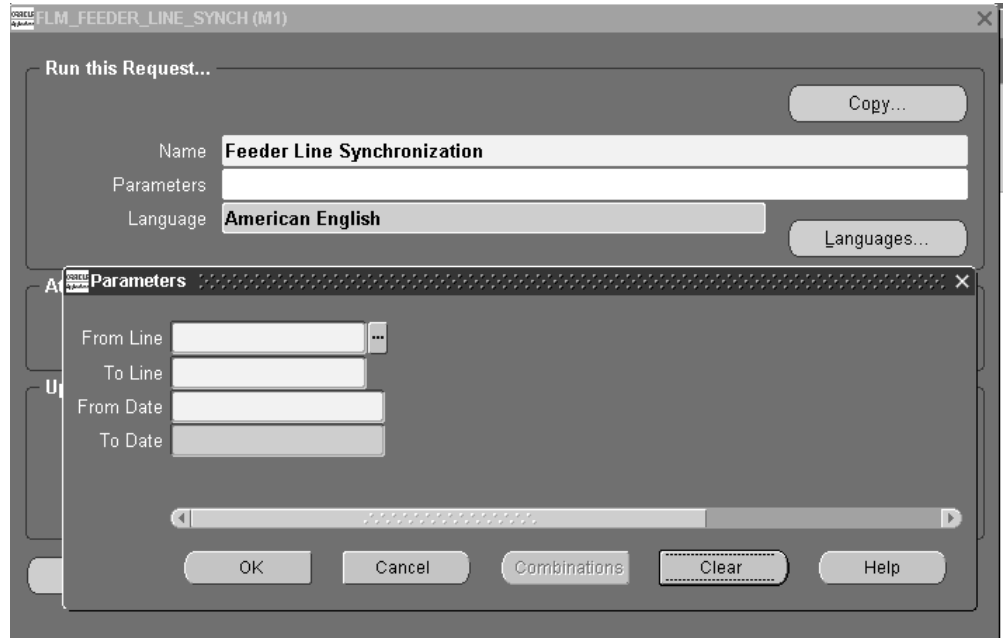
- ☐ Create flow routings and routing networks for both the parent and subassembly

items.

- ☐ Define the operation sequence for the subassembly on the bill of material of the parent item. This must be equal to the event sequence on the parent routing of the event consuming the subassembly.
- ☐ Define the variable lead time on the organization item master for each item to equal the TAKT time of each line, respectively.
- ☐ Define the fixed lead time on the organization item master for each item to equal the total product cycle time (TPCT) or the TAKT time of the line multiplied by the number of operations along the longest primary path of the line.
- ☐ Define the maximum hourly rate on the line definition for both the parent and feeder line the same as the standard daily running rate for each line, respectively.

To create flow schedules that synchronize feeder lines with parent assembly lines:

1. Navigate to the Feeder Line Synchronization window.
The Parameters window appears.
2. Select a parent line or range of Lines From and To.



3. Select a range of Dates From and To.

This is the date range of the start dates of the parent assembly schedule; it is used to determine which parent schedules are synchronized.

4. Choose OK.
5. Choose Submit Request.

Outbound Broadcast Sequencing

This chapter describes Outbound Broadcast Sequencing—a process that enables the creation of replenishment signals on a Flow line where the supplier has special requirements, preferences, or conditions.

This chapter covers the following topics:

- Overview of Outbound Broadcast Sequencing
- Setting Up Outbound Broadcast Sequencing

Overview of Outbound Broadcast Sequencing

Outbound Broadcast Sequencing is a process that enables the creation of replenishment signals to supply components for scheduled assemblies, dependent on the production sequence of the respective parent assembly. Although generally associated with supplier components, this process works with all methods of kanban supply.

Kanban signals control the replenishment of component material to a Flow line. Kanban planning has calculated container sizes, or quantity based on a predictable cycle, ensuring the availability of all necessary components. In some business processes the predictability of components is not possible, or the schedule of assemblies determines the component requirements. Outbound Broadcast Sequencing provides a method for sourcing components. For these components, the purchase requisition is generated based on firm Flow schedules. This functionality is implemented when you set up pull sequences with the Auto Request flag enabled.

A concurrent program is used to calculate kanban requirements and generate kanban cards by specifying production lines, date ranges, and other criteria. The Outbound Broadcast Sequencing process sends the appropriate date for creation of the purchase requisition for the relevant components, depending upon the Source type specified in the item's pull sequence.

Specifying a Range of Completion Dates and Build Sequences

You can specify a range of completion dates and build sequences as the Autoreplenishment Program supports selection of schedules by specifying a range of

completion dates and build sequences.

- The parameters From Completion Date, To Completion Date, From Build Sequence, and To Build Sequence enable you to specify the range of completion dates and build sequences.
- The list of values in the From Completion Date field lists all the dates starting from the current date and the list of values in the To Completion Date field lists all dates greater than or equal to From Completion Date.
- From Completion Date defaults to the current date.
- To Completion Date defaults to the From Completion Date.
- To Build Sequence initially defaults to From Build Sequence.
- If the value of the From Build Sequence parameter is greater than To Build Sequence, then no schedule is picked.
- If both the From Completion Date and From Build Sequence are null, then the From Completion Date is taken as the current date.

Note: You cannot launch multiple requests for autoreplenish concurrent program simultaneously as multiple replenishments cannot be done for the same schedule.

Related Topics

Overview of Kanban Replenishment, *Oracle Inventory User's Guide*.

Defining Kanban Pull Sequences, *Oracle Inventory User's Guide*

Setting Up Outbound Broadcast Sequencing, page 9-2

Process Flow, page 9-3

Setting Up Outbound Broadcast Sequencing

Outbound Broadcast Sequencing functionality is set up when you create your pull sequences.

- Pull Sequence window

Enable the Auto Request box, this is used to identify this type of pull sequence replenishment.

The size and number of kanban cards are not used for kanban calculation, if the Auto Request box is enabled.

Related Topics

Process Flow, page 9-3

Overview of Kanban Replenishment, *Oracle Inventory User's Guide*

Defining Kanban Pull Sequences, *Oracle Inventory User's Guide*

Process Flow

The following steps are performed when Outbound Broadcast Sequencing kanbans program is initiated:

- Retrieves Flow schedules in the date range specified.
 - Selects Flow schedules not previously processed.
 - Considers schedules with components flagged as Item Attribute--Release Time Fence, Kanban Item--Do Not Plan, and Supply Type—Pull.
 - Considers pull sequences where item and point of use matches the bill of material, component, location, and the Auto Request flag is enabled.
 - The basis type of the flow schedule assembly is considered.
- Calculates required date of the components based on the Flow schedule completion date and the routing where the component is used. Minimum order quantity data is not used.
- Creates non-replenishable kanban cards for the components using the pull sequence information.
- For kanban card supply type Supplier, the purchase requisition is created with the reference to the Flow schedule information.

Related Topics

Changing and Viewing Kanban Details, page 11-25

Viewing Kanban Demands, page 11-24

Defining Kanban Cards, *Oracle Inventory User's Guide*

Fixed Component Usage, page 2-7

Using the Kanban Calculation Program, *Oracle Master Scheduling/MRP and Oracle Supply Chain Planning User's Guide*

Flow Execution Workstation

This chapter describes the Flow Execution Workstation—used to track the flow of shop floor work, view schedule and line operation information, and complete schedules.

This chapter covers the following topics:

- Overview of the Flow Execution Workstation
- Setting Your Preferences
- Initializing the Flow Execution Workstation
- Viewing Schedule Information
- Viewing Event Information
- Completing Line Operations
- Completing Schedules
- Viewing Resources
- Attaching and Viewing Instructions
- Viewing Detail Properties
- Viewing Components and Replenishing Kanban Cards
- Updating Component Requirements
- Assigning Lot and Serial Numbers on the Workstation
- Integration with Oracle Quality
- Customized Properties Region
- Engineering Change Order Alerts

Overview of the Flow Execution Workstation

The Flow Execution Workstation is a web based workstation providing a comprehensive set of features for tracking the flow of work, completing assemblies, and

viewing details of component and resource information. The workstation has regions for schedules, events, component details, property information, and instructions. The features include the ability to:

- View current shop floor manufacturing work
- View a line operation's open schedules and events
- View detailed information on the components and resources required for line operations and events
- View kanban information for components and request replenishment
- View operating instructions, diagrams, and other information attached to assemblies and events
- Complete line operations and schedules
- Assign lot and serial number information
- Specify alternate or rework routing paths while performing completions
- Customize personal display of information
- Check eligible Oracle Bills of Material engineering change orders (ECOs) for line operation or event component changes
- Create a region to display your own customized parameters not included in the workstation pages.
- Integration with Oracle Quality pages to enter quality data and new plans

Setting Your Preferences

Data display on the workstation is controlled by preferences selected in the Workstation Parameters page. Parameters and the corresponding behavior are defined for organization, production line, or line operation. The preference for a parameter setting determines the data entry mode, the display of regions and fields on your pages, ECO checking, and the ability to enable other workstation locations.

When you enter parameters in the Flow Execution Workstation, the system first searches for records with the most detail specified at the line operation level. If the operation level parameter is not defined, it searches one level up at the line level. If production line is not specified, the system uses the organization level defaults.

To set or view preferences in the Flow Execution Workstation:

1. In the Flow Manufacturing Sequencing and Execution Home page, select the Setup

tabbed region.

2. Select Workstation Preferences to display the Select Preferences page.
3. Specify preference parameters.

You are required to select an organization, line and operation are optional:

- Organization: If only organization is specified, the data preference is defined at the organization level only.
- Production Line: If organization and line are specified, the data preference is defined for a specific organization and production line combination.
- Line Operation: If organization, production line, and line operation are specified; the data preference is defined for a specific organization, line, and line operation combination.

The screenshot shows the Oracle Flow Manufacturing interface. At the top, there's a navigation bar with 'Home', 'Logout', 'Help', 'Preferences', and 'Diagnostics'. Below this is a sub-navigation bar with 'Home', 'Sequencing Tasks', 'Workstation', and 'Setup'. The 'Workstation' tab is active, showing a sub-menu with 'Sequencing Attributes', 'Sequencing Rules', 'Production Lines', and 'Workstation Parameters'. The 'Workstation Parameters' page is displayed, featuring a 'Select Level' section with three input fields: 'Organization' (S1), 'Production Line' (Laptops), and 'Line Operation' (LOP1). Each field has a magnifying glass icon. There are 'Update' buttons next to the 'Organization' and 'Line Operation' fields. A note indicates that '*' denotes a required field. On the right, a 'Workstation Parameters' section explains that parameters can be set at three levels: organization, line, or operation. It states that if only organization is specified, it's an organization-level parameter; if only line is specified, it's an organization-level specific line parameter; and if only operation is specified, it's an organization-level specific line, specific operation parameter. It also mentions that if a level is specified but doesn't exist, a new one will be created and its values initialized from its upper-level parameters, or defaulted from pre-defined values if no upper-level parameter exists. The bottom of the page includes 'About this Page', 'Privacy Statement', and a copyright notice for Oracle (c) 2006.

4. Choose Update to display the Update Workstation Parameters page.
If the parameter or parameter combination specified does not exist as a preference, a new one is created using default values. You can change the values for either new or existing preferences on the Workstation Parameters page.
5. Set your preference for the Workstation Enabled box.
This parameter specifies if the workstation is used at this particular organization, line, or operation. If set to:
 - Enabled: The workstation displays applicable Flow schedules and the operator is required to choose Line Operation Completion.
 - Disabled: A workstation is not used at this specific organization, line, or operation.

Update Workstation Parameters

Organization **S1** Production Line **LINE1** Line Operation

Settings

☒ Workstation Enabled

Schedule Selection Mode: **Work Queue**

Automatic Schedule Transition: **No**

View All Schedules: **No**

Automatic ECO Check: **No**

ECO Days Prior: **5**

ECO Days After: **5**

Backflush Option: **Actual / Primary Path**

Instruction Frame Height: **800**

Pick Release Operation: **OP10**

Label Start Operation: **OP60**

Label End Operation:

Cancel **Apply**

6. In the Settings region, Schedule Selection Mode, select how schedules display on the workstation. Your choices are:
 - **Work Queue:** You can provide search criteria, and display the resulting list of schedules and their corresponding events.
 - **User Entered:** You can search, enter, or scan (using a mobile device) a specific schedule number to open the Line Operation page.

7. If the Schedule Selection Mode is set for Work Queue mode, set the Automatic Schedule Transition field.

Used to specify navigation options for line operations. Your choices are:

- **Yes**—after choosing Complete Line Operation, the next available schedule displays on the page.
- **No**—after choosing Complete Line Operation, the Work Queue page displays listing the available schedules for the next transaction.

Note: The Used Entered mode does not use this setting because the next available schedule is always user specified.

8. Use the View All Schedules field to filter the schedule display for a particular workstation.
 - **Yes:** All uncompleted schedules for the current line operation are displayed.
 - **No:** Only completed schedules for the previous line operation are displayed.
9. In the Automatic ECO Check, indicate if you want the system to automatically check for eligible Engineering Change Orders. Otherwise, you can check for ECOs manually. Select Yes or No.

10. When you check for ECOs, you can indicate a range of days to check before and after the ECO Effective Date. In the ECO Days Prior and ECO Days After, enter the number of days.

Only Implemented ECOs in the date range are considered. See: Engineering Change Order Alerts, page 10-34

11. The Backflush Option parameter sets the method for material backflushing:

- Actual/Primary—Components are backflushed only for completed operations or operations on the primary path of the routing.
- All—All components on all events are backflushed.

See: Updating Component Requirements, page 10-31

12. In the Instruction Frame Height field, set the view size of attachments on the pages. Enter a pixel value to increase or decrease the size of the display. See: Attaching and Viewing Instructions, page 10-24

13. If this preference is defined at the line level, you can define specific transactions performed at line operation completion. These fields are available only at the line level preference.

- Pick Release Operation: Operation where the Pick Release process is called. See: Component Pick Release, *Oracle Work in Process User's Guide*
- Label Start and End Operation: Label printing capabilities are available at the time of completion for various types of manufacturing information. If you have defined label types at the flow line Start or flow line operation, labels are printed. See: Compliance Labeling Setup, *Oracle Warehouse Management User's Guide*.

14. In the Line Operation and Event tabbed regions—select how you want the page regions, tabbed regions, and instructions to display on the pages. In Page Regions, choose the regions that display on the Line Operation and Event pages.

Select categories from the Available box—and move to the Selected box. For example:

- In the Line Operation tabbed region, if the Events and Tabs are in the Selected box—the Line Operation page displays an Events region, and tabbed region.
- If the Events, Instructions, and Tabs are in the Selected box—the Line Operation page displays the Events region, the Instruction region, and tabbed region.
- In the Event tabbed region, if the Components and Tabs are in the Selected box—the Event page displays a Components region, and tabbed region.

See: Viewing Schedule Information, page 10-13 and Viewing Event Information, page 10-16

The screenshot shows a window titled "Page Regions" with two tabs: "Line Operation" and "Event". The "Line Operation" tab is selected. The window is divided into two main sections: "Available" and "Selected".

Available Section:

- A list box containing: Instructions, Components, Resources, Schedule Properties, Customized Properties.
- A "Description" text area with a scroll bar.

Selected Section:

- A list box containing: Events, Tabs.
- A "Description" text area with a scroll bar.

Between the sections are the following controls:

- Move: A right-pointing arrow icon.
- Move All: A right-pointing arrow icon with the text "Move All" below it.
- Remove: A left-pointing arrow icon.
- Remove All: A left-pointing arrow icon with the text "Remove All" below it.

On the right side of the Selected list box are four arrow icons:

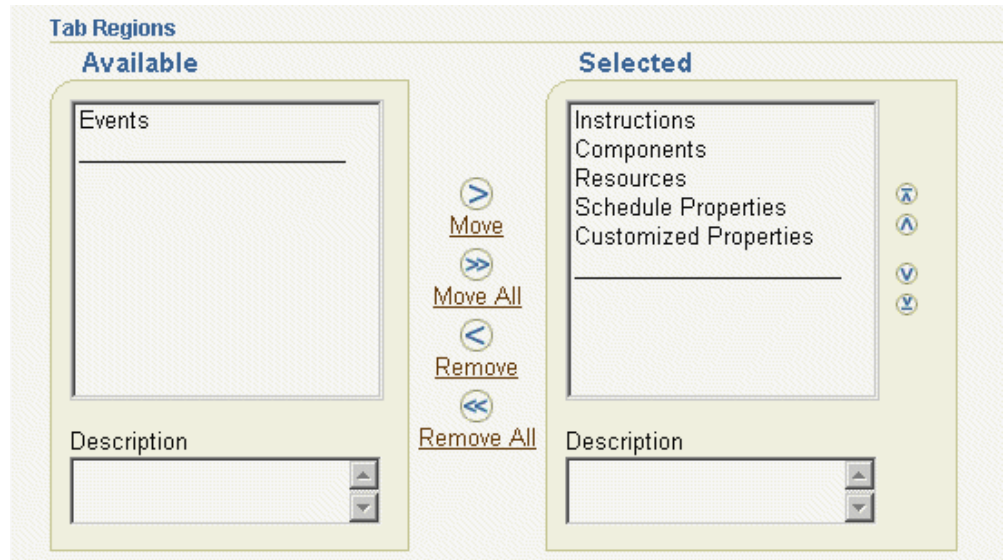
- Up arrow
- Down arrow
- Up arrow
- Down arrow

The regions available for line operations and events page regions are Instructions, Components, Resources, Schedule Properties, and Customized Properties, Tabs, Events (for Line Operation only). The order listed in the Selected box, is the order they display on the page. Use the Arrow tool to change the order.

Regions are displayed only in one mode. That is, a region is only displayed either as a region on a page, or a region on a tabbed region. It is not displayed in both areas.

15. In Tab Regions, choose the regions to display on the Line Operation and Event pages.

Select Regions from the Available box—and move to the Selected box. The order listed in the Selected box, is the order they display on the tabbed region. Use the Arrow tool to change the order.



16. In the Instructions boxes, select entity attachments to automatically display in the Instructions region.

Select attachment sources from the Available box—and move to the Selected box. The order listed in the Selected box, is the order they display on the page. Use the Arrow tool to change the order. See: Attaching and Viewing Instructions, page 10-24, and Attachments, *Oracle E-Business Suite User's Guide*

Note: All Available attachments are viewable. The Selected attachments automatically display.

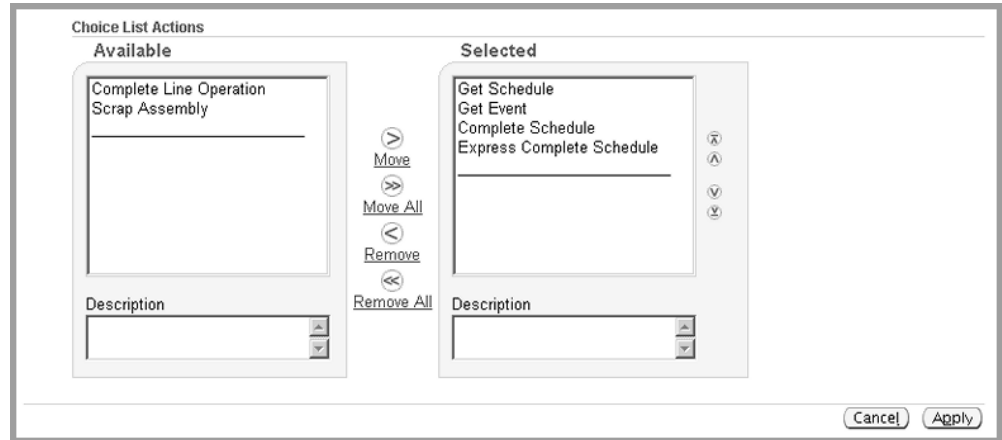
The 'Instructions' dialog box is divided into two main sections: 'Available' and 'Selected'. The 'Available' section on the left contains a list of items: 'Bills of Material', 'Bills Of Material Component', 'Routing', 'Event', 'Standard Event', 'Sales Order Header', and 'Sales Order Line'. Below this list is a 'Description' field. The 'Selected' section on the right contains a list with 'Line Operation' and 'Standard Line Operation'. Below this list is also a 'Description' field. Between the two sections are several action buttons: 'Move' (single arrow), 'Move All' (double arrow), 'Remove' (single arrow), and 'Remove All' (double arrow). On the far right of the 'Selected' section, there are four small icons: a magnifying glass, a list icon, a checkmark, and a close button.

17. In the Schedule Actions tabbed region, select how you want your actions to appear on the Work Queue, Line Operation and Event pages. This preference specifies whether the listed actions are viewed on the page as buttons in the Button Actions region, or as a drop-down list of values in the Choice Actions region. For the Choice actions, the order the actions are listed in the selected box determines the order the actions are listed in all choice lists.

The values in the Scheduled Actions region are Complete Line Operation, Complete Schedule, Get Event, Get Schedule, Express Complete Schedule, and Scrap Assembly. Select categories from the Available box—and move to the Selected box.

Actions are displayed only in one mode on a page, that is, either as a button or drop-down list.

The 'Schedule Actions' dialog box is divided into two main sections: 'Available' and 'Selected'. The 'Available' section on the left contains a list of actions: 'Complete Schedule', 'Express Complete Schedule', 'Get Event', 'Get Schedule', and 'Scrap Assembly'. Below this list is a 'Description' field. The 'Selected' section on the right contains a list with 'Complete Line Operation'. Below this list is also a 'Description' field. Between the two sections are several action buttons: 'Move' (single arrow), 'Move All' (double arrow), 'Remove' (single arrow), and 'Remove All' (double arrow). On the far right of the 'Selected' section, there are four small icons: a magnifying glass, a list icon, a checkmark, and a close button.



18. Choose Apply to save your work.

Initializing the Flow Execution Workstation

The Flow Execution Workstation is initialized by selecting an organization, production line, and line operation. The preference determines if the workstation displays a listing of Flow schedules (the Work Queue)—or you manually enter or scan information (the User Entered mode). See: Setting Your Preferences, page 10-2

The Work Queue mode opens the Work Queue Search page where you can select search criteria. From these results, select a schedule from the Work Queue list to access the Line Operation page. The User Entered mode navigates directly to the Line Operation page for the line and operation entered.

The Line Operation page enables you to view individual schedules, operations, events, and instructions. You can complete schedules and line operations, and enter quality results.

To view and enter Flow schedule details in the Workstation:

1. Navigate to the Flow Execution Workstation.

The Flow Execution Workstation Startup page displays for selecting schedule parameters.

ORACLE[®]
Flow Manufacturing

Close Window Preferences

Home Sequencing Tasks **Workstation** Setup

Flow Execution Workstation

* Indicates required field

Start (2)

Parameters

* Organization S1

* Production Line Laptops

* Line Operation

Start (2)

2. Select parameters for Organization, Production Line, and Line Operation.

The Workstation preference for the organization, line, and operation determines the entry mode: If the mode is:

- Work Queue: You can provide search criteria, and display the resulting list of schedules and corresponding events. See: Work Queue Mode, page 10-2
- User Entered: You can search, enter, or scan a specific schedule number to open the Line Operation page. See: User Entered Mode, page 10-13

Work Queue Mode:

To access Flow Execution Workstation pages in Work Queue mode:

1. Navigate to the Flow Execution Workstation, and select parameters to display the Flow Execution Workstation Startup page.
2. The Work Queue Criteria page displays. Select any criteria for filtering your flow schedule search.

You can define your search by selecting values, or a range, in the following fields—Completion Date, Assembly, Schedule Group, and Build Sequence.

ORACLE®

Flow Manufacturing

[Home](#)
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[Help](#)
[Preferences](#)
[Diagnostics](#)

[Home](#)
[Sequencing Tasks](#)
[Workstation](#)
[Setup](#)

Logged In As FLMDEV

Flow Execution Workstation

* Indicates required field

Start (2)

Parameters

* Organization

S1

* Production Line

Laptops

* Line Operation

LOP1

Work Queue Criteria

From Completion Date

17-Apr-2006 00:00:00

(example: 17-Apr-2006 19:45:00)

To Completion Date

From Assembly

To Assembly

Schedule Group

From Schedule Number

To Schedule Number

From Build Sequence

To Build Sequence

Start (2)

3. Choose Start to display the Work Queue page.

The work queue consists of a list of schedules and events for a selected schedule. You can perform completions of line operations and schedules from the work queue.

Note: You can handle schedules that have operations with no events in the Flow Execution Workstation. You can identify the schedules with no events in the Flow Execution Workstation by an icon and a message column. By default these columns are hidden and you have to enable them for display through personalization.

ORACLE Flow Manufacturing

Home Logout Help Preferences Diagnostics

Home Sequencing Tasks Workstation Setup

Workstation > Line/Line Operation LINE1/OP10
Logged In As FLMDEV

Work Queue (S1)

Flow Schedules

Select Schedule: Complete Line Operation (5) Action Complete Schedule Go Previous 1-5 Next 5

Select Schedule Number	Assembly	Assembly Description	Build Seq	Planned Quantity	Completion Date
☒ 199108	DB-ASSEMB1	ASSEMBLY 1		100	24-Mar-2006 00:00:00
☐ 197725	MGR-PLAINAS	MGR-PLAINAS	13		10/24-Jun-2006 00:00:00
☐ 197719	MGR-PLAINAS	MGR-PLAINAS	10		9/25-Jun-2006 23:59:00
☐ 197718	MGR-PLAINAS	MGR-PLAINAS	11		10/26-Jun-2006 23:59:00
☐ 197717	MGR-PLAINAS	MGR-PLAINAS	12		10/28-Jun-2006 23:59:00

Events: 199108

Event Seq	Code	Description	Department
1010	FLEF	Fabrication	FABR

Return to Workstation (0)

In the Flow Schedules region, this page displays Schedule Number, Assembly Part Number, Assembly Description, Build Sequence, Planned Quantity, and Completion Date.

- Planned Quantity field value is the quantity created when the schedule is created; it does not change.
- Completion Date field value is the scheduled completion date of the flow schedule; it does not change with the actual completion of line operation or flow schedule.

The fields in the Events region are Event Sequence, Operation Code, Event Description, and Department. See: Viewing Event Information, page 10-16

- To navigate to the Line Operation page, select the schedule link for specific schedule.

You can also select a schedule by enabling the radio button in the Select column, and selecting the Get Schedule action. See: Viewing Schedule Information, page 10-13

- To navigate to the Event page, select a schedule by enabling the radio button in the Select column—click on the appropriate event link.

You can also select an event by enabling the radio button in the Select column, and selecting the Get Event action. See: Viewing Event Information, page 10-16

- To complete schedules or line operations from this page—select a record and choose Complete Line Operation, Complete Schedule, or Express Complete Schedule.

When a schedule is completed, it is no longer shown on the work queue. See: Completing Line Operations, page 10-18, and Completing Schedules, page 10-19

User Entered Mode:

To access Flow Execution Workstation pages in User Entered mode

1. Navigate to the Flow Execution Workstation, and select parameters to display the Flow Execution Workstation Startup page.
2. Choose Start to display the Enter Schedule page.

ORACLE Flow Manufacturing

Close Window Preferences Diagnostics

Home Sequencing Tasks Workstation Setup

Workstation

Enter Schedule (S1)

* Indicates required field

* Schedule Number 104237

Return to Workstation (0)

Get Schedule (Z) Get Event (B)

Line/Line Operation Laptops/LOP1

3. In the Schedule Number field, search, enter, or scan a specific schedule number.
4. Choose either Get Schedule or Get Event to access the schedules and events for this Flow schedule.

See: Viewing Schedule Information, page 10-13 and Viewing Event Information, page 10-16

Viewing Schedule Information

The Line Operation page displays schedule information in the General region. This includes fields for Schedule Number, Assembly Description, Build Sequence, Assembly Item, Scheduled Quantity, and Open Quantity.

ORACLE® Flow Manufacturing

Home Logout Help Preferences Diagnostics

Home Sequencing Tasks Workstation Setup

Workstation > Enter Schedule (M1) >

Line/Line Operation Case/ASSY
Logged In As FLMDDEV

Schedule: 32801 (M1)

Complete Line Operation (5) Action Complete Schedule Go

General

Schedule Number	32801	Assembly	SB65301	Scheduled Qty	1
Build Sequence	4875	Assembly Description	Casing Subassembly - Bronze	Open Qty	1

Update Schedule

Events Instructions Components Resources Detail Properties Customized Properties

Event Seq	Code	Description	Department	Effective Date	Disable Date
1000	ASSY	Assemble Case	VISION PAD	13-Jun-2000 00:00:00	

Return to Enter Schedule (0) Complete Line Operation (5) Action Complete Schedule Go

Other information is displayed as a region or tab, depending on the preference setting. This includes events, instructions, components, resources, detail properties, and customized properties. See: [Setting Your Preferences](#), page 10-2

The Line Operation page enables you view and perform transactions for Flow schedules including:

- Updating schedule information, See: [Updating Flow Schedule Information](#), page 10-15
- View the schedules corresponding event information. See: [Viewing Event Information](#), page 10-16
- View details of resources attached to events, for this line operation and schedule. See: [Viewing Resources](#), page 10-23
- View detail information about the line schedule. See: [Viewing Detail Properties](#), page 10-28
- View and replenish components using Kanban cards See: [Viewing Components and Replenishing Kanban Cards](#), page 10-29
- Complete schedules and line operations. See: [Completing Line Operations](#) , page 10-18, and [Completing Schedules](#), page 10-19
- Enter quality data results for mandatory or optional collection plans, at specific points on a routing. See: [Integration with Oracle Quality](#), page 10-33
- View attached instructions and documents pertaining to the assembly and schedule. See: [Attaching and Viewing Instructions](#), page 10-24
- View a region of custom parameters created to fit your business needs. See: [Customized Properties Region](#), page 10-33

Updating Flow Schedule Information

You can update values on your flow schedule on the Update Schedule page. The schedule information appears in the Update Schedule header region of the page, the changeable fields appear in the Schedule Details region.

For assemblies under lot control, the Assembly Lot region appears for entering lot numbers. For assemblies under serial control, the Assembly Serial region appears. Both of these region enable you to select and generate numbers for the Scheduled Quantity value listed.

ORACLE® Flow Manufacturing

Home Logout Preferences Diagnostics

Home Sequencing Tasks **Workstation** Setup

Workstation > Work Queue (S1) > Schedule: 206039 (S1) > Line/Line Operation Laptops/LOP1
Logged In As FLMDEV

Update Schedule (S1)

Schedule Number **206039** Assembly **assy ser_ctrl** Scheduled Qty **1** Cancel Apply
Build Sequence Assembly Description **ser_ctrl** Open Qty **1**

* Indicates required field

Schedule Details

* Scheduled Qty Alternate Routing

Build Sequence Sales Order

Subinventory Order Line

BOM Revision Schedule Group

BOM Revision Date Kanban Card

(example: 26-Apr-2006 19:45:00)

Routing Revision

Routing Revision Date

Alternate BOM

Assembly Serial

Select Serial Entry:

Select From To Attributes

Assembly Lot

Select Lot Entry:

Select Lot Number Lot Attributes Qty

To update schedule information

1. On the Schedule page, choose Update Schedule to display the Update Schedule page.

2. Change the appropriate information. You can update the following fields: Schedule Quantity, Build Sequence, Subinventory, BOM Revision, BOM Revision Date, Routing Revision, Routing Revision Date, Alternate Routing, Sales Order, Order Line, Schedule Group, and Kanban Card.
3. For lot controlled assemblies, the Assembly Lot region appears to select lot numbers.
 - Select the list of values of generated lot numbers in the Select From and To fields.
 - If lot numbers have not been generated, choose Generate Lot Numbers to access the request program.
4. For serialized assemblies, the Assembly Serial region appears to select serial numbers.
 - Select the list of values of generated serial numbers in the Select From and To fields.
 - If serial numbers have not been generated, choose Generate Serial Numbers to access the Generate Serial Numbers request program.
5. Choose Apply to save your work.

Viewing Event Information

The Event page displays the event sequence information in the General region. This includes the corresponding Flow schedule, the item Assembly number, Assembly Description, Build Sequence, and Schedule Quantity.

Other information is displayed as a region or tab, depending on the preference setting. This includes instructions, components, resources, detail properties, and customized properties. See: Setting Your Preferences, page 10-2

ORACLE® Flow Manufacturing

Home Logout Help Preferences Diagnostics

Home Sequencing Tasks **Workstation** Setup

Workstation > Work Queue (S1) > Line/Line Operation DB-LINE1/OP10
Logged In As FLMDEV

Event: FLEF (S1)

Complete Line Operation (5) Action Complete Schedule Go

Schedule Number **199108** Assembly **ASSEMBLY1** Scheduled Qty **100**
Build Sequence Assembly Description **ASSEMBLY 1** Open Qty **88**

Event Description
Fabrication

Instructions
View All Instructions

Instruction Seq	Source	Entity Value	Description	Category	Data Type	File Name
No results found.						

Components
Find ECO Update Components

Event Seq	Item Seq	Component	Description	Ref	Revision	Basis	UOM	Qty	Effectivity Date	Disable Date	Kanban
1010	10	COMP1	COMP1		A	Item	Ea	1	03-Apr-2006 14:15:24	25-May-2088 14:15:24	

The Event page enables you view and perform transactions for Flow schedules. Navigate between consecutive events by choosing Next. This enables you to display sequential events in order performed, until the line operation is completed.

- View details of resources attached to events, for this line operation and schedule. See: Viewing Resources, page 10-23
- View detail information about the line schedule. See: Viewing Detail Properties, page 10-28
- View and replenish components using Kanban cards. See: Viewing Components and Replenishing Kanban Cards , page 10-29
- Update Component Information, see: Updating Component Requirements, page 10-31
- View a region of custom parameters created to fit your business needs. See: Customized Properties Region, page 10-33
- Complete schedules and line operations. See: Completing Line Operations , page 10-18, and Completing Schedules, page 10-19
- Enter quality data results for mandatory or optional collection plans, at specific points on a routing. See: Integration with Oracle Quality, page 10-33
- Check ECOs to insure you have the correct changes for work as it progresses. See: Engineering Change Order Alerts, page 10-34.

- Display a table of all attached instructions and documents pertaining to the production line. You can then view each individual document. See: Attaching and Viewing Instructions, page 10-24

Completing Line Operations

You can perform line operation completions from either the Work Queue, Line Operation, or Event pages.

Completions for line operations with one destination, and no existing mandatory Oracle Quality plans, are performed automatically when you choose Complete Operation. If the line operation has a network of choices—that is primary, alternate, and rework operations—the Line Operation Completion page appears to select the path. This page also appears if quality plan data is required for the operation. You can then access the Enter Results page on the Quality Workbench.

To perform a completion for a line operation:

1. Navigate to the Flow Execution Workstation.
 - If your preference is Work Queue mode, you can complete line operations from the Work Queue page, the Line Operation page, or Event page. Select a record for completion.
 - If your preference is in the User Entered mode, completions are performed in the Line Operation or Event page.
2. Choose Complete Line Operation.

For line operations with one destination and no existing quality plans, the completion transaction is performed. The Work Queue mode no longer displays the completed line operation. A confirmation message displays.

For line operations with several routing options or existing quality plans, the Line Operation Completion page displays.

ORACLE® Flow Manufacturing

Home Logout Help Preferences Diagnostics

Home Sequencing Tasks **Workstation** Setup

Workstation > Work Queue (S1) > Schedule: 197718 (S1) > Line/Line Operation DB-LINE1/OP10

Information
Mandatory QA plans must be completed before operation completion.

Complete Line Operation: 197718 (S1)

Schedule Number **197718** Assembly **MGR-PLAINAS** Scheduled Qty **10** Cancel Apply
Build Sequence **11** Open Qty **10**
Assembly Description **MGR-PLAINAS**

Applicable Quality Plans
Expand All Collapse All

Focus Plan Name	Information	Mandatory	Completed	Attachments	Duplicate	Update	Delete	Add Child Data
TEST S1	-test for FLM flows	Yes	X					

Completion Destination
Choose Destination ☒ OP20(DB-TEST) - Primary(100.0%)

Cancel Apply

3. If quality information needs to be entered, a message appears. The collection plan information displays in the Applicable Quality Plans region including Plan Name, Information, Mandatory flag, and status information.

Access the Enter Results page on the Quality Workbench by choosing Add Child Data. See: *Entering Quality Results, Oracle Quality User's Guide*

4. If this line operation has a choice of destinations, choose the destination. Options are primary, alternate, and rework operations.
5. Choose Apply to save your work.

A confirmation message appears when the line operation is completed.

Related Topics

Completing Schedules, page 10-19

Integration with Oracle Quality, page 10-33

Completing Schedules

You can perform assembly completion directly from the workstation. Completions are performed on the Work Queue, Line Operation, and Event pages. There are two types of completions:

- **Complete Schedule**—Used when information is required to perform the completion transaction, or when the schedule assembly or component details need to be reviewed prior to completion.

- Express Complete Schedule—Used when default information enables an automatic completion transactions.

You can also assign lot and serial numbers at the time of completion. For items under lot and serial control, the Assembly Lot and Serial region appears on this page. The information here is defaulted from recorded information. This region enables you to access the Serial Detail or Lot Detail pages, and generate lot and serial numbers, and assign lot and serial numbers. Items with dual unit of measure can also be used as assembly and component items.

To perform a completion for a schedule:

1. Navigate to the Flow Execution Workstation.
 - If your preference is Work Queue mode, you can complete schedules from the Work Queue, Line Operation, or Event pages.
 - If your preference is in the User Entered mode, completions are performed in the Line Operation or Event pages.
2. Select a record for completion.
3. Select Express Complete Schedule if required values are set as defaults for Quantity, Unit of Measure, and Subinventory. The completion transaction is performed.
4. Select Complete Schedule if you need to enter required information such as Quantity, Unit of Measure, Subinventory, or other values.

The Complete Schedule: Assembly Details page appears.

Warning
Low-level Diagnostic Logging is turned on. This may temporarily reduce performance.

Complete Schedule: Assembly Details (S1)

Schedule Number: 422329 Assembly: FLM_DUOM_LASBYND Scheduled Qty: 24
Build Sequence Assembly Description: lot assembly with no default Open Qty: 5

Personalize Stack Layout (MainRF)
* Indicates required field

Assembly Details

Personalize "Assembly Details"

Quantity: 5
UOM: Ea
Subinventory: Stores
BOM Revision: A
BOM Revision Date: 26-Dec-47 11:22:30:00
Routing Revision: A
Routing Revision Date:
Alternate BOM:
Alternate Routing:
Sales Order:
Order Line:
☐ Reserved
Reason:
Comment:

Assembly Lot and Serial

Personalize "Assembly Lot and Serial"
Personalize Flow Layout
Personalize "Lot"

Select Lot: Lot Details Actions Generate Lot Number Go

Select	Lot	Parent Lot	Qty	Transaction Qty	Avail	UOM	Quantity	Secondary Qty	Avail	UOM	Action
☐	LAN01		24	24		Ea			0	DZ	
			Total	24							

Table Diagnostics Table Diagnostics Table Diagnostics
Personalize Stack Layout (FormValuesRF)

Diagnostic Console

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5. Select the values required for this transaction, you can change the following fields:
 - Quantity: Defaults to the schedule number quantity
 - Unit of Measure
 - Subinventory: Select a Completion Subinventory. If a completion subinventory is specified on the routing, it is defaulted but can be changed.
 - BOM Revision
 - BOM Revision Date
 - Routing Revision
 - Routing Revision Date
 - Alternate BOM
 - Alternate Routing
6. If you have selected a Sales Order and Order Line, you can enable the Reserve option to create a reservation.
7. Optionally, you can select a value in the Reason field, and enter text in the Comment field.

8. For items under lot control, the Assembly Lot and Serial region appears with a Lot area and any existing recorded information. Select a record in the table and choose Lot Details to display the Lot Detail page.

This page displays the lot's detail information, attributes, and descriptive flexfield data.

9. If lot numbers have not been generated, you have the option to create your records with the following choices for generation in the Actions box:

- Generate Lot Number
- Generate Lot and Serial Numbers
- Generate Serial Numbers

10. Select generated lot numbers in the Select Lot field. Existing parent lot information appears.

11. Enter the appropriate values in the Primary and Secondary Quantity fields.

12. Choose Re Calculate to calculate the Available Quantity.

13. The Assembly Lot and Serial region appears for items under serial control with a Serial area and any existing recorded information. Select a record in the table and choose Serial Details to display the Serial Detail page.

This page displays the serial number's detail information, attributes, and descriptive flexfield data.

14. If serial numbers have not been generated, you have the option to create your records with the following choices for generation in the Actions box:

- Generate Lot Number
- Generate Lot and Serial Numbers
- Generate Serial Numbers

15. Select generated serial numbers in the Select Serial From and To fields, and enter the quantity.

16. Select Finish to save your work and create the completion transaction.

17. Select Continue to access quality plans, or change component backflush information.

The Complete Schedule: Quality and Component Details page appears. You can also assign lot and serial number records on this page.

Complete Schedule: Quality and Component Details (S1)

Schedule Number: 202009 Assembly: DB-FLMASSV1 Scheduled Qty: 10
 Build Sequence: 10 Assembly Description: DESKTOP FLM600A Open Qty: 4

Transaction Details

Applicable Quality Plans

Plan Name	Information	Mandatory	Completed	Attachments	Duplicate	Update	Delete	Add Child Data
No applicable plans.								

Backflush Components

Select Component: Remove Enter Multiple Lot and Serial Remove All

Select	Status	Op Seq	Component	Description	Subinv	Locator	Revision	Lot From	To Parent	Required	Transaction UOM	Reason	Reference
☒	✓	30	DB-FLMSA3	DB-FLMSA3	RJP					6	6	Ea	
☐	✓	30	DB-FLMSA4	DB-FLMSA4	RJP					24	24	Ea	
☐	✓	30	DB-FLM-COM	DB-FLM-COMP3	RJP					6	6	Ea	
☐	✓	-30	DB-FLMSAP	Hard Drive Phantom	RJP					8	8	Ea	
☐	✓	10	DB-FLMSA1	DB-FLMSA1	RJP					2	2	Ea	
☐	✓	10	DB-FLM-COM	DB-FLM-COMP1	RJP					12	12	Ea	
☐	✓	20	DB-FLMSA2	DB-FLMSA2	FGI					8	8	Ea	

Add 5 Rows

DB-FLMSA3: Additional Information

Context Value

Cancel Back Finish

18. To access available quality plans, choose Add Child Data in the Applicable Quality Plans region. See: *Entering Quality Results, Oracle Quality User's Guide*
19. To assign lot or serial numbers for applicable components, enter the lot and serial information. To assign multiple ranges of lot and serials, select Enter Multiple Lot and Serial.

Lot and serial numbers defaulted from existing recorded information. For recording lot and serial information, see: *Assigning Lot and Serial Numbers on the Workstation*, page 10-32
20. To change component and backflush information, you can update fields in the Backflush Components region, see: *Updating Component Requirements*, page 10-31
21. Choose Finish to save your work and create the completion transaction.

Related Topics

Completing Line Operations, page 10-18

Work Order-less Completions, *Oracle Work in Process User's Guide*

Generating Serial Numbers, *Oracle Inventory User's Guide*

Lot Control, *Oracle Inventory User's Guide*

Viewing Resources

The Resources region provides the details of resources attached to events for a selected

schedule. Resources include employees, machines, outside processing services, and physical space. A list of resources is displayed in view-only mode and includes information for event sequence, resource sequence, type, code, description, basis, usage, and availability.

ORACLE Flow Manufacturing

Home Logout Preferences Personalize Page

Workstation > Work Queue (S1) > Line/Line Operation Laptops/LOP1

Schedule: 136701 (S1)

Complete Schedule (3) Complete Line Operation (5)

General

Schedule Number **136701** Assembly **MC24713**
 Build Sequence **3725** Assembly Description **Plastic Base - Black**
 Scheduled Quantity **1**

Enter Quality Results (4) Find ECO View All Instructions

Events Instructions Components Resources Detail Properties Customized Properties

Event Seq	Resource Seq	Type	Code	Description	UOM	Basis	Usage	Available 24 Hours	Schedule
10	10	Person	LBR-FAB1	Fabrication Labor Grade 1	HR	Item	0.002	No	Yes
10	20	Person	LBR-FAB2	Fabrication Labor Grade 2	HR	Item	0.033333	No	Yes
10	30	Machine	SHEAR	Shearing Machine	HR	Lot	0.033333	Yes	Prior
15	10	Person	LBR-FAB	Fabrication Labor	HR	Item	0.1	No	Yes

Events Instructions Components Resources Detail Properties Customized Properties

Return to Work Queue (0) Complete Schedule (3) Complete Line Operation (5)

This feature enables you to track what is actually being used at the line operation, and to view the time an assembly spends at an operation. The information is derived from the Resources and Operation Resources windows in Oracle Bills of Material.

Related Topics

Defining a Resource, *Oracle Bills of Material User's Guide*

Assigning Operation Resources, *Oracle Bills of Material User's Guide*

Attaching and Viewing Instructions

The Flow Execution Workstation provides the ability to display attached instructions and documents pertaining to your production. This can include procedures, diagrams, and assembly instructions. Instructions are viewable from the Schedule and Event pages. The attachment displays in a frame, the size can be changed in the Workstation Parameters page. If more than one attachment exists, documents are displayed in a table format. You can specify the order of display in the table.

Instructions and documents are attached to entities in their specific product application. For example, an instruction for an item is attached in the Master Item window in Oracle Inventory. Instructions for a standard line operation are attached in the Standard Line Operation window in Oracle Bills of Material. See: Attachments, *Oracle E-Business Suite User's Guide*

Setup:

In specific product applications, attach the instructions and documents.

You can attach instructions to the following entities for display in the Flow Execution Workstation:

- Item
- Bills of Material
- Bills Of Material Component
- Routing
- Event
- Standard Event
- Sales Order Header
- Sales Order Line
- Line Operation
- Standard Line Operation

See: System Items, *Oracle Inventory User's Guide*

To set up instruction views in the Flow Execution Workstation:

1. In the Flow Execution Workstation, navigate to the Workstation Parameters page for the parameter preference. See: Setting Your Preferences, page 10-2
2. In the Instruction Frame Height field, you have the option to change the size of the view. This is a pixel value.
3. In the Instructions boxes, select where to display the instructions and attachments.

Select attachment source from the Available box—and move to the Selected box. The order listed in the Selected box, is the order they display on the page. Use the Arrow tool to change the order. This enables you to specify the priority view of the attachments. For example, at the line operation level, you can set your display at that line operation, but also all the documents attached to the events under that line operation.

Instructions

Available

- Item
- Bills of Material
- Bills Of Material Component
- Routing
- Event
- Standard Event
- Sales Order Header
- Sales Order Line

Description

Selected

- Line Operation
- Standard Line Operation

Description

Move, Move All, Remove, Remove All

4. Choose Apply to save your work.

To view instructions:

1. Navigate to the Schedule or Event page.

If only one instruction attachment exists at the schedule or event level, it is shown in a frame automatically on the page.

ORACLE Flow Manufacturing

Home Logout Preferences Personalize Page

Home Sequencing Tasks Workstation Setup

Workstation > Work Queue (S1) > Line/Line Operation Laptops/LOP1

Event: FA12 (S1)

Back Finish Complete Schedule (3) Complete Line Operation (5)

General

Schedule Number 136701 Assembly MC24713 Enter Quality Results (4) Find ECO View All Instructions

Build Sequence 3725 Assembly Description Plastic Base - Black

Scheduled Quantity 1

Instructions

Source Standard Operation Entity Value FA12 Attach Seq 10 Description OMS

1 A) Attach fan extension cable

2 E) Complete any necessary verifications/signoffs on data sheet. F) Tape data sheet on the front of the unit and place unit in test kanban.

3 Drivers are extremely sensitive to mechanical shock and vibration. HANDLE WITH CARE

4 Blank plate must be centered evenly in bezel plate opening. Re-adjust, if necessary.

5 Fan Power Cable must be routed underneath SCSI Cable #3 to terminator, ensure cable has adequate free play.

6 Verify that fan exhausts to rear of assembly.

7 Use a wrist strap at a grounded work station.

Out felt to approximately 1 inch.

CAUTION

2. If more than one attachment exists, a table displays the list of instructions. Select a record to navigate to the instruction frame.
3. Select View All Instructions to display a list of all instructions attached to a schedule.

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Flow Manufacturing

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[Workstation](#) > [Work Queue \(S1\)](#) > [Event: EV10 \(S1\)](#)

Line/Line Operation ks_line1/LOP1

View All Instructions (S1)

Instructions for Event EV10

Select	Attach Seq	Source	Entity Value	Description	Category	Data Type	File Name
☐	10	Standard Operation	EV10	Standard Event EV10 Attach 10	Miscellaneous	Short Text	Undefined
☐	10	Standard Operation	LOP1	Attach Seq 10 for Standard LineOp LOP1	Miscellaneous	File	OMSLOP1.gif
☐	20	Standard Operation	EV10	Standard Event EV10 Attach 20	Miscellaneous	Long Text	Undefined
☐	30	Standard Operation	EV10	Standard Event EV10 Attach 30	Miscellaneous	Web Page	www.yahoo.com
☒	40	Standard Operation	EV10	Standard Event EV10 Attach 40	Miscellaneous	File	oms_electronics_medium.gif

Details

Source

Standard Operation

Entity Value

EV10

Attach Seq

40

Description

Standard Event EV10 Attach 40

ELECTRONICS

ASSEMBLY #	OPERATION
REF ENG#	SEQUENCE
169080-91D	-

084.

025.

053.

028.

025.

MFG 87 NO TIP .080 BEAD OF ACS 429 UNDER

PDAS, LEADS TO BE FREE OF ACS 429.

Viewing Detail Properties

The Detail Properties region displays all the line, schedule, bill of material, and assembly information about a particular Flow schedule. The information in this region is derived from the data created when the Flow schedule is created.

Instructions		Components		Resources		Detail Properties		Customized Properties	
Line	Laptops	Line Operation	LOP1						
Schedule Number	104237	Build Sequence	21						
Assembly	MC24713	Assembly Description	Plastic Base - Black						
Scheduled Completion Date	2004.Jan.22 00:00:00	Scheduled Quantity	16						
BOM Revision	A	BOM Revision Date	2004.Jan.22 00:00:00						
Routing Revision	A	Routing Revision Date	2004.Jan.22 00:00:00						
Alternate BOM Designator		Alternate Routing Designator							
Schedule Group	Day	Sales Order							
Completion Subinventory	FGI	Completion Locator							
Project		Task							
Unit Number									
Assembly Item Properties									
Late Demands Penalty									
Invoice UOM									
Graphical Link for Web Reqs									
Material Over-Capacity Penalty									
Charge Master Description									
Service Level									
Japanese Desc									
Instructions		Components		Resources		Detail Properties		Customized Properties	
Return to Enter Schedule (0)				Complete Schedule (3)		Complete Line Operation (5)			

Existing descriptive item flexfield data also displays. You can use the System Items Flexfield for reporting item information. See: *System Items, Oracle Inventory User's Guide*

Viewing Components and Replenishing Kanban Cards

You can view and replenish components in the Components region. This region displays field information for Event Sequence, Item Sequence, Component Description, Reference, Revision, Unit of Measure, Quantity, Effectivity Date, and Disable Date.

The Replenish Kanban Card page can be accessed for a specific component by selecting the Kanban Cards Exist icon. The Replenish Kanban Card page displays both eligible and ineligible kanban cards defined for a particular component.

You can update components on the Update Components page, accessed from the Event page. The Components region displays as Recorded Components when the backflush transaction is recorded.

Instructions Components Resources Detail Properties Customized Properties										
Event Seq	Item Seq	Component	Description	Ref	Revision	UOM	Qty	Effectivity Date	Disable Date	Kanban
1010	10	CM00033	600MB Internal 24X CD-ROM Drive	A101	A	Ea	1	2004.Apr.18 11:31:33	2086.Jun.10 11:31:33	
1010	20	CM00034	600MB Internal 32X CD-ROM Drive	A101	A	Ea	1	2004.Apr.18 11:31:33	2086.Jun.10 11:31:33	
1010	30	CM00035	12GB 4mm Tape Drive		A	Ea	1	2004.Apr.18 11:31:33	2086.Jun.10 11:31:33	
1010	40	CM93501	Mounting Bracket Left Hand		A	Ea	1	2004.Apr.18 11:31:33	2086.Jun.10 11:31:33	
1010	70	AT-35T-F-ASY	Flywheel Assy		A	Ea	1	2004.Apr.18 11:31:33	2086.Jun.10 11:31:33	
1010	80	CM23597	Formed Sheet Metal Shell Bottom		A	Ea	1	2004.Apr.18 11:31:33	2086.Jun.10 11:31:33	
1020	50	CM00036	7GB 8mm Tape Drive		A	Ea	1	2004.Apr.18 11:31:33	2086.Jun.10 11:31:33	
1020	60	CM00037	CD Writer		A	Ea	1	2004.Apr.18 11:31:33	2086.Jun.10 11:31:33	
Instructions Components Resources Detail Properties Customized Properties										

To replenish Kanban cards:

1. Navigate to the Components region, select the component for replenishing by choosing the Kanban Cards Exist icon.

The Replenish Kanban Cards page displays.

2. In the Cards Available for Replenishment region, select components by checking the Select box for the card numbers you want to replenish.
3. Choose Replenish.

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Flow Manufacturing

Home Logout Preferences Diagnostics

Home Sequencing Tasks **Workstation** Setup

Workstation > Work Queue (S1) > Schedule: 153344 (S1) Line/Line Operation ks_line1/LOP1

Replenish Kanban Cards (S1)

Component **CM23597** Subinventory **FloorStock**
Description **Formed Sheet Metal Shell Bottom** Locator

Cards Available for Replenishment

Select Kanban Cards: **Replenish**

Select All | Select None

Select Card Number	Size	Supply Status	Source Type	Activity Date
☐ 864	200	Full	Supplier	2003.Dec.10 20:38:53
☐ 865	200	New	Supplier	2003.Oct.21 17:26:04

Cards Currently Not Available for Replenishment

Card Number	Size	Card Status	Supply Status	Source Type
No data exists.				

When you transaction is performed, the record you selected displays in the Cards Currently Not Available for Replenishment Region.

Updating Component Requirements

You can update your component requirements for a flow schedule on the Update Components page accessed from the Event page. This page enables you to assign lot and serial numbers, add or delete components, change subinventory locations, and quantity values.

When the backflush transaction is recorded, the Components region appears as Recorded Components region for the operation you are viewing.

To update components on a flow schedule:

1. In the Event page, select Update Components to display the Update Components page. Schedule information displays in the header region of the page. The Backflush Components region is the area used to change component information.
2. To add components, choose Add Another Row

ORACLE® Flow Manufacturing

Home Logout Preferences Diagnostics

Home Sequencing Tasks Workstation Setup

Workstation > Work Queue (S1) > Event: FLEF (S1) >

Line/Line Operation DB-LINE1/OP10
Logged In As FLMDEV

Update Components (S1)

Schedule Number 199108 Assembly ASSEMBLY1 Scheduled Qty 100
Build Sequence Assembly Description ASSEMBLY 1 Open Qty 88

Cancel Apply

Backflush Components

Select Component: Assign Lot and Serial Delete

Select	Op Seq	Component	Subinventory	Locator	Qty	Lot UOM Number	Serial From To Parent
☐	1010	DB-FLM-COMP1	Consumable		1	Ea	
☐	1010	03CR Item2	FloorStock		100		
☐	1010	03Lot Controlled	RIP		500	Ea	

Add Another Row

Cancel Apply

3. You can change or select information in the following fields: Component, Subinventory, Locator, and Quantity.
4. Single lot and serial number or single range serial numbers can be assigned through the Update Components page. Select Assign Lot and Serial Number to create multiple, or out of range, lot and serial number records for applicable components. When lot or serial tracking for components is enabled, the lot and serial number columns are updateable. See: Assigning Lot and Serial Numbers on the Workstation, page 10-32
5. Select Apply to save your work.

Assigning Lot and Serial Numbers on the Workstation

You can assign lot and serial number records for components. The Component Lot Serial Update page is accessed from the Update Components page.

To assign lot and serial numbers on the workstation:

1. From the Update Components, select Assign Lot and Serial. The Assign Lot and Serial Number page appears.

If lot or serial numbers are assigned, the applicable numbers display in the Select Lot and Serial Number Entry boxes. You can delete these records or change the quantity.

ORACLE® Flow Manufacturing

Home Logout Preferences Diagnostics

Home Sequencing Tasks Workstation Setup

Workstation > Work Queue (S1) > Event: AS11 (S1) > Update Components (S1) >

Line/Line Operation: Laptops/LOP1
Logged In As: FLMIDEV

Assign Lot and Serial Number: comp lot_ctrl (S1)

Schedule Number 205409 Assembly assy_ctrl Scheduled Qty 1
Build Sequence Assembly Description assy_ctrl Open Qty 1

Cancel Apply

Lot	Attributes	Qty
No results found.		

Add Another Row

Cancel Apply

2. For items under lot control, the Lot box appears. Choose Add Another Row to select lot numbers.
 - Select a value from the list of values in the Select Lot Number box, and enter a quantity.
 - Optionally, choose the Attributes icon to access the Lot Attributes page to view or change information.
3. For items under serial control, the Serial box displays. Choose Add Another Row to select serial numbers.
 - Select a value from the list of values in the Select Serial Number box, or enter a range of numbers. Enter a quantity.
 - Optionally, choose the Attributed icon to access the Serial Number Attributes page to view or change information.

4. Choose Apply to save your work.

Related Topics

Setting Up Lot Control, *Oracle Inventory User's Guide*

Maintaining Item Lot Information, *Oracle Inventory User's Guide*

Setting Up Serial Number Control, *Oracle Inventory User's Guide*

Assigning Serial Numbers, *Oracle Inventory User's Guide*

Integration with Oracle Quality

If collection plans exist for line operations or schedules, you can access Oracle Quality to enter quality data. You can enter results for mandatory collection plans, or optional collection plans, at specific points on a routing. Line operations and schedules cannot be completed if there are remaining mandatory collections plans. If the collection plan is mandatory, the Line Operation Completion page displays when you choose Complete Schedule or Complete Operation. To access available Quality plans, choose Add Child Data in the Applicable Quality Plans region.

When the collection plan data is entered, and the completion transaction is saved, the quality data is also saved. See: *Entering Quality Results, Oracle Quality User's Guide*

Note: Add Child Data is enabled when Oracle Quality is enabled for the organization, and a collection plan exists for the line operation.

Customized Properties Region

You can create a region of new parameters, customized to fit your business needs. These are fields that do not already exist on the Schedule and Detail Properties pages, and parameters not delivered with the Flow Execution Workstation.

Instructions	Components	Resources	Detail Properties	Customized Properties
------------------------------	----------------------------	---------------------------	-----------------------------------	---------------------------------------

Name	Value
Priority	This is a high priority customer, no delays accepted
Customer Note	Product needs to be packed in yellow box

Instructions	Components	Resources	Detail Properties	Customized Properties
------------------------------	----------------------------	---------------------------	-----------------------------------	---------------------------------------

[Return to Work Queue \(0\)](#)
[Complete Schedule \(3\)](#)
[Complete Line Operation \(5\)](#)

This region has Name and Value fields. For example, you may want to add a Name designated as Priority, with a corresponding Value as specified as High Priority.

The custom parameters are added using an applications program interface (API)—Flow Manufacturing, Execution Utility.

Engineering Change Order Alerts

You can check for engineering change orders (ECOs) related to your assembly. Changes include adding, changing, or disabling components. The information is accessed from the Schedule and Event pages. The ECOs that are eligible for display include:

- Bill of Material ECOs, not routing ECOs
- Status is Implemented
- Components are used at current line operation or event only

ORACLE® Flow Manufacturing										Home Logout Preferences Personalize Page	
										Home Sequencing Tasks Workstation Setup	
Workstation > Work Queue (S1) > Schedule: 136702 (S1) >										Line/Line Operation Laptops/LOP1	
Component Changes (S1)											
				Assembly MC24713 Assembly Description Plastic Base - Black Alternate Bom Designator		Date Range From 06-Apr-2004 10:33:56 Date Range To 01-Sep-2004 08:24:00					
Operation Seq	Item Seq	Action	Component	Description	ECO	Effectivity Date	Old Quantity	New Quantity			
15	120	Add	75100001	Printer Mechanism	S1-1008	02-Jun-2004 13:55:47	1	1			
Return to Schedule (0)											

This functionality is set up in the Automatic ECO Check field in the Workstation Preferences page. See: Setting Your Preferences, page 10-2

Eligible ECOs can be flagged for a range of dates. This enables you to have the correct changes for work as it progresses on the shop floor. For example, if you have a value of 2 in the ECO Days Prior field, and a value of 3 in and ECO Days After field—the system considers ECOs in a range of 5 days. This is 2 days before today's date, and 3 days after the scheduled completion date.

Graphical Kanban Workbench

This chapter covers the following topics:

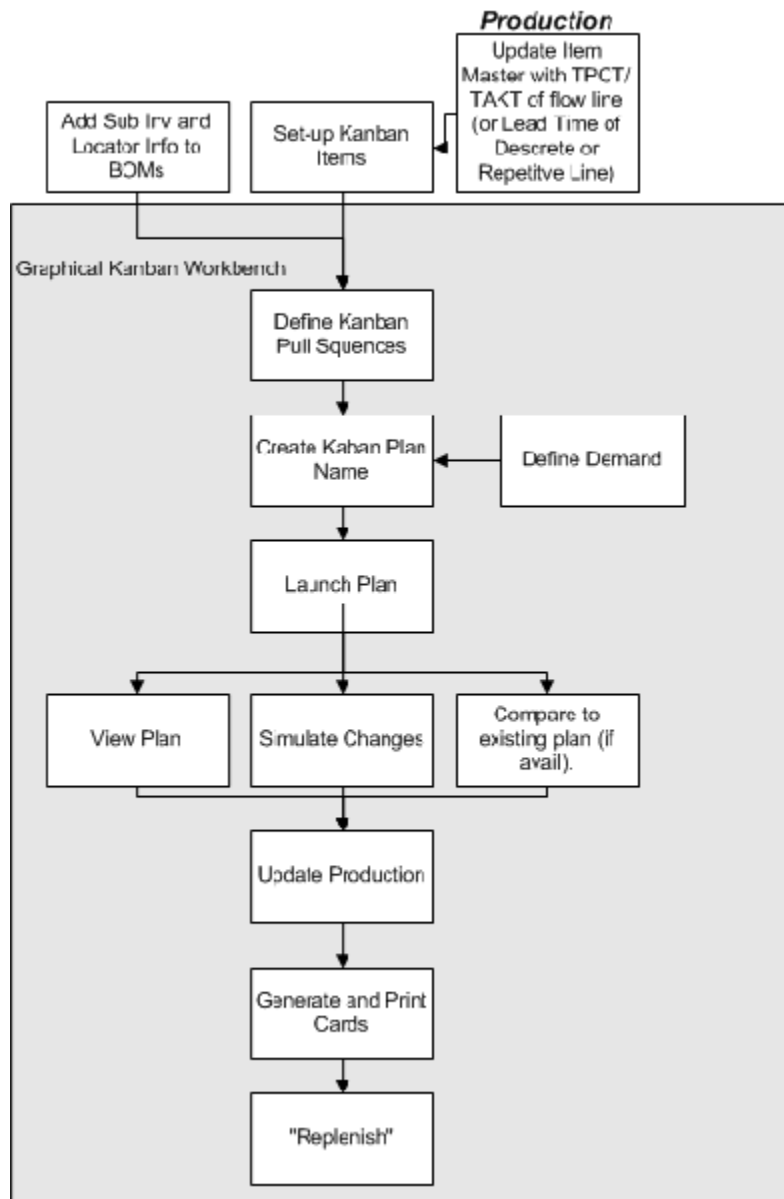
- Overview of the Graphical Kanban Workbench
- Accessing the Production Region
- Adding an Item to the Tree
- Modifying and Creating Pull Sequences on the Graphical Network
- Pull Sequence Window
- Using a Template to Create a Pull Sequence
- Accessing the Planning Region
- Viewing Kanban Demands
- Changing and Viewing Kanban Details
- Viewing and Changing Summary Pull Sequences
- Launching a Kanban Plan
- Updating Production
- Purging Kanban Cards and Pull Sequences

Overview of the Graphical Kanban Workbench

Flow Manufacturing a kanban pull replenishment system to signal material requirements and pull material from its defined source, as needed, to meet demand. The objective of the kanban replenishment system is to pull the material, signaling with a replenishment signal.

The term kanban refers to a visual replenishment signal, such as a card or an empty container for material. In a kanban system, each workstation on the flow line can have several containers, each holding the same quantity of material. Typically, an empty container is a signal to replenish. The following diagram demonstrates the flow of the kanban planning system. Oracle Flow Manufacturing uses the Graphical Kanban

Workbench to create these transactions.



The Graphical Kanban Workbench combines kanban setup, kanban planning, pull sequence definition, and planning simulation in one interface. The graphical interface includes the following features:

- Graphical view the pull sequence network
- Graphical view the demand pattern for monitoring requirements
- You can define pull sequence, and have the capability to mass add pull sequences for items

- You can add items under locator control and specify point of use, point of supply, or both as locator sources
- You can perform simulation replans and updates to production
- You can view comparison of kanbans by variances in size, or the number of cards between the two kanban plans being analyzed

Note: The Bills of Material Workday calendar is used during kanban planning.

See: Overview of Kanban Replenishment, *Oracle Inventory User's Guide*.

Navigating on the Graphical Kanban Workbench

The Graphical Kanban Workbench is comprised of two panes. The left pane is the tree hierarchy. There are two tree tabs located to the left of the tree hierarchy that allow you to toggle between the Production and Planning views. The View By list box enables you to change the way you view data in the tree.

The right pane displays the Production or Planning regions in a graphical view of the kanban network. The right pane contains the data that is associated with the selected item on the tree. Or it displays the template windows used for creating templates for your kanban network.

Color Representation

The arrow on the replenishment chain canvas represents the pull sequence between point of use and a point of supply. One or more cards are generated for a pull sequence. A color on the arrow is used to easily identify the card status on the pull sequence. This information is also viewable on the View Kanban Demands and Compare Kanban Plans windows. The following values are used in the graphical replenishment network in the right pane:

- Black: Cards have not been generated for the pull sequence
- Green: There are cards in the pull sequence that are in the active status
- Blue: Some cards in this pull sequence are on hold, and none are active
- Red: All of the cards are canceled

Workbench Toolbar

The right pane canvas displays a toolbar when you are in the Production region. It contains icons used for creating and modifying your graphical network.



This toolbar contains icons for:

- *Connector*: Enables you to create links between entities on the canvas
- *Pointer*: Enables you to move existing entities on the canvas
- *Organization*: A point of supply that is in another organization
- *Supplier*: A point of supply that is a supplier
- *Subinventory*: A point of user or point of supply that is a subinventory
- *Production*: A point of supply that is a flow line, repetitive line, or discrete production area

Tools Menu and Pop-up Menus

The Graphical Kanban Workbench enables you to access other manufacturing functionality. These options are available, depending upon the currently active window and other conditions, in the Tools menu and Pop-up menus.

Tools Menu

Pull Sequence: Accesses the Pull Sequences Summary window to view, update, and define the source of replenishment for a kanban item, see: *Defining Kanban Pull Sequences, Oracle Inventory User's Guide* .

Kanban Cards: Accesses the Kanban Cards Summary window to view, define, and update kanban cards, see: *Defining Kanban Cards, Oracle Inventory User's Guide*

Print Kanban Cards: Accesses the request window for printing kanban cards for a specific item or for all items. see: *Printing Kanban Cards, Oracle Inventory User's Guide*

Generate Kanban Cards: Accesses the Generate Kanban Cards window, see: *Generating Kanban Cards, Oracle Inventory User's Guide*,

Forecast/Schedule: Accesses the Item Forecast Entries window where you can review information for items in a forecast set. See: *Defining a Forecast, Oracle Master Scheduling/MRP and Oracle Supply Chain Planning User's Guide*

Launch Kanban Plan: Accesses the Kanban Planner request window for entering plan information for kanban calculations. See: *Launching a Kanban Plan*, page 11-33 and *Using the Kanban Calculation Program, Oracle Master Scheduling/MRP and Oracle Supply Chain Planning User's Guide*

Update Production: Accesses the Kanban Details or Update Production Kanbans window, depending on whether a plan or item is selected on the navigation tree. These

windows enable you to review adjustments to your replenishment chain, and update the changes in production plans and kanban cards. See: *Defining a Forecast, Oracle Master Scheduling/MRP and Oracle Supply Chain Planning User's Guide*

Pop-up Menus

You can access windows and functions in the pop-up menus by right-clicking on an icon in the right pane, or a detail in the tree hierarchy.

These options are available, depending upon the currently active window and where your cursor is located when you right-click in the right pane:

- *Location*: Accesses the Location window for the replenishment source for a kanban location
- *Delete Location*: Enables you to remove unconnected locations
- *New*: Enables you to add a new items' replenishment chain, see: *Adding an Item to the Tree*, page 11-9
- *Pull Sequence*: Accesses the Pull Sequences Summary window to view, update, and define the source of replenishment for a kanban item.
- *Kanban Cards*: Accesses the Kanban Cards Summary window to view, define, and update kanban cards
- *Print Kanban Cards*: Accesses the request window for printing kanban cards for a specific item or for all items
- *Generate Kanban Cards*: Accesses the Generate Kanban Cards window
- *Delete Pull Sequence*: Enables you to remove the pull sequence; a pull sequence can only be deleted if the color is black, signifying that there are no cards associated with it
- *Place Kanban Cards On Hold*: Enables you to change the kanban cards to inactive by changing the status to hold
- *Activate Kanban Cards*: Enables you to change the status of kanban cards from hold back to active
- *Cancel Kanban Cards*: Cancels all the cards associated with a pull sequence; items can be received using the card but not replenished
- *Delete Kanban Cards*: Deletes all the kanban cards associated with a pull sequence, this option is only available if all cards have been canceled indicated by the color red of the link
- *Forecast/Schedule*: Accesses the Item Forecast Entries window where you can review

information for items in a forecast set

- *Launch Kanban Plan:* Accesses the Kanban Planner request window for entering plan information for kanban calculations
- *Update Production:* Accesses the Kanban Details or Update Production Kanbans window, depending on whether a plan or item is selected on the navigation tree. These windows enable you to review adjustments to your replenishment chain, and update the changes in production plans and kanban card

Accessing the Production Region

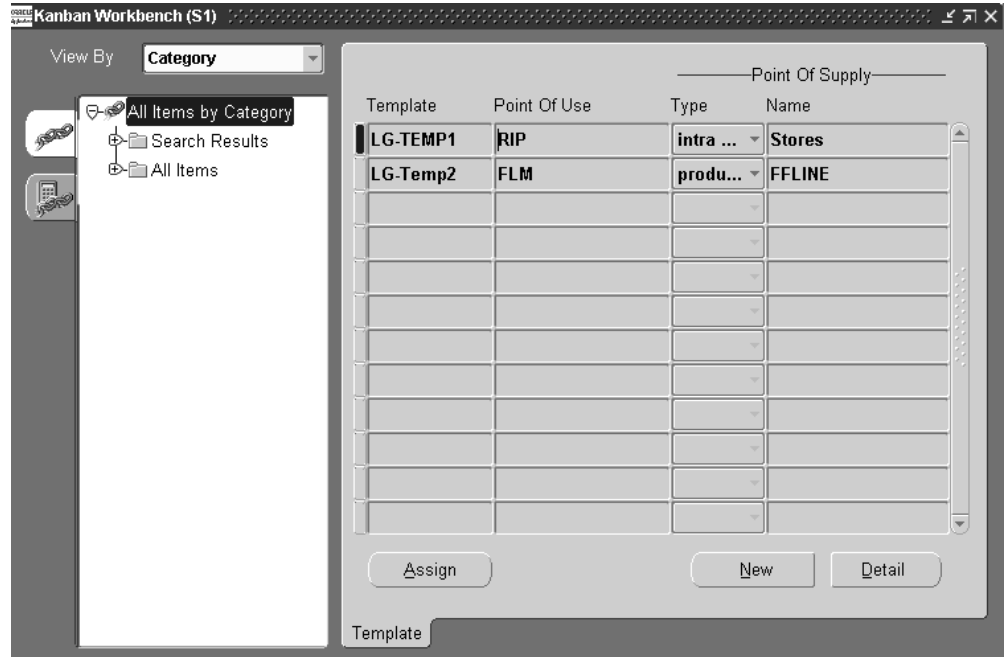
The Production region tree displays all the items that already have pull sequences defined, and the right pane consists of the detail kanban chain diagram.

This graphical detail view shows the replenishment chain. The replenishment chain for an item shows all pull sequences that belong to that item.

To view kanban production information on the Graphical Kanban Workbench:

1. Select Graphical Kanban Workbench on the Flow menu.
2. In the View By list box, select viewing criteria for the kanban data.

Your choices are: Category, Buyer Code, Planner Code, Supplier, Location, Line Operation, and Process. In the View by Line Operation or Process mode, the hierarchical tree displays all flow lines and you can expand the tree to see line operations and processes on the selected line. In the line operation view, an Unreplenished node at the Line level provides a listing of bill of material components without defined pull sequences.

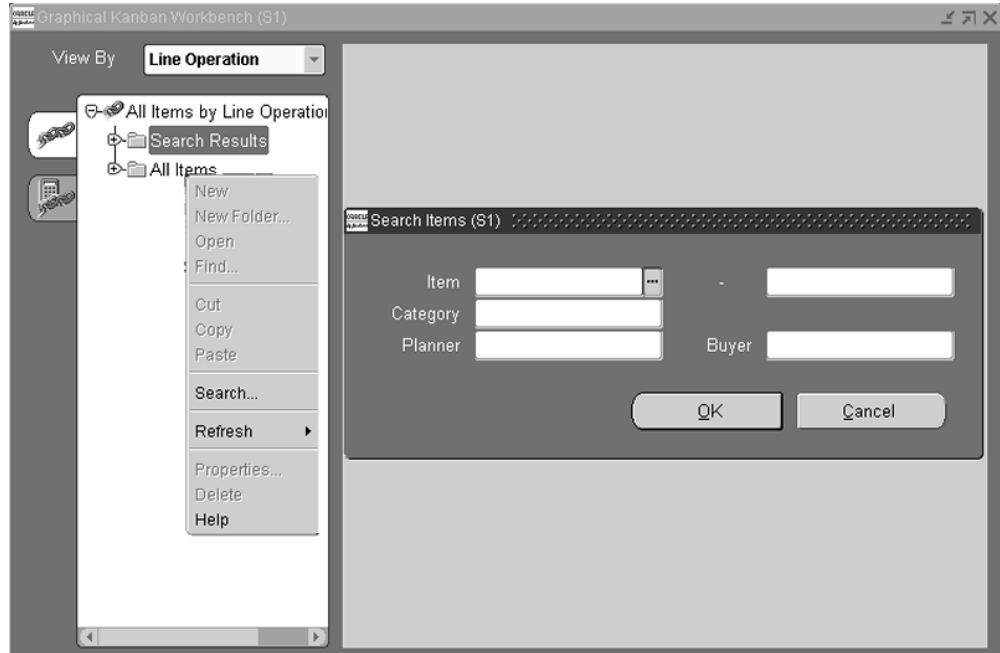


3. Select the Production tree tab.

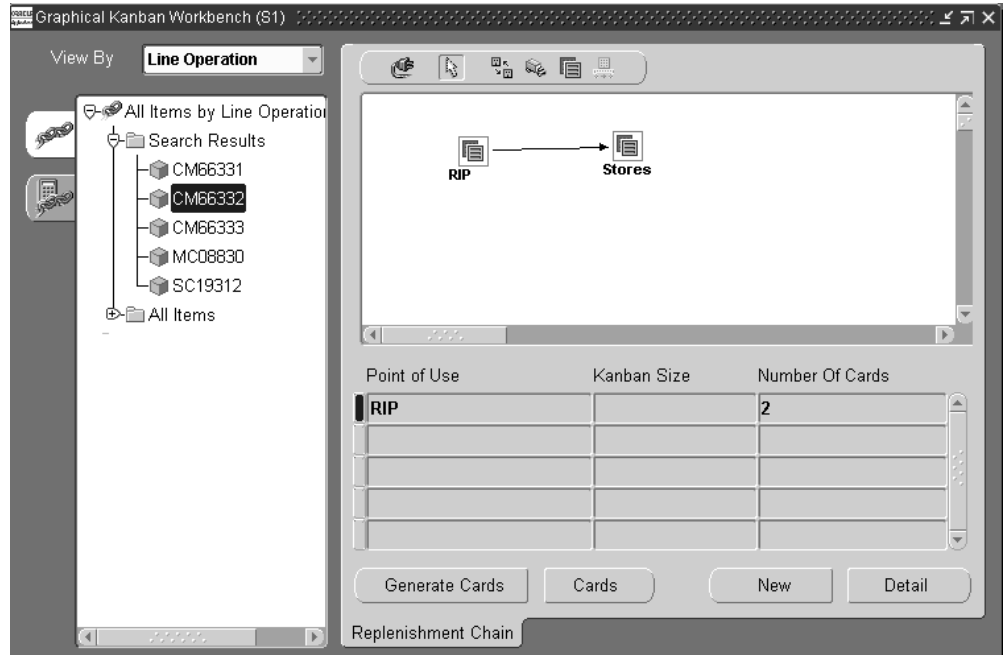
In the initial view, the workbench displays the Template window used for creating new kanban networks. See: Using a Template to Create a Pull Sequence, page 11-14

In the All Items by node, you can search for records using the Search Results node, or view all the records in the All Items node.

4. If you want to find a specific record, or range of records, select the Search Results node and right-click with your mouse to display the pull-down menu. Select the Search menu option.



5. Enter search criteria; you can search by item, category, planner, buyer, or any combination of these values.
6. Choose OK to display the results of your query under the Search Results node.



7. Select a node detail on the tree.

The tree displays all the items that have the pull sequence defined for the specified plan. The graphical view the pull sequence network displays in the right pane.

You can add, change, delete, or move the icons in the graphical network. See: *Modifying and Creating Pull Sequences on the Graphical Network*, page 11-10

8. There are four buttons on this window that enable you to navigate to other kanban windows.
 - *Generate Cards*: Accesses the Generate Kanban Cards window, see: *Generating Kanban Cards, Oracle Inventory User's Guide*
 - *Cards*: Accesses the Kanban Cards Summary window to view, define, and update kanban cards, see: *Defining Kanban Cards, Oracle Inventory User's Guide*
 - *New*: Accesses the Pull Sequence window to fill in fields to create a new pull sequence, see: *Pull Sequence Window*, page 11-12
 - *Detail*: Accesses the Pull Sequence window to view, update, and define this pull sequence, see: *Pull Sequence Window*, page 11-12.

Adding an Item to the Tree

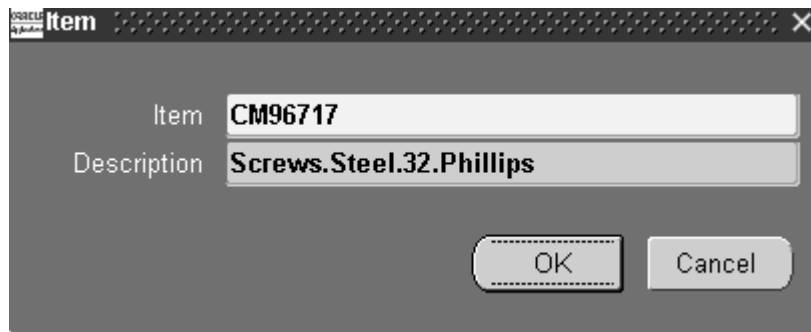
The tree hierarchy displays all the items that have a replenishment chain. You can add

new items using the Item window.

To add an item to the replenishment chain:

1. On the tree hierarchy, select the top level node where you want to add another item.
2. Right click with you mouse on your selection to display the pop-up menu.
3. Select New.

The Item window displays.



4. Enter an item, or select from the list of values.
5. Choose OK

The item appears at the bottom of the list for the top level item, the right canvas is blank.

Modifying and Creating Pull Sequences on the Graphical Network

Pull Sequences are used to view, update, and define the source of replenishment for a kanban planned item in a kanban location. You can use the Graphical Kanban Workbench to create, move, and delete pull sequences.

To create pull sequences for items defined on the tree:

1. Select a node detail on the tree.

The tree hierarchy only displays items that already have pull sequences defined. The replenishment chain displays in the right pane.

2. Select a kanban source type icon from the toolbar, and select a location to place the pull sequence on the right pane canvas See: Workbench Toolbar, page 11-3

The Location window displays with different fields, depending on your replenishment source:

- If you select Organization, select the source organization and subinventory. If the organization, subinventory, or item is under locator control, you must also enter a stock locator. See: *Assigning a Completion Subinventory and Locator*, *Oracle Bills of Material User's Guide*

The screenshot shows a dialog box titled "Location". It has three input fields: "Organization" (a dropdown menu), "Subinventory" (a dropdown menu), and "Locator" (a text input field). Below these fields are two buttons: "OK" and "Cancel".

- If you select Supplier, you can optionally select the supplier and supplier site. If you do not select a supplier, Oracle Purchasing will choose the supplier based on sourcing rules when it creates a purchase order/blanket release.

The screenshot shows a dialog box titled "Location". It has two input fields: "Supplier" (a dropdown menu) and "Supplier Site" (a text input field). Below these fields are two buttons: "OK" and "Cancel".

- If you select Subinventory, select the source subinventory, and stock locator if applicable.

Additional Information: If you choose a subinventory with no locator control—you can specify a locator on the item master, the bill of material, and the source of the pull sequence—but the system uses the locator value for planning only. It is not transacted at the location level.

- If you select Production, optionally enter the line code.

3. Select the Connector icon to join the objects together to form a replenishment chain. Select the location and drag from the point of supply to the point of use.

Connect the objects together—starting with the sub-inventory/locator closest to your production. When you create the link, the Pull Sequence window appears. Enter details about the Point of Use and in the Point of Supply. See: Pull Sequence Window , page 11-12.

4. Save your work.

Pull Sequence Window

Use the Pull Sequence window to view, update, and define the source of replenishment for a kanban item in a kanban location. A pull sequence groups information for an item defining a kanban location, source information, and planning parameters. An item can have multiple pull sequences that make up a replenishment chain. The kanban calculation program determines the optimal number of kanban cards or size according to the values you enter in the Pull Sequence window for size, cards, demand, lead time to replenish. See: Defining Kanban Pull Sequences, *Oracle Inventory User's Guide*

To view and modify pull sequence detail information:

1. Navigate to the Pull Sequence window.
2. Select a value in the Point of Use Subinventory field and a Locator, if applicable.

This is the location where material is used.

The screenshot shows the 'Pull Sequence' window with the following fields and values:

- Item Name:** CM23591
- Molded Plastic Shell - Top** (Title)
- Point Of Use Subinventory:** (Empty)
- Locator:** (Empty)
- Point of Supply:**
 - Type:** Inter Org
 - Supplier:** (Empty)
 - Site:** (Empty)
 - Organization:** (Empty)
 - Subinventory:** (Empty)
 - Locator:** (Empty)
 - Line Code:** (Empty)
- Kanban:**
 - Planning Only:** (Checked)
 - Auto Request:** (Unchecked)
 - Calculate:** Kanban Size
 - Size:** (Empty)
 - Number Of Cards:** (Empty)
 - Minimum Order Qty:** (Empty)
- Planning:**
 - Lead Time:** (Empty)
 - Lot Multiplier:** (Empty)
 - Safety Stock Days:** (Empty)

At the bottom, there is a button labeled 'OK' and a button labeled 'Cancel'.

3. In the Point of Supply region, indicate the type of pull sequence and the source.

Depending on the source type indicated in the Type field, you must supply information in the following fields for, Supplier, Site, Organization, Subinventory, and Line Code. See Changing and Viewing Kanban Details, page 11-25

Additional Information: If you choose a subinventory with no locator control—you can specify a locator on the item master, the bill of material, and the source of the pull sequence—but the system uses the locator value for planning only. It is not transacted at the location level.

4. The Kanban region is used to select parameters for kanban creation. The Planning Only box is used if you want the program to perform the kanban calculations, but you will use a manual system to execute the kanbans.
5. The Auto Request box is used to indicate if this kanban record is flagged for the Outbound Broadcast Sequencing functionality. See: Overview of Outbound

6. The Calculate, Number of Cards, Size, and Minimum Order Qty fields are used to define kanban card calculations. See: *Defining Kanban Pull Sequences, Oracle Inventory User's Guide*
7. In the Planning region, enter the Lead Time for this location, and optionally enter Lot Multiplier, and number of Safety Stock Days.
8. Choose Generate Cards if you want to create kanban cards for this pull sequence. See: *Generating Kanban Cards, Oracle Inventory User's Guide*.
9. Save your work.

Using a Template to Create a Pull Sequence

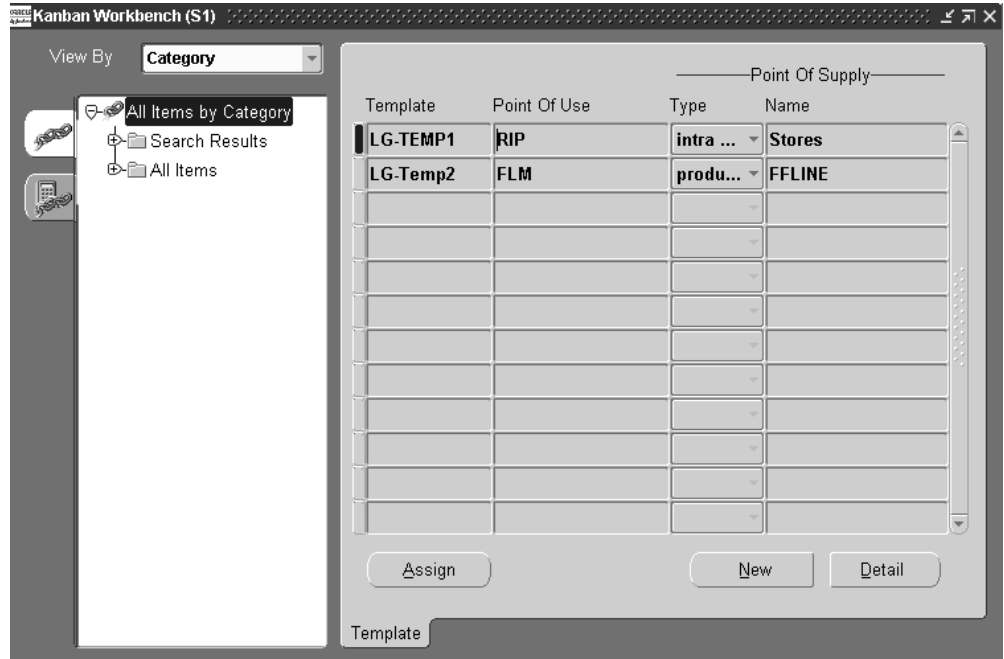
Templates enable you to mass add pull sequences for many items. There are several options available when creating pull sequences from templates— you can use an existing template and assign other items, create an entirely new template and assign items, or you can change the details of an existing template. After creation, the templates are accessible on the Production region of the Graphical Kanban Workbench when adding a new pull sequence.

Locator Control

You can add items under locator control when retrieving items for the pull sequences. When retrieving items you can specify point of use, point of supply, or both as locator sources in your criteria. Locator records can be retrieved either from the Bill of Material or from the Pull Sequences.

To create pull sequences for items using an existing template:

1. Navigate to the Graphical Kanban Workbench.
2. Select All Items in the Tree Hierarchy.



The right pane displays the Template window. This window displays any existing template information in a multiple row format including:

Template: The name of the existing template

Point of Use: The subinventory where material is used

Point of Supply— Type and Name: Point of Supply stock areas, from material stores, a supplier, a production area, or another organization

3. Select the template you want to use in the multi row table.
4. To add items to this template, choose Assign.

The Template Assignment window displays.

Add Items Criteria

Product Family ...

Parent Assembly

Component -

Backflush Subinventory **FloorStock**

Category -

Item Attributes

Buyer Code

Planner Code

Make/Buy

Release Time Fence

Locator

Point Of Use **Bill Of Material**

Point Of Supply

8. Select values in the Locator region if you want to use locator records in the selection criteria.

Point of Use is where the item is used on the flow line. Your choices are:

- Bill Of Material
- Pull Sequence
- Default—the search uses both bill of material and pull sequence records after using the other selection criteria entered on the window

Point of Supply is the stock location. You can search for locator records by the row, rack, bin, project, or task.

9. Choose Find.

The items that met your criteria appear in the Template Assignment window. You can deselect items by deselecting the check box. Optionally, you can add a Point of

Use Locator or Point of Supply Locator

	Item	Point of Use Locator	Point of Supply Locator	Description
☒	eng_item1		3.2.1..	3.5 Disk Drive
☒	mfg_item1		3.2.1..	Blank Board
☒	mfgComp2		3.2.1..	P200 Mhz CPU
☒	engComp1		3.2.1..	Resistor
☒	engAssembly		3.2.1..	CPU Chasis
☒				
☒				
☐				
☐				
☐				
☐				
☐				
☐				
☐				
☐				

Retrieve OK Cancel

10. Choose OK to save this retrieval.

The Template window displays again.

To create a new template and assign items:

1. Navigate to the Graphical Kanban Workbench.
2. Select All Items in the Tree Hierarchy.

The right pane displays the Template window. This window displays any existing template information in a multiple row format.

3. Choose New

The Pull Sequence template window displays.

4. Enter a unique value in the Template name field.
5. Add information to the fields on this window, see: Pull Sequence Window, page 11-12
6. Save your work.

The Template window displays again.
7. Assign items to this template, see: Pull Sequence Window, page 11-12

To change the details of an existing template:

1. Navigate to the Graphical Kanban Workbench.
2. Select All Items in the Tree Hierarchy.
3. Select the template in the multi row table, and choose Detail.

The Pull Sequence template window displays with the name of this template in the Template name field.

You can change any of the fields on this window, see: Pull Sequence Window, page 11-12

4. Save your work.

The Template window displays again.

Accessing the Planning Region

The Planning region is used for kanban planning. The tree hierarchy displays the kanban plan and all the items that already have pull sequences defined and demand from this plan. The right pane displays detailed information for the specified plan. In this region you can view kanban demands, compare kanbans, perform simulation replans, add, and delete pull sequences or icons on the graphical network.

To create a new plan:

1. Select the Planning tree tab on Graphical Kanban Workbench on the Flow menu.
2. In the View By list box, select viewing criteria for the kanban data.

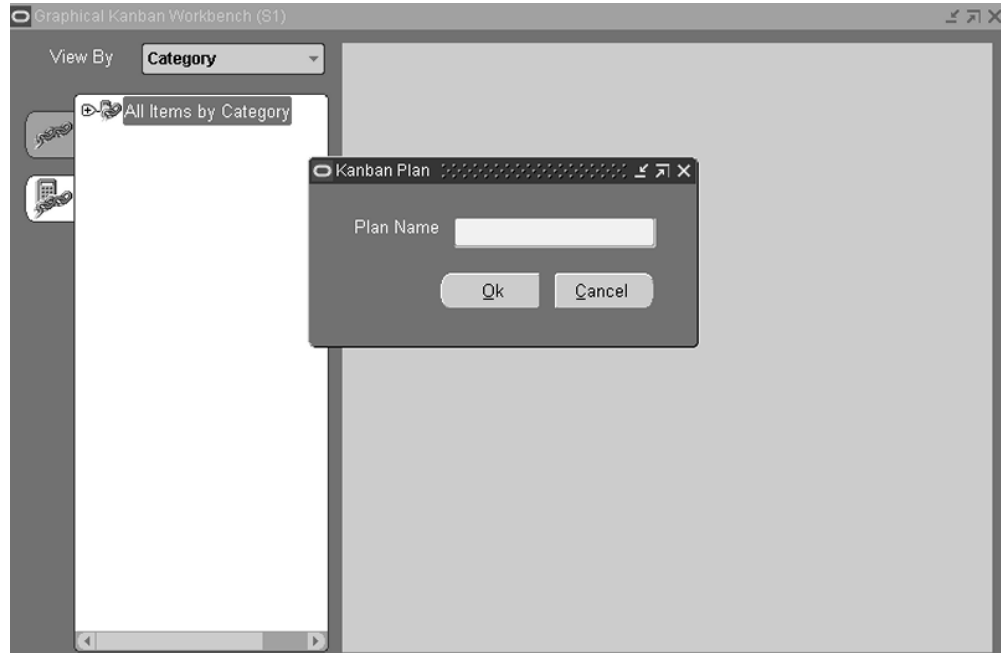
Your choices are: Category, Buyer Code, Planner Code, Supplier, and Location.

3. To create a new plan, select the Item node and right-click to display the popup menu. Choose New.

The Kanban Plan window displays.

4. Enter a unique value in the Plan Name field, and choose OK.

The plan Detail pane displays.

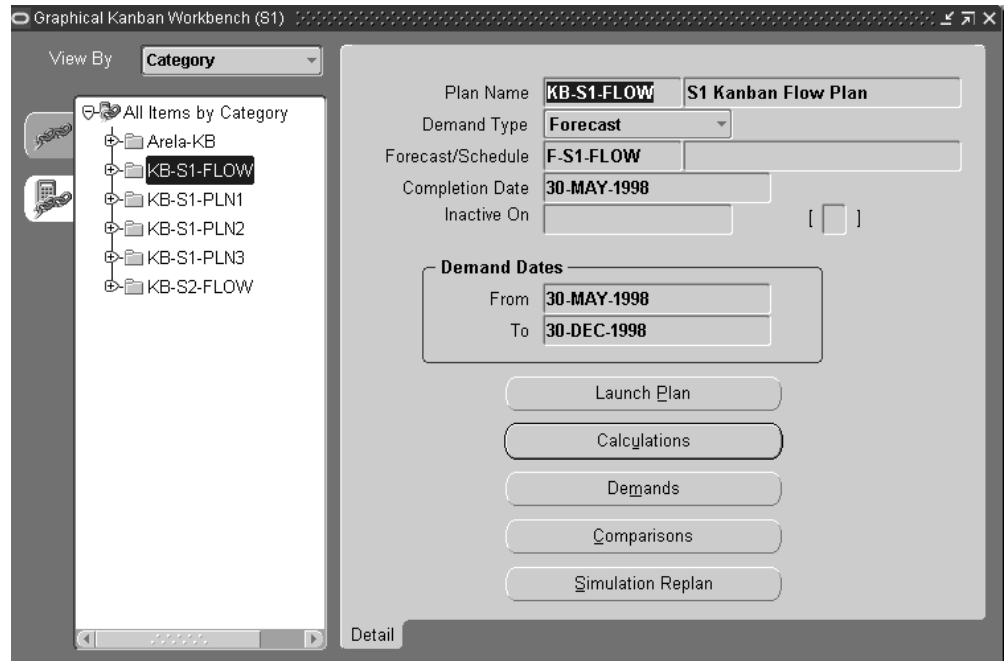


5. Optionally, enter a description of this plan in the Description field.
6. In the Demand Type field, select the demand source. Your choices are Forecast, MDS, MPS, or Actual Production.
7. In the Forecast/Schedule field select the name of the specific forecast or schedule you want to use.
8. Optionally, you can enter a date in the Inactive On field to indicate when this plan will not be active.
9. Save your work.

To view plan detail information:

1. Select the Planning tree tab on Graphical Kanban Workbench, and select a plan node on the tree.

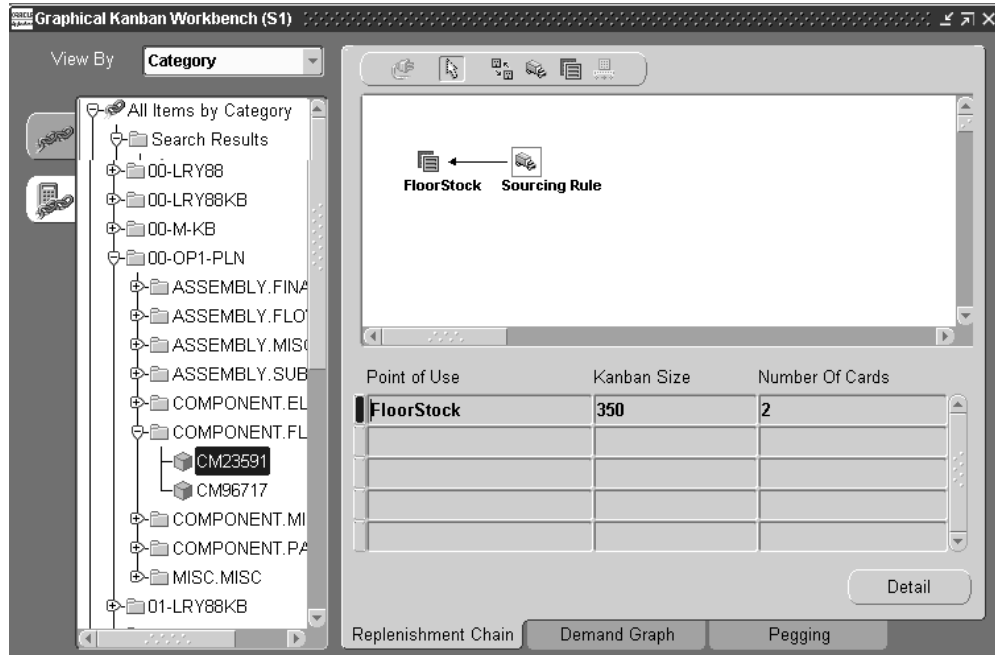
The plan Detail pane displays with the specific information for this plan including plan name, demand type, forecast/schedule, and Inactive On date, and demand dates. If this plan is completed, a value displays in the Completion Date field.



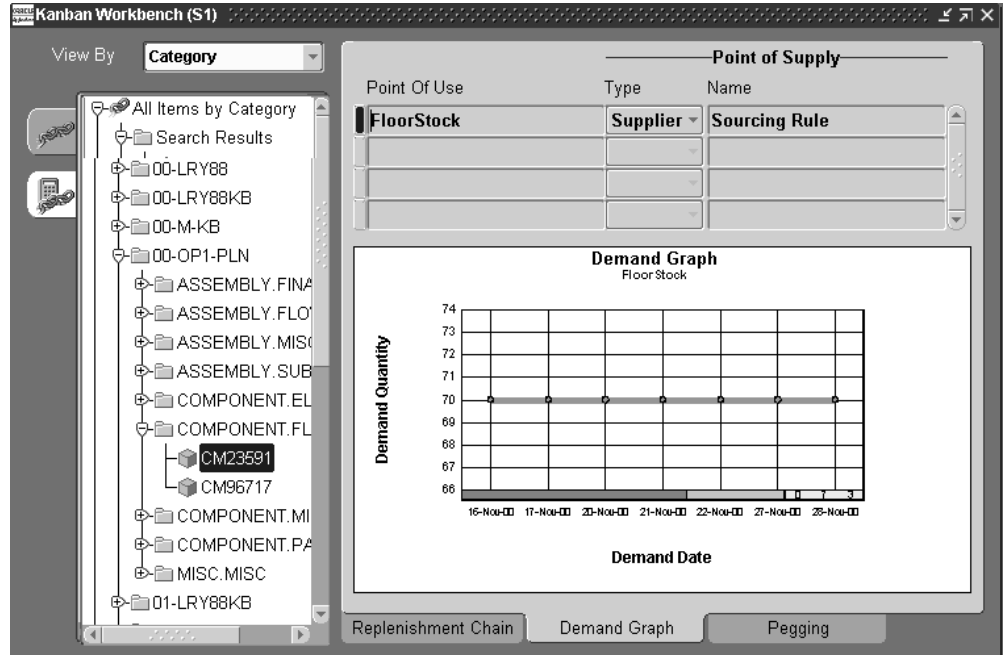
2. Choose Launch Plan to access the kanban calculation concurrent program to determine the number of kanban cards or sizes according to your parameters. See: Launching a Kanban Plan, page 11-33, and Using the Kanban Calculation Program, *Oracle Master Scheduling/MRP and Oracle Supply Chain Planning User's Guide*
3. Choose Calculations to access the Kanban Details window to view the pull sequences that resulted from the calculations performed in the plan.
4. Choose Demands to access the View Kanban Demand window, see: Viewing Kanban Demands, page 11-24
5. Choose Comparisons to access the Compare Kanban Plans window, see: Viewing and Updating Kanban Calculations, *Oracle Master Scheduling/MRP and Oracle Supply Chain Planning User's Guide*
6. Choose Simulation Replan to submit a request to the kanban calculation program to generate a new simulation plan based on the new values entered.

To view plan replenishment information on the Graphical Kanban Workbench:

1. Select the Planning tree tab.

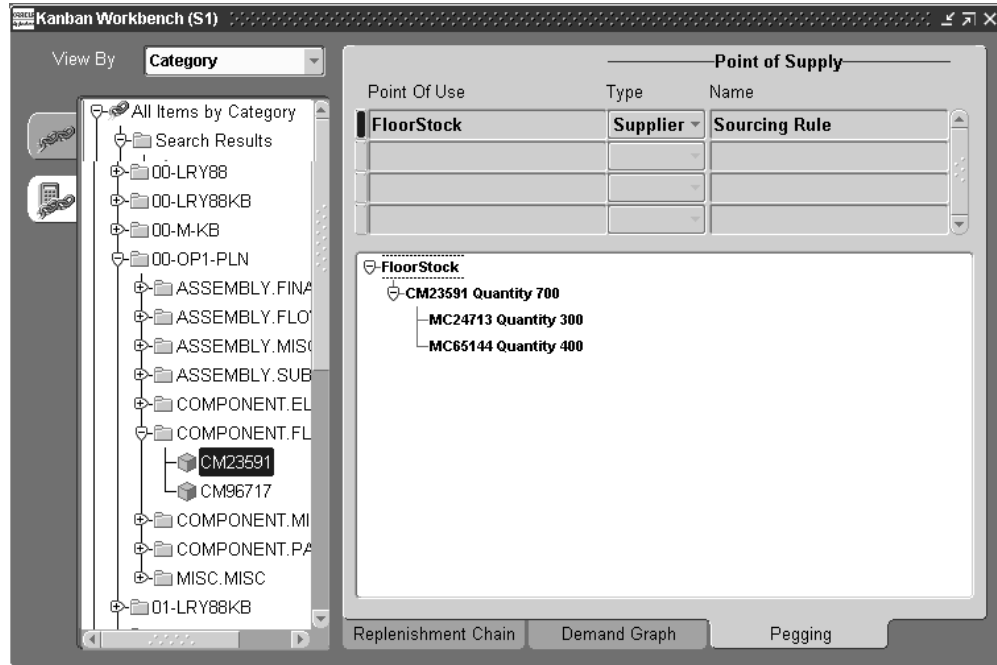


2. In the View By list box, select viewing criteria for the kanban data.
Your choices are: Category, Buyer Code, Planner Code, Supplier, and Location.
3. Select a node detail on the tree.
The tree displays all the items that have the planning pull sequence for the specified plan.
The Replenishment Chain tab displays. This is a graphical view the replenishment chain for the specified plan selected on the tree. The table displays the calculated size and the number of cards to each pull sequence. See: .
See: Calculation Formula, *Oracle Master Scheduling/MRP and Oracle Supply Chain Planning User's Guide*
4. You can modify pull sequence information in this pane by Choosing Detail.
See: Pull Sequence Window, page 11-12
5. Select the Demand Graph tab.
The graphical view of the demand pattern of a simulation kanban item displays.
This can be used to monitor increases in demand.



6. Select the Pegging tab.

You can use the graphical pegging feature to analyze the impact of changing a supply or demand order at any level of your bill of material. Pegging traces supply information for an item to its corresponding end demand details.



Viewing Kanban Demands

The View Kanban Demand window displays the demand distribution for the specified plan. This window is accessed from the Plan Detail window, displaying the detail information pertaining to the chosen plan. You can also view this information in a graphical format in the Demand Graph tab of the Graphical Kanban Workbench.

View Kanban Demand (S1)					
Plan Name		00-OP1-PLN			
Forecast/Schedule		2000-OP1			
		2000 OP1 KANBANS PLAN			
		YEAR 2000 OP1 FCST			
Item	Subinventory	Locator	Demand Item	Demand Subinventory	Dema
CM23591	FloorStock		MC24713		
CM23591	FloorStock		MC24713		
CM23591	FloorStock		MC24713		
CM23591	FloorStock		MC24713		
CM23591	FloorStock		MC24713		
CM23591	FloorStock		MC24713		
CM23591	FloorStock		MC24713		
CM23591	FloorStock		MC24713		
CM23591	FloorStock		MC24713		
CM23591	FloorStock		MC24713		

Changing and Viewing Kanban Details

The Kanban Detail window enables you to view and update source, quantity and size, and planning information. You can also change or view pull sequence information in a graphical format on the Kanban Graphical Workbench.

To edit or view kanban detail information:

- 1. Navigate to the Kanban Detail window.

The Kanban Detail window displays, initially in the Source tab view.
- 2. Select a pull sequence.

You can change subinventory, location, description, and source type.
If the subinventory or item is under locator control, you must also enter a stock locator.

Kanban Details (S1)

Plan Name: **00-OP1-PLN** **2000 OP1 KANBANS PLAN**

Source Kanban Planning

	Item	Subinventory	Locator	Calculate	Size
☒	CM23591	FloorStock		Kanban Size	4000
☐	CM96717	FloorStock		Kanban Size	50000
☐					
☐					
☐					
☐					
☐					
☐					
☐					
☐					

Simulation Replan Update Production Select All Demand

4. In the Planning tabbed region, the Lead Time for this location displays. Optionally enter information in the Allocation Percent, Fixed Lot Multiplier, and Number Of Safety Stock Days fields.

Kanban Details (S1)

Plan Name: **00-OP1-PLN** **2000 OP1 KANBANS PLAN**

Source Kanban Planning

	Item	Subinventory	Locator	Lead Time	Fixed Lot Multiplier
☒	CM23591	FloorStock		5	
☐	CM96717	FloorStock		1	
☐					
☐					
☐					
☐					
☐					
☐					
☐					
☐					

Simulation Replan Update Production Select All Demand

5. Save your work.
6. There are four buttons on this window that enable you to do the following:

- *Simulation Replan* submits a request for kanban replanning.
- *Demand* displays the View Kanban Demand window, see: Viewing Kanban Demands, page 11-24
- *Update Production* updates your production information with this kanban.
- *Select All* selects all the details on this window.

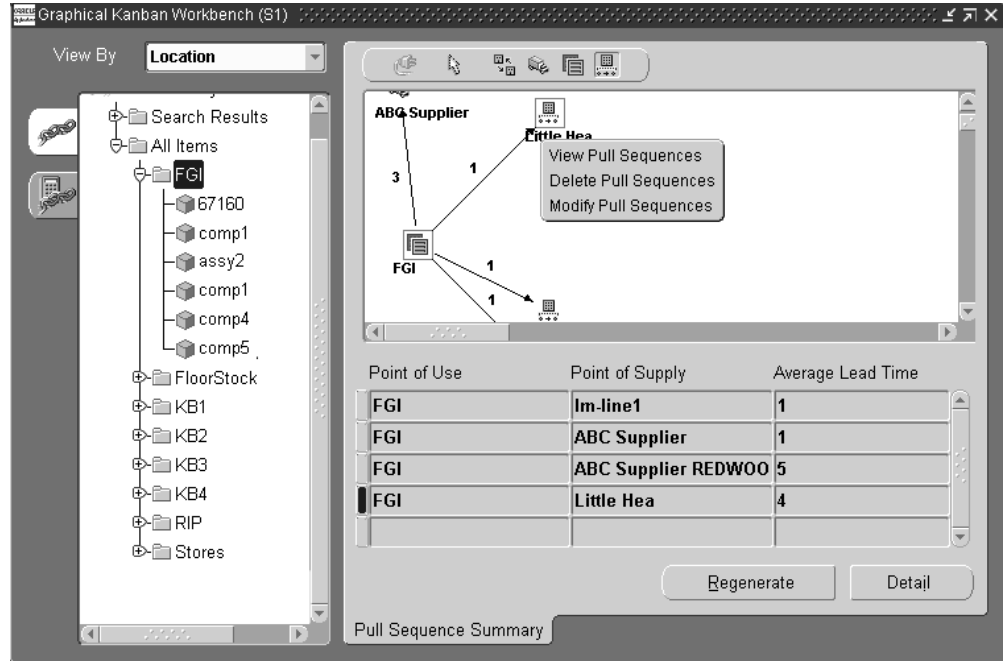
Viewing and Changing Summary Pull Sequences

You have the ability to view a summary of all pull sequences of the item numbers assigned to planners, suppliers, subinventory locations, or buyers. The Pull Sequence Summary window groups pull sequences and displays the Point of Use, Point of Supply, and Average Lead Time. This enables you to navigate to windows to view the number of pull sequences for the selected criteria, make changes or delete multiple pull sequences simultaneously.

The summary pull sequence view is available both in the Production and the Planning areas. In the Planning Area you can add and delete mass pull sequences. However, in the Production area, changes are not allowed to pull sequences that have released or canceled cards.

To view, delete, or modify summary pull sequence information:

1. Navigate to the Graphical Kanban Workbench, and select viewing criteria in the View by list box.
2. In the View By list box, select viewing criteria for the kanban data.
3. The pulls sequence summary view is available for are: Buyer Code, Planner Code, Supplier, Location.



4. Select a record on the hierarchal tree to display the pull sequences on the graphical pane.
5. Select a pull sequence on the graphical pane, right-click with your mouse to display the pop-up menu.

The number next to the pull sequence indicates how many records are in the summary view. The menu choices enable you to view, delete, or modify pull sequences.

6. Select View Pull Sequences to display all pull sequences the view you selected.
The Pull Sequences window displays.

Pull Sequences (S1)

Point Of Use **FGI** Point Of Supply **ABC Supplier**

Item	Point of Use Locator	Point of Supply Locator	Number Of Cards	Kanban Size
-comp1			3	3
-comp4			3	3
-comp5			3	3

OK Cancel

The Point of Use and Point of Supply Subinventory display. The multi-row region of the window displays all items used in the pull sequence, their associated point of use and point of supply subinventory locators, the number of kanban cards for each item, and the kanban card size.

- Choose OK to close this window and return to the Graphical Kanban Workbench.
- Select Delete Pull Sequences to display the Pull Sequences window.

The multi-row region of the window displays all items used in the pull sequence, their associated point of use and point of supply subinventory locators, the number of kanban cards for each item, and the kanban card size.

Pull Sequences (S1)

Point Of Use **FGI** Point Of Supply **ABC Supplier**

	Item	Point of Use Locator	Point of Supply Locator	Number Of Cards	Kanban Size
☐	comp1			3	3
☒	comp4			3	3
☒	comp5			3	3
☐					
☐					
☐					
☐					
☐					
☐					
☐					
☐					
☐					
☐					
☐					
☐					
☐					

Delete Cancel

9. Mark the check box next to the items you want to delete.
10. Choose Delete.
11. Select Modify Pull Sequences.

The Change Pull Sequences window displays. This window has six tabbed regions including Kanban Size, Number of Cards, Minimum Order, Lead Time, Lot Multiplier, and Safety Stock. The multi-row region of the window displays all items used in the pull sequence, the corresponding subinventory, and the calculation method.

Item	Subinventory	Locator	Calculate	Current	New
☒ comp1	FGI		Kanb...	3	3
☒ comp4	FGI		Kanb...	3	3
☒ comp5	FGI		Kanb...	3	3
☐					
☐					
☐					
☐					
☐					

Changes

☒ Percent (+/-)
 ☐ Value (+/-)
 ☐ Constant

12. For each item you are changing, ensure that the check box next to it is marked.
13. Select the tabbed region needed for the specific changes.
 Each tabbed region contains a Current and New field for the corresponding changed values. For example, if you are viewing the Kanban Size tab, the number in the Current field is the kanban size. Similarly, if you are in the Number of Cards tab, the value in the Current field is the number of kanban cards for this item in the pull sequence.
14. Enter the new value for the modification you are creating.
15. Optionally, in the Changes region, you can enter a global change for all records on the window by adding a value or percentage the current value. Mark the Percentage, Value, or Constant radio button.
16. Enter the value you want to add or subject to the current value.
 The value you enter adds or subtracts to the current value. If you mark Percentage, a percent is added. If you mark Value, that number is added. And if you mark Constant, the value is changed to the new number.
 For example, the value in the Current field for Safety Stock is 3 for the first item, and 2 for the second item. If you mark the Value radio button, and enter 3—when you apply your modification, the first item changes to 6 and the second item changes to 5.
17. Choose Apply to create your global change on this window.

18. Choose OK to save this record.

Launching a Kanban Plan

The kanban calculation program determines the optimal number of kanban cards or size according to the values entered in the Pull Sequence window. The calculation program uses the launch parameters entered including the plan name and demand date ranges. The kanban calculation program is launched from the Planning region. After you select a plan in the navigation tree, you can choose Launch Kanban Plan from the Tools or popup menu. You can also choose Launch Plan from the Plan Detail window on the workbench.

Additional Information: The Bills of Material Workday Calendar is used for kanban planning calculations.

Note: Kanban calculation does not support independent demand for components.

See: Using the Kanban Calculation Program, *Oracle Master Scheduling/MRP and Oracle Supply Chain Planning User's Guide*.

Updating Production

Once you have made adjustments to your replenishment chain in the planning region, you need to update production to reflect the changes in your production plans and kanban cards.

If you select a plan on the navigation tree, the Kanban Details window is accessed. You can review kanban data for records in the plan by Source, Kanban, and Planning detail views.

If you select an item on the navigation tree, the Update Production Kanbans window is accessed. You can review all items kanban data for a particular plan.

To update production with replenishment information:

1. Navigate to the Graphical Kanban Workbench and select the record you want to update on the tree,
2. Right click to display the popup menu. Choose Update Production.

You can also select Update Production from the:

- Tools menu

- Compare Kanban Plans window
- Kanban Details window

Select	Item	Point Of Use	Point Of Supply	Cards	Size
☒	CM23591	RIP		3	3
☒	CM23591	FloorStock		2	701
☐					
☐					
☐					
☐					
☐					
☐					
☐					
☐					

☒ Generate Print Cards

Deselect All Select All Update Production Cancel

3. If a plan was selected on the navigation tree, the Find Kanban Details window displays. Enter search criteria and choose Find to display the Kanban Details window.

You can review kanban data for records in the plan by Source, Kanban, and Planning detail views.

See: Changing and Viewing Kanban Details, page 11-25

4. If an item was selected on the navigation tree, the Update Production Kanbans window displays.

This window displays eligible records for the plan, and the applicable forecast or schedule.

5. Select the items you want to update on either the Kanban Details or Update Production Kanbans window.
6. If you are accessing the Update Production Kanbans window, you have the option to print new kanban cards after you update production. Enable the Generate Print Cards box for this print option. See: Printing Kanban Cards, *Oracle Inventory User's Guide*.
7. Choose Update Production.

If there are no conflicting pull sequences, a message displays with the number of kanbans to be updated.

8. Choose Yes to update production.

The Graphical Kanban Workbench displays.

To resolve pull sequence conflicts:

1. If there is a discrepancy with conflicting pull sequences, the Conflict Resolution window displays when you choose Update Production on the Kanban Details or Update Production Kanbans windows.

This window has two regions, New Pull Sequences and Deleted Pull Sequences. Both regions display fields for Part Number, Description, Point of Use, a Planning Area and Production Area. The Planning and Production areas display Point of Supply, Cards, and Size. Conflicts or mismatches with pull sequences result when:

- In the New Planning Sequence region, if the Point of Supply is different for the pull sequence in the Planning Area and Production Area.
- In the Deleted Pull Sequence region, if the corresponding planning sequence has an active kanban.

You can verify these conditions in the Replenishment tab detail page.

Conflict Resolution - Plan Vs Production

New Pull Sequences

Resolve Conflict	Part Number	Description	Point Of Use	Planning Area Point Of Supply	Cards	Size
☒	CM23591	Molded Plastic St	RIP		3	
☐						
☐						
☐						

Deleted Pull Sequences

Resolve Conflict	Part Number	Description	Point Of Use	Planning Area Point Of Supply	Cards	Size
☒	CM23591	Molded Plastic St	FloorStock		2	70
☐						
☐						
☐						

Proceed Cancel

2. Enable the Resolve Conflict box to proceed with updating production.
3. Choose Proceed.
A message displays with the number of kanbans to be updated.
4. Choose Yes to update production.
This Production pull sequence is deleted with its corresponding kanban cards, and the second Production pull sequence is updated.
The Graphical Kanban Workbench displays.

Purging Kanban Cards and Pull Sequences

Kanban cards are created based on the information of the corresponding pull sequence. Over time, unused cards and pull sequences can accumulate in your database and may affect the performance of the kanban system. You can purge unused pull sequences and kanban cards by submitting a concurrent program. The concurrent function purges according to the parameters specified, and creates a report.

To purge kanban cards and pull sequences:

1. Navigate to the Purge Pull Sequences window.
The Parameters window displays.
2. In the Type field, select whether you want a summary or detailed report generated.

Both formats of the report list the number of cards, the number of pull sequences deleted, and exception messages. The detailed version also lists locator, item, and the number of cards deleted per category.

3. In the Item From and To fields, you can select a range of pull sequence item numbers.
4. In the Subinventory From and To fields, you can select a range of point of use subinventories for the pull sequences you want to purge.

The screenshot shows a 'Parameters' dialog box with the following fields and values:

Field	Value
Type	Summary
Item From	
To	
Subinventory From	
To	
Source Type	Supplier
Line	
Supplier	
Organization	
Subinventory	
Delete Cards	Cancelled Cards Only

At the bottom of the dialog are four buttons: OK, Cancel, Clear, and Help.

5. In the Source Type field, select the source type of the pull sequence.
You choices are inter-organization, supplier, intra-organization, and production.
6. Optionally, you can add more search criteria for the source type selected.
This information includes:
 - Flow line for production pull sequences
 - Supplier for supplier pull sequences
 - Organization for inter-organization pull sequences
 - Subinventory for both inter-org and intra-org pull sequences
7. Select a kanban card delete option in the Delete Cards field.
Your choices are Canceled cards only to delete all cancelled kanban cards. Or Canceled and Active-New cards to delete any card with a status of active and a supply status of new.
8. Choose OK to display the Purge Pull Sequence request window.

9. Choose Submit to create this request.

All the pull sequences and kanban cards that satisfy the criteria are deleted and the total number of records purged is displayed on the report

Pull sequences are deleted if:

- The point of use subinventory and locator defined on the pull sequence does not exist on any bill of material or WIP Supply location in the organization item attributes (unreferenced pull sequences). Pull sequences that are unreferenced are printed as exceptions on the report.
- The point of use subinventory and locator defined on the pull sequence does not exist as the point of supply on any other pull sequence.
- Kanban cards do not exist for the pull sequence.

Reports and Processes

This chapter covers the following topics:

- Schedule Report
- Linearity Report

Schedule Report

The Schedule Report displays the scheduled production of all assemblies for a line within a given date range. The purpose of the schedule report is to provide personnel on the shop floor ready access to the production plan and sequence of assemblies.

The report includes line name, scheduled completion date, build sequence, assembly name, UOM, quantity, schedule group, schedule number, order number, order line, and order requested date.

Setting the Display BOM to Yes explodes the Bills of Material and shows the following for all components of each assembly: operation sequence, component number, component UOM, quantity per revision, ECO number, and whether the component is optional. The Display Optional Items Only parameter is linked with the Display BOM parameter. Setting the Display Optional Items Only to Yes results in the report displaying only optional components for the BOM.

Report Parameters

From Line

Select the single line or the first line in the range you wish to review. The list of values displays all active lines in the organization. This is an optional parameter.

To Line

Select the final line in the range to be displayed, if you are requesting data for a range of lines. The list of values displays all active lines in the organization. Tab through this field to request date for a single line. This is an optional parameter.

From Date

Choose the start date from which you wish to view line data, if you want a date other than the default value of the system date. The date must be greater than the system date. This is a required parameter.

To Date

Choose the end date through which you wish to view line data, if you want a date other than the default value of the system date. This is a required parameter.

Schedule Group

Select a specific schedule group to limit the data shown. The default is to show data for all schedule groups within the line(s) selected. This is an optional parameter.

Display BOM

Enter Yes to include the Bill of Material for the first level of all components for each assembly in the report. Phantom items are displayed and the Bill of Material will explode until it reaches a non-phantom item. The default is No.

Display Optional Items Only

This field is active only when Display BOM is set to Yes. Enter Yes to display only optional items in the BOM.

Linearity Report

The Linearity Report provides a performance monitoring tool that compares planned and actual production on a daily basis for the selected date range. The report displays the planned production rate, the actual production rate, the variance, and the linearity index for selected lines and, optionally, schedule groups and assemblies over a given time range.

Variance can be either positive or negative. The linearity index tracks the absolute value of variance as deviation. The formula for linearity index is:

$$\text{linearity index} = \{1 - [\text{total deviations}/\text{total planned rate}]\} * 100$$

The higher the linearity index, the more closely matched the actual production rate is to the planned rate.

The Linearity Report can be run with two different sort options and two levels of detail. The report options include: Linearity Report by Week (Summary), Linearity Report by Week (Detail), Linearity Report by Line (Summary), and Linearity Report by Line (Detail). Detail reports include information by assembly in addition to the information provided on Summary reports. The sort order for Week-based reports is: week of > line > schedule group. The sort order for Line-based reports is: line > schedule group > week of.

Summary reports include the following information: date from to date to, week of, line, schedule group, planned production, actual production, variance, linearity index, day #,

summary for line, summary for week, and weekly total.

Detail reports include the following information: date from to date to, week of, line, schedule group, assembly, planned production, actual production, variance, linearity index, day #, summary for schedule group, summary for line, summary for week, summary for schedule group, and weekly total.

Report Parameters

Report Option

Select Summary or Detail. If Detail is selected, the report will include information for individual assemblies. The default value is Summary.

Sort Option

Select whether the data should be sorted by Week or by Line. The default value is Week. The sort order for Week-based reports: week of > line > schedule group. The sort order for Line-based reports: line > schedule group > week of.

From Line

Select the single line or the first line in the range you wish to review. The list of values displays all active lines in the organization. This is an optional parameter.

To Line

Select the final line in the range to be displayed, if you are requesting data for a range of lines. The list of values displays all active lines in the organization. This is an optional parameter.

From Date

Choose the start date from which you wish to view line data, if you want a date other than the default value of the system date. This is a required parameter.

To Date

Choose the end date through which you wish to view line data, if you want a date other than the default value of the system date. This is a required parameter.

Schedule Group

Select a specific schedule group to limit the data shown. The default is to show data for all schedule groups within the line(s) selected. This is an optional parameter.

Windows and Navigation Paths

Windows and Navigator Paths

Although your system administrator may have customized your navigator, typical navigational paths include the following windows and pages.

Oracle Application	Description
FLM	Oracle Flow Manufacturing
FLMSE	Oracle Flow Manufacturing Sequencing and Execution

Window Name	Navigation Path
Add Item Routing	FLM > Product Sync > Graphical Line Designer > Pop-Up Menu > New
Alternates	FLM > Setup > Bills > Alternates
Attribute Controls	FLM > Setup > Items > Attribute Controls
Bill of Materials Parameters	FLM > Setup > Bills > Bill of Materials Parameters
Bills	FLM > Bills > Bills FLM > Reports > Bills
BOM Comparisons	FLM > Bills > BOM Comparisons

Window Name	Navigation Path
Category Sets	FLM > Setup > Items > Categories > Category Sets
Catalog Groups	FLM > Setup > Items > Catalog Groups
Category Accounts	FLM > Setup > Items > Categories > Category Accounts
Category Codes	FLM > Setup > Items > Categories > Category Codes
ChangeOrganization	FLM > Other > ChangeOrganization
Compare Kanban Plans	FLM > Kanbans > Graphical Kanban Workbench > Planning Region > Comparisons
Comparison	FLM > Reports > Comparison
Default Category Set	FLM > Setup > Items > Categories > Default Category Set
Delete Groups	FLM > Delete Groups
Delete Groups	FLM > Reports > Delete Groups
Delete Items	FLM > Products & Parts > Delete Items
Department	FLM > Lines > Department
Department Classes	FLM > Setup > Bills > Department Class
Disable Import of Configurator Model	FLM > Other > Configurator > Disable Import of Configurator Model
Documents	FLM > Products & Parts > Documents FLM > Bills > Documents
Event	FLM > Product Sync > Graphical Line Designer > Line Operation or Process tab > Toolbar > Event icon

Window Name	Navigation Path
Feeder Line Synchronization	FLM > Line Scheduling > Feeder Line Synchronization
Flow Routings	FLM > Product Synchronization > Flow Routings
Flow Workstation	FLM > Production > Flow Workstation
Generate Kanban Cards	FLM > Kanbans > Kanban Setup > Generate Kanban Cards
Graphical Kanban Workbench	FLM > Kanbans > Graphical Kanban Workbench
Graphical Line Designer	FLM > Product Sync > Graphical Line Designer
Import	FLM > Bills > Import
Indented Bills	FLM > Bills > Indented Bills
Item	FLM > Kanbans > Graphical Kanban Workbench > Pop-Up Menu > New
Item Types	FLM > Setup > Items > Item Types
Item Where Used	FLM > Bills > Item Where Used
Kanban Cards	FLM > Kanbans > Kanban Setup > Kanban Cards
Kanban Details	FLM > Kanbans > Graphical Kanban Workbench > Planning Region > Calculations
Launch	FLM > Kanbans > Kanban Planning > Launch
Line Operation	FLM > Product Sync > Graphical Line Designer > Line Operation tab > Toolbar > Line Operation icon
Lines	FLM > Lines > Lines

Window Name	Navigation Path
Location	FLM > Kanbans > Graphical Kanban Workbench > Toolbar > Source Type icon
Locations	FLM > Setup > Bills > Locations
Mass Changes	FLM > Bills > Mass Changes
Master Items	FLM > Products & Parts > Master Items
Mixed Model Map Workbench	FLM > Product Synchronization > Mixed Model Map Workbench
Names	FLM > Kanbans > Kanban Planning > Names
Network Connection	FLM > Graphical Line Designer > Line Operation or Process tab > Toolbar > Connection icon
Organization Items	FLM > Products & Parts > Organization Items
Populate Configurator Models	FLM > Other > Configurator > Populate Configurator Models
Print Kanban Cards	FLM > Kanbans > Kanban Setup > Print Kanban Cards
Process	FLM > Graphical Line Designer > Process tab > Toolbar > Process icon
Product Family	FLM > Products & Parts > Product Family
Product Family Members	FLM > Products & Parts > Product Family Members
Productivity	FLM > Reports > Productivity
Pull Sequence	FLM > Kanbans > Kanban Setup > Pull Sequence FLM > Graphical Kanban Workbench > Replenishment Chain tab > Detail

Window Name	Navigation Path
Refresh Configurator Models	FLM > Other > Configurator > Refresh Configurator Models
Related Routings	FLM > Graphical Line Designer > Pop-Up Menu > New > Add Item Routing > Related Routings
Requests	FLM > Other > Requests
Resource Groups	FLM > Setup > Bills > Resource Groups
Resources	FLM > Lines > Resources
Routing Details	FLM > Graphical Line Designer > Detail
Routing Revisions	FLM > Graphical Line Designer > Tools > Routing Revisions
Routings	FLM > Reports > Routings
Scheduling	FLM > Reports > Scheduling
Scheduling Rules	FLM > Setup > Scheduling Rules
Scheduling Workbench	FLM > Line Scheduling > Scheduling Workbench
Setup	FLM > Reports > Setup
Standard Events	FLM > Product Synchronization > Standard Events
Standard Line Processes	FLM > Product Synchronization > Standard Line Processes
Standard Processes	FLM > Product Synchronization > Standard Processes
Status Codes	FLM > Setup > Items > Status Codes

Window Name	Navigation Path
Switch to Primary	FLM > Product Sync > Graphical Line Designer > Tools > Switch to Primary
Template Assignment	FLM > Kanbans > Graphical Kanban Workbench > Assign > Add Items Criteria
Templates	FLM > Setup > Items > Templates
View Kanban Demand	FLM > > Kanbans > Graphical Kanban Workbench > Demand Graph tab
WIP Accounting Classes	FLM > Setup > Bills > WIP Accounting Classes
WIP Parameters	FLM > Setup > Bills > WIP Parameters
Work Order-less Completions	FLM > Production > Work Order-less Completions
Workbench	FLM > Kanbans > Kanban Planning > Workbench

Page Name	Navigation Path
Adjust and Submit Sequencing Task	FLMSE > Create Sequencing Task > Next > Enter Production Lines > Next FLMSE > Create Sequencing Task > Next > Enter Production Lines > Next > Filter Demands > Next
Assign Lot and Serial Number	FLMSE > Event > Update Components > Assign Lot and Serial
Assign Production Lines to Rule	FLMSE > Setup tab > Sequencing Rules > Assign Lines
Assign Rule to Production Lines	FLMSE > Setup tab > Production Lines > Update > Assign Rule

Page Name	Navigation Path
Complete Line Operation	FLMSE > Workstation tab > Start > Enter Schedule > Line Operation > Complete Line Operation FLMSE > Workstation tab > Start > Work Queue Criteria > Start > Work Queue > Complete Operation FLMSE > Workstation tab > Start > Work Queue Criteria > Start > Work Queue > Event > Complete Line Operation
Complete Schedule: Assembly Details	FLMSE > Work Queue > Complete Line Schedule FLMSE > Work Queue > Line Operation > Complete Line Schedule FLMSE > Work Queue > Event > Complete Line Schedule
Complete Schedule: Quality and Component Details	FLMSE > Complete Schedule: Assembly Details > Continue
Create Attributes	FLMSE > Setup tab > Sequencing Attributes > Create Attribute
Create Constraint for Rule: Enter Details	FLMSE > Setup tab > Sequencing Rules > Create Rule > Create Constraint > Continue
Create Constraint for Rule: Specify Priority and Attribute	FLMSE > Setup tab > Sequencing Rules > Create Rule > Create Constraint
Create Rule	FLMSE > Setup tab > Sequencing Rules > Create Rule
Enter Production Lines	FLMSE > Create Sequencing Task > Next
Enter Sequencing Task Parameters	FLMSE > Create Sequencing Task
Enter Schedule	FLMSE > Workstation tab > Start

Page Name	Navigation Path
Event	FLMSE > Workstation tab > Start > Enter Schedule > Get Event FLMSE > Workstation tab > Start > Work Queue Criteria > Start > Work Queue > Get Event
Filter Demands	FLMSE > Create Sequencing Task > Next > Enter Production Lines > Next
Line Operation	FLMSE > Workstation tab > Start > Enter Schedule > Get Schedule FLMSE > Workstation tab > Start > Work Queue Criteria > Start > Work Queue > Get Schedule
Move Demands	Adjust and Submit Sequencing Task > Move
Parameters	FLMSE > Workstation tab
Production Lines	FLMSE > Setup tab > Production Lines
Rule detail	FLMSE > Setup tab > Sequencing Rules > rule Name
Select Preferences	FLMSE > Setup tab > Workstation Preferences
Sequencing Attributes	FLMSE > Setup tab > Sequencing Attributes
Sequencing Rules	FLMSE > Setup tab > Sequencing Rules
Sequencing Tasks	FLMSE > Sequencing Tasks
Split Demands	Adjust and Submit Sequencing Task > Split
Task Detail	FLMSE > Sequencing Tasks > Sequencing Task Name Task Summary > View
Task Summary	FLMSE > Sequencing Tasks > Sequencing Task Name

Page Name	Navigation Path
Update Components	FLMSE > Event > Update Components
Update Constraint for Rule: Specify Priority and Attributes	FLMSE > Setup tab > Sequencing Rules > Create Rule > Update
Update Demands	Adjust and Submit Sequencing Task > Update
Update Line Task Parameters	FLMSE > Create Sequencing Task > Next > Enter Production Lines > Next > Adjust and Submit Sequencing Task > Update FLMSE > Create Sequencing Task > Next > Enter Production Lines > Next > Filter Demands > Next Adjust and Submit Sequencing Task > Update
Update Production Line	FLMSE > Setup tab > Production Lines > Update
Update Schedule	FLMSE > Line Operation > Update Schedule FLMSE > Event > Update Schedule
Work Queue Criteria	FLMSE > Workstation tab > Start
Work Queue	FLMSE > Workstation tab > Start > Work Queue Criteria > Start
Workstation Parameters	FLMSE > Setup tab > Workstation Preferences > Edit

Line Scheduling Calculation Examples

This appendix contains examples of the calculations used to create flow schedules in the Line Scheduling Workbench.

This appendix covers the following topics:

- Scheduling Calculations and Examples
- Roll Flow Schedules
- Tolerance Fences
- Schedule Completion Time
- Feeder Line Synchronization Calculations

Scheduling Calculations and Examples

Scheduling rules use the following routines for calculations:

- Sequence is determined by the criteria in the rule.
- The first flow schedule created is slated to the first available date in the scheduling horizon.
- This process of attaining the line rate for each day continues until the end of the scheduling horizon.

Note: For No Level Load and Level Daily Rate algorithms, unscheduled orders generated from planned orders have quantities initially changed by order modifiers, if they exist. Order modifiers (fixed order quantity, minimum order quantity and maximum order quantity) are used to limit lot size.

Note: If items have the rounding attribute set to yes, flow schedule

quantities are non-fractional, that is, quantities are rounded to whole numbers.

No Level Loading Algorithm

If you choose a scheduling rule that has no level loading, the level daily rate and the mix model algorithms are not used in scheduling the sales orders. The criteria defined in the scheduling rule is used to prioritize and sequence the sales orders. Orders are scheduled on the line with capacity constraint considerations. After using the criteria in the scheduling rule to sequence the sales orders, scheduling begins with the first sales order in the sequence. The scheduling process searches for available capacity beginning with the first date of the scheduling date range. If the entire order quantity cannot be scheduled on a date, the remaining quantity is scheduled on subsequent days where capacity is available.

Level Daily Rate Algorithm

If you choose a scheduling rule that uses level daily rate for leveling, the demand ratio of the item is used to calculate the quantity scheduled. The demand ratio of an item equals the total demand of the item divided by the grand total demand.

The quantity scheduled for an item on a particular day is determined by the available capacity of the line on the day and the demand ratio of the item. The calculations performed each day are as follows:

- The total demand for each item is calculated from all unscheduled sales orders.
- The grand total demand is calculated for all items.

The demand ratio is calculated. For each day in the scheduling horizon, the production quantity for each item is calculated by multiplying the available capacity on the line by the demand ratio of the item.

Scheduling using level daily rate attempts to schedule, for each day, each item at a rate in regard to the demand ratio of the item. The criteria selected is used to prioritize schedules; scheduling continues until capacity is reached or until all demand is scheduled.

Mixed Model Algorithm

In scheduling rules using a mixed model algorithm for leveling, sales orders are scheduled with the goal of achieving a balanced production mix across all items. In this case, flow schedules are created with unit quantity and scheduled in order to create the total demand of each item.

The key difference between the mixed model algorithm and the level daily rate is that

the calculation is at the unit level, whereas in level daily rate the calculation is at the daily level. There are two key steps in the process of establishing the production pattern:

- The total demand for each item is calculated from unscheduled sales orders.
- The production pattern is established using the total demand of each item and spreading out the demand to show the demand mix of all items.

The criteria defined in the scheduling rule prioritize and sequence the sales orders, and the orders are scheduled on the line based on the established production pattern.

During the scheduling process, the earliest available capacity is used to schedule the items on the line. Daily capacity constraints are used in the calculation.

Roll Flow Schedules

Rolling flow schedules forward adds under-completion quantities to the schedule, and subtracts overcompletions quantities from the schedule – past planned quantities are not reduced or increased. A report is generated from this process showing the effect of moving flow schedule quantities to future dates. Select your flow schedules by date ranges, lines, or categories. Choose Close All Past Due Schedules, Roll with Details, or Spread Evenly Without Details.

Tolerance Fences

Tolerance fences indicate how much flexible capacity the line has available. You enter a value in the Day in Advance field on the Tolerance Fence window in Oracle Work in Process. This value indicates the number of days of notice needed before increasing the line capacity by the tolerance percentage. The Days in Advance value should be greater than or equal to the fixed lead time value.

This feature is activated by checking the Flex Tolerance check box on the Line Scheduling Workbench.

Scheduling with tolerance fences begins after the last day indicated from the Days in Advance value. For example, if Days in Advance is 5, scheduling at the new capacity begins on the sixth day. Other factors in the calculations include:

- Capacity is added incrementally. The standard rate is used first to schedule all items. If the items cannot be scheduled by using the standard rate, the program considers the first tolerance fence and attempts to schedule within that capacity. If there is still not enough capacity, the second tolerance fence is considered, and so on.
- When line scheduling start and stop times are within the line definition start and stop times, the extra capacity provided by the tolerance fence is prorated based on the portion of the day being scheduled. For example-capacity is 100 for a line, the

tolerance fence percentage is 10 percent, and the standard line definition is from 7:00 to 15:00 hours (8 hours). If the scheduled start and stop times are 7:00 to 11:00 hours (4 hours), the flexible tolerance fence capacity is 5 percent.

4 divided by 8 = .5

.5 multiplied by 10 = 5

- Decimal calculation results for the new line capacity are rounded up.
- All schedules created within the tolerance fence, but over the standard running line capacity, are scheduled to complete at the stop time of the line.

Tolerance fences provide temporary increases to capacity above the normal operating rate to handle rises in orders. If you want to permanently increase the standard operating rate to correspond with a change in demand—update the line rate in the line design.

Note: The lead time of the item being scheduled is not part of the Days in Advance total. You must perform the scheduling operation sufficiently in advance to accommodate the lead time.

Schedule Completion Time

When the line scheduling start and stop times are outside the line definition start and stop times, the calculations are as follows:

- The first schedule completion time is the sum of the line start time and quantity divided by the line rate.
- The other schedule completion times are the sum of the previous completion times and quantity divided by the line rate.
- The last schedule completion time ends at the end time of the line.

When the line scheduling with start and stop times are within the line definition start and stop times:

- The first schedule completes at scheduling start time
- The other schedule completion times are the sum of the previous completion time and quantity divided by the line rate.
- The last schedule completion time ends at the scheduling end time

Start times = [completion time - (fixed lead time + (quantity-1) x variable lead time)]

Feeder Line Synchronization Calculations

The feeder line synchronization program calculates the date and time when the subassembly is delivered to the particular line operation. This date is then used as the completion date for the subassembly line. To calculate this time, the program uses the following formula:

$$S = P - \sum_{\text{last operation}}^{\text{consuming operation}} f_n$$

$$= P - (30 \text{ min} + 60 \text{ min}) = P - 90 \text{ minutes}$$

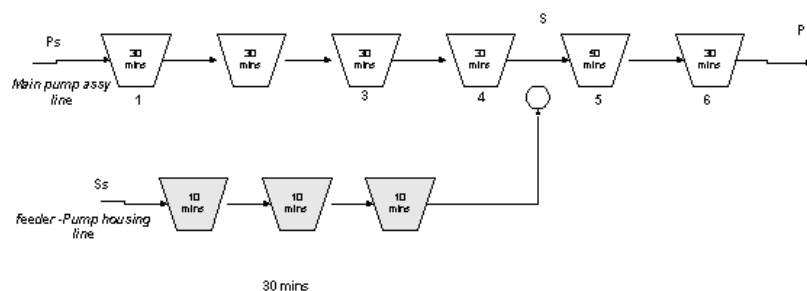
The values are defined as follows:

- S is the subassembly completion time.
- P is the parent item completion time.
- f_n is the operation time rounded up to the closest multiple of takt. Summation is done by following the routing network primary paths from the point of consumption to the consuming operation.

takt = parent line takt = 1/maximum hourly rate from parent line definition

Example: Feeder Line Synchronization

The following graphic illustrates an example of a flow line producing pumps. On the main line is the final pump assembly. The subassembly line produces pump housings and feeds the line at Operation 5.



The values are defined as follows:

- P is the completion time of the final pump assembly as scheduled

- S is when the subassembly, the pump housing, is consumed at Operation 5
- T is the takt time, 30 minutes

When you schedule the parent line, schedules are created with P as the completion time. The completion time of the subassembly is shown in the following calculation:

$$S = P - \sum_{\text{last operation}}^{\text{consuming operation}} f_n$$

$$Ss = S - (F + (Q-1) \times V)$$

Mixed Model Map Workbench Calculations

Mixed Model Map Workbench Calculations

Mixed Model Map Workbench uses the following calculations:

Calculate

The Calculate function on the Mixed Model Map Workbench performs calculations using parameters specified for a plan. Calculate options include:

- **No IPK**

Resource requirements are calculated using the IPK value as zero on the line operation.

IPK Needed = 0

Resource Needed per operation

= Weighted Resource Usage

divided by Operation TAKT for the operation

- **Based on One Resource**

The IPK needed on the line operation is calculated assuming one resource at each operation.

Resource Needed per operation = 1

$$IPKNeeded_{Operation} = \frac{\max_{Re\ source} (Weighted\ Re\ source\ Usage_{Re\ source, Operation}) - OperationTAKT_{Operation}}{\max_{Re\ source} (Weighted\ Re\ source\ Usage_{Re\ source, Operation})} * \frac{HoursPerDay_{Line}}{OperationTAKT_{Operation}}$$

- **Based on Assigned IPK**

Resources are calculated using the Assigned IPK at the line operation.

IPK Needed for the operation = IPK assigned or the operation

Resource Needed at each line operation

$$Re\ sourceNeeded_{Re\ source, Operation} = \frac{(HoursPerDay_{Line} / OperationTAKT_{Operation} - IPKAssigned_{Operation}) * Weighted\ Re\ source\ Usage_{Re\ source, Operation}}{HoursPerDay_{Line}}$$

- **Based on Assigned Resource**

The IPK Needed on the line operation is calculated using all the Assigned resources.

IPK Needed for the operation

$$IPKNeeded_{Operation} = \frac{\max_{Re\ source} \left(\frac{Weighted\ Re\ source\ Usage_{Re\ source, Operation}}{Re\ sourceAssigned_{Re\ source, Operation}} \right) - OperationTAKT_{Operation}}{\max_{Re\ source} \left(\frac{Weighted\ Re\ source\ Usage_{Re\ source, Operation}}{Re\ sourceAssigned_{Re\ source, Operation}} \right)} * \frac{HoursPerDay_{Line}}{OperationTAKT_{Operation}}$$

Resource needed calculation:

$$Re\ sourceNeeded_{Re\ source, Operation} = \frac{(HoursPerDay_{Line} / OperationTAKT_{Operation} - IPKNeeded_{Operation}) * Weighted\ Re\ source\ Usage_{Re\ source, Operation}}{HoursPerDay_{Line}}$$

Average Daily Demand Calculation

Total Demand multiplied by (1 + Boost %

Divided by Demand Days

Line TAKT

Rate the line produces one item to fulfill demands:

Working Hours Per Day (per line)

divided by Average Daily Demand for the item.

Operation TAKT

Rate the operation produces one item to fulfill operation demands. A single line can be used to manufacture multiple items, different items can have different routings, and some operations might be unique for a particular item.

Operation TAKT

= Working Hours per day for the line

divided by the item Process Volume

Process Volume

The number of items an operation produces in manufacturing day. A single operation can be used to manufacture multiple items, different items can use different routings, and some operations might be unique for a particular item. Process Volume is not effected by Assigned Resources and IPK.

Process volume

= Average Daily Demand for the item

multiplied by Net Planning Percent for the operation

divided by Reverse Cumulative Yield for the operation

Machine, Labor, Elapsed, Weighted Time

This includes:

- Total Machine Resources usage time in an operation
- Total Labor Resources usage time in an operation
- Total Machine and Labor Resources usage time in an operation
- Weighted Time is the weighted average usage time across all items produced in a line.

Machine, Labor, Elapsed, and Weighted Time are not effected by Assigned Resources, only usages are considered.

$$WeightedTime_{Operation} = \frac{\sum_{Item} Time_{Item} * ProcessVolume_{Item,Operation}}{\sum_{Item} ProcessVolume_{Item,Operation}}$$

$$Weighted Resource Usage_{Resource,Operation} = \frac{\sum_{Item} ResourceUsage_{Resource,Item,Operation} * ProcessVolume_{Item,Operation}}{\sum_{Item} ProcessVolume_{Item,Operation}}$$

Usage, IPK, Assigned Example

For example:

Usage is 1

Assigned is 0.5, and the calculation is based on Assigned Resources

IPK is used to fulfill the required resource

The outputs will then be used by the next operation to keep the line flowing.

If the IPK is 20, 20 extra components are needed at the end of the resource per day, to keep the line flowing

See: In Process Kanban Assignment, page 5-18

Takt Time for Assigned

Takt time for assigned is the time per unit that each operation will achieve given the number of resources assigned. (Also called the Operational Cycle Time.)

Takt Time for Assigned is calculated using the following formula:

Machine Weighted Time / Machines Assigned

If machines assigned is equal to 0, then takt time for assigned is 0.

Operation TAKT As Assigned

Example 1:

Operation TAKT is 15 minutes

Usage is 1, Assigned is 2

Resource TAKT is 7.5 minutes

The work can be completed 2 times faster, that is—15 divided by 2 = 7.5 minutes

Example 2:

If Operation TAKT is 15 minutes

Usage is 2, Assigned is 3

Resource TAKT is 10 minutes

The work can be finished $3/2$ times faster, that is—15 divided by $(3/2) = 10$ minutes

Line TAKT As Assigned

The ratio of Line TAKT As Assigned, and Line TAKT, is the maximum ratio of the Operation TAKT As Assigned and Operation TAKT in the line.

- When Operation TAKT As Assigned is greater than Operation TAKT, the current setup results in slower time to complete the operation.
- When Operation TAKT As Assigned is less than Operation TAKT, the current setup results in faster time to complete operation.

Color Representation on the Mixed Model Map Workbench

Lines

Line TAKT As Assigned

- Red—Resource overcapacity or IPK overcapacity is greater than 0
- Yellow—Line TAKT As Assigned is greater than Line TAKT

Operations

Operation, Operation TAKT As Assigned

- Red—Resource overcapacity or IPK overcapacity is greater than 0
- Yellow—Operation TAKT As Assigned is greater than Operation TAKT

Operation Detail

Resource, Usage, Needed, IPK Needed

- Red—IPK Needed is greater than IPK Assigned
- Yellow—IPK Needed is greater than 0, and Maximum Resource Needed or Assigned is less than Usage

Resources

Resource, Needed, Relocation Needed

- Red—Resource Needed is greater than Resource Assigned
- Yellow—Relocation Needed = Yes

Resource Detail

Line, Operation, Resource Needed, IPK Needed

- Red—Resource Needed is greater than Resource Assigned
- Red—IPK Needed is greater than IPK Assigned

Resource Assignments

Department, Unit, Total Unit

- Red—Unit is less than Assigned
- Red—Total Units is greater than Total Assigned

Percentage Needed

The percentage difference between Assigned and Needed based on the Needed value:

Assigned subtracted by Needed divided by Needed multiplied by 100

- Green—resource is overassigned
- Red—resource is underassigned

Percentage Assigned

The percentage difference between Needed and Assigned based on the Assigned value:

Needed subtracted by Assigned divided by Assigned multiplied by 100

- Green—resource is overassigned
- Red—resource is underassigned

Percentage number 999 signifies a value is zero or null.

Absolute Value

Needed (blue) and Assigned (green) values are displayed next to each other for comparison.

Oracle Flow Manufacturing Client Extensions

This appendix provides information about the Flow Manufacturing client extensions.

This appendix covers the following topics:

- Custom Kanban Quantity Calculation
- Custom Schedule
- Flow Sequencing Customization and Extension
- Flow Execution Workstation Customization and Extension

Custom Kanban Quantity Calculation

Calculates kanban size and quantity.

Functional Overview

Package Name: MRP_PUB_KANBAN_QTY_CAL

Procedure Name: CALCULATE_KANBAN_QUANTITY

Custom Kanban Quantity Calculation enables the extension of Kanban Quantity Calculation to suit the need of different business requirements. The Kanban planning engine will use the resulting kanban quantity and size returned by this client extension. The table below provides the specifications:

Parameter	Usage	Type	Required	Derived	Optional
p_version_number	IN	NUMBER	x	-	-

Parameter	Usage	Type	Required	Derived	Optional
p_average_demand	IN	NUMBER	x	-	-
p_minimum_order_quantity	IN	NUMBER	x	-	-
p_fixed_lot_multiplier	IN	NUMBER	x	-	-
p_safety_stock_days	IN	NUMBER	x	-	-
p_replenishment_lead_time	IN	NUMBER	x	-	-
p_kanban_flag	IN	NUMBER	x	-	-
p_kanban_size	IN/OUT	NUMBER	-	-	-
p_kanban_number	IN/OUT	NUMBER	-	-	-
p_return_status	OUT	VARCHAR2	-	-	-

Custom Schedule

The Custom Schedule enables you to extend the line scheduling algorithms that are seeded with the product.

Functional Overview

Package Name: MRP_CUSTOM_LINE_SCHEDULE

Procedure Name: CUSTOM_SCHEDULE

The Custom Schedule enables the extension of the Line Scheduling algorithm to suit the need of different business requirements. The following table provide specifications:

Parameter	Usage	Type	Required	Derived	Optional
p_api_version_number	IN	NUMBER	x	-	-
p_rule_id	IN	NUMBER	x	-	-
p_line_id	IN	NUMBER	x	-	-
p_org_id	IN	NUMBER	x	-	-
p_flex_tolerance	IN	NUMBER	x	-	-
p_scheduling_start_date	IN	DATE	x	-	-
p_scheduling_end_date	IN	DATE	x	-	-
x_return_status	OUT	VARCHAR2	-	-	-
x_msg_count	OUT	NUMBER	-	-	-
x_msg_data	OUT	VARCHAR2	-	-	-

Procedure

Selected flow schedules are converted to flow schedules before calling the scheduling and sequencing procedure. All rows that are to be sequenced or scheduled in this call are contained in the table wip_flow_schedules. All of these rows have the column Request_id set to USERENV('SESSIONID'). The procedure should operate on these rows, sequence and schedule them based on user defined algorithms, and update the rows in the flow schedule table.

New rows that are created should have the following columns populated.

- wip_entity_id
- Scheduled_flag
- Inventory_item_id

- Organization_id
- Line_id
- Last_update_date
- Last_update_by
- Creation_date
- Created_by
- Class_code
- Planned_quantity
- Quantity_completed
- Scheduled_start_date
- Scheduled_completion_date
- Status
- Schedule_number

The calling functions set a save point, before a call to the Custom Schedule. In case of a failure message from this procedure, the program will rollback to the save point and return an error message to the calling routines.

Package Name: MRP_CUSTOM_LINE_SCHEDULE

Procedure Name: IS_VALID_DEMAND

This procedure is being called from Line Scheduling Workbench before inserting the rows into wip_flow_schedule. It allows the user to customize the filtering of the demand to be inserted into wip_flow_schedules. This is called before the scheduling engine is invoked.

Parameter	Usage	Type	Required	Derived	Optional
p_api_version_number	IN	NUMBER	x	-	-
p_rule_id	IN	NUMBER	x	-	-
p_line_id	IN	NUMBER	x	-	-

Parameter	Usage	Type	Required	Derived	Optional
p_org_id	IN	NUMBER	x	-	-
p_demand_type	IN	NUMBER	x	-	-
p_demand_id	IN	NUMBER	x	-	-
p_valid_demand	OUT	BOOLEAN	-	-	-
x_return_status	OUT	VARCHAR2	-	-	-
x_msg_count	OUT	NUMBER	-	-	-
x_msg_data	OUT	VARCHAR2	-	-	-

- p_demand_type**
 To identify the demand type. 2 for sales order, 100 for planned order.
- p_demand_id**
 The identifier for the demand.
 For sales order, p_demand_id = sales order line id in oe_order_lines_all.
 For planned order, p_demand_id = transaction_id in mrp_recommendations.
- p_valid_demand**
 The demand will be inserted into wip_flow_schedules, if the p_valid_demand is TRUE, otherwise it will be ignored.

Flow Sequencing Customization and Extension

The Flow Sequencing Customization and Extension is used by Flow Sequencing.

Functional Overview

Package Name: FLM_SEQ_CUSTOM

Procedure Name: GET_ATTRIBUTE_VALUE

The Flow Sequencing Customization and Extension enables the extension of all seeded

attribute types to suit the need of different business requirements. The Sequencing engine uses the resulted attribute value returned by this client extension. There are three different types of customized attributes: Item, Sales Order and Planned Order.

Parameter	Usage	Type	Required	Derived	Optional
p_api_version_number	IN	NUMBER	Y	-	-
p_org_id	IN	NUMBER	Y	-	-
p_id	IN	NUMBER	Y	-	-
p_attribute_id	IN	NUMBER	Y	-	-
p_attribute_type	IN	NUMBER	Y	-	-
p_other_id	IN	NUMBER	-	-	Y
p_other_name	IN	VARCHAR2	-	-	Y
x_value_number	OUT	NUMBER	-	-	-
x_value_name	OUT	VARCHAR2	-	-	-
x_return_status	OUT	VARCHAR2	-	-	-
x_msg_count	OUT	NUMBER	-	-	-
x_msg_data	OUT	VARCHAR2	-	-	-

- **p_id**

Values passed depending on the type of the customized attributes:

- Item attribute, p_id = inventory_id from MTL_SYSTEM_ITEMS table
- Sales Order attribute, p_id = line_id from OE_ORDER_LINES_ALL table
- Planned Order attribute, p_id = transaction_id from

MRP_RECOMMENDATIONS table

- **p_attribute_type**

Attribute Type. Value passed depending on the type of the customized attributes:

- Item attribute, p_attribute_type = 11
- Sales Order attribute, p_attribute_type = 12
- Planned Order attribute, p_attribute_type = 13

- **x_value_num**

Return attribute value of type number or date. The Julian number is returned for date type.

Functional Overview

Package Name: FLM_SEQ_CUSTOM

Procedure Name: NAME: POST_PROCESS_DEMAND

This procedure provides post-process demands for a task. The demands for a specific sequencing run are stored in FLM_SEQ_TASK_DEMANDS table identified by seq_task_id. The sequencing engine uses the demands stored to create flow schedules. To modify the demands, the API needs to either modify the quantity values (to increase or decrease demand), or delete the row (to remove the demand).

Parameter	Usage	Type	Required	Derived	Optional
p_api_version_number	IN	NUMBER	Y	-	-
p_seq_task_id	IN	NUMBER	Y	-	-
x_return_status	OUT	VARCHAR2	-	-	-
x_msg_count	OUT	NUMBER	-	-	-
x_msg_data	OUT	VARCHAR2	-	-	-

Flow Execution Workstation Customization and Extension

The Flow Execution Workstation Customization and Extension is used by the Flow Execution Workstation.

Functional Overview

Package Name: FLM_EXECUTION_UTIL

Procedure Name: GET_ATTRIBUTES

The procedure retrieves the information to display on the custom Property region of the Flow Execution Workstation, providing additional information for completing the work at a particular line operation or event. As an input parameter, the procedure takes wip_entity_id, operation sequence id, and operation type (line operation or event). This indicates the operation displaying the information. It returns a table of name value pairs to be rendered on custom Property region of Flow Execution Workstation.

Parameter	Usage	Type	Required	Derived	Optional
p_api_version_number	IN	NUMBER	Y	-	-
p_wip_entity_id	IN	NUMBER	Y	-	-
p_op_seq_id	IN	NUMBER	Y	-	-
p_op_type	IN	NUMBER	Y	-	-
p_cust_attr_tab	OUT	FLM_CUST_ATTRIBUTE_TBL	-	-	-
x_return_status	OUT	VARCHAR2	-	-	-
x_msg_count	OUT	NUMBER	-	-	-
x_msg_data	OUT	VARCHAR2	-	-	-

- **p_op_seq_id**
Operation Sequence ID of this operation/event

- **p_op_type**
Operation Type: 1=Event, 2=Process, 3=Line Operation
- **p_cust_attrb_tab**
PL/SQL table to keep attributes. The table is based on record type
FLM_CUST_ATTRIBUTE
- **x_msg_data**
Standard Oracle API output parameter

Parameter	Type
attribute_name	VARCHAR2(240)
attribute_value	VARCHAR2(2000)

Glossary

allocation percent

See *kanban allocation percent*.

assemble-to-order (ATO)

An environment where you open a final assembly order to assemble items per customers order. Assemble-to-order is also an item attribute that you can apply to standard, model, and option class items.

assembly scrap

A process which allows you to scrap both scheduled and unscheduled flow assemblies from any flow operation. Components, their associated costs, and all labor and machine resources used at all events prior to the scrap line operation are automatically backflushed.

ATO

See **assemble-to-order**.

basis

Indicates whether the component quantity is required per item or per lot.

cumulative yield

Product of the yields at each operation, process, or event on a flow line.

dual card kanban

A demand pull signal that uses a *move and produce* communication method. Generally, move cards are collected and when the produce lot size is reached, the produce card is used to create the replenishment. This procedure is generally used when a minimum order quantity is required as a result of long set up times or economic order cost.

efficiency

Productivity measure that focuses on actual performance against a standard. Expressed

in a percentage figure, it is calculated by dividing actual resource time charged to a task by the standard resource requirements for the same task.

elapsed time

The clock time between start and completion.

event

An event is an identifiable point in time among a set of related activities. In Flow Manufacturing, events are the lowest level of activities in a flow routing. Resources are assigned to events. Events can be grouped into processes and operations.

feeder line

A production line designed to feed subassemblies to a line producing higher level assemblies.

feeder line synchronization

A concurrent process enabling you to synchronize subassembly flow schedule sequences with the parent assembly line flow schedule.

flexible tolerance fences

Used to represent the flexibility of supplier capacity. Shown as the percent increase or decrease over daily production rate available, from a given supplier, for a set amount of time. Flexible tolerance fences provide temporary increases to capacity to handle increases in orders. The line rate is adjusted by the tolerance fence percentage before scheduling rules are applied. Tolerance fences are set up when creating line definition.

fixed component usage

Component usage quantity that is fixed irrespective of the build quantity of the assembly item.

flow line

The physical location where you manufacture a flow assembly, usually associated with a routing. You can build many different assemblies on the same line at the same time. Also known as assembly line.

Flow Manufacturing

Manufacturing philosophy utilizing production lines and schedules instead of work orders to drive production. Mixed models are grouped into families and produced on lines, balanced, to the TAKT time.

flow routing

A sequence of manufacturing events that you perform to manufacture an assembly. A

routing consists of an item, a series of events, processes and/or operations, a operation sequences, operation effective dates, and a flow routing network.

flow schedule

Represents the volume and mix of products to be produced. Scheduling can be done based on customer orders, with an objective of matching the customer orders as closely as possible while establishing an achievable pace and consistent flow of products. Schedules do not produce work orders.

just-in-time

A system for producing and delivering the right items, at the right time, in the right amounts, at the right place. The key elements are flow, pull, standard work and TAKT time.

IPK (In Process Kanban)

Designated as a work in process holding area for work in between operations. If the operation time for one operation is shorter in length than the next operation—the IPK is a buffer area where the item is designated until the next operation is available. The calculation is:

$$(\text{elapsed weighted time} - \text{line takt time}) * \text{hours/day} * \text{time conversion factor} \div \text{elapsed weighted time} * \text{line TAKT time}$$

kanban

A method of production that uses standard containers or lot sizes with a single card attached to each. It is a pull system, work centers signal with a card that they wish to withdraw parts from feeding operations or suppliers. The Japanese word *kanban*, loosely translated, means *card*, *billboard*, or *sign*.

kanban allocation percent

Percent of independent demand for the kanban item that is to be supplied from a given pull sequence.

kanban card

The visual replenishment signal. It corresponds to a kanban bin. Replenishable kanban cards are based on pull sequences; non-replenishable cards can be created manually.

kanban chain

A series of kanban pull sequences that model the replenishment network on the shop floor.

kanban location

The designated location where a kanban bin is stored, and where the replenishment is delivered.

kanban size

The number of items in each kanban bin.

Lean manufacturing

Refers to the elimination of wastes in a process—these are tasks that do not add value. Lean production aims to eliminate waste in every area of production including customer relations, product design, supplier networks, and production floor management.

line balancing

Organizing work on the production line so that resources can be synchronized to daily demand.

line operations

Regrouping of events on a flow routing to achieve approximate TAKT time. Line operations are line specific and are derived and defined during line balancing.

mixed model production

Enables a mix of products to be built on one production line. Mixed models are grouped into families and built on lines balanced to the TAKT time.

mixed model map

Tool used to design balanced lines. The projected volume and mix of demand for a group of products is used to calculate weighted average work content times. These averages are compared to TAKT time to regroup events into balanced operations, and reallocate resources.

net planning percent

Percent of product that passes through a process or line operation. It is the sum of the product of the network percentages at each operation along each path.

non-replenishable kanban

Used to replenish a kanban location for one instance. This card is used typically for custom products, onetime customer orders, or sudden spikes in demand.

operation time

The total of setup and run time for a specific task. In Flow Manufacturing, operation time includes the machine time, labor time, and elapsed time for events, processes, and line operations on your flow routing.

operation yield

The percent of material that passes through an operation, process or event on a flow line without being scrapped.

operational cycle time

See *TAKT time*.

operational method sheet (OMS)

Operational method sheets (OMS) are documents that describe the operation to be performed for an assembly. The information in the OMS often includes graphical representation of the process, material needed, and detailed work instructions. You can use attachments in Oracle Flow Manufacturing to attach OMS's to line operations in the flow routing.

phantom

An item or component which is never stocked and is used as a part in building the final item.

point of use (POU)

Inventory located at a specific operation on a flow line where it will be used. Material is pulled from these locations using a kanban signal. These locations are supplied from either raw material stores or directly from the supplier.

process

A planned series of actions or operations (such as. mechanical, electrical, chemical, inspection, or test) that advances a material or procedure from one stage of completion to another. In Flow Manufacturing, processes are generic activities on a flow routing that often consist of several events performed in a specific sequence. They are specific to a line, and are often defined during the analysis on a flow line.

process volume

The number of items an operation has added value per day. Calculated process volumes represent the projected number of work units by process/line operation. These numbers are influenced by the demand mix, routing resource requirements, routing networks (alternate paths, rework loop percentages), net planning percentages, and yields.

product family

A group of products with similar characteristics, often used in production planning. Flow product families often have similar product synchronization.

product synchronization

Process of defining events, processes, and operations and assigning them to a flow routing in a specific sequence they are performed.

production lines

Manufacturing work areas where you manufacture families of products.

pull sequence

A body of information that defines the kanban location, source information, and planning parameters required to calculate the kanban size for a given kanban bin. Replenishment chains are created by multiple pull sequences.

replenishable kanban

A replenishable kanban card cycles through the system until it is put on temporary hold, or completely removed from the replenishment cycle.

reverse cumulative yield

Product of the yields at each operation, process, or event on a flow line—starting with the last operation, process, or event.

roll flow schedules

A function where you can move the flow schedules you did not complete into the next available day, or use overcompletion quantities for future schedules.

routing network

A process-map of your processes and operations where you specify the primary path, alternate paths, feeder lines and rework loops within your flow line. Defines the flow of work from one line operation to the next.

scheduling rule

Basic rules are used in a scheduling system. Scheduling rules specify the amount of calendar time to allow for a move, queue, and load calculation.

TAKT

A German word for beat. It is a statement of hours divided by demand. The flow line and every process/line operation is measured against required TAKT:

- Line TAKT is the rate the line produces one item to fulfill the demands ($\text{Hours per Day} / \text{Daily Demand}$)
- Operation TAKT is the rate the operation produces one item to fulfill the operation demands ($\text{Hours per Day} / \text{Process Volume}$)

total product cycle time (TPCT)

The total time along the longest path of your flow routing. Calculated by taking the sum of the elapsed times, along the longest primary path on the routing network.

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